

APPENDIX A UFC 1-200-01 GLOSSARY

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ACRONYMS

AA&E	Arms, Ammunition, and Explosives
ABA	Architectural Barriers Act
AE	Ammunition and Explosives
AEC	Architect / Engineer / Construction
AFCEC	Air Force Civil Engineer Center
AFMAN	Air Force Manual
AHJ	Authority Having Jurisdiction
BO	Building Official
BIA	Bilateral Infrastructure Agreement
CCR	Criteria Change Request
CECW-EC	Corps of Engineers Civil Works, Chief of Engineering and Construction Division
CP	Cathodic Protection
CPC	Corrosion Prevention & Control
CFR	Code of Federal Regulations
CTR	Component Technical Representative
DAFMAN	Department of the Air Force Manual
DCS	Deputy Chief of Staff
DoDD	DoD Directive
DoDI	DoD Instruction
DoDM	DoD Manual
DUSD (I&E)	Deputy Under Secretary of Defense for Installations and Environment
DWG	Discipline Working Group

EI&E	Energy, Installations, and Environment
ESC	Environmental Severity Classification
ESEP	Engineering Senior Executive Panel
ESQD	Explosives Safety Quantity Distance
E.O.	Executive Order
FASS	Facilities Systems Safety
FC	Facilities Criteria
HQ	Headquarters
HQMC	Headquarters, U.S. Marine Corps
HQUSACE	Headquarters, U.S. Army Corps of Engineers
HNFA	Host Nation Funded Construction Agreements
HVAC	Heating, Ventilation, and Air Conditioning
IBC	International Building Code®
ICC	International Code Council
ICCET	ISO Corrosivity Category Estimation Tool
IEBC	International Existing Building Code®
IECC	International Energy Conservation Code®
IE&E	Installations, Energy, and Environment
IE&L	Installations, Environment, and Logistics
IFC	International Fire Code®
IFGC	International Fuel Gas Code®
IgCC	International Green Construction Code®
IMC	International Mechanical Code®
IPC	International Plumbing Code®
IPMC	International Property Maintenance Code®

IRC	International Residential Code®
LPG	Liquified Petroleum Gas
MCM	Metal Composite Materials
MDF	Medium Density Fiberboard
NAF	Naval Air Facility
NAVFAC	Naval Facilities Engineering Systems Command
NDAA	National Defense Authorization Act
NEC	National Electrical Code®
O&M	Operations and Maintenance
OEBDG	Overseas Environmental Baseline Guidance Document (DoDM 4715.05)
OSHA	Occupational Safety and Health Administration
OASD	Office of the Assistant Secretary of Defense
OUSD	Office of the Under Secretary of Defense
P.L.	Public Law
RFP	Request for Proposal
RMF	Risk Management Framework
SAP	Special Access Program
SCI	Sensitive Compartmented Information
SOFA	Status of Forces Agreements
SOH	Safety and Occupational Health
TOC	Total Ownership Costs
USACE	U.S. Army Corps of Engineers
USC	United States Code
UFC	Unified Facilities Criteria
UFGS	Unified Facilities Guide Specifications

USD (AT&L)

Under Secretary of Defense for Acquisition, Technology, and Logistics

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DEFINITION OF TERMS

Critical Asset: A specific facility, system, infrastructure, or resource whose failure or degradation would have a serious, immediate, or cascading impact on a Mission Essential Function (MEF) or Primary Mission Essential Function (PMEF).

Discipline Working Group: Representatives from the DoD components responsible for the unification and maintenance of criteria documents. (MIL-STD-3007)

Engineering Senior Executive Panel: Panel established by the DoD Installations Policy Board to implement the UFC and UFGS system for DoD. The ESEP consists of a representative from the Office of the Secretary of Defense and the three Service Chiefs of Engineering. (MIL-STD-3007)

Facilities Criteria: A criteria document that is not adopted by all services and will be used only by services indicated in the document

Mission Critical Facility: One whose loss, degradation, or disruption would immediately and directly compromise a DoD mission, resulting in failure of a mission, loss of life, severe operational degradation, loss of significant combat capability, and/or loss of required command and control.

Mission Essential Facility: One that is necessary for successful mission execution, but its temporary loss does not immediately result in mission failure. Its degradation creates major impediments, reduces operational efficiency, increases mission risk, and/or requires rapid restoration.

Mission Support Facility: One that provides indirect support to mission execution. Its loss degrades quality of life, slows mission execution, causes administrative or support delays, but does not directly cause mission failure.

Technical Representative: Author of a particular criteria document or the working-level representative from another participating organization for a particular document. (MIL-STD-3007)

APPENDIX B UFC 1-200-01 REFERENCES

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS

<https://www.ashrae.org>

ANSI/ASHRAE/IESNA Standard 90.1, *Energy Standard for Buildings Except Low-Rise Residential Buildings* (Refer to chapter 2 of UFC 1-200-02, for applicable publication date)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS

<https://www.asme.org>

AMERICAN SOCIETY OF SAFETY PROFESSIONALS

<https://www.assp.org>

ANSI/ASSP Z590.3, *Prevention through Design: Guidelines for Addressing Occupational Hazards and Risks in Design and Redesign Processes*

AMERICAN WELDING SOCIETY

<https://www.aws.org/>

AWS D1.1/D1.1M 2020

Structural Welding Code

CODE OF FEDERAL REGULATIONS

<https://www.osha.gov/laws-regs/regulations/standardnumber/1910>

29 CFR 1910, *Occupational Safety and Health Standards*

INTERNATIONAL CODE COUNCIL

<https://www.iccsafe.org>

IBC, *International Building Code*®, 2024

IEBC, *International Existing Building Code*®, 2024

IECC, *International Energy Conservation Code*® (Refer to Chapter 2 of UFC 1-200-02 for applicable edition)

IgCC, *International Green Construction* , (Refer to UFC 1-200-02 for applicable edition)

IMC, *International Mechanical Code*®, 2024

IPC, *International Plumbing Code*®, 2024

IRC, *International Residential Code*®, 2024

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION

<https://www.iso.org/>

ISO 9223, *Corrosion of Metals and Alloys – Corrosivity of Atmospheres – Classification, Determination and Estimation*

NATIONAL FIRE PROTECTION ASSOCIATION

<https://www.nfpa.org>

NFPA 1, *Fire Code*

NFPA 54 (ANSI Z223.1), *National Fuel Gas Code*

NFPA 58, *Liquefied Petroleum Gas Code*

NFPA 70, *National Electrical Code*®

NFPA 101, *Life Safety Code*®

NFPA 285, *Standard Fire Test Method for Evaluation of Fire Propagation Characteristics of Exterior Wall Assemblies Containing Combustible Components*

UNIFIED FACILITIES CRITERIA

<https://www.wbdg.org/dod/ufc>

UFC 1-200-02, *High Performance and Sustainable Building Requirements*

UFC 1-201-01, *Non-Permanent DoD Facilities in Support of Military Operations*

UFC 1-201-02, *Assessment of Existing Facilities for Use in Military Operations*

UFC 1-300-02, *Unified Facilities Guide Specifications (UFGS) Format Standard*

FC 1-300-09N, *Navy and Marine Corps Design Procedures*

UFC 3-101-01, *Architecture*

UFC 3-110-03, *Roofing*

UFC 3-120-01 Design: Sign Standards

UFC 3-120-10, *Interior Design*

UFC 3-130-04 Arctic and Subarctic Buildings

UFC 3-190-06, *Protective Coatings and Paints*

UFC 3-201-01, *Civil Engineering*

UFC 3-201-02, *Landscape Architecture*

UFC 3-210-10, *Low Impact Development*

UFC 3-220-01, *Geotechnical Engineering*

UFC 3-230-01, *Water Storage and Distribution*

UFC 3-230-03, *Water Treatment*

UFC 3-240-01, *Wastewater Collection and Treatment*

UFC 3-301-01, *Structural Engineering*

UFC 3-301-02, *Design of Risk Category V Structures, National Strategic Military Assets*
UFC 3-401-01, *Mechanical Engineering*
UFC 3-410-01, *Heating, Ventilating, and Air Conditioning Systems*
UFC 3-420-01, *Plumbing Systems*
UFC 3-430-05, *Natural Gas and Liquefied Petroleum Gas (LPG) Distribution Pipelines*
UFC 3-460-01, *Design: Petroleum Fuel Facilities*
UFC 3-490-06, *Elevators*
UFC 3-501-01, *Electrical Engineering*
UFC 3-520-01, *Interior Electrical Systems*
UFC 3-530-01, *Interior and Exterior Lighting Systems and Controls*
UFC 3-540-01, *Engine-Driven Generator Systems for Prime and Standby Power Applications*
UFC 3-550-01, *Exterior Electrical Power Distribution*
UFC 3-560-01, *Operation and Maintenance: Electrical Safety*
UFC 3-570-01, *Cathodic Protection*
UFC 3-580-01, *Telecommunications Interior Infrastructure Planning and Design*
UFC 3-600-01, *Fire Protection Engineering for Facilities*
UFC 3-701-01, *DoD Facilities Pricing Guide*
UFC 4-010-01, *DoD Minimum Antiterrorism Standards for Buildings*
UFC 4-010-05, *SCIF/SAPF Planning, Design, and Construction*
UFC 4-010-06, *Cybersecurity of Facility-Related Control Systems*
UFC 4-021-01, *Design and O&M: Mass Notification Systems*
UFC 4-510-01, *Design: Military Medical Facilities*
UFC 4-711-01, *Family Housing*
UFC 4-721-10, *Navy and Marine Corps Unaccompanied Housing*

UNITED STATES ACCESS BOARD

Architectural Barriers Act (ABA) Standards
<https://www.access-board.gov/files/aba/ABASTandards.pdf>

UNITED STATES AIR FORCE

<https://www.e-publishing.af.mil>

DESR6055.09_AFMAN91-201, *Explosives Safety Standards*

DAFMAN 91-203, *Air Force Occupational Safety, Fire and Health Standards*

UNITED STATES ARMY

<https://armypubs.army.mil/>

DA PAM 385-64, *Ammunition and Explosives Safety Standards*

DA PAM 385-16, *System Safety Management Guide*

UNITED STATES DEPARTMENT OF DEFENSE

Deputy Secretary of Defense Memorandum for Secretaries of the Military Departments, et al.

Subject: Access for People with Disabilities, October 31, 2008

<https://www.access-board.gov/aba/background/dod-policy-memo.html>

DoDM 4715.05, *Overseas Environmental Baseline Guidance Document*

<https://www.esd.whs.mil/Directives/issuances/dodm/>

DoD 5200.08-R, *Physical Security Program*

<https://www.esd.whs.mil/DD/DoD-issuances/OUUSD>

DoDI 6055.01, *DoD Safety and Occupational Health (SOH) Program*

<https://www.esd.whs.mil/DD/DoD-issuances/>

DoDM 6055.09, *DoD Ammunition and Explosives Safety Standards*

<https://www.esd.whs.mil/DD/DoD-issuances/>

DoDI 1125.03, *Vending Facility Program for the Blind on DoD-Controlled Federal Property*

<https://www.esd.whs.mil/DD/DoD-issuances/>

DoDM 5100.76, *Physical Security of Sensitive Conventional Arms, Ammunition and Explosives (AA&E)*

<https://www.esd.whs.mil/DD/DoD-issuances/>

DoDM 5200.01, *DoD Information Security Program*

<https://www.esd.whs.mil/DD/DoD-issuances/>

DoDM 5205.07 Volume 3, *DoD Special Access Program (SAP) Security Manual: Physical Security*

<https://www.esd.whs.mil/DD/DoD-issuances/>

MIL-STD-3007G, *Standard Practice for Unified Facilities Criteria, Facilities Criteria and Unified Facilities Guide Specifications*, 1 November 2019

<https://www.wbdg.org/FFC/FEDMIL/milstd3007g.pdf>

OUUSD Memorandum. *Subject: Department-Wide Policy for Nursing and Lactation Rooms*, 1

November 2016 https://www.wbdg.org/FFC/DOD/DOD_NursingMothersMemo_110116.pdf

OUUSD Memorandum for Assistant Secretary(s) of the Army (IE&E), Navy (EI&E), Air Force (IE&L), Directors of Defense Agencies, Directors of Defense Activities, Director, Washington Headquarters Service. *Subject: Floodplain Management on Department of Defense Installations*, 11 February 2014 https://www.wbdg.org/FFC/DOD/DUSDIE_Memo_FloodplainMgmt.pdf

UNITED STATES NAVY

NAVSEA OP 5, *Ammunition and Explosives Safety Ashore*
<https://nossa.dc3n.navy.mil/nrws3/> (must be registered user)

UNITED STATES OFFICE OF PERSONNEL MANAGEMENT

OPM Guide for Establishing a Federal Nursing Mother's Program
<https://www.opm.gov/policy-data-oversight/worklife/reference-materials/nursing-mother-guide.pdf>

APPENDIX A UFC 1-200-02 GLOSSARY

A-1 ACRONYMS

≥	greater than or equal to
>	greater than
≤	less than or equal to
<	less than
AFCEC	Air Force Civil Engineer Center
AFARS	Army Federal Acquisition Regulation Supplement
AFFARS	Air Force Federal Acquisition Regulation Supplement
AHJ	Authority Having Jurisdiction
APF	Appropriated Funding
ASHRAE	American Society of Heating, Refrigerating and Air Conditioning Engineers
ASTM	American Society of Testing and Materials
BEAP	Base Exterior Architecture Plan
BSI	Business Systems Integration Directorate (EI&E Office)
CF	Cubic Feet
CFR	Code of Federal Regulations
CONUS	Continental United States
DFARS	Defense Federal Acquisition Regulation Supplement
DOE	Department of Energy
DoD	Department of Defense
DoDI	Department of Defense Instruction
ECA	Energy Compliance Analysis
EEIM	Enterprise Energy Information Management
EISA	Energy Independence and Security Act
EMS	Environmental Management System

EPA	Environmental Protection Agency
EPAct	Energy Policy Act
EPEAT	Electronic Product Environmental Assessment Tool
ERC	Estimated Replacement Cost
FAR	Federal Acquisition Regulation
FEMP	Federal Energy Management Program
GBCI	Green Business Certification Incorporated
GBI	Green Building Initiative
GPF	Gallons per Flush
GSA	General Services Administration
GSF	Gross Square Feet
GSM	Gross Square Meters
HPSB	High Performance Sustainable Building
HQUSACE	Headquarters, U.S. Army Corps of Engineers
HVAC	Heating, Ventilation and Air Conditioning
IAQ	Indoor Air Quality
ICC	International Code Council
IECC	International Energy Conservation Code
IESNA	Illuminating Engineering Society of North America
KWh	Kilowatt Hour
LCC	Life-Cycle Cost
LCCA	Life-Cycle Cost Analysis
LCCE	Life-Cycle Cost Effective
LEED	Leadership in Energy and Environmental Design
LPF	Liters per Flush
MIL-STD	Military Standard

MOU	Memorandum of Understanding
NAF	Non-Appropriated Funding
NAVFAC	Naval Facilities Engineering Systems Command
NIST	National Institute of Standards and Technology
NMCARS	Navy Marine Corps Acquisition Regulation Supplement
OCONUS	Outside Continental United States
ODS	Ozone Depleting Substances
O/M	Operations and Maintenance
OMB	Office of Management and Budget
PL	Public Law
RCRA	Resource Conservation and Recovery Act
RPA	Real Property Accountability
SDHW	Solar Domestic Hot Water
SF	Square Foot
SHPO	State Historic Preservation Office
SMACNA	Sheet Metal and Air Conditioning Contractors' National Association
SNAP	Significant New Alternatives Policy
TPC	Third-Party Certification
UFC	Unified Facilities Criteria
UFGS	Unified Facilities Guide Specifications
USGBC	U.S. Green Building Council
US	United States
USACE	U.S. Army Corps of Engineers
USC	U.S. Code
USDA	United States Department of Agriculture
WBDG	Whole Building Design Guide

Acquisition: A multidisciplinary process encompassing a variety of functions that involve all facets of life-cycle management, including determination of need, planning; design; construction, lease, or purchase; sustainment, modernization, and disposal of military installations and facilities as well as other goods and services. Acquiring supplies and services is a process governed by Federal, Defense, and service branch specific Acquisition Regulations (FAR, DFARS, and AFFARS, AFARS, or NMCARS) and Public Law (PL). Together there are more than 4,000 documents controlling the acquisition process. The acquisition process depends upon: (1) who the customer is, (2) the source of the item to be acquired, and (3) what type of funds are to be used, Appropriated (APF) or Non-Appropriated (NAF). In general, the Military Construction Program process is funded by Appropriated Funds. However, projects for Non-Appropriated Funds activities (commissaries, bowling centers, etc.) use NAF funds for building construction, upgrades, special equipment, and furnishings. APF funding is used most typically for demolition and environmental work on NAF activity sites. A combination of APF and NAF funds may be used.

Alternative Water Sources: Non-potable water from sources such as harvested rainwater (refer to UFC 3-210-10), treated wastewater, air handler condensate capture, grey water, or reclaimed water. The use of alternative water sources must comply with applicable codes and standards.

Authority Having Jurisdiction (AHJ): The term "Authority Having Jurisdiction" (AHJ) as used in the codes and standards referenced in this UFC means the Service office of responsibility, i.e., U.S. Army, HQ USACE/CECW-CE; U.S. Navy, NAVFACENGCOM HQ Code CHE; U.S. Marine Corps, HQMC Code LFF-1; and U.S. Air Force, AFCEC. The enforcement of the codes and standards as they pertain to facility projects can be delegated to the local Service Office's Chief Engineer's Technical Representative at the discretion of the Service's aforementioned office.

Building: Per UFC 1-300-08, "A roofed and floored facility enclosed by exterior walls and consisting of one or more levels that is suitable for single or multiple functions."

Charrette: An intensive creative work session in which a design team focuses on a particular design problem and arrives at a collaborative solution with stakeholders from the project area. A charrette can be a breakthrough event that helps create a meaningful master plan or facility design. Properly executed, this technique can produce a master plan or facility design that is more useful, better understood, and more quickly produced than one formed by any other method.

Commercial and Multi-Family High-Rise Residential Buildings: All buildings, other than low-rise residential buildings.

Commissioning: Per ASHRAE 90.1-2019: Quality-focused process for enhancing the delivery of a project for verifying and documenting that the building and its systems, controls, and building envelope are planned, designed, installed, tested, and include plans for operation and maintenance to meet specified requirements.

Component: One of the Military Departments, Defense Agencies, or DoD Field Activities, per USD (AT&L) Memorandum dated 29 May 2002.

Comprehensive Replacement: Comprehensive replacement to a building includes significant opportunities for improvement in: energy and water efficiency (such as HVAC, lighting, building envelope and other building components); indoor air quality; other requirements in this UFC; and additions that are part of the comprehensive replacement. /1/

DD Form 1391: A programming document used by the Department of Defense to submit requirements and justifications in support of funding requests for military construction to Congress.

Employee Work Areas: Per International Building Code 2015: “All or any portion of a space use by employees and only for work. Corridors, toilet rooms, kitchenettes and break rooms are not employee work areas.”

Energy Compliance Analysis (ECA): The ECA must identify specific energy conservation criteria that applies to the project, the software used to prepare the necessary calculations, a summary of all input to and output from the calculations, and the calculated energy consumption of the proposed design.

Energy Modeling: The process by which conceptual designs, including size; material choices; factors such as site and solar orientations; daylighting percentages; and energy system choices (solar water heat, underfloor vs. overhead air distribution systems) are analyzed to show how to optimize these factors for efficient building operation and resource consumption.

Energy Optimization: The process of evaluating strategies to comply with Federal, DoD, and service component energy requirements. The process is an element of integrated design and includes strategy identification, energy modeling, LCCA, and strategy selection based on results.

Energy Resilience: Per DoDI 4170.11, “The ability to prepare for and recover from energy disruptions that impact mission assurance on military installations.”

Environmental Management System (EMS): A framework that helps to achieve environmental goals through consistent control of operations. The assumption is that this increased control will improve environmental performance.

Environmentally Preferable Products: Products or services having a lesser or reduced effect on human health and the environment when compared with competing products or services serving the same purpose. This comparison may consider raw materials acquisition, production, manufacturing, packaging, distribution, reuse, operation, maintenance, or product or service disposal.

Estimated Replacement Cost (ERC): ERC is the cost listed in the project’s DD1391 “Economic Analysis.” ERC is the cost of replacing the current physical plant with modern facilities built at today’s construction costs using today’s construction standards.

Facility: Per UFC 1-300-08, a building, structure, or linear structure out to an imaginary line surrounding a facility at a distance of five feet from the foundation that, barring specific direction to the contrary such as a utility privatization agreement, denotes what is included in the basic record for the facility (examples: landscaping, sidewalks, utility connections). This imaginary line is commonly referred to as the “five-foot line.”

Historic Properties: Properties that are included in the National Register of Historic Places or that meet the criteria for the National Register.

Life Expectancy: The terms life expectancy and study period are used interchangeably for LCCA in 10 CFR 436, EISA 2007, and NIST Handbook 135. Per 10 CFR 436 “study period is the expected life of the retrofit or new building.”

Life-Cycle Costing (LCC): An important economic analysis used in the selection of alternatives that impact both pending and future costs. It compares initial investment options and identifies the cost alternatives for a 40-year period. As applied to building design energy conservations measures, the process is mandated by law and is defined in 10 CFR Part 436, Subpart A: *Methodology and Procedures for Life-Cycle Cost Analysis*. The National Institute of Standards and Technology has established the Building Life-Cycle Cost (BLCC) computer program to perform LCC analyses. The program incorporates user entered data for and compares the following: Sunk Costs, First Costs, Salvage Value, Future Investment, Residual Value, Annually Recurring Fixed Costs, Annually Recurring Escalating Costs, and Energy (Fuel Costs) Escalation Rates.

Life-Cycle Cost Analysis (LCCA): Assessment of the direct, indirect, recurring, nonrecurring, and other related costs incurred or estimated to be incurred in the design, development, production, operation, maintenance, support, and final disposition of a major system over its anticipated useful life span. LCCA considers all costs (capital, operating, and decommissioning expenses for the duration of a project) for various alternative approaches, including inflation and discount rates.

Life-Cycle Cost Effectiveness (LCCE): LCCE is a state of having a LCC that is equal to or lower than a baseline OR having a lower LCC than all alternatives, depending on the applicable criteria. This includes a documented statement of costs to be incurred to complete all stages of a project from planning through acquisition, maintenance, operation, remediation, disposition, long-term stewardship, and disposal. The results of a LCCA.

Low-Rise Residential Buildings: All buildings three stories or less in height above grade that include sleeping accommodations where the occupants are primarily permanent in nature (30 days or more).

Planning: Initiated by a using entity whose facility needs are driven by mission requirements. The insertion of facility projects into the official budget is based on the requirement for mission support. As mission priorities change, projects can be moved ahead of their initial scheduled fiscal year or alternatively, be dropped altogether.

Process load: The load on a building resulting from energy consumed in support of a manufacturing, industrial, or commercial process. Process loads do not include energy consumed maintaining comfort and amenities for the occupants of the building (including space conditioning for human comfort).

Programming: The data collection process done by the military project management team in order to achieve a conceptual design in enough detail to prepare a viable cost estimate. Programming data is reported on DD Form 1391 for design-bid-build and design-build projects and is scrutinized from Installation-level through the Office of Management and Budget (OMB), Congress, and signed by the President before it is approved as a project.

Project Definition: As the military programming process often predates actual design by two to five years or more, when the design services portion of a project is funded, the project design team must validate stated project requirements and personnel assignments based upon updated mission criteria. This process is sometimes referred to as Project Definition to distinguish it from the usual private sector A/E reference to “programming.”

Project Scope: Refers to the physical size, cost, and mission of a project. With historical data on facilities for various missions, experienced programmers will formulate the conceptual design for the DD Form 1391 based upon the required size of a facility to accommodate mission functions. Once the scope for a project has been enacted by Congress, changing the scope of a project will delay a project until the revision has been justified and approved. This is true whether or not the costs have also changed. Reductions of project scope (by no more than 25%) will require Congressional notification, review, and authorization before proceeding. Increasing the project scope is not authorized without reprogramming. See USC Title 10 Section 2853 for more information.

Real Property Asset: An individual building, structure, linear structure or land parcel, owned by the Department of Defense.

Receptacle Load: The load on a building resulting from energy consumed by all equipment plugged into electrical outlets.

Recommissioning: Per DOE Guidance 42 USC 8253(f), “The process of commissioning a previously commissioned facility or system after expiration of the project development and warranty phase. The primary goal of recommissioning is to optimize facility performance, in accordance with design or operating needs, over the useful life of the facility.”

Resilience: The ability to prepare for and recover from disruptions that impact mission assurance on military installations.

Resource-efficient solutions: Solutions that minimize energy, water, natural, and human resources required.

Retrocommissioning: Per DOE Guidance 42 USC 8253(f), “The retroactive commissioning of equipment or a system that was not commissioned at the time of installation or during the warranty phase. Typically, retrocommissioning is performed long after the facility is constructed and placed into service.”

Service: When used to describe a DoD entity, one of the three Military Departments under the DoD, per USD (AT&L) Memorandum dated 29 May 2002.

Stand-Alone Additions: Stand-alone additions are scoped to function like stand-alone new buildings. Although the addition is attached to an existing building, all of its new systems operate separately from the existing building.

Structure: Per UFC 1-300-08, a facility, other than a building or linear structure (road, fence, etc.), that is constructed on or in the land.

Sustainable Site: Based on the selection process, a site is considered sustainable when it uses less energy, water, and natural resources, generates less waste, and minimizes the impact on land compared to conventional design, construction, and maintenance techniques.

Sustainment, Restoration, and Modernization (SRM): Per the Office of the Assistant Secretary of Defense for Sustainment, Construction:

- Sustainment activities include scheduled and unscheduled inspection, maintenance and repair to ensure facilities remain in good working order throughout their service lives.
- Restoration activities include repair and replacement efforts to renovate facilities damaged by inadequate sustainment, excessive age, natural disaster, fire, accident, or other causes.
- Modernization activities include implementation of new or higher standards; accommodating new functions; or replacing building components that typically last more than 50 years.

Third-Party Certification (TPC): TPC is the generic term for a third-party product that provides either certification of the third-party vendor’s proprietary product requirements (examples: LEED, Green Globes), or a validation program by the third-party vendor that this UFC’s requirements have been met (examples: Guiding Principles Assessment, Guiding Principles Compliance.)

Total Ownership Cost (TOC): Total of all direct and indirect costs associated with an asset or acquisition over its entire life cycle.

Tracking: Method by which each Service internally validates HPSB Guiding Principles compliance, with the use of HPSB Checklist.

APPENDIX B UFC 1-200-02 REFERENCES

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS (ASHRAE)

<https://www.ashrae.org/>

NOTE: Include all references Errata, except Errata for Addenda. Include Interpretations, except Interpretations for Addenda. Do not include Addenda. Exception: include ASHRAE 90.1-2019 Addendum ch for energy consumption reduction value as cited in paragraph titled "[Commercial and Multi-Family High-Rise Residential Buildings](#)."

ANSI/ASHRAE/IESNA Standard 90.1-2019 (ASHRAE 90.1), Energy Standards for Buildings Except Low Rise Residential Buildings, 2019

INTERNATIONAL CODE COUNCIL

<https://codes.iccsafe.org/search/>

International Energy Conservation Code (IECC), 2015

International Green Construction Code (IgCC), 2018

International Energy Conservation Code (IECC), 202

MILITARY STANDARDS

MIL-STD-3007G Standard Practice Unified Facilities Criteria, Facilities Criteria And Unified Facilities Guide Specifications, 2019

<https://www.wbdg.org/FFC/FEDMIL/milstd3007g.pdf>

NATIONAL INSTITUTE OF BUILDING SCIENCES (NIBS)

Whole Building Design Guide website

<https://www.wbdg.org/>

NATIONAL INSTITUTE OF STANDARDS AND TECHNOLOGY

<https://www.wbdg.org/nist/criteria>

NIST Handbook 135, Life-Cycle Costing Manual for the Federal Energy Management Program

The Annual Supplement to NIST Handbook 135 and SP 709, Energy Price Indices and Discount Factors for Life-Cycle Cost Analysis

SHEET METAL AND AIR CONDITIONING CONTRACTOR'S NATIONAL ASSOCIATION (SMACNA)

<https://www.smacna.org/>

SMACNA *Indoor Air Quality (IAQ) Guidelines for Occupied Buildings Under Construction*, 2nd Edition, 2008

UNITED STATES CODE

<https://www.govinfo.gov/app/collection/uscode>

USC Title 10 Section 2853, *ADMINISTRATION OF MILITARY CONSTRUCTION AND MILITARY FAMILY HOUSING – Authorized Cost and Scope of Work Variations*

USC Title 16 Section 106, *National Historic Preservation Act*

42 U.S.C. § 6962, Federal Procurement

UNITED STATES CODE OF FEDERAL REGULATIONS

<https://www.govinfo.gov/collection/cfr-index>

10 CFR Part 433, *Energy Efficiency Standards for New Federal Commercial and Multi-Family High-Rise Residential Buildings*

10 CFR Part 436, Subpart A: *Methodology and Procedures for Life-Cycle Cost Analysis*

36 CFR Part 800, *Protection of Historic Properties*

UNITED STATES DEPARTMENT OF AGRICULTURE

BioPreferred Program

<http://www.biopREFERRED.gov/>

UNITED STATES DEPARTMENT OF DEFENSE

<http://www.esd.whs.mil/dd/>

DoD 4160.21-M, *Defense Materiel Disposition Manual*, 28 May 2004

DoDI 4165.14, *Real Property Inventory (RPI) and Forecasting*, 17 January 2014

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APPENDIX A UFC 1-201-01 GLOSSARY

A-1 ACRONYMS.

AFCEE	Air Force Center for Engineering and the Environment
BIA	Bilateral Infrastructure Agreement
DoD	Department of Defense
HQUSACE	Headquarters, U.S. Army Corps of Engineers
HNFA	Host Nation Funded Construction Agreements
NAVFAC	Naval Facilities Engineering Command
SOFA	Status of Forces Agreements
UFC	Unified Facilities Criteria
U.S.	United States

APPENDIX B UFC 1-201-01 REFERENCES

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APPENDIX A UFC 3-101-01 GLOSSARY

A-1 ACRONYMS

AAMA	American Architectural Manufacturer Association
ASCE	American Society of Civil Engineers
ACI	American Concrete Institute
AF	Air Force
AFCEC	Air Force Civil Engineer Center
AFMAN	Air Force Manual
AHJ	Authority Having Jurisdiction
AICUZ	Air Installation Compatible Use Zones
ANSI	American National Standards Institute
AR	Army Regulations
ASA	Acoustical Society of America
ASHRAE	American Society of Heating, Refrigeration and Air Conditioning Engineers
ASTM	American Society of Testing and Materials
AT	Antiterrorism
BIA	Brick Industry Association
BIM	Building Information Modeling
BLCC	Building Life Cycle Cost
C	Celsius
CADD	Computer-aided Design and Drafting
CBD	Chronic Beryllium Disease
CCB	Construction Criteria Base
CCR	Criteria Change Request
CFR	Code of Federal Regulations

CNEL	Community Noise Equivalent Level
CONUS	Continental United States
CSA	Canadian Standards Association
DNL	Day Night Average Sound Level
DoD	Department of Defense
ECB	Engineering and Construction Bulletins
EIFS	Exterior Insulation Finish System
EIMA	EIFS Industry Members Association
EPA	Environmental Protection Agency
EPG	Electronic Project Generator
ESC	Environmental Severity Classification
F	Fahrenheit
FACD	Functional Analysis Concept Development
FEC	Facilities Engineering Command
FGI	Facilities Guidelines Institute
ft	Foot or feet
HEMP	High-Altitude Electromagnetic Pulse
HEPA	High-Efficiency Particulate Air
Hg	Mercury
HNFA	Host Nation Funded Construction Agreements
HPSB	High Performance and Sustainable Building Requirements
HQUSACE	Headquarters, U.S. Army Corps of Engineers
GCC	Geographic Combatant Commander
GSA	General Services Administration
IBC	International Building Code

ICD	Intelligence Community Directive
ICS	Intelligence Community Standard
IESNA	Illuminating Engineering Society of North America
IFS	Installation Facilities Standards
in	Inch or inches
IP	Inch-pound
ISO	International Organization for Standardization
Kg	Kilogram
LCCA	Life Cycle Cost Analysis
m	Meters
mm	Millimeters
MPI	Master Painters Institute
MWR	Moral, Welfare and Recreation
NACE	National Association of Corrosion Engineers
NASA	National Aeronautics and Space Administration
NAVFAC	Naval Facilities Engineering Systems Command
NAVRAMP	Navy's Radon Assessment and Mitigation Program
NC	Noise Criteria
NCPC	National Capital Planning Commission
NCR	National Capital Region
NFRC	National Fenestration Rating Council
NIC	Noise Isolation Class
NIH	National Institute of Health
NIST	National Institute of Standards and Technology
NRCA	National Roofing Contractors Association

NS	Net Savings
OITC	Outdoor Indoor Transmission Class
OSHA	Occupational Safety and Health Administration
Pa	Pascal (SI unit of pressure)
PBS	Public Buildings Services
PCB	Polychlorinated Biphenyls
PCR	Planning Charrette Report
psi	Pound per square inch
PTS	Performance Technical Specifications
PVC	Polyvinyl Chloride
RAMP	Radon Assessment and Mitigation Program
RC	Room Criteria
RCRA	Resource Conservation and Recovery Act
RF	Radio Frequency
RFP	Request for Proposal
SAPF	Special Access Program Facility
SC	Shading Coefficient
SCIF	Sensitive Compartmented Information Facility
SDSFIE	Spatial Data Standard for Facilities, Infrastructure and Environment
SEI	Structural Engineering Institute
sf	square feet
SHGC	Solar Heat Gain Coefficient
SI	International System of Units (Metric System)
SIR	Savings to Investment Ratio
sm	square meters

SMACNA	Sheet Metal and Air Conditioning Contractors' National Association
SOFA	Status of Forces Agreement
STC	Sound Transmission Coefficient
TSCA	Toxic Substances Control Act
TM	Technical Manual(s)
UEPH ARMY	Unaccompanied Enlisted Personnel Housing
UFC	Unified Facilities Criteria
UFGS	Unified Facilities Guide Specifications
USACE	U.S. Army Corps of Engineers
USGBC	U.S. Green Building Council
VT	Visible Transmittance
WBDG	Whole Building Design Guide
WDMA	Window and Door Manufacturer Association
WRB	Water Resistant Barrier

Air Barrier. The term “air barrier” refers to the “continuous air barrier” the combination of interconnected materials, assemblies, and sealed joints and components of the building envelope that minimize air leakage into or out of the building envelope specified by ASHRAE 90.1.

Building System and Subsystems. An assembly of dimensionally and functionally pre-coordinated subsystems which, when combined, produces an essentially complete building. A subsystem is one of many building components designed and manufactured to be integrated with other subsystems to produce an entire building system.

Building Information Model (BIM). A BIM is a three-dimensional digital model of a facility. It incorporates all disciplines and facilitates coordination, calculations of relevant building parameters, and the development of construction documents.

Corrosion. Corrosion is the deterioration of a material or its properties due to a reaction of that material with its chemical environment and is inclusive of the deterioration of all materials, which can be caused through sun exposure, mold and mildew, wind, and other environmental elements.

Environmental Severity Classification (ESC). Environmental severity is defined as the corrosivity of the local environment of a given location or region. Environmental severity contributes directly to the occurrence of corrosion. The effects of corrosion and the rate at which they occur are consequences of the corrosion system, which is comprised of a material or physical system, the environment, and operational conditions.

Noise Isolation Class (NIC). NIC is a method for field transmission ratings. The higher the number the better the noise control. NIC rates a partition's ability to block airborne noise transfer. NIC testing is preferable to STC testing as it is specified on more specific space types (such as spaces with operable walls, unaccompanied housing sleeping rooms to each other or to the adjacent corridor, and the actual work spaces in administration facilities). For a field STC test, the individual transmission loss measurements are modified based upon the reverberation time, the size of the room, and the size of the tested partition. The NIC does not include these modifications and simply measures the transmission loss between 125 and 4,000Hz. The value of this rating is that it better tests the performance of the assembly in the field, though it is highly dependent on field conditions of the tested space.

Outdoor-Indoor Transmission Class (OITC). OITC is defined as the A-weighted sound level reduction of a test specimen (or a space) in the presence of an idealized mixture of transportation noises; aircraft takeoff, freeway, and railroad pass-by. It assigns a single number rating to measure Sound Transmission Loss (TL) data obtained in accordance with ASTM E-90. The higher the number, the better the control.

Sound Transmission Class (STC). STC is a single number developed under laboratory conditions that represents the effectiveness of materials or construction to

retard the transmission of air-borne sound. It is not as effective for measurement in low frequency noise sources such as mechanical equipment.

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APPENDIX A UFC 3-110-03 GLOSSARY

A-1 ACRONYMS.

ACM	Asbestos containing material
AFCEC	Air Force Civil Engineer Center
ANSI	American National Standards Institute
APP	Atactic Polypropylene
ASCE	American Society of Civil Engineers
ASHRAE	American Society of Heating, Refrigerating and Air Conditioning Engineers
ASSP	American Society of Safety Professionals
ASTM	American Society of Testing and Materials
BASH	Bird/wildlife aircraft strike hazard
BIA	Bilateral Infrastructure Agreement
BUR	Built-up roof
C	Celsius
CCR	Criteria Change Request
CERL	Construction Engineering Research Laboratories
CONUS	Continental United States
COTS	Commercial Off-the-Shelf
CRREL	Cold Regions Research and Engineering Laboratory
CSPE	Chlorosulfonated Polyethylene
CQM	Construction Quality Management
DD 1391	Department of Defense Form 1391
DFT	Dry Film Thickness
DoD	Department of Defense
EFVM	Electric field vector mapping

EPDM	Ethylene Propylene Diene Monomer
EPS	Molded expanded polystyrene board.
ESC	Environmental Severity Classification
F	Fahrenheit
FCI	Facility Condition Index
FM	Factory Mutual
FRT	Fire Retardant Treated
ft	Foot or feet
g	Gram or Grams
ga	Gauge
HQUSACE	Headquarters, U.S. Army Corps of Engineers
HNFA	Host Nation Funded Construction Agreements
hr	Hour or Hours
HVAC	Heating, ventilation, and air conditioning
IBC	International Building Code
IEBC	International Existing Building Code
IgCC	International Green Construction Code
in	Inch or Inches
IR	Infrared
IRMA	Inverted Roof Membrane Assembly
LCCA	Life cycle cost analysis
LWCC	Lightweight cellular concretes
LWIC	Lightweight insulating concrete
m	Meter
mm	Millimeter

MB	Modified bitumen
MBMA	Metal Building Manufacturers Association
MREC	Mesh Reinforced Elastomeric Roof Coating
NAVFAC	Naval Facilities Engineering Systems Command
NBS	National Bureau of Standards
NDL	No Dollar Limit
NMCI	Navy/Marine Corps Intranet
NRCA	National Roofing Contractors Association
O&M	Operations and Maintenance
OSHA	Occupational Safety and Health Administration
PE	Professional engineer
PFAS	Personal Fall Arrest System
PMMA	Polymethylmethacrylate
PV	Photovoltaic
PVC	Polyvinyl chloride
PVDF	Polyvinylidene fluoride (also PVF)
PVF	Polyvinylidene fluoride (also PVDF)
QPPF	Qualified Person for Fall Protection
RA	Registered architect
ROI	Return on investment
RRO	Registered roofing observer
RRC	Registered roofing consultant
SBS	Styrene Butadiene Styrene
SHPO	State Historic Preservation Office
SMACNA	Sheet Metal and Air Conditioning Contractors National Association

SME	Subject matter expert
SMS	Sustainment Management System
SOFA	Status of Forces Agreements
SPF	Spray-applied polyurethane foam
SPFA	Spray Polyurethane Foam Alliance
SPRI	Single Ply Roofing Industry
SSMR	Standing Seam Metal Roof
TPC	Third-Party Certifiers
TPO	Thermoplastic Polyolefin
UFC	Unified Facilities Criteria
UFGS	Unified Facilities Guide Specification
UL	Underwriters Laboratories Incorporated
U.S.	United States
USACE	United States Army Corps of Engineers
UV	Ultraviolet
VOC	Volatile Organic Compounds
XPS	Extruded polystyrene board

A-2 DEFINITION OF TERMS.

See UFC 3-101-01 and the NRCA Roofing Manual for a more complete list of definitions.

Aggregate: (1) Crushed stone, crushed slag, or water-worn gravel used for surfacing a built-up roof. (2) Any granular mineral material.

Asphalt: A dark brown to black elastomeric cementitious material whose predominating constituents are bitumen's that occur in nature or are obtained in petroleum processing.

Atactic: A chain of molecules in which the position of the side methyl groups is more or less random. (Amorphous; Low Crystallinity)

Ballast: Loose aggregate, concrete pavers, or other material designed to prevent wind uplift or flotation of a loose-laid roof system.

Base sheet: A saturated or coated felt placed as the first ply in a multi-ply bituminous roofing membrane.

Batten: (1) Raised rib, in a metal roof, or a separate part or formed portion in a metal roofing panel. (2) One of a series of strips laid in a matrix under tile roof applications.

Bitumen: (1) A class of amorphous, black or dark colored, (solid, semisolid, or viscous) cementitious substances natural or manufactured, composed principally of high molecular weight hydrocarbons, soluble in carbon disulfide, and found in asphalts, tars, pitches, and asphaltites; (2) A generic term used to denote any material composed principally of bitumen; (3) In the roofing industry there are two basic bitumens: asphalt and coal-tar pitch. Before application, they are either (a) heated to a liquid state, (b) dissolved in a solvent, or (c) emulsified.

Bituminous (adj.): Containing or treated with bitumen. Examples: bituminous concrete, bituminous felts and fabrics, bituminous pavement.

Bond: The adhesive and cohesive forces holding two roofing components in intimate contact.

Boot: A bellows type covering to exclude dust, dirt, and moisture, forming a flexible closure.

British thermal unit (BTU): Heat energy required to raise the temperature of one pound of water by 1°F (= 1055 joules).

Building Code: Published regulations and ordinances established by a recognized agency describing design loads, procedures, and construction details for structures. Usually applying to designated political jurisdiction (city, county, and state). Building codes control design, construction, and quality of materials, use and occupancy,

location and maintenance of buildings and structures within the area for which the code was adopted (see Model Codes).

Built-Up Roofing (BUR): A continuous, semiflexible membrane consisting of plies of saturated felts, coated felts, fabrics, or mats assembled in place with alternate layers of bitumen, and surfaced with mineral aggregate, bituminous material, or a granule surfaced sheet (abbreviation, BUR).

Cant strip: A beveled strip used under flashings to modify the angle at the point where the roofing or waterproofing membrane meets any vertical element.

Cap sheet: A granule-surfaced coated felt used as the top ply of a built-up roofing membrane.

Chalking: A powdery residue on the surface of a material resulting from degradation or migration of an ingredient, or both.

Chlorosulfonated polyethylene (CSPE): Family of polymers that are produced by polyethylene reacting with chlorine and sulfur dioxide. Present polymers contain 25-43% chlorine and 1.0-1.4% sulfur. They are used in both vulcanized and nonvulcanized forms. Most membranes based on CSPE are nonvulcanized. ASTM designation for this polymer is CSM. Best known by the DuPont Tradename "Hypalon."

Coal-Tar Pitch: A dark brown to black cementitious material produced by the destructive distillation of coal.

Condensation: The conversion of water vapor or other gas to liquid as the temperature drops or atmospheric pressure rises.

Copolymer: A mixed polymer, the product of polymerization of two or more substances at the same time.

Corrosion: The deterioration of a material or its properties due to a reaction of that material with its chemical environment; it is inclusive of the deterioration of all materials, which can be caused through sun exposure, mold and mildew, wind, and other environmental elements.

Counterflashing: Formed metal or elastomeric sheeting secured on or into a wall, curb, pipe, rooftop unit, or other surface, to shield the upper edge of a base flashing and its associated fasteners.

Coverage: The surface area to be continuously covered by a specific quantity of a particular material.

Covering: The exterior roof and wall covering for a metal building system.

Cricket: A relatively small, elevated area of a roof constructed to divert water from a horizontal intersection of the roof with a chimney, wall, expansion joint or other projection.

Curb: A raised member used to support roof penetrations for features such as skylights, hatches, and equipment.

Cure: To change the properties of a polymeric system into a more stable, usable condition by the use of heat, radiation, or reaction with chemical additives. NOTE: Cure may be accomplished, for example, by removal of solvent or cross-linking.

Deck: The structural substrate to which the roofing or waterproofing system (including insulation) is applied.

Design loads: The “live load” and “environmental load” (such as superimposed loads) that a structure is designed to resist (with appropriate safety factor) plus “dead load” (such as weight of permanent loads).

Dry (n.): A material that contains no more water than one would find at its equilibrium moisture content.

Eave: The line along the sidewall formed by the intersection of the planes of the roof and wall.

Elastomer: A macromolecular material that returns rapidly to its approximate initial dimensions and shape after subsequent release of stress.

Environmental Severity Classification (ESC): Environmental severity is defined as the corrosivity of the local environment of a given location or region. Environmental severity contributes directly to the occurrence of corrosion. The effects of corrosion and the rate at which they occur are consequences of the corrosion system, which is comprised of a material or physical system, the environment, and operational conditions.

EPDM: A synthetic elastomer based on ethylene, propylene, and a small amount of a non-conjugated diene to provide sites for vulcanization. EPDM features excellent heat, ozone and weathering resistance, and low temperature flexibility.

Fabric: A woven cloth of organic or inorganic filaments, threads, or yarns.

Fabrication: (1) The manufacturing process performed in a plant to convert raw material into finished metal building components. The main operations are cold-forming, cutting, punching, welding, cleaning, and painting; (2) the creation of large panels of rubber from smaller calendar width sheets as in EPDM.

Fascia: A decorative trim or panel projecting from the face of a wall, serving as a weather closure at gable and end wall.

Felt: A flexible sheet manufactured by the interlocking of fibers through a combination of mechanical work, moisture, and heat, without spinning, weaving, or knitting. Roofing felts are manufactured from vegetable fibers (organic felts), glass fibers (glass fiber felts) or polyester fibers (synthetic fiber mats).

Field: The “job site,” “building site,” or general market area.

Fixity: Fixity is the location in which a single metal roof panel is attached to the building structure and is the point from which all thermal expansion radiates. The line of fixity is the line created from the individual points of fixity.

Flashing: The system used to seal membrane edges at adjacent perpendicular surfaces (walls), expansion joints, drains, gravel stops, and other places where the membrane is interrupted or terminated. Base flashing covers the edges of the membrane. Cap or counterflashing shields the upper edges of the base flashing.

Flood coat: The top layer of bitumen used to hold the aggregate on an aggregate surfaced roofing membrane.

Fluid-applied elastomer: An elastomeric material, fluid at ambient temperature, that dries or cures after application to form a continuous membrane.

Galvalume: Trade name for steel coated with aluminum-zinc alloy for corrosion protection (55% AlZn).

Galvanized steel: Steel coated with zinc for corrosion resistance.

Granule: See Mineral Granules.

Gravel: Coarse, granular aggregate, with pieces larger than sand grains, resulting from the natural erosion of rock.

Gutter: A channel member installed at the eave of the roof for the purpose of carrying water from the roof to the drains or down spouts.

Humidity: The amount of moisture contained in the atmosphere. Generally expressed percent relative humidity. (The ratio of the vapor pressure to the saturation pressure for given conditions times 100.)

Hydrocarbons: An organic chemical compound containing mainly the elements carbon and hydrogen. Aliphatic hydrocarbons are straight chain compounds of carbon and hydrogen. Aromatic hydrocarbons are carbon-hydrogen compounds based on the cyclic or benzene ring. They may be gaseous (CH₄, ethylene, butadiene), liquid (hexene, benzene), or solid (Natural rubber, naphthalene, cispolybutadiene).

Hydrokinetic Metal Roof System: This is a metal roof system in which the metal standing seam, eaves and valleys accommodate expansion and are constructed in a way which allows some water infiltration. The infiltrating water is drained through weeps

in the roof assembly and with a watertight underlayment membrane under the roof. The fixity will typically be at the ridge in this roof system.

Hydrostatic Metal Roof System: The metal standing seam, laps at the eave and valley are construction with positive attachment and non-curing sealant to provide a watertight assembly in this metal roof system. Fixity will typically be at the eave and valleys in this roof system.

Inorganic (adj.): Comprising matter other than hydrocarbons and their derivatives, or matter not of plant or animal origin.

Insulation: See Thermal Insulation.

Lap: Dimension by which a felt covers an underlying felt in BUR membrane. “Edge” or side lap indicates the transverse cover; “End” lap indicates the cover at the end of the roll. These terms also apply to single-ply membranes.

Loose-laid Membrane: An unadhered roofing membrane anchored to the substrate only at the edges and penetrations through the roof and ballasted against wind uplift by loose aggregate or pavers.

Mastic: Caulking or sealant normally used in sealing roof panel laps.

Membrane: A flexible or semi-flexible roof covering or waterproofing whose primary function is the exclusion of water.

Metal flashing: See Flashing—frequently used as through-wall, cap, or counterflashing.

Mineral granules: Natural or synthetic aggregate, ranging in size from 500 μ m (1 μ m = 10⁻⁶m) to 1/4 in. diameter, used to surface BUR or modified bitumen cap sheets, asphalt shingles, and some cold process membranes.

Model Codes: Codes established to provide uniformity in regulations pertaining to building construction. Example: International Building Code published by the ICC.

Monomer: A simple molecule which is capable of combining with a number of like or unlike molecules to form a polymer.

Mopping: Application of hot bitumen with a mop or mechanical applicator to the substrate or to the plies of a built-up or modified-bitumen roof. There are four types of mopping: (1) solid—a continuous coating; (2) spot—bitumen is applied in roughly circular areas, generally about 460 mm (18 in.) in diameter, leaving a grid of unmopped, perpendicular area, (3) strip—bitumen is applied in parallel bands, generally 200 mm (8 in.) wide and 300 mm (12 in) apart; (4) sprinkle—bitumen is shaken on the substrate from a broom or mop in a random pattern.

Neoprene: Synthetic rubber (polychloroprene) used in liquid or sheet-applied elastomeric roofing membranes or flashing.

Organic (adj.): Composed of hydrocarbons or their derivatives; or matter of plant or animal origin.

Organic coating: Coatings that are generally inert or inhibited. May be temporary (for example slushing oils) or permanent (such as paints, varnishes and enamels).

Panel clip: Independent clip used to attach roof panels to substructure.

Parapet: Portion of wall above the roofline.

Penetration: The consistency of a bituminous material expressed as the distance in tenths of a millimeter (0.1 mm) that a standard needle or cone vertically penetrates a sample of material under specified conditions of loading, time, and temperature.

Perlite: An aggregate used in lightweight insulating concrete and in preformed perlite insulating board, formed by heating and expanding siliceous volcanic glass.

Phenolic Plastics: Plastics based on resins made by the condensation of phenols, such as phenol and cresol, with aldehydes.

Pitch: See Coal-tar pitch.

Plastic: A material that contains as an essential ingredient one or more organic polymeric substances of large molecular weight. It is solid in its finished state and at some stage in its manufacture or processing into finished articles can be shaped by flow.

Ply: A layer of felt in a roofing membrane; a four-ply membrane should have at least four plies of felt at any vertical cross section cut through the membrane.

Polyester Fiber: Generic name for a manufactured fiber in which the fiber-forming substance is any long chain synthetic polymer composed of an ester of a dihydric alcohol and terephthalic acid. Scrims made of polyester fiber are used for fabric reinforcement.

Polyisobutylene: The polymerization product of isobutylene varying in consistency from a viscous liquid to a rubberlike solid, with corresponding variation in molecular weight from 1,000 to 400,000.

Polyisocyanurate: Thermoset polymer formed by polymerization of isocyanate; rigid foam insulation meeting ASTM C 1289 *Standard Specification for Faced Rigid Cellular Polyisocyanurate Thermal Insulation Board*; a thermal insulation similar in appearance to polyurethane foam, but with improved fire resistance or rating.

Polymer: A macromolecular material formed by the chemical combination of monomers having either the same or different chemical composition. Plastics, rubbers, and textile fibers are all high molecular weight polymers.

Polyvinyl Chloride (PVC): A synthetic thermoplastic polymer prepared from vinyl chloride. PVC can be compounded into flexible and rigid forms through the use with plasticizers, stabilizers, filler, and other modifiers; rigid forms used in pipes; flexible forms used in manufacture of sheeting.

Ponding: Water in low or irregular roof areas that remains longer than 48 hours after the cessation of rainfall, under conditions conducive to evaporation.

Preventive Maintenance: The regular, scheduled, inspection for and the repair of normal, expected breakdown of materials and equipment.

Prime coat: First liquid coat applied in a multiple coat system.

Primer (bituminous): A thin liquid bitumen applied to a surface to improve the adhesion of heavier applications of bitumen and to absorb dust.

Protected Membrane Roof (PMR): Roof assembly with insulation on top of membrane instead of vice versa, as in conventional roof assembly (also known as inverted or upside-down roof assembly).

Puncture Resistance: Index of a material's ability to withstand the action of a sharp object without perforation.

R-Factor: Resistance to heat flow. The summation of individual thermal resistances in an assembly.

Rake: The sloped edge of a roof at the first or last rafter.

Re-covering: The process of covering an existing roof system with a new roof.

Reglet: A groove in a wall or other surface adjoining a roof surface for the insertion and attachment of counterflashing.

Relative Humidity: The ratio of the mass per unit volume (or partial pressure) of water vapor in an air-vapor mixture to the saturated mass per unit volume (or partial pressure) of the water vapor at the same temperature, expressed as a percentage.

Replacement: The removal of all roof system components down to the structural deck followed by installation of a completely new roofing system.

Reroofing: Replacement or re-cover of an existing roof system.

Retrofit: The modification of an existing building or facility to include new systems or components.

Ridge: Highest point on the roof of the building, a horizontal line running the length of the building.

Roof Curb: An accessory used to mount and level units (such as air conditioning and exhaust fans) on the sloped portion of the building roof.

Roof Slope: The angle a roof surface makes with the horizontal, measured in a ratio such as $\frac{1}{2}$:12.

ROOFER Sustainment Management System (SMS): The ROOFER SMS is a practical decision-making tool to help manage an installation's or base's low-slope membrane and asphalt shingle roofing assets. It includes procedures for collecting inventory and inspection information, evaluating roof condition, identifying repair/replacement strategies, prioritizing projects, and developing work plans. Micro ROOFER, software provides data storage and analysis, and generates management reports. Contact the Service-specific SME to gain access to ROOFER SMS.

Roofing System: A combination of interacting components designed to weatherproof, and normally to insulate, a building's or structure's top surface; does not include the roof deck.

Rubber: A material capable of quickly recovering from large deformations, normally insoluble in boiling solvent such as benzene, methyl ethyl ketone, and ethanol toluene azeotrope. A rubber in its modified state retracts within 1 mm to less than 1.5 times its original length after being stretched to twice its length.

Sealant: Any material used to close up cracks or joints to protect against leaks. Lap sealant is applied to exposed lap edges in E/P systems.

Sealing Washer: A metal-backed rubber washer assembled on a screw to prevent water from migrating through the screw hole.

Seam Strength: Strength of a seam of material measured either in shear or peel modes, reported either in absolute units, for example, pounds per inch of width—or as a percent of the sheeting strength.

Service Life: Anticipated useful life of a building, building component or building subsystem (for example, roof system).

Shingle: (1) A small unit of prepared roofing designed for installation with similar units on overlapping rows on inclines normally exceeding 25%; (2) To cover with shingles, and (3) To apply any sheet material in overlapping rows like shingles.

SI: The international symbol for the metric unit (Le Systeme International d'Unites).

Sill: The bottom horizontal framing member of an opening such as a window or door.

Skylight: A roof accessory to admit light, normally mounted on a curbed, framed opening.

Slope: Tangent of the angle between the roof surface and the horizontal plane, expressed as a ratio. (See also roof slope)

Soffit: The underside covering of any exterior overhanging section of a roof, gable or sidewall.

Square: A roof area of 100 ft² (9.29 m²), or enough material to cover 100 ft² (9.29 m²) of deck.

Stainless Steel: An alloy of steel which contains a high percentage of chromium. Also may contain nickel or copper. Has excellent resistance to corrosion.

Standing Seam: Watertight seam type featuring an upturned rib, which may also be structural. It is made by turning up the edges of two adjacent metal panels and then folding them over in one of a variety of ways.

Susceptibility: When not otherwise qualified, the degree of change in viscosity with temperature.

Thermal Insulation: A material designed to reduce the conductive heat flow.

Thermoplastic Elastomers: Polymers capable of remelt, but exhibiting elastomeric properties; related to elasticized polyolefins. They have a limited upper temperature service range.

U-Factor: The heat flow across an entire assembly; for example, from air within a building to outside air; the inverse of R-Factor. Also known as U-value.

Uplift: Wind load on a building which causes a load in the upward direction.

Vapor Barrier: See Vapor retarder.

Vapor Pressure: The pressure exerted by a vapor that is in equilibrium with its solid or liquid form.

Vapor Retarder: A material that resists the transmission of water vapor.

Vent: Opening designed to convey water vapor or other gas from inside a building or a building component to the atmosphere.

Viscosity: Index of a fluid's internal resistance to flow, measures in centistokes (cSt) for bitumens. (Water has a viscosity of roughly 1 cSt, light cooking oil 100 cSt.)

Waterproofing: Treatment of a surface or structure to prevent the passage of water under hydrostatic pressure.

APPENDIX B UFC 3-110-03 REFERENCES GOVERNMENT

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<https://federalregister.gov/presidential-documents/executive-orders>

29 CFR Part 1910, *Occupational Safety and Health Standards*; Subpart D *Walking-Working Surfaces* and Subpart I *Personal Protective Equipment*

<https://www.ecfr.gov/current/title-29/subtitle-B/chapter-XVII/part-1910>

DEPARTMENT OF DEFENSE

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www.acq.osd.mil/eie/imr/mc/Downloads/DoD_Facility_Inspection_Policy.pdf

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UFC 3-101-01, *Architecture*

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UFC 3-420-01, *Plumbing Systems*

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UFGS 07 18 16, *Cold Liquid-Applied Reinforced Polymethyl Methacrylate (PMMA) Roofing*

UFGS 07 22 00, *Roof and Deck Insulation*

UFGS 07 31 13, *Asphalt Shingles*

UFGS 07 31 26, *Slate Shingles*

UFGS 07 32 13, *Concrete Tile Roofing Replacement or Repair*

UFGS 07 32 14, *Clay Tile Roofing Replacement or Repair*

UFGS 07 41 13, *Metal Roof Panels*

UFGS 07 41 13.16, *Copper Roof Panels*

UFGS 07 41 63, *Fabricated Roof Panel Assemblies*

UFGS 07 42 13, *Metal Wall Panels*

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UFGS 07 51 13, *Built-Up Asphalt Roofing*

UFGS 07 52 00, *Modified Bituminous Membrane Roofing*

UFGS 07 53 23, *Ethylene Propylene Diene Monomer Roofing*

UFGS 07 54 19, *Polyvinyl Chloride (PVC) Roofing*

UFGS 07 54 23, *Thermoplastic Polyolefin (TPO) Roofing*

UFGS 07 55 00, *Protected Membrane Roofing (PMR)*

UFGS 07 56 00.60, *Mesh Reinforced Elastomeric Coating (MREC) Roofing*

UFGS 07 57 13, *Sprayed Polyurethane Foam (SPF) Roofing*

UFGS 07 60 00, *Flashing and Sheet Metal*

UFGS 07 61 14, *Steel Standing Seam Roofing*

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UFGS 07 62 13, *Copper Sheet Metal Flashing and Trim*

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www.erd.usace.army.mil/Locations/CRREL/

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EM 385-1-1, *Safety and Health Requirements Manual*

www.usace.army.mil/Missions/Safety-and-Occupational-Health/Safety-and-Health-Requirements-Manual/

NON-GOVERNMENT

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

www.astm.org/

ASTM A153/A153M, *Standard Specification for Zinc Coating (Hot Dip) on Iron and Steel Hardware*

ASTM A653/A653M, *Standard Specification for Steel Sheet, Zinc Coated (Galvanized) or Zinc-Iron Alloy Coated (Galvannealed) by the Hot Dip Process*

ASTM C406/C406M, *Standard Specification for Roofing Slate*

ASTM C1289, *Standard Specification for Faced Rigid Cellular Polyisocyanurate Thermal Insulation Board*

ASTM D4263, *Standard Test Method for Indicating Moisture in Concrete by the Plastic Sheet Method*

ASTM D4541, *Standard Test Method for Pull-off Strength of Coatings Using Portable Adhesion Testers*

ASTM D4586/D4586M, *Standard Specification for Asphalt Roof Cement, Asbestos-Free*

ASTM D5957, *Standard Guide for Flood Testing Horizontal Waterproofing Installations*

ASTM E1592, *Standard Test Method for Structural Performance of Sheet Metal Roof and Siding Systems by Uniform Static Air Pressure Difference*

ASTM E2140, *Standard Test Method for Water Penetration of Metal Roof Panel Systems by Static Water Pressure Head*

ASTM F2170, *Standard Test Method for Determining Relative Humidity in Concrete Floor Slabs Using in situ Probes*

AMERICAN SOCIETY OF CIVIL ENGINEERS (ASCE)

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ASCE 7, *Minimum Design Loads and Associated Criteria for Buildings and other Structures*

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS, INC., (ASHRAE)

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INTERNATIONAL CODE COUNCIL (ICC)

www.iccsafe.org

IBC, *International Building Code*

IEBC, *International Existing Building Code*

IgCC, *International Green Construction Code*

METAL BUILDING MANUFACTURERS ASSOCIATION (MBMA)

www.mbma.com

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NATIONAL ROOFING CONTRACTORS ASSOCIATION (NRCA)

www.nrca.net

NRCA Repair Manual for Low-Slope Membrane Roof Systems

NRCA Construction Details

NRCA Guidelines for Roof-mounted Photovoltaic Systems

NRCA Roofing Manual

NRCA Roofing and Waterproofing Manual

NRCA Vegetative Roof Systems Manual

NATIONAL SLATE ASSOCIATION

slateassociation.org/

Slate Roofs: Design and Installation Manual

SHEET METAL AND AIR CONDITIONING CONTRACTORS NATIONAL ASSOCIATION (SMACNA)

www.smacna.org

Architectural Sheet Metal Manual

SINGLE PLY ROOFING INDUSTRY (SPRI)

www.spri.org

ANSI/SPRI ES-1, Test Standard for Edge Systems Used with Low-slope Roofing Systems

ANSI/SPRI FX-1, Standard Field Test Procedure for Determining the Withdrawal Resistance of Roofing Fasteners

SPRAY POLYURETHANE FOAM ALLIANCE

www.sprayfoam.org

SPFA-127, Maintenance Manual for Spray Polyurethane Foam Roof Systems

UNDERWRITER'S LABORATORY

www.shopulstandards.com/

UL 1897, Standard for Uplift Tests for Roof Covering Systems

UL 2218, Impact Resistance of Prepared Roof Covering Materials/Standard Test Method for Determining Forming Limit Curves

APPENDIX B UFC 3-120-01 GLOSSARY

AASHTO	American Association of State Highway and Transportation Officials
ABAAS	Architectural Barriers Act Accessibility Standards for Defense Department Facilities
ACP/ECF	Access Control Point/Entry Control Facility
IBC	International Building Code
ICC	International Code Council
IPS	Installation Planning Standards
ISA	International Symbol of Accessibility
ITE	Institute of Transportation Engineers
MUTCD	Manual on Uniform Traffic Control Devices
NFPA	National Fire Protection Association
SDDCTEA	Military Surface Deployment and Distribution Command Transportation Engineering Agency
UFC	Unified Facilities Criteria
UFGS	Unified Facilities Guide Specifications
DoD	Department of Defense
FHWA	Federal Highway Administration
PMS	Pantone Matching System
USDOT	U.S. Department of Transportation
ABA	Architectural Barriers Act
mm	millimeter

APPENDIX B UFC 3-120-01 REFERENCES

AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS (AASHTO)

A Policy on Geometric Design of Highways and Streets (Green Book)

Roadside Design Guide

FEDERAL HIGHWAY ADMINISTRATION

Manual on Uniform Traffic Control Devices (MUTCD), <http://mutcd.fhwa.dot.gov/>

Standard Alphabets for Highway Signs and Pavement Markings

Standard Highway Signs, http://mutcd.fhwa.dot.gov/ser-shs_millennium.htm

INSTITUTE OF TRANSPORTATION ENGINEERS (ITE)

Traffic Control Devices Handbook

INTERNATIONAL CODE COUNCIL (ICC)

International Building Code (IBC)

ICC/ANSI 117.1, *Accessibility Standard*

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 101, *Life Safety Code*

SURFACE DEPLOYMENT AND DISTRIBUTION COMMAND TRANSPORTATION ENGINEERING AGENCY (SDDCTEA)

http://www.tea.army.mil/pubs/nr/DynaListDOD.asp?Cat_id=4&Sub_id=14&Topic_id=0&Cat_Name=DOD%20Programs%20for%20National%20Defense&Topic_Name=Pamphlets%2C%20Manuals%2C%20and%20Directories

DoD supplement to the MUTCD

Pamphlet 55-14, *Traffic Engineering for Better Signs and Markings*

Pamphlet 55-15, *Traffic Engineering for Better Entry Control Facilities*

Pamphlet 55-17, *Better Military Traffic Engineering*

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UFGS 01 58 00, *Project Identification*

UFGS 10 14 01, *Exterior Signage*

UFGS 10 14 00.20, *Interior Signage*

UNITED STATES ACCESS BOARD

<http://www.access-board.gov/guidelines-and-standards/buildings-and-sites/about-the-aba-standards/aba-standards>

Architectural Barriers Act Accessibility Standard for Department of Defense Facilities
(ABAAS)

APPENDIX A UFC 3-120-10 GLOSSARY

A-1 ACRONYMS.

AE	Architect/Engineer
A/E/C	Architecture/Engineering/Construction CAD Standard
AFCEC	Air Force Civil Engineer Center
AFCFS	Air Force Corporate Facilities Standards
AFNAFPO	Air Force Non-Appropriated Funds Purchasing Office
BIA	Bilateral Infrastructure Agreement
BPA	Blanket Purchase Agreement
BIM	Building Information Modeling
BOD	Beneficial Occupancy Date
BVD	Best Value Determination
CAD	Computer-Aided Design
CCD	Construction Completion Date
CCR	Criteria Change Request
CE	Cost Engineering
CFCI	Contractor Furnished Contractor Installed
CID	Comprehensive Interior Design
CIDA	Council for Interior Design Accreditation
DB	Design-Build
DBB	Design-Bid-Build
DD	Department of Defense Forms Management Program is administered by the Directives Division (DD) of the Executive Services Directorate
DoD	Department of Defense
FAR	Federal Acquisition Regulations
FF&E	Furniture, Fixtures and Equipment

FLC	Fleet Logistics Command
FSC	Federal Supply Category
FSRM	Facilities Sustainment, Restoration and Modernization
FY	Fiscal Year
GFCI	Government Furnished Contractor Installed
GFGI	Government Furnished Government Installed
GSA	General Services Administration
HNC	U.S. Army Engineering and Support Center Huntsville
HNFA	Host Nation Funded Construction Agreements
HVAC	Heating, Ventilation and Air-Conditioning
HQUSACE	Headquarters, U.S. Army Corps of Engineers
M3	Minimum Model Matrix
MILCON	Military Construction
MPI	Master Painter's Institute
NAF	Non-Appropriated Funds
NASA	National Aeronautics and Space Administration
NAVFAC	Naval Facilities Engineering Command
NFPA	National Fire Protection Association
NCIDQ	National Council for Interior Design Qualification
O&M	Operations and Maintenance
OMB	Office of Management and Budget
OSHA	Occupational Safety and Health Administration
P&D	Planning and Design
POC	Point of Contact
POE	Post Occupancy Evaluation

PTS	Performance Technical Specifications
PxP	Project Execution Plan
RFP	Request for Proposal
SID	Structural Interior Design
SIN	Special Item Number
SOFA	Status of Forces Agreement
TPC	Third Party Certification
UFC	Unified Facilities Criteria
UFGS	Unified Facilities Guide Specifications
UNICOR	Federal Prison Industries, Inc. (also known as FPI)
U.S.	United States
VE	Value Engineering
WBDG	Whole Building Design Guide

APPENDIX A UFC 3-120-10 REFERENCES

Note: The most recent edition of referenced publication applies, unless otherwise specified.

A/E/C CAD STANDARD

<https://cadbimcenter.erdcdren.mil/default.aspx?p=a&i=7&t=1>

CAD BIM TECHNOLOGY CENTER

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FF&E SOURCES OF SUPPLY

Ability One, <https://www.abilityone.gov/>

Federal Acquisition Regulations (FAR) PART 8, <https://www.acquisition.gov/?q=browsefar>

Federal Prison Industries, Inc. (UNICOR),
https://www.unicor.gov/Shopping/viewCat_m.asp?iStore=UNI

GSA Advantage, <https://www.gsaadvantage.gov>

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<https://www.gsaelibrary.gsa.gov/ElibMain/ScheduleList%3bjsessionid=www.gsaelibrary.gsa.gov-881f:403294b2:c56e211b4174c7c3>

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UNITED STATES DEPARTMENT OF THE AIR FORCE

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https://rfpwizard.mrsi.erdcdren.mil/MRSI/content/Furniture/furniture_home_page/Library/USACE%20Standard%20FFE%20Nomenclature_Feb%2024.pdf

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https://rfpwizard.mrsi.erdcdren.mil/MRSI/content/Furniture/furniture_home_page/Library/HNC%20Centralized%20Furnishings%20Program%20-%20Ancillary%20Buys_Feb%2024.pdf

HNC Furnishings Program, <https://mrsi.erdcdren.mil/furniture/>

UNITED STATES DEPARTMENT OF THE NAVY

NAVFAC Building Information Management and Modeling (BIM) & Facility Electronic Operation and Maintenance Support Information (eOMSI) Program, <https://www.wbdg.org/navy/bim-eomsi>

NAVFAC Collateral Equipment (FF&E) Criteria & Templates, <https://www.wbdg.org/navy/ffe-criteria-templates>

WHOLE BUILDING DESIGN GUIDE, A PROGRAM OF THE NATIONAL INSTITUTE OF BUILDING SCIENCES

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APPENDIX A UFC 3-130-01 GLOSSARY

A-1 INTRODUCTION.

The following acronyms and terms used in this document and are specific to cold regions engineering, design, and construction. ASTM D7099 also provides an excellent reference for these and additional terms.

Additionally, a glossary of permafrost and other terms related to ground ice are available in the Glossary of Permafrost and Related Ground-Ice Terms from IPA (<https://www.permafrost.org/publication/glossary-of-permafrost-and-related-ground-ice-terms/>).

A-2 ACRONYMS.

ACE	Air convection embankments
ACI	American Concrete Institute
AFCEC	Air Force Civil Engineer Center
AHJ	Authority Having Jurisdiction
ANSI	American National Standards Institute
ASTM	American Society for Testing and Materials
ASCE	American Society of Civil Engineers
ASHRAE	American Society of Heating, Refrigerating, and Air Conditioning Engineers
BIA	Bilateral Infrastructure Agreement
CGF	Cloud-Gap-Filled
CAC	Common Access Card
CATEX	Categorical Exclusion
CCR	Criteria Change Request
CIMP	Coupled Model Intercomparison Projects
CoCoRaHS	The Community Collaborative Rain, Hail, and Snow Network
COOP	Cooperative Observer Program
CRREL	Cold Regions Research and Engineering Laboratory
CX	Categorical Exclusion

°C	Degrees Celsius
°F	Degrees Fahrenheit
DEM	Digital elevation model
DoDI	Department of Defense Instructions
DoR	Designer of Record
DSM	Digital surface model
ERDC	Engineer Research and Development Center
EWD	Engineering Weather Data
FEMA	Federal Emergency Management Agency
ft	Feet
GHG	Greenhouse gas
GIS	Geographic information system
GPCP	Global Precipitation Climatology Project
GSOD	Global Surface Summary of Day
h	Hours
HEC-HMS	Hydrologic Engineering Center's Hydrologic Modeling System
HQUSACE	Headquarters, U.S. Army Corps of Engineers
HNFA	Host Nation Funded Construction Agreements
IBC	International Building Code
ICRP	Installation Climate Resilience Plan
in.	Inches
in. ²	Square inches
IPA	International Permafrost Association
IPCC	Intergovernmental Panel on Climate Change
ISO	International Organization for Standardization
kg	Kilograms

km	Kilometers
kPa	Kilopascal
km ²	Square kilometers
lb	Pounds
m	Meters
MERRA-2	Modern-Era Retrospective Analysis for Research and Applications, Ver. 2
mm	Millimeters
mph	Miles per hour
NAVFAC	Naval Facilities Engineering Command
NCEI	National Centers for Environmental Information
NEPA	National Environmental Policy Act
NFS	Non–frost susceptible
NOAA	National Oceanic and Atmospheric Administration
NOHRSC	National Operational Hydrologic Remote Sensing Center
NSIDC	National Snow and Ice Data Center
NWS	National Weather Service
OCDS	Operational Climatic Data Summary
OSHA	Occupational Safety and Health Administration
PCASE	Pavement-Transportation Computer Assisted Structural Engineering
PRISM	Parameter-Elevation Regressions on Independent Slopes Model
RCP	Representative Concentration Pathway
SCA	Snow-covered area
SLDT	Structural Load Data Tool
SNAP	Scenarios Network for Alaska and Arctic Planning
SNODAS	Snow Data Assimilation System
SNOTEL	Snowpack Telemetry

SOFA	Status of Forces Agreements
SWE	Snow water equivalent
TI	Temperature index
UFC	Unified Facilities Criteria
USDA	U.S. Department of Agriculture
USGS	U.S. Geological Survey
UAF	University of Alaska Fairbanks
WBAN	Weather Bureau Army Navy

A-3 DEFINITION OF TERMS.

A-3.1 General.

Installation: With a capital “I,” this refers to any DOD post, base, station, camp, fort, ranges or training areas.

A-3.2 Regions.

Arctic: The northern region in which the mean temperature for the warmest month is less than 50°F (10°C) and the mean annual temperature is below 32°F (0°C). In general, Arctic land areas coincide with the tundra region north of the limit of trees.

Subarctic: The region adjacent to the Arctic in which the mean temperature for the coldest month is below 32°F (0°C), the mean temperature for the warmest month is above 50°F (10°C), and there are less than four months having a mean temperature above 50°F (10°C). In general, Subarctic land areas coincide with the circumpolar belt of dominant coniferous forest.

Seasonal frost: Those areas of the earth having significant ground freezing during the winter but without development of permafrost. In seasonal frost areas, seasonal temperatures cause frost that affects earth materials and keep these frozen during only the winter; the occurrence of ground temperatures below 0°C appear only part of the year.

A-3.3 Soil and Frost Terms.

For common cold regions’ soil terminology, see ASTM D7099, *Standard Terminology Relating to Frozen Soil and Rock*.

Active layer: The top layer of ground subject to annual freezing and thawing with a permafrost layer below.

Aggradation: Progressive increase in the thickness of permafrost, taking place over a period of years due to a change in climate or terrain conditions.

Annual frost zone: The top layer of ground subject to annual freezing and thawing. In Arctic and Subarctic regions where annual freezing penetrates to the permafrost table, the active layer, suprapermfrost, and the annual frost zone are identical.

Closed system: A condition in which no source of free water is available during the freezing process beyond that originally contained in the voids of soil.

Creep: Extremely slow, continuous strain deformation of materials under stress, at rates so slow as to usually be imperceptible except by observations of high precision or of extended duration.

Degradation: Progressive lowering of the permafrost table, occurring over a period of years.

Excess ice: Ice in the ground that exceeds the total volume of the pores that the ground would have under natural unfrozen conditions.

Free water: That portion of the pore water that is free to move between interconnected pores under the influence of gravity.

Freeze–thaw cycle: A freezing followed by thawing of a material over time.

Freezing (of ground): The changing of phase from water to ice in soil or rock.

Freezing front: The advancing boundary between frozen (or partially frozen) and unfrozen ground.

Freezing point: The temperature at which a pure liquid solidifies under atmospheric pressure; the temperature at which a ground material starts to freeze.

Freezing-point depression: The number of degrees by which the freezing point of an earth material is depressed below 0°C (32°F).

Freezing pressure: The positive pressure developed at ice–water interfaces in a soil as it freezes.

Frost action: A general term for the alternate freezing and thawing of moisture in materials, such as soil. It also covers the effects on these materials and on structures of which they are a part of or with which they are in contact. The term “frost” is often used to refer to frost action in general.

Frost boil: The breaking of a limited section of a highway or airfield pavement under traffic and ejection of soft, semiliquid subgrade soil. This is caused by the melting of the segregated ice formed by frost action. May also occur on unpaved ground surfaces, resulting in a small soil mound.

Frost creep: The ratchet-like downslope movement of particles because of frost heaving and subsequent ground settling on thawing. The heaving is normal to the slope and the settling more vertical.

Frost heave: The upward or outward movement of a ground surface (or objects on or in the ground) due to ice formation in the underlying soil. As the freeze front progresses downward, moisture migrates to the front, producing accelerated expansion and heaving.

Frost penetration: The movement of the freezing front into the ground during freezing.

Frost slough: A shallow slide that occurs when the stability of frost-loosened and moisture-saturated fine-grained soils on slopes is reduced during thaw.

Frost-stable ground: Soil or rock in which little or no segregated ice forms during seasonal freezing.

Frost-susceptible soil: Soil that will experience significant ice segregation when the requisite moisture and freezing conditions are present.

Frozen fringe: The zone in a freezing, frost-susceptible soil between the warmest isotherm at which ice exists in pores and the isotherm at which the warmest ice lens is growing.

Frost table: The surface, usually irregular, that represents the level, at any time in spring and summer, to which thawing of seasonal frozen ground has penetrated.

Frost thrust: A force due to frost action.

Frozen zone: A range of depth within which the soil is frozen.

Ground ice: A body of soil-free ice within frozen ground, including ice in pores, cavities, voids, or other openings in soil or rock, including massive ice; a general term referring to all types of ice in freezing and frozen ground.

Ground settlement: The downward movement of the ground, lowering the ground surface, resulting from the melting of ground ice more than that contained in pore fillings.

Hard-frozen ground: Frozen soil or rock which is firmly cemented by ice.

Heat capacity: The amount of heat required to raise the temperature of a unit mass of a substance by one degree. It is commonly expressed in Joules per kilogram per degree Kelvin.

Heterogeneously frozen soil: A soil with part of its water frozen as macroscopic ice occupying space more than the original voids in the soil.

Homogeneously frozen soil: A soil in which water is frozen within the material voids without macroscopic segregation of ice.

Ice lens: A lens-shaped body of ice ranging in thickness from hairline to 0.3 m. Ice layers more than 0.3 m in thickness are better termed massive ice beds.

Ice lenses: Lenticular formations of ice in soil occurring parallel to each other, normal to the direction of heat loss, and commonly in repeated layers.

Ice-rich permafrost: Perennially frozen ground that contains ice more than that required to fill pore spaces.

Intrusive ice: Ice formed from water intruded or injected under pressure into soils and rocks.

Ice segregation: The growth of ice within soil more than the amount produced by in-place conversion of the original void moisture to ice. Ice segregation occurs most often as distinct lenses, layers, veins, and masses, commonly, but not always, oriented normal to the direction of heat loss.

Ice wedge: A wedge-shaped ice mass, usually with its apex pointing downwards, in permafrost. Usually associated with fissures on trough-type polygons.

Ice wedge polygon: Any polygon surrounded by troughs underlain by ice wedges.

Latent heat of fusion: The amount of heat required to melt all the ice (or freeze all the pore water) in a unit mass of soil or rock.

Non-frost-susceptible materials: Cohesionless materials such as crushed rock, gravel, sand, slag, and cinders in which there is no significant ice segregation under normal freezing conditions.

Normal period: The time of the year when there is no alteration in strength of foundation materials because of frost action. In seasonal frost areas, it extends from mid- or late spring to mid- or late fall.

Open system: A condition where more free water than originally contained in the voids of the soil is available to move within the soil and to form segregated ice in frost-susceptible soil.

Percent heave: The ratio, expressed as a percentage, of the amount of heave to the depth of frozen soil.

Permafrost: Perennially frozen ground. A thermal condition in soil or rock where temperatures below 32°F persist over at least two consecutive winters and the intervening summer; moisture in the form of water and ground ice may or may not be present.

Permafrost base: The lower boundary surface of permafrost, above which temperatures are below 32°F, and below which temperatures are above 32°F.

Permafrost, continuous: Permafrost occurring everywhere beneath the exposed land surface throughout a geographic region, except for widely scattered locations.

Permafrost, discontinuous: Permafrost occurring in some areas beneath the ground surface throughout a geographic region where other areas are free of permafrost.

Permafrost table: An irregular surface within the ground that represents the upper boundary surface of permafrost.

Residual thaw layer: A layer of thawed or unfrozen ground between the permafrost and the annual frost zone. This layer does not exist where annual frost (active layer) extends to permafrost.

Seasonally active permafrost: The uppermost layer of the permafrost that undergoes seasonal phase changes due to the lowered thawing temperature and freezing-point depression of its pore water.

Segregated ice: Ice formed by the migration of pore water to the freezing plane, where it forms into discreet lenses, layers, or seams, ranging in thickness from hairline to greater than 10 m (32 ft). The ice is formed by ice segregation.

Snowdrift: An accumulation of wind-blown snow, often much thicker than the surrounding snow cover.

Snowline: The boundary of a highland region in which snow never melts.

Snowmelt: Melting of the snow cover; the period during which the melting of the snow cover occurs at the end of winter.

Solifluction: The perceptible, slow downslope flow of saturated unfrozen soil over a base of impervious or frozen material. Movement takes place primarily when melting of segregated ice or infiltration of surface runoff concentrates excess water in the surface soil, which then behaves like a viscous fluid. Solifluction features lobes, stripes, sheets, and terraces.

Suprapermafrost: The entire layer of ground above the permafrost table.

Sporadic permafrost: A subzone of the zone of discontinuous permafrost: In North American usage, it is permafrost underlying less than 30% of the exposed land surface; in Russian usage, it is permafrost underlying from 3% to 20% of the exposed land surface.

Tangential adfreeze shear: Tangential shear between frozen ground or ice and another material to which it is bonded by freezing.

Thaw consolidation: The process by which a reduction in volume and increase in density of a soil mass occurs, following thaw, in response to the escape of water under the weight of the soil itself, an applied load, or both. Thaw consolidation may proceed for many years.

Thaw-sensitive permafrost: Perennially frozen ground that, on thawing, will experience significant thaw settlement and suffer loss of strength to a value significantly lower than that of a similar material in an unfrozen condition.

Thaw settlement: The differential downward movement of the ground surface resulting from the escape of water on melting of excess ice in the soil and the thaw consolidation of the soil mass.

Thaw slumping: A type of mass movement caused by the conversion of ice into water in a soil by ground thaw, creating the kind of landslide that most closely resembles the more temperate climate earth flow with a well-developed breakaway scarp front.

Thaw-stable frozen soils: Frozen soils that do not, on thawing, show loss of strength below normal, long-time thawed values nor produce detrimental settlement.

Thaw-unstable frozen soils: Frozen soils that show, on thawing, significant loss of strength below normal, long-time thawed values or significant settlement as a direct result of the melting of excess ice in the soil.

A-3.3.1 Temperature-related Terms.

Average annual temperature: The average of the average daily temperatures for a particular year.

Average daily temperature: The average of the maximum and minimum temperatures for one day or the average of several temperature readings taken at equal time intervals for one day, hourly.

Average monthly temperature: The average of the average daily temperatures for a particular month.

Breakup period: The period of the spring thaw during which the ground surface is excessively wet and soft and ice is disappearing from streams and lakes. Duration of the breakup period usually varies usually from 1 to 6 weeks, depending on regions or local climatic conditions.

Degree-days: The degree-days for any one day equal the difference between the average daily air temperature and 32°F. The degree-days are negative when the average daily temperature is below 32°F (freezing degree-days) and positive when above 32°F (thawing degree-days). Degree-days may be computed in either Fahrenheit or Celsius units; this manual uses Fahrenheit degree-days.

Design freezing index: For design of permanent pavements, the design freezing index is the average air freezing index of the three coldest winters in the latest 30 years of record. If 30 years of record are not available, substitute the air freezing index for the coldest winter in a 10-year period. For design of foundations for average permanent structures, compute the design freezing index for the coldest winter in 30 years of record, or estimate to correspond with this frequency if the number of years of record is limited. Use the latest available periods of record. To avoid the necessity for adopting a new and only slightly different freezing index each year, the design index at a site with continuing construction need not change more than once in five years, unless the more recent temperature records indicate a significant change.

Design thawing index: The design thawing index is computed on the same frequency and basis as the design freezing index, except that summer thaw conditions are used.

Freeze-up period: The period during which the ground surface freezes and an ice cover is forming on streams and lakes. The duration of the freeze-up period varies from 1 to 3 months, depending on regional or local climatic conditions.

Freezing index: The number of degree-days between the highest and lowest points on a curve of cumulative degree-days versus time for one freezing season. It is a measure of the combined duration and magnitude of below-freezing temperatures occurring during any given freezing season. The index determined for air temperatures at

approximately 4.5 ft above the ground surface is commonly designated as the air freezing index, while that determined for temperatures at, or immediately below, the ground surface is known as the surface freezing index.

Freezing season: That period during which the average daily temperature is below 32°F.

Frost-melting period: An interval of the year during which ice in the ground is returning to a liquid state. It ends when all the ice in the ground melts or when freezing starts again. Although in the generalized case there is only one frost-melting period, beginning during the general rise of the air temperatures in the spring, one or more significant frost-melting intervals may take place during a winter season.

Geothermal gradient: The temperature gradient in the ground below the zone of annual temperature fluctuations, produced by the continuous flow of heat from the Earth's hot interior toward the Earth's cool surface.

Mean annual temperature: The average value taken from several years of annual temperature averages.

Mean daily temperatures: The average taken from the averages of the daily temperatures for a given day for several years.

Mean freezing index: The freezing index determined from mean temperatures.

Mean monthly temperature: The average taken from the averages of the monthly temperatures for a given month for several years.

Mean thawing index: The thawing index determined from mean temperatures.

Period of weakening: An interval of the year that starts at the beginning of the frost-melting period and ends when the subgrade strength has returned to normal-period values or when freezing starts again. In seasonal frost areas, the period of weakening may be longer than the frost-melting period; but in permafrost areas, the periods coincide.

Strudel scour: An Arctic phenomenon that occurs every spring when the fresh-water rivers melt and form off-river deltas when river discharge drains through holes in the sea ice cover. Strudel scours constitute a major design consideration for subsea pipelines or other structures near rivers.

Thawing index: The number of degree-days between the lowest and highest points on a curve of cumulative degree-days versus time for one thawing season. It is a measure of the combined duration and magnitude of above-freezing temperatures during any given thawing season. The index determined for air temperatures at 4.5 ft above the ground is commonly designated as the air thawing index, while the temperature at or immediately below the ground surface is known as the surface thawing index.

Thermal regime: The pattern of temperature variations found in the ground with time and at a certain depth from the surface.

Thermal erosion: The erosion of ice-rich permafrost by the combined thermal and mechanical action of moving water or air (sublimation).

Thermal expansion (or contraction) coefficient: The volume change per unit volume of a substance due to a one-degree change in its temperature.

Thermal pile(s): Structural piling modified to passively remove heat from the ground whenever the ambient air temperature is lower than the ground temperature.

Total annual freezing index: The cumulative number of degree-days, calculated by adding all the negative mean daily air temperatures (in degrees Celsius) for a specific station during a calendar year.

Total annual thawing index: The cumulative number of degree-days, calculated by adding all the positive mean daily air temperatures (in degrees Celsius) for a specific station during a calendar year.

Wind chill: The excess rate of removal of body heat from exposed skin by moving air compared to still air at low temperatures. It is often expressed as a lower equivalent air temperature that is a function of actual air temperature and wind speed.

A-3.4 Terrain Terms.

Frost mound: A localized mound-shape on the land surface caused by frost action with or without hydrostatic pressure.

Icing: A surface ice mass formed by freezing of successive sheets of water from seepage, flow from a spring, or emergence from below river or lake ice through fractures.

Muskeg: Poorly drained organic terrain consisting of a mat of living vegetation overlying an extremely compressible mixture of partially decomposed peat, varying in thickness from a few inches to many feet.

Patterned ground: A general term describing ground patterns that result from frost action, such as polygons, circles and nets, stripes, and solifluction features.

Pingo (hydrolaccolith): A large ice-cored frost mound, often 100 ft high or more, consisting of a core of massive ice, produced primarily by injection of water and covered with soil and vegetation.

Strudel scour: A localized, seasonal scour formation that occurs in the spring when melting fresh water in rivers and streams flows over the surface and drains through holes of frozen shore-fast ice.

Thermokarst: The irregular topography resulting from differential thaw settlement or caving of the ground because of excess ice melting in thaw-unstable permafrost; the process by which characteristic landforms result from the thawing of ice-rich permafrost.

Tundra: Treeless terrain of grasses and shrubs characteristic of the Arctic, found at both high latitudes and high altitudes.

APPENDIX B UFC 3-130-01 SUPPLEMENTAL RESOURCES

The following references are reliable sources for information related to infrastructure planning, design, and construction in the Arctic and Subarctic. These sources provide direction for specific applications not addressed in this UFC or provide additional information to guide or aid the designer in the various phases of the design. This list is provided for the convenience of the designer and may not include references for all specific applications relevant to all projects. The designer is responsible for ensuring the design conforms to all criteria relevant to the project.

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APPENDIX A UFC 3-130-04 GLOSSARY

A-1 ACRONYMS.

AFCEC	Air Force Civil Engineer Center
AHJ	Authority Having Jurisdiction
AHU	Air-handling unit
ACI	American Concrete Institute
ASCE	American Society of Civil Engineers
ASHRAE	American Society of Heating, Refrigerating and Air-Conditioning Engineers
BIA	Bilateral Infrastructure Agreements
cm	Centimeter
CCR	Criteria Change Request
CDC	Child development center
CRREL	Cold Regions Research and Engineering Laboratory
°C	Degrees Celsius
°F	Degrees Fahrenheit
DoR	Designer of Record
ft	Feet
FPE	Fire Protection Engineer
HNFA	Host Nation Funded Construction Agreements
HQUSACE	Headquarters, U.S. Army Corps of Engineers
in.	Inches
m	Meter
mm	Millimeters
NAVFAC	Naval Facilities Engineering Systems Command
NFPA	National Fire Protection Association

SLDT	Structural Load Data Tool
SOFA	Status of Forces Agreements
TRB	Transportation Research Board
UFC	Unified Facilities Criteria

A-2 DEFINITION OF TERMS.

UFC 3-130-01, Appendix B, provides a list of general terms for Arctic and Subarctic engineering. This appendix includes terms specific to UFC 3-130-04.

Aerodynamic shade: An area protected from wind and strong air movement. An area of relatively calm conditions conducive to the settling of snow.

Arctic entrance: A type of vestibule. The exterior door of the vestibule is closed to shut out the cold air, high wind, or drifting snow before the inside door is opened to enter the building.

Downwind: In the direction that the wind is blowing.

Ice dam: An ice buildup on building eaves that occurs when snow melted by escaping building heat flows to the eaves and freezes when air temperature drops below 32°F (0°C).

Ice fog: Results when a dew point air temperature below 32°F (0°C) meets an available supply of moisture. Ice fog can seriously impair visibility for extended periods.

R-value: A material's ability to resist heat flow. Otherwise known as *thermal resistance*.

Snow fences: Lath-type fences designed to stop windblown snow and protect a downwind facility from snowdrifts.

Snow transport (Q): The mass of snow transported by the wind over a specified time and width across the wind within the first 16 ft (5 m) above the surface, per meter of width across the wind.

Thermal break: An area of low thermal conductivity between two areas of higher thermal conductivity. The opposite of a thermal bridge.

Thermal bridge: An area of high thermal conductivity between two areas of high thermal conductivity. An example is a through-metal connection in a building envelope component.

Upwind: In the direction from which the wind is blowing.

Whiteout: An optical phenomenon occurring in the Arctic and Subarctic regions in which the snow-covered ground blends with a uniformly white sky, blotting out shadows, clouds, horizon, and so on, and destroying all sense of depth, direction, or distance.

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APPENDIX A UFC 3-130-05 GLOSSARY

A-1 ACRONYMS

ADEC	Alaska Department of Environmental Conservation
AFB	Air Force Base
AFCEC	Air Force Civil Engineer Center
ASCE	American Society of Civil Engineers
BIA	Bilateral Infrastructure Agreements
Btu	British thermal unit
C	Celsius
cm	Centimeter
CCR	Criteria Change Request
DoR	Designer of Record
DDC	Direct Digital Control
F	Fahrenheit
ft	Feet
FGS	Final Governing Standards
GNWT	Government of the Northwest Territories
HQUSACE	Headquarters, US Army Corps of Engineers
HDPE	High Density Polyethylene
HNFA	Host Nation Funded Construction Agreements
hr	Hour
IFS	Installation Facility Standard
kPa	Kilopascal
m	Meter
mph	Miles per hour
NAVFAC	Naval Facilities Engineering Systems Command

OEBGD	Overseas Environmental Baseline Guidance Document
POL	Petroleum, oil, and lubricant
PVC	Polyvinyl Chloride
pcf	Pounds per cubic ft
psi	Pound per square inch
s	Seconds
SOFA	Status of Forces Agreements
UFC	Unified Facilities Criteria
W	Watt
WBDG	Whole Building Design Guide

APPENDIX B UFC 3-130-05 REFERENCES

AMERICAN SOCIETY OF CIVIL ENGINEERS

<https://ascelibrary.org>

Cold Regions Utilities Monograph, <https://doi.org/10.1061/9780784401927>

AMERICAN SOCIETY OF HEATING, REFRIGERATION AND AIR-CONDITIONING ENGINEERS

<https://www.ashrae.org/>

District Heating Guide

Guide for Resilient Thermal Energy Systems Design in Cold and Arctic Climates, A. Zhivov (ed.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

<https://www.astm.org>

ASTM F2620, *Standard Practice for Heat Fusion Joining of Polyethylene Pipe and Fittings*

AMERICAN WATER WORKS ASSOCIATION

<https://www.awwa.org/>

AWWA C502, *Dry-Barrel Fire Hydrants*

INTERNATIONAL CODE COUNCIL

<http://www.iccsafe.org/>

International Building Code

STATE OF ALASKA

<https://dec.alaska.gov/commish/regulations/>

Standard 18 AAC 72, *Wastewater Treatment and Disposal*, ADEC

Standard 18 AAC 74, *Water and Wastewater Operator Certification and Training*, ADEC

Standard 18 AAC 80, *Drinking Water*, ADEC

UNIFIED FACILITIES CRITERIA

<https://www.wbdg.org/dod/ufc>

Consult active UFCs for all aspects of design, including but not limited to:

UFC 1-200-01, *DoD Building Code*

UFC 1-200-02, *High Performance and Sustainable Building Requirements*

UFC 2-100-01, *Installation Master Planning*

UFC 3-130-01, *Arctic and Subarctic Engineering*

UFC 3-130-02, *Arctic and Subarctic Site Assessment and Selection*

UFC 3-130-03, *Arctic and Subarctic Foundations for Freezing and Thawing Conditions*

UFC 3-130-04, *Arctic and Subarctic Buildings*

UFC 3-201-01, *Civil Engineering*

UFC 3-230-01, *Water Storage, Distribution, and Treatment*

UFC 3-230-02, *Operation and Maintenance: Water Supply Systems*

UFC 3-240-01, *Wastewater Collection and Treatment*

UFC 3-410-02, *Direct Digital Control for HVAC and Other Building Control Systems*

UFC 3-430-05, *Natural Gas and Liquefied Petroleum Gas (LPG) Distribution Pipelines*

UFC 3-501-01, *Electrical Engineering*

UFC 3-550-01, *Exterior Electrical Power Distribution*

UFC 3-600-01, *Fire Protection Engineering for Facilities*

UFC 3-410-01, *Heating, Ventilating, and Air Conditioning Systems*

UFC 4-010-06, *Cybersecurity of Facility-Related Control Systems*

UNIFIED FACILITIES GUIDE SPECIFICATIONS

<https://www.wbdg.org/dod/ufgs>

UFGS 23 07 00, *Thermal Insulation for Mechanical Systems*

UFGS 33 11 00, *Water Utility Distribution Piping*

UFGS 33 30 00, *Sanitary Sewerage*

UNITED STATES GEOLOGICAL SERVICE

<https://waterdata.usgs.gov/nwis>

USGS, *Water Data for the Nation*

OTHER

“Calculation of Heat Loss from Pipes,” *Utilities Delivery in Arctic Regions*, Report No. EPS 3-WP-77-1, D.E. Thornton, Environmental Protection Service, Environment Canada, Edmonton, Alberta, 1977

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Electrical Grounding in Cold Regions, Cold Regions Technical Digest No. 87-1, K. Henry, U.S. Army Cold Regions Research and Engineering Laboratory, Hanover, NH, 1987

“Environmental Engineering Failures in Alaska,” *ASCE Cold Regions Engineering 2009. The 14th Conference on Cold Regions Engineering*, D.H. Schubert J.A. Crum, J. Olofsson, G.V. Jones, L.A. Woolard, and A. Ronimus, Duluth, MN, 31 Aug–2 Sep 2009

Fundamentals of Heat Transfer, S.S. Kutateladze, Harper & Row, New York, NY, 1963.

Geologic Hazards of the Fairbanks Area: Alaska Division of Geological & Geophysical Surveys Special Report 15, T.L. Péwé, 1982

Losses from the Fort Wainwright Heat Distribution System, Special Report 81-14, G.L. Phetteplace, W. Willey, and M.A. Novick, U.S. Army Cold Regions Research and Engineering Laboratory, Hanover, NH, 1981

Prevention of Freezing and Other Cold Weather Problems at Wastewater Treatment Facilities, Special Report 85-11, S.C. Reed, D.S. Pottle, W.B. Moeller, C.R. Ott, R. Peirent, and E.L. Niedringhaus, U.S. Army Cold Regions Research and Engineering Laboratory, Hanover, NH, 1985

Two-Dimensional Analysis of Natural Convection and Radiation in Utilidors, CRREL Report 99-7, P. Richmond, U.S. Army Cold Regions Research and Engineering Laboratory, Hanover, NH, 1999

Underground Utility Lines, Technical Translation TT-1221, G.V. Prokhaer, National Research Council, Ottawa, Ontario, 1959

Water Supply Systems in Frozen Ground, L.R. Janson, in *Proceedings International Permafrost Conference*, 403–433, National Academy of Science, Washington, D.C., 1966

APPENDIX A UFC 3-190-06 GLOSSARY

A-1 ACRONYMS

AFCEC	Air Force Civil Engineer Center
ASHRAE	American Society of Heating, Refrigerating and Air-Conditioning Engineers
ASTM	ASTM International
BIA	Bilateral Infrastructure Agreement
CCR	Criteria Change Request
CCS	Coating Condition Survey
CQC	Contractor Quality Control
DoD	Department of Defense
EPA	United States Environmental Protection Agency
ESC	Environmental Severity Classification
FAA	Federal Aviation Administration
HQUSACE	Headquarters, U.S. Army Corps of Engineers
HNFA	Host Nation Funded Construction Agreements
L	Liter
MPa	Megapascal
MPI	Master Painters Institute
NACE	NACE International – The Corrosion Society
NAVFAC	Naval Facilities Engineering Systems Command
NAVSEA	Naval Sea Systems Command
NICAAP	NACE International Institute Contractor Accreditation Program
OSHA	United States Occupational Safety and Health Administration
PCCP	SSPC Painting Contractor Certification Programs
PDS	Product Data Sheet

psi	Pounds per square inch
SDS	Safety Data Sheet
SOFA	Status of Forces Agreements
SSPC	SSPC – The Society for Protective Coatings
TDS	Technical Data Sheet
TSP	Tri Sodium Phosphate
UFC	Unified Facilities Criteria
UFGS	Unified Facilities Guide Specification
U.S.	United States
UV	Ultraviolet
VOC	Volatile Organic Compounds

A-2 DEFINITION OF TERMS

Coating: (1) A liquid, liquefiable, or mastic composition that is converted to a solid protective, decorative, or functional adherent film after application as a thin layer [ASTM D 16]; (2) Generic term for paint, lacquer, and enamel (from SSPC Glossary)

Corrosion: The deterioration of a material or its properties due to a reaction of that material with its chemical environment. (from 10 USC §2228(f))

Corrosion prone locations: Locations with one or more of the following characteristics (1) Exterior exposed metallic elements at a location with an ESC of C3, C4, or C5, (2) Exterior exposed nonmetallic elements at a location with an ESC of C4 or C5, (3) Locations where microenvironmental factors (for example, prevailing winds, ventilation, waterfront environments, industrial emissions, deicing salt application, possible chemical splash/spillage, adverse weather events such as flooding or wind-driven rain, and penetrations of the building envelope) may create a locally corrosive environment regardless of ESC, (4) Humid locations identified in ANSI/ASHRAE/IES 90.1 as climate zones 0A, 1A, 2A, 3A, 3C, 4C, and 5C, and (5) High humidity interior areas (for example, bathrooms, locker rooms, laundry rooms, pools, and trainers).

Environmental Severity. Describes the corrosivity of the local environment of a given location or region.

Environmental Severity Classification (ESC): A classification of geographical macro environments developed by DoD based on the International Standards Organization Corrosivity Classification Method. Additional details are available at <https://www.wbdg.org/ffc/dod/cpc-source/environmental-severity-classification>.

Paint: (1) Any pigmented liquid, liquefiable, or mastic composition designed for application to a substrate in a thin layer that is converted to an opaque solid film after application. Used for protection, decoration, identification, or to serve some other functional purposes; (2) Application of a coating material. (from SSPC Glossary)

APPENDIX B UFC 3-190-06 REFERENCES

ASHRAE

<https://www.ashrae.org/>

ANSI/ASHRAE/IES Standard 90.1: Energy Standard for Buildings Except Low-Rise Residential Buildings

ASTM INTERNATIONAL

<https://www.astm.org/>

ASTM C881/C881M-20, *Standard Specification for Epoxy-Resin-Base Bonding Systems for Concrete*

ASTM D3274-09(2017), *Standard Test Method for Evaluating Degree of Surface Disfigurement of Paint Films by Fungal or Algal Growth, or Soil and Dirt Accumulation*

ASTM D4214-07(2015), *Standard Test Methods for Evaluating the Degree of Chalking of Exterior Paint Films*

ASTM D4258-05(2017), *Standard Practice for Surface Cleaning Concrete for Coating*

ASTM D4259-18, *Standard Practice for Preparation of Concrete by Abrasion Prior to Coating Application*

ASTM D4260-05(2017), *Standard Practice for Liquid and Gelled Acid Etching of Concrete*

ASTM D4261-05(2018), *Standard Practice for Surface Cleaning Concrete Masonry Units for Coating*

ASTM D4263-83(2018), *Standard Test Method for Indicating Moisture in Concrete by the Plastic Sheet Method*

ASTM F1869-16a, *Standard Test Method for Measuring Moisture Vapor Emission Rate of Concrete Subfloor Using Anhydrous Calcium Chloride*

ENVIRONMENTAL PROTECTION AGENCY

<https://www.epa.gov/>

EPA 402-K-01-001, *Mold Remediation in Schools and Commercial Buildings*

FEDERAL AVIATION ADMINISTRATION

<https://www.faa.gov/>

FAA Advisory Circular 70/7460-1L, *Obstruction Marking and Lighting*

NACE INTERNATIONAL

<https://www.nace.org>

NACE SP0178, *Design, Fabrication, and Surface Finish Practices for Tanks and Vessels to Be Lined for Immersion Service*

Joint SSPC/NACE standards are listed under the SSPC heading

NAVSEA

<https://www.navsea.navy.mil/>

NAVSEA Standard Item 009-32, *Cleaning and Painting Requirements; accomplish*

U.S. AIR FORCE

Air Force Manual (AFMAN) 91-203, *Air Force Occupational Safety, Fire, and Health Standards*

U.S. ARMY

EM 385-1-1, *Safety and Health Requirements*

SSPC – THE SOCIETY FOR PROTECTIVE COATINGS

www.sspc.org

SSPC-ACS 1/NACE No. 13, *Industrial Coating and Lining Application Specialist Qualification and Certification*

SSPC-Guide 6, *Guide for Containing Surface Preparation Debris Generated during Paint Removal Operations*

SSPC-Guide 7, *Guide to the Disposal of Lead-Contaminated Surface Preparation Debris*

SSPC-Guide 11, *Protecting Edges, Crevices, and Irregular Steel Surfaces by Stripe Coating*

SSPC-SP 1, *Solvent Cleaning*

SSPC-SP 2, *Hand Tool Cleaning*

SSPC-SP 3, *Power Tool Cleaning*

SSPC-SP 5/NACE No. 1, *White Metal Blast Cleaning*

SSPC-SP 6/NACE No. 3, *Commercial Blast Cleaning*

SSPC-SP 7/NACE No. 4, *Brush-Off Blast Cleaning*

SSPC-SP 10/NACE No. 2, *Near White Metal Blast Cleaning*

SSPC-SP 11, *Power Tool Cleaning to Bare Metal*

SSPC-SP 13/NACE No. 6, *Surface Preparation of Concrete*

SSPC-SP 15, *Commercial Grade Power Tool Cleaning*

SSPC-SP 16, *Brush-Off Blast Cleaning of Coated and Uncoated Galvanized Steel, Stainless Steels, and Non-Ferrous Metals*

SSPC-SP 17, *Thorough Abrasive Blast Cleaning of Non-Ferrous Metals*

SSPC-SP WJ-1/NACE WJ-1, *Waterjet Cleaning of Metals -- Clean to Bare Substrate*

SSPC-SP WJ-2/NACE WJ-2, *Waterjet Cleaning of Metals -- Very Thorough Cleaning*

SSPC-SP WJ-3/NACE WJ-3, *Waterjet Cleaning of Metals -- Thorough Cleaning*

SSPC-SP WJ-4/NACE WJ-4, *Waterjet Cleaning of Metals -- Light Cleaning*

SSPC-SP 5 (WAB)/NACE WAB-1, *White Metal Wet Abrasive Blast Cleaning*

SSPC-SP 6 (WAB)/NACE WAB-3, *Commercial Wet Abrasive Blast Cleaning*

SSPC-SP 7 (WAB)/NACE WAB-4, *Brush-Off Wet Abrasive Blast Cleaning*

SSPC-SP 10 (WAB)/NACE WAB-2, *Near White Metal Wet Abrasive Blast Cleaning*

SSPC-SP CAB-1, *Thorough Blast Cleaning of Concrete*

SSPC-SP CAB-2, *Intermediate Blast Cleaning of Concrete*

SSPC-SP CAB-3, *Brush Blast Cleaning of Concrete*

SSPC-TU 3, *Overcoating: Existing Coatings Systems Applied to Steel Substrates*

SSPC-TU 6, *Chemical Stripping of Organic Coatings from Steel Structures*

UNIFIED FACILITIES CRITERIA

<http://www.wbdg.org/ffc/dod/unified-facilities-criteria-ufc>

UFC 1-200-01, *DoD Building Code (General Building Requirements)*

UFC 1-200-02, *High Performance and Sustainable Building Requirements*

UFC 3-230-01, *Water Storage and Distribution*

UFC 3-460-01, *Design: Petroleum Fuel Facilities*

UFD 5-570-01, *Cathodic Protection*

UNIFIED FACILITIES GUIDE SPECIFICATIONS

<https://www.wbdg.org/ffc/dod/unified-facilities-guide-specifications-ufgs>

UFGS 02 83 00, *Lead Remediation*

UFGS 09 90 00, *Paints and Coatings*

UFGS 09 96 00, *High-Performance Coatings*

UFGS 09 67 23.15, *Fuel Resistive Resinous Flooring, 3-Coat System*

UFGS 09 67 23.16, *Fuel Resistive Resinous Flooring, 5-Coat System*

UFGS 09 96 59, *High-Build Glaze Coatings*

UFGS 09 97 02, *Painting: Hydraulic Structures*

UFGS 09 97 13.00 40, *Steel Coatings*

UFGS 09 97 13.15, *Low VOC Polysulfide Interior Coating of Welded Steel Petroleum Fuel Tanks*

UFGS 09 97 13.16, *Interior Coating of Welded Steel Water Tanks*

UFGS 09 97 13.17, *Three Coat Epoxy Interior Coating of Welded Steel Petroleum Fuel Tanks*

UFGS 09 97 13.25, *Maintenance, Repair, and Coating of Tall Antenna Towers*

UFGS 09 97 13.26, *Coating of Steel Waterfront Structures, Zero VOC, (SZC) Splash Zone Coating*

UFGS 09 97 13.27, *Exterior Coating of Steel Structures*

UFGS 33 52 80, *Liquid Fuels Pipeline Coating Systems*

UFGS 32 17 23, *Pavement Markings*

APPENDIX A UFC 3-201-01 GLOSSARY

A-1 ACRONYMS.

AASHTO	American Association of State Highway and Transportation Officials
ACE	Annual Chance Event
ADT	Average Daily Traffic
AFCEC	Air Force Civil Engineer Center
AIROPS	Airfield Operations Bird/Animal Aircraft Strike Hazard
AREMA	American Railway Engineering and Maintenance-of-Way Association
ASR	Alkali-Silica Reaction
AT&L	Acquisition, Technology, and Logistics
AWWA	AMERICAN WATER WORKS ASSOCIATION
BASH	Bird/Animal Aircraft Strike Hazard
BFE	Base Flood Elevation
BIA	Bilateral Infrastructure Agreement
DFE	Design Flood Elevation
DoD	Department of Defense
DoR	Designer of Record
DOT	Department of Transportation
DRSL	DoD Regional Sea Level
EISA	Energy Independence and Security Act
EO	Executive Order
EPA	Environmental Protection Agency
EWL	Extreme Water Level
FEMA	Federal Emergency Management Agency
FIRM	Flood Insurance Rate Map
FIS	Flood Insurance Study

HQUSACE	Headquarters, U.S. Army Corps of Engineers
HNFA	Host Nation Funded Construction Agreements
IBC	International Building Code
IDF	Intensity-Duration-Frequency
LID	Low Impact Development
NAVFAC	Naval Facilities Engineering Command
NEPA	National Environmental Policy Act
NOAA	National Oceanic and Atmospheric Administration
OEBGD	Overseas Environmental Baseline Guidance Document
SDDCTEA	Surface Deployment and Distribution Command Transportation Engineering Agency
SOFA	Status of Forces Agreements
TR-55	Technical Release 55
UFC	Unified Facilities Criteria
UFGS	Unified Facilities Guide Specifications
U.S.	United States
WBDG	Whole Building Design Guide

A-2**DEFINITION OF TERMS.**

Climate-Informed Science Approach: The elevation and flood hazard area that result from using the best-available, actionable hydrologic and hydraulic data and methods that integrate current and future changes in flooding and tidally-influenced inundation based on climate science. This approach will also include an emphasis on whether the action is a critical action as one of the factors to be considered when conducting the analysis.

Non-Tidally Influenced Locations: Locations that are not subject to oceanic astronomical tidal influence.

Tidally Influenced Locations: Coastal and inland locations with oceanic astronomical tidal influence.

Use the following order of precedence for additional definitions:

1. The applicable legal definition.
2. The definitions provided in the applicable building code.
3. The definitions provided in the applicable standard.

APPENDIX B UFC 3-201-01 REFERENCES

A-1 FEDERAL GOVERNMENT.

CODE OF FEDERAL REGULATIONS

44 CFR Section 65.10, *Mapping of Areas Protected by Levee Systems*

EXECUTIVE ORDERS

EO 11988, *Floodplain Management* (May 24 1977), 42 FR 26951, 3 CFR, 1977

EO 11990, *Protection of Wetlands* (May 24 1977), 42 FR 26961, 3 CFR, 1977

EO 13690, *Establishing a Federal Flood Risk Management Standard and a Process for Further Soliciting and Considering Stakeholder Input* (January 30, 2015), 3 CFR, 2016

EO 14030, *Climate Related Financial Risk*, (May 20, 2021)

PUBLIC LAW

PL 115-232, Public Law 115-232, The National Defense Authorization Act for Fiscal Year 2019, Section 2805: Updates And Modifications To Department Of Defense Form 1391, Unified Facilities Criteria, And Military Installation Master Plans as amended by PL 116-92

PL 116-92, Public Law 116-92, The National Defense Authorization Act for Fiscal Year 2020, Section 2804 Amendment Of Unified Facilities Criteria To Promote Military Installation Resilience, Energy Resilience, Energy And Climate Resiliency, And Cyber Resilience

PL 117-81, Public Law 117-81, The National Defense Authorization Act for Fiscal Year 2022, Section 2805 Flood Risk Management For Military Construction

DEPARTMENT OF TRANSPORTATION

AC 150/5320-5, *Airport Drainage Design*¹
https://www.faa.gov/regulations_policies/advisory_circulars/

HDS-5, *Hydraulic Design of Highway Culverts*
https://www.fhwa.dot.gov/engineering/hydraulics/library_listing.cfm

Manual on Uniform Traffic Control Devices
<https://mutcd.fhwa.dot.gov/ser-pubs.htm>

SHSM, *Standard Highway Signs and Markings*

¹ Requirement for Airfield Drainage criteria and Best Practice for storm drainage systems other than airfields

FEDERAL EMERGENCY MANAGEMENT AGENCY (FEMA)

FEMA GD34, *Guidance for Flood Risk Analysis and Mapping*

<https://www.fema.gov/guidelines-and-standards-flood-risk-analysis-and-mapping>

NATIONAL OCEANIC AND ATMOSPHERIC ADMINISTRATION, FEDERAL GEODETIC CONTROL COMMITTEE

Standards and Specifications for Geodetic Control Networks

https://www.ngs.noaa.gov/FGCS/tech_pub/1984-stds-specs-geodetic-control-networks.pdf

DEPARTMENT OF DEFENSE

DoDI 4715.05, *Overseas Environmental Baseline Guidance Document*

Regional Sea Level Database

Public: <https://drsl.serdp-estcp.org/Site>

FOUO: <https://sealevelscenarios.serdp-estcp.org>

DTM 22-003, Directive-type Memorandum (DTM) 22-003, "Flood Hazard Area Management for DoD Installations"

UNIFIED FACILITIES CRITERIA

<https://www.wbdg.org/ffc/dod/unified-facilities-criteria-ufc>

Consult active UFCs for all aspects of design, including but not limited to:

UFC 1-200-01, *DoD Building Code*

UFC 2-100-01, *Installation Master Planning*

UFC 3-101-01, *Architecture*

UFC 3-210-10, *Low Impact Development*

UFC 3-220-01, *Geotechnical Engineering*

UFC 3-230-01, *Water Storage and Distribution*

UFC 3-230-03, *Water Treatment*

UFC 3-240-01, *Wastewater Collection and Treatment*

UFC 3-250-01, *Pavement Design for Roads and Parking Areas*

UFC 3-250-04, *Standard Practice for Concrete Pavements*

UFC 3-260-01, *Airfield and Heliport Planning and Design*

UFC 3-260-02, *Pavement Design for Airfields*

UFC 3-301-01, *Structural Engineering*

UFC 3-420-01, *Plumbing Systems*

UFC 3-460-01, *Design: Petroleum Fuel Facilities*

UFC 3-530-01, *Interior and Exterior Lighting Systems and Controls*

UFC 3-600-01, *Fire Protection Engineering for Facilities*

UFC 4-022-01, *Security Engineering: Entry Control Facilities/Access Control Points*

UFC 4-022-02, *Selection and Application of Vehicle Barriers*

UFC 4-152-01, *Design: Piers And Wharves*

UFC 4-860-01, *Railroad Design and Rehabilitation*

U.S. ARMY CORPS OF ENGINEERS (HQUSACE)

EM 1110-2-1913, *Design and Construction of Levees*

EM 1110-2-2104, *Strength Design for Reinforced-Concrete Hydraulic Structures*

EM 1110-2-2502, *Floodwalls and Other Hydraulic Retaining Walls*

EM 1110-1-1904, *Settlement Analysis*

EM 1110-2-2504, *Sheet Pile Walls*

ETL 1110-2-583, *Engineering and Design: Guidelines for Landscape Planting and Vegetation Management at Levees, Floodwalls, Embankment Dams, and Appurtenant Structures*

EM 1110-2-1619, *Risk-Based Analysis for Flood Damage Reduction Studies*

ER 1105-2-101, *Risk Assessment for Flood Risk Management Studies*

ER 1110-2-1150, *Engineering and Design for Civil Works Projects*

ETL 1110-2-299, *Overtopping of Flood Control Levees and Floodwalls*

U.S. ARMY SURFACE DEPLOYMENT AND DISTRIBUTION COMMAND - TRANSPORTATION ENGINEERING AGENCY

<https://www.sddc.army.mil/sites/TEA/Functions/SpecialAssistant/TrafficEngineeringBranch/Pages/pamphlets.aspx>

SDDCTEA Pamphlet 55-8, *Traffic Engineering Study Reference*

SDDCTEA Pamphlet 55-15, *Traffic and Safety Engineering for Better Entry Control Facilities*

SDDCTEA Pamphlet 55-17, *Better Military Traffic Engineering*

U.S. DEPARTMENT OF AGRICULTURE - FOREST SERVICE

USDA 0625 1808-SDTDC, *Low-Water Crossings: Geomorphic, Biological, and Engineering Design Considerations*

<https://www.fs.usda.gov/t-d/pubs/pdf/LowWaterCrossings/LoWholeDoc.pdf>

**U.S. DEPARTMENT OF AGRICULTURE - NATURAL RESOURCES
CONSERVATION SERVICE**

TR-55, Technical Release 55, *Urban Hydrology for Small Watersheds*

A-2 NON-GOVERNMENT.

AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS

<https://www.transportation.org/>

AASHTO GBF, *Guide for the Development of Bicycle Facilities*

AASHTO GDHS, *A Policy on Geometric Design of Highways and Streets*

AASHTO GPF, *Guide for the Planning, Design and Operation of Pedestrian Facilities*

AASHTO LRFDBDS, *LRFD Bridge Design Specifications*

AASHTO RSDG-4, *Roadside Design Guide*

AASHTO VLVLR, *Guidelines for Geometric Design of Very Low-Volume Local Roads*
($ADT \leq 400$)

AMERICAN CONCRETE INSTITUTE

<https://www.concrete.org/>

ACI 522R, *Report on Pervious Concrete*

AMERICAN RAILWAY ENGINEERING AND MAINTENANCE-OF-WAY ASSOCIATION (AREMA)

<http://www.arema.org>

Manual for Railway Engineering (Volumes 1 and 2)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

<https://www.astm.org/>

ASTM C76, *Standard Specification for Reinforced Concrete Culvert, Storm Drain, and Sewer Pipe*

ASTM C76M, *Standard Specification for Reinforced Concrete Culvert, Storm Drain, and Sewer Pipe (Metric)*

AMERICAN SOCIETY OF CIVIL ENGINEERS

<https://www.asce.org/>

ASCE 24-14, *Flood Resistant Design and Construction*

AMERICAN WATER WORKS ASSOCIATION (AWWA)

<https://www.awwa.org/>

AWWA M27, *External Corrosion for Infrastructure Sustainability*

**INNOVATIVE PAVEMENT RESEARCH FOUNDATION, AIRPORT CONCRETE
PAVEMENT TECHNOLOGY PROGRAM**

<http://www.iprf.org>

ACPA Document No. JP007P, *Best Practices for Airport Portland Cement Concrete
Pavement Construction*

INTERLOCKING CONCRETE PAVEMENT INSTITUTE (ICPI)

<https://www.icpi.org>

*Permeable Interlocking Concrete Pavements Manual - Design, Specification,
Construction, Maintenance*

INSTITUTE OF TRANSPORTATION ENGINEERS

<https://www.ite.org/>

ITE LP-674B, *Highway Capacity Manual*

INTERNATIONAL CODE COUNCIL

<https://www.iccsafe.org/>

IBC, *International Building Code*

NATIONAL SOCIETY OF PROFESSIONAL SURVEYORS

<https://www.nsps.us.com/>

Model Standards for Topographic Surveys

WATER ENVIRONMENT FEDERATION

<https://www.wef.org/>

WEF MOP FD-4, *Design of Wastewater and Stormwater Pumping Stations*, 1993

WEF MOP FD-6, *Existing Sewer Evaluation and Rehabilitation*, Third Edition

A-3 INDUSTRY PUBLICATIONS.

Pumping Station Design, 3rd Edition, edited by Garr M. Jones with Robert L. Sanks,
George Tchobanoglous, and Bayard Bosserman, Butterworth-Heinemann, 2008.

APPENDIX A UFC 3-201-02 GLOSSARY

A-1 ACRONYMS AND ABBREVIATIONS.

AASHTO	American Association of State and Highway Transportation Officials
ABA	Architectural Barriers Act
ADP	Area Development Plan
AHJ	Authority Having Jurisdiction
ANSI	American National Standards Institute
ASA IE&E 1V1/	Assistant Secretary of the Army Installations, Energy and Environment
BASH	Bird/Wildlife Aircraft Strike Hazard
BIA	Bilateral Infrastructure Agreement
BoD	Basis of Design
CRM	Cultural Resources Manager
CRZ	Critical Root Zone
DBH	Diameter at Breast Height
DoDI	DoD Instruction
DoR	Designer of Record
EPA	Environmental Protection Agency
FGS	Final Governing Standards
HNFA	Host Nation Funded Construction Agreements
IDGD	Installation Design Guidance Documents
IDP	Installation Development Plan
IgCC	International Green Construction Code
INRMP	Integrated Natural Resource Management Plan
IPM	Integrated Pest Management
IPS	Installation Planning Standards

LID	Low Impact Development
MILCON	Military Construction
NEPA	National Environmental Policy Act
NFPA	National Fire Protection Association
NHPA	National Historic Preservation Act
NHO	Native Hawaiian Organization
OASD EI&E	Office of Assistant Secretary of Defense for Energy, Installations, and Environment
O&M	Operations and Maintenance
PoR	Program of Requirements
POV	Privately Owned Vehicle
R	Solar Reflectivity
RFP	Request for Proposals
SAES	Statement of Architectural and Engineering Services
SOFA	Status of Forces Agreements
SoW	Scope of Work
SRI	Solar Reflectance Index
UFC	Unified Facilities Criteria
USDA	United States Department of Agriculture
WBDG	Whole Building Design Guide

A-2 DEFINITION OF TERMS

Authority Having Jurisdiction (AHJ): The component office of responsibility, i.e., U.S. Army, HQ USACE/CECW-CE; U.S. Navy, NAVFACENGCOM HQ Code CHE; U.S. Marine Corps, HQMC Code LFF-1; and U.S. Air Force, AFCEC. The enforcement of the codes and standards as they pertain to facility projects can be delegated to the local Components Office's Chief Engineer's Technical Representative at the discretion of the components aforementioned office.

Bioremediation: The treatment of pollutants or waste (as in an oil spill, contaminated groundwater, or an industrial process) by the use of microorganisms (such as bacteria) that break down the undesirable substances.

Bird/Aircraft Strike Hazard (BASH): The conflict between birds, airfields, and flight paths that poses safety issues for the flying environment. Problem species can be drawn away from active airfield areas by managing habitats on or near airfields, including enhancing habitat elsewhere.¹

Building: "A roofed and floored facility enclosed by exterior walls and consisting of one or more levels that is suitable for single or multiple functions and that protects human beings and their properties from direct harsh effects of weather such as rain, wind, sun, etc."²

Certified Seed: "Certification is the process by which a state seed certifying agency gives official recognition to seeds produced of a cultivar or named variety under a limited generation system which ensures genetic purity, identity, and a given minimum level of quality. Progeny of breeder, foundation or registered seed, handled under procedures acceptable to the Department of Agriculture and Forestry to maintain satisfactory genetic purity and identity."³

Charrette: "An intensive creative work session in which a design team focuses on a design problem and arrives at a collaborative solution with stakeholders from the project area. A charrette can be a breakthrough event that helps creates a meaningful master plan or facility design. Properly executed, this technique can produce a master plan or facility design that is more useful, better understood, and more quickly produced than one formed by any other method."⁴

Complete streets: Complete streets "are designed and operated to enable safe access for all users, including pedestrians, bicyclists, motorists, and transit riders of all ages and abilities."⁵

Cultural landscape: "Cultural landscapes are geographical areas (including both cultural and natural resources and the wildlife or domestic animals therein), associated with a historic event, activity or person or exhibiting other cultural or aesthetic values. There are primarily four types of cultural landscape: designed landscapes, ethnographic landscapes, historic sites, and vernacular landscapes."⁶

Cultural resource (CR): Per DoDI 4715.16:

- Historic properties (any prehistoric or historic district, site, building, structure, or object as defined by part 800 of Reference (k) included in, or eligible for inclusion in, the National Register of Historic Places, whether such eligibility has been formally determined), including artifacts, records, and material remains related to such a property or resource.
- Cultural items as defined in Reference (m).
- American Indian, Eskimo, Aleut, or Native Hawaiian sacred sites as defined in Reference (l).
- Archaeological resources as defined in section 470 aa-mm of Reference (x).
- Archaeological artifact collections and associated records as defined in part 79 of Reference (k).⁷

Embodied energy: The total energy required to manufacture or produce a material, transport it to a project site, and install it.

Hardscape: Inorganic or inanimate landscape elements, primarily referring to pavements and other non-vegetative materials used for paths and gathering areas.

Integrated Cultural Resources Management Plan: "A plan that defines the process for the management of cultural resources on DoD Installations."⁷

Installation Design Guidance Documents (IDGD): Installation-level plans and guidelines including, but not limited to the Installation Master Plan (IMP), Installation Development Plan (IDP), Area Development Plan (ADP), Base Exterior Architectural Plan (BEAP), and Installation Planning Standards (IPS), Air Force's Installation Facilities Standards (IFS), Army's Installation Design Guides (IDG), and Navy's Installation Appearance Plans (IAP).

Landscape area: Pervious area with landscape improvements that is not covered by built improvements and impervious surfaces (including buildings, other structures, or paved surfaces such as including parking areas and walkways).

Large tree: Type 1 and 2 shade trees as classified by the *American Standard for Nursery Stock* (ANSI Z60.1).

Lawn: A managed grass space.

Native plant: A species that is endemic to a geographic or ecological region, or a cultivar of that species.

Landscape establishment period: A period lasting a minimum of 365 days from final acceptance of project, during which plant material is irrigated and observed for viability. The irrigation system, if used is maintained and repaired as needed. This period may be adjusted by the Reviewing Government Landscape Architect. Runs concurrently with *landscape warranty period*.

Landscape warranty period: A period lasting a minimum of 365 days from final acceptance of project, during which plant material must meet 95% survival rate and the irrigation system is in good working order. Runs concurrently with *landscape establishment period*.

Phytoremediation: the treatment of pollutants or waste (as in contaminated soil or groundwater) by the use of green plants that remove, degrade, or stabilize the undesirable substances (such as toxic metals).

Resiliency: The capacity to anticipate, prepare for, respond to, recover from, and adapt to hazards and the risks they pose, including severe weather events and the effects of climate change.

Site commissioning: “Site Commissioning is a systematic process of verifying that systems within the built environment, beyond a building’s skin, perform in accordance with design intent and the property owner’s operational needs.”⁸

Site furnishings: Functional and aesthetic amenities that enhance a site for human comfort.

Site improvements: All site modification that will result in permanent paving features or landforms. Does not refer to enclosed buildings or airfield pavements.

Small Tree: Type 3 small upright trees and Type 4 small spreading trees as classified by the *American Standard for Nursery Stock* (ANSI Z60.1).

Turf: The upper stratum of soil bound by grass and plant roots into a thick mat.

Viewshed: The natural environment that is visible from one or more viewing points

Xeriscaping: Xeriscaping is landscaping and gardening that reduces or eliminates the need for supplemental water from irrigation. This term was coined by Denver Water in 1981 and is associated with seven design principles elaborated on at utility’s website: <https://www.denverwater.org/residential/rebates-and-conservation-tips/remodel-your-yard/xeriscape-plans/xeriscape-principles>

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² DoDI 4165.14, “Real Property Inventory (RPI) and Forecasting,” 17 January 2014, <http://www.esd.whs.mil/Portals/54/Documents/DD/issuances/dodi/416514p.pdf?ver=2017-11-14-112325- 230>.

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⁴ UFC 1-200-02, *High Performance and Sustainable Building Requirements, with Change 2*, 7 June 2018, <https://www.wbdg.org/ffc/dod/unified-facilities-criteria-ufc/ufc-1-200-02>.

⁵ Smart Growth America – National Complete Streets Coalition, <https://smartgrowthamerica.org/program/national-complete-streets-coalition/>

⁶ National Park Service, *The Secretary of the Interior’s Standards for the Treatment of Historic Properties and Guidelines for the Treatment of Cultural Landscapes*, 12 July 1995, <https://www.nps.gov/tps/standards/four-treatments/landscape-guidelines/terminology.htm>.

⁷ DoDI 4715.16, “Cultural Resources Management,” 18 September 2008, <http://www.esd.whs.mil/Portals/54/Documents/DD/issuances/dodi/471516p.pdf?ver=2017-11-21-114100-670>

⁸ General Services Administration, *Site Commissioning White Paper*, July 2017, https://www.gsa.gov/cdnstatic/2017-13-07_SiteCommissioning_AccSingle_WhitePaper.pdf.

APPENDIX B UFC 3-201-02 REFERENCES

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UFC 3-210-10, Low Impact Development

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UFC 4-010-06, Cybersecurity of Facility-related Control Systems

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U.S. Department of Agriculture (USDA)

National Invasive Species Information Center (NISIC), <https://www.invasivespeciesinfo.gov/index.shtml>

Plant Hardiness Zone Map, <http://planthardiness.ars.usda.gov/PHZMWeb/>

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American Association of State and Highway Transportation Officials (AASHTO)

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https://bookstore.transportation.org/collection_detail.aspx?ID=116

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<https://www.americanhort.org/page/standards>

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Xeriscape Principles, <https://www.denverwater.org/residential/rebates-and-conservation-tips/remodel-your-yard/xeriscape-plans/xeriscape-principles>

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<https://www.irrigation.org/IA/Advocacy/Standards-Best-Practices/Landscape-Irrigation-BMPs/IA/Advocacy/Landscape-Irrigation-BMPs.aspx?hkey=93b546ad-c87a-41b8-bf70-8c4fd2cff931>

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Smart Growth America

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APPENDIX A UFC 3-210-10 GLOSSARY

A-1 ACRONYMS.

ARC	Antecedent Runoff Condition
Bio	Biological
BMP	Best Management Practice
CN	Curve Number
CWA	Clean Water Act
DoD	Department of Defense
DoDAAC	Department of Defense Activity Address Code
EISA	Energy Independence and Security Act
EPA	United States Environmental Protection Agency
GHCN	Global Historical Climate Network
GSOD	Global Summary of the Day
HN	Host Nations
HSG	Hydrologic Soil Group
I _a	Initial Abstraction
IMP	Integrated Management Practice
LID	Low Impact Development
NCEI	National Centers for Environmental Information
NEH	National Engineering Handbook of Hydrology
NPDES	National Pollutant Discharge Elimination System
NOAA	National Oceanic and Atmospheric Administration
NRCS	USDA Natural Resources Conservation Service (formerly SCS)
O&M	Operations and Maintenance
OMSI	Operation and Maintenance Support Information
PGDER	Prince George's County Department of Environmental Resources

SCS	USDA Soil Conservation Service
SWM	Stormwater Management
T_c	Time of Concentration
TR-55	NRCS Technical Release 55
UFC	Unified Facilities Criteria
USDA	United States Department of Agriculture
WMO	World Meteorological Organization

95th Percentile Rainfall Event: The 95th percentile rainfall event represents a precipitation amount which 95 percent of all rainfall events for the period of record do not exceed. In more technical terms, the 95th percentile rainfall event is defined as the measured precipitation depth accumulated over a 24-hour period for the period of record that ranks as the 95th percentile rainfall depth based on the range of all daily event occurrences during this period.

The 24-hour period is typically defined as 12:00:00 am to 11:59:59 pm. In general, a 30-year period of rainfall data is recommended for such an analysis, with a required minimum period of 10 years. Small rainfall events that are 0.1 inch or less (that is, ≤ 0.1 inches) are excluded from the percentile analysis because this rainfall generally does not result in any measurable runoff due to absorption, interception, and evaporation by permeable, impermeable, and vegetated surfaces.

Building: DODI 4165.14, Enclosure 2, defines a building as: A roofed and floored facility enclosed by exterior walls and consisting of one or more levels that is suitable for single or multiple functions and that protects human beings and their properties from direct harsh effects of weather such as rain, wind, sun, etc.

Federal Facility: Section 401(8) of EISA states: The term “Federal facility” means any building that is constructed, renovated, leased, or purchased in part or in whole for use by the Federal Government.

Low Impact Development (LID): LID is a stormwater management strategy designed to maintain site hydrology and mitigate the adverse impacts of stormwater runoff and nonpoint source pollution.

LID actively manages stormwater runoff by mimicking a project site’s pre-development hydrology using design techniques that infiltrate, store, and evaporate runoff close to its source of origin. LID strategies provide decentralized hydrologic source control for stormwater runoff. In short, LID seeks to manage the rain, beginning at the point where it falls. The LID features are distributed small-scale controls that closely mimic the hydrological behavior of the pre-project sites for a design storm event.

Pre-Development: pre-project conditions that exist at the beginning of design. Where phased development occurs, the existing conditions at the time prior to the first phase being submitted will establish pre-development conditions.

APPENDIX B UFC 3-210-10 REFERENCES

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UFC 3-201-01, *Civil Engineering*

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UFC 3-420-01, *Plumbing Systems*

UFC 4-010-01, *DoD Minimum Antiterrorism Standards for Buildings*

UFC 4-010-06, *DoD Cybersecurity Of Facility-Related Control Systems*

APPENDIX A UFC 3-220-01 GLOSSARY

A-1 ACRONYMS.

AFCEC	Air Force Civil Engineering Center
BIA	Bilateral Infrastructure Agreement
DoD	Department of Defense
DoR	Designer of Record – The responsible designer whose seal is placed on the design documents (Navy, Marine Corps)
EoR	Engineer of Record – Same as DoR (Army, Air Force)
ft	foot
HQUSACE	Headquarters, U.S. Army Corps of Engineers
HNFA	Host Nation Funded Construction Agreements
NAVFAC	Naval Facilities Engineering Systems Command
SOFA	Status of Forces Agreements
sq m	square meter
sf	square foot
UFC	Unified Facilities Criteria
U.S.	United States
USACE	United States Army Corps of Engineers

APPENDIX B UFC 3-220-01 REFERENCES

AMERICAN CONCRETE INSTITUTE

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ACI 351.3R, Foundations for Dynamic Equipment

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NATIONAL CENTER FOR EARTHQUAKE ENGINEERING RESEARCH

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APPENDIX C UFC 3-220-01 SUPPLEMENTAL RESOURCES

INTRODUCTION

The following references are reliable sources for information related to geotechnical engineering and design. These sources provide direction for specific geotechnical applications not addressed in this UFC or provide additional information to guide or aid the designer in the various phases of the design. This list is provided for the convenience of the designer and may not include references for all specific applications relevant to all projects. The designer is responsible for insuring the design conforms to all criteria relevant to the project.

UNIFIED FACILITIES CRITERIA

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APPENDIX A UFC 3-230-01 GLOSSARY

A-1 ACRONYMS.

AFCEC	Air Force Civil Engineering Center
AT&L	Acquisition, Technology, and Logistics
AWWA	American Water Works Association
B&S	Bell & Spigot
BIA	Bilateral Infrastructure Agreement
CCB	Construction Criteria Base
DDC	Direct Digital Controls
DoD	Department of Defense
DoR	Designer of Record
e.g.	<i>Exempli Gratia</i> (one or more possible examples)
EPA	Environmental Protection Agency
FAA	Federal Aviation Administration
FGS	Final Governing Standards
fPVC	Fusible Polyvinyl Chloride
HQUSACE	Headquarters, U.S. Army Corps of Engineers
HNFA	Host Nation Funded Construction Agreements
i.e.	<i>Id Est</i> (clarifies, more precisely)
IPC	International Plumbing Code
NASA	National Aeronautics and Space Administration
NAVAIR	Naval Air Systems Command
NAVFAC	Naval Facilities Engineering Command
NEMA	National Electrical Manufacturers Association
NEPA	National Environmental Policy Act
OEBGD	Overseas Environmental Baseline Guidance Document

ORP	Oxidation Reduction Potential
OSHA	Occupational Safety and Health Administration
PE	Polyethylene
PVC	Polyvinyl Chloride
SOFA	Status of Forces Agreements
UFC	Unified Facilities Criteria
UFGS	Unified Facilities Guide Specifications
U.S.	United States
WBDG	Whole Building Design Guide

A-2 DEFINITION OF TERMS.

Distribution Mains: All pipelines of the potable water distribution system, except the service lines (e.g., water, fire, irrigation).

Pipe Size: The nominal internal diameter of the pipe.

Surge pressure: The maximum hydraulic transient pressure increase (also known as water hammer) above the anticipated operating pressure in the system as the result of sudden changes in velocity of the water column. Two types of surge pressures are recurring (cyclic) surge pressure and occasional (emergency or transient) surge pressure.

Working pressure: The maximum sustained operating pressure applied to the pipe exclusive of surge pressures.

For additional definitions refer to the definitions given in the applicable standard.

APPENDIX B UFC 3-230-01 REFERENCES

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AMERICAN WATER WORKS ASSOCIATION

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AWWA M19, *Emergency Planning for Water Utilities*

AWWA M22, *Sizing Water Service Lines and Meters*

AWWA M27, *External Corrosion Control for Infrastructure Sustainability*

AWWA M31, *Distribution Systems Requirements for Fire Protection*

AWWA M32, *Computer Modeling of Water Distribution Systems*

AWWA M42, *Steel Water Storage Tanks*

AWWA M51, *Air-Valves: Air-Release, Air/Vacuum and Combination*

AWWA M55, *PE Pipe - Design and Installation*

AWWA M58, *Internal Corrosion Control of Water Distribution Systems*

AWWA C510, *Double Check-Valve Backflow Prevention Assembly*

AWWA C511, *Reduced-Pressure Principle Backflow Prevention Assembly*

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AWWA D100, *Welded Carbon Steel Tanks for Water Storage*

AWWA D102, *Coating Steel Water-Storage Tanks*

AWWA D103, *Factory-Coated Bolted Carbon Steel Tanks for Water Storage*

AWWA D104, *Automatically Controlled, Impressed Current Cathodic Protection for the Interior Submerged Surfaces of Steel Water Tanks*

AWWA D106, *Sacrificial Anode Cathodic Protection Systems for the Interior Submerged Surfaces of Steel Water Storage Tanks*

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AWWA D110, *Wire- and Strand-Wound, Circular, Prestressed Concrete Water Tanks*

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DEPARTMENT OF DEFENSE (DOD)

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UFC 3-240-01, *Wastewater Collection and Treatment*

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UFC 3-301-01, *Structural Engineering*

UFC 3-410-01, *Heating, Ventilating, And Air Conditioning Systems*

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UFC 3-520-01, *Interior Electrical Systems*

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INTERNATIONAL CODE COUNCIL

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NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

<http://www.nema.org>

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NATIONAL FIRE PROTECTION ASSOCIATION

<http://www.nfpa.org>

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PLASTIC PIPE INSTITUTE

<http://plasticpipe.org>

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APPENDIX A UFC 3-240-01 GLOSSARY

A-1 ACRONYMS.

AFCEC	Air Force Civil Engineering Center
AHJ	Authority Having Jurisdiction
ASCE	American Society of Civil Engineers
ASTM	American Society for Testing and Materials
BEQ	Bachelor Enlisted Quarters
BIA	Bilateral Infrastructure Agreements
BOD	Biochemical Oxygen Demand
BOQ	Bachelor Officer Quarters
COD	Chemical Oxygen Demand
DAF	Dissolved Air Flotation
DoD	Department of Defense
DoR	Designer of Record
FGS	Final Governing Standards
HQUSACE	Headquarters, U.S. Army Corps of Engineers
HNFA	Host Nation Funded Construction Agreements
IPC	International Plumbing Code
IPSDC	International Private Sewage Disposal Code
LET	Load Equalization Tank
NAVFAC	Naval Facilities Engineering Command
NPDES	National Pollution Discharge Elimination System
MOP	Manual of Practice
OEBGD	Overseas Environmental Baseline Guidance Document
OSHA	Occupational Safety and Health Administration
OWHT	Oily Waste Holding Tanks

OWTP	Oily Waste Transfer Pumps
OWWO	Oily waste and Waste Oil
O&G	Oil and Grease
POTW	Public Owned Treatment Works
RBC	Rotating Biological Contactors
RBP	Rapid Bioassay Procedures
RMF	Risk Management Framework
SBR	Sequencing Batch Reactor
SOFA	Status of Forces Agreements
TSS	Total Suspended Solids
UFC	Unified Facilities Criteria
UFGS	Unified Facilities Guide Specifications
U.S.	United States
VSS	Volatile Suspended Solids
WEF	Water Environment Federation
WET	Whole Effluent Toxicity
WOT	Waste Oil Tanks
WRRF	Water Resource and Recovery Facility

APPENDIX B UFC 3-240-01 REFERENCES

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AMERICAN NATIONAL STANDARDS INSTITUTE

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WATER ENVIRONMENT FEDERATION

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APPENDIX B UFC 3-250-01 GLOSSARY

B-1 ACRONYMS

AASHTO	American Association of State Highway and Transportation Officials
AC	Asphalt concrete
ADT	Average Daily Traffic
AOS	Apparent Opening Size
ASR	Alkali-Silica Reaction
ASTM	American Society for Testing and Materials
CBR	California Bearing Ratio
CD	Compact Disk
CPE	Corrugated Polyethylene
DoD	Department of Defense
DoT	Department of Transportation
ESAL	Equivalent Single Axle Load
FASSI	Frost Area Soil Support Index
HEMTT	Heavy Expanded Mobility Tactical Truck
HET	Heavy Equipment Transport System
HMMWV	High Mobility Multipurpose Wheeled Vehicle
LCF	Lime-Cement-Fly Ash
LL	Liquid Limit
NFS	Non-Frost Susceptible
OGM	Open Graded Material
PCASE	Pavement-Transportation Computer Assisted Structural Engineering
PCC	Portland Cement Concrete
PCI	Pavement Condition Index
PI	Plasticity Index

PVC	Polyvinyl Chloride
RCCP	Roller-Compacted Concrete Pavement
RDM	Rapid Draining Material
UFC	Unified Facilities Criteria
UFGS	Unified Facility Guide Specifications
U.S.	United States
USACE	United States Army Corp of Engineering
USCS	Unified Soil Classification System

B-2 DEFINITION OF TERMS

Apparent Opening Size (AOS): A measure of the opening size of a geotextile. AOS is the sieve number corresponding to the sieve size at which 95 percent of the single-size glass beads pass the geotextile (O95) when tested in accordance with ASTM D4751.

Average Daily Temperature: The average of the maximum and minimum temperatures for a day, or the average of several temperature readings taken at equal time intervals, generally hourly, during a day.

Base Course: Course containing all granular unbound, chemical- or bituminous-stabilized material between the pavement surfacing layer and the un-treated, chemical or bituminous stabilized subgrade.

Bound Base: Chemical-stabilized or bituminous-stabilized soil used in the base and subbase course, consisting of a mixture of mineral aggregates and soil with one or more commercial stabilizing additives. Bound base is characterized by a significant increase in compressive strength of the stabilized soil compared with the untreated soil. In frost areas bound base usually is placed directly beneath the pavement surfacing layer where its high strength and low deformability make possible a reduction in the required thickness of the pavement surfacing layer or the total thickness of pavement and base, or both. If the stabilizing additive is Portland cement, lime, or lime-cement-fly ash (LCF), the term bound base is applicable only if the mixture meets the requirements for cement-stabilized, lime-stabilized, or LCF-stabilized soil set forth within this UFC and in UFC 3-250-11.

Boulder Heave: The progressive upward migration of a large stone present within the frost zone in a frost-susceptible subgrade or base course. This is caused by adhesion of the stone to the frozen soil surrounding it while the frozen soil is undergoing frost heave. The stone will be kept from an equal, subsequent subsidence by soil that will have tumbled into the cavity formed beneath the stone. Boulders heaved toward the surface cause extreme pavement roughness and may eventually break through the surface, necessitating repair or reconstruction.

Coefficient of Permeability: A measure of the rate at which water passes through a unit area of material in a given amount of time under a unit hydraulic gradient.

Choke Stone: A small size stone used to stabilize the surface of an OGM. For a choke stone to be effective, the ratio of d₁₅ of the coarse aggregate to the d₁₅ of the choke stone must be less than 5, and the ratio of the d₅₀ of the coarse aggregate to d₅₀ of the choke stone must be greater than 2.

Composite Pavement: Existing pavement to be overlaid with rigid pavement is composed of an all-bituminous or flexible overlay on a rigid base pavement.

Cumulative Damage: The process by which each application of traffic load or each cycle of climatic change produces a certain irreversible damage to the pavement. The pavement deteriorates continuously under successive load applications or climatic cycles.

Degree-Days: The Fahrenheit degree-days for any given day equal to the difference between the average daily air temperature and 32 degrees Fahrenheit. The degree-days are minus when the average daily temperature is below 32 degrees Fahrenheit (freezing degree-days) and plus when above (thawing degree-days). Figure 19-1 shows sample curves obtained by plotting cumulative degree-days against time.

Design Freezing Index: The average air freezing index of the three coldest winters in the latest 30 year of record. If 30 years of record are not available, the air freezing index for the coldest winter in the latest 10 year period may be used. To avoid the necessity of adopting a new and only slightly different freezing index each year, the design freezing index at a site with continuing construction need not be changed more than once in 5 year unless the more recent temperature records indicate a significant change in thickness design requirements for frost. The design freezing index is illustrated in Figure 19-1.

Drainage Layer: A layer in the pavement structure that is specifically designed to allow rapid horizontal drainage of water from the pavement structure. The layer is also considered to be a structural component of the pavement and may serve as part of the base or subbase.

Effective Porosity: The effective porosity is defined as the ratio of the volume of voids that will drain under the influence of gravity to the total volume of a unit of aggregate. The difference between the porosity and the effective porosity is the amount of water that will be held by the aggregate. For materials such as the RDM and OGM, the water held by the aggregate will be small; thus, the difference between the porosity and effective porosity will be small (less than 10 percent). The effective porosity may be estimated by computing the porosity from the unit dry weight of the aggregate and the specific gravity of the solids which then should be reduced by 5 percent to allow for water retention on the aggregate.

Flexible Base Pavement: Existing pavement to be overlaid is composed of bituminous concrete, base, and subbase courses.

Flexible Overlay: A flexible pavement (either all-bituminous or bituminous with base course) used to strengthen an existing rigid or flexible pavement.

Freezing Index: The number of degree-days between the highest and lowest points on a curve of cumulative degree-days versus time for one freezing season. It is used as a measure of the combined duration and magnitude of below-freezing temperatures occurring during any given freezing season. The index determined for air temperature about 4.5 ft (1.4 m) above the ground is commonly designated as the air freezing index, while that determined for temperatures immediately below a surface is known as the surface freezing index.

Frost Action: The general term for freezing and thawing of moisture in materials and the resultant effects on these materials and on structures of which they are a part, or with which they are in contact.

Frost Boil: The breaking of a small section of a highway or airfield pavement under traffic with ejection of soft, semi-liquid subgrade soil. This is caused by the melting of the segregated ice formed by the frost action. This type of failure is limited to pavements with extreme deficiencies of total thickness of pavement and base over frost-susceptible subgrades, or pavements having a highly frost-susceptible base course.

Frost Heave: is the raising of a surface due to ice formation in the underlying soil.

Frost Melting Period: The interval of the year when the ice in the base, subbase, or subgrade materials is returning to a liquid state. It ends when all the ice in the ground has melted or when freezing is resumed. In some cases there may be only one frost-melting period, beginning during the general rise of air temperatures in the spring. However, one or more significant frost-melting intervals may occur during a winter season.

Frost Susceptible Soil: Soil in which significant detrimental ice segregation will occur when the requisite moisture and freezing conditions are present.

Geocomposite Edge Drain: A manufactured product using geotextiles, geogrids, geonets, or geomembranes in laminated or composite form, which can be used as an edge drain in place of trench-pipe construction.

Geotextile: A permeable textile used in geotechnical projects. For this UFC geotextile will refer to a nonwoven needle punch fabric that meets the requirements of the apparent opening size (AOS), grab strength and puncture strength specified for the particular application.

Granular Unbound Base Course: A base course containing no agents that impart higher cohesion by cementing action. Mixtures of granular soil with Portland cement, lime, or fly ash, in which the chemical agents have merely altered certain properties of the soil such as plasticity and gradation without imparting significant strength increase, also are classified as granular unbound base.

Ice Segmentation: The growth of ice as distinct lenses, layers, veins and masses in soils, commonly but not always oriented normal to the direction of heat loss.

Hardstand: An area for parking heavy vehicles, both wheeled and tracked, for significant periods of time. Hardstands are usually constructed of rigid concrete but may be constructed of compacted stone or compacted earth for short periods of time.

Hazen's Effective Particle Diameter: The Hazen's effective particle diameter is the particle size, in millimeters, which corresponds to 10 passing on the grain-size distribution curve. This parameter is one of the major parameters in determining the permeability of a soil.

Mean Daily Temperature: The mean of the average daily temperatures for a given day in each of several years.

Mean Freezing Index: The freezing index determined on the basis of mean temperatures. The period of record over which temperatures are averaged is usually a minimum of 10 year, preferably 30, and should be the latest available. The mean freezing index is illustrated in Figure 19-1.

Non-Frost Susceptible Materials: Cohesionless materials such as crushed rock, gravel, sand, slag, and cinders that do not experience significant detrimental ice segregation under normal freezing conditions. Non-Frost-susceptible materials also include cemented or otherwise stabilized materials that do not evidence detrimental ice segregation, loss of strength upon thawing, or freeze-thaw degradation.

Open Graded Material (OGM): A granular material having a very high permeability (greater than 5,000 ft/day (1,500 m/day)) which may be used for a drainage layer. Such a material will normally require stabilization for construction stability or for structural strength to serve as a base in a flexible pavement.

Open Storage Areas: Permanent exterior storage areas used by any type of wheeled vehicular traffic or having non-pneumatic loadings in excess of 200 psi (1.38 MPa), covered or uncovered, that are separate from the interior area of a building.

Overlay Pavement: A pavement constructed on an existing base pavement to increase load-carrying capacity or correct a surface defect.

Pavement Pumping: The ejection of water and soil through joints, cracks, and along edges of pavements caused by downward movements of sections of the pavement. This is actuated by the passage of heavy axle loads over the pavement after free water has accumulated beneath it.

Pavement Structure: Pavement structure is the combination of subbase, base, and surface layers constructed on a subgrade.

Period of Weakening: The interval of the year that starts at the beginning of a frost-melting period and ends when the subgrade strength has returned to normal summer values, or when the subgrade has again become frozen.

Permeable Base: An open-graded granular material with most of the fines removed (e.g., less than 10 percent passing the No. 16 sieve) to provide high permeability (1,000 ft/day or more) for use in a drainage layer.

Porosity: The amount of voids in a material, expressed as the ratio of the volume of voids to the total volume.

Rapid Draining Material (RDM): A granular material having a sufficiently high permeability 1,000 to 5,000 ft/day (300 to 1,500 m/day) to serve as a drainage layer and also having the stability to support construction equipment and the structural strength to serve as a base or a subbase.

Rigid Base Pavement: An existing rigid pavement is one on which an overlay is to be placed.

Rigid Overlay: A rigid pavement used to strengthen an existing flexible or rigid pavement.

Separation Layer: A layer provided directly beneath the drainage layer to prevent fines from infiltration or pumping into the drainage layer and to provide a working platform for construction and compaction of the drainage layer.

Stabilization: Stabilization refers to either mechanically or chemically stabilizing the drainage layer to increase the stability and strength to withstand construction traffic and design traffic. Mechanical stabilization is accomplished by the use of a choke stone and compaction. Chemical stabilization is accomplished by the use of either Portland cement or asphalt.

Subbase Course: The layer of supporting material between the base course and the subgrade.

Subsurface Drainage: The process of collecting and removing water from the pavement structure. Subsurface drainage systems are categorized into two functional categories: one for draining surface infiltration water, and the other for controlling groundwater.

APPENDIX B UFC 3-250-01 REFERENCES

UNIFIED FACILITIES CRITERIA

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ASTM D4751 - Standard Test Method for Determining Apparent Opening Size of a
Geotextile

ASTM D4833/D4833M - Standard Test Method for Index Puncture Resistance of
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ASTM D6951/D6951M - Standard Test Method for Use of the Dynamic Cone
Penetrometer in Shallow Pavement Applications

APPENDIX A UFC 3-301-01 GLOSSARY

A-1 ABBREVIATIONS.

g	Gravitational Acceleration
µm	Micrometer (micron)
3-D	Three Dimensional
AASHTO	American Association of State Highway and Transportation Officials
ACI	American Concrete Institute
AEC	Architecture/Engineering/Construction
AFCEC	Air Force Civil Engineer Center
AHJ	Authority Having Jurisdiction (See MIL-STD 3007G, Nov 2019)
AISC	American Institute of Steel Construction
AISI	American Iron and Steel Institute
ANSI	American National Standards Institute
AREMA	American Railway Engineering and Maintenance-of-Way Association
ASCE	American Society of Civil Engineers
ASM	American Society for Metals
ASME	American Society of Mechanical Engineers
ASSE	American Society of Safety Engineers
ASTM	American Society for Testing and Materials, now ASTM International
ATFP	Anti-Terrorism Force Protection
ATCT	Air Traffic Control Tower
AWWA	American Water Works Association
BIA	Brick Industry Association (formerly Brick Institute of America)
BPON	Basic Performance Objective Equivalent to New Building Standards
BSE	Basic Safety Earthquake
CEFAPP	CERL Equipment Fragility and Protection Procedure

CERL	Construction Engineering Research Laboratory (formerly USACERL)
CP	Collapse Prevention
CRREL	Cold Regions Research and Engineering Laboratory
DC	Damage Control
DoD	Department of Defense
DoE	Department of Energy
EGL	Energy Grade Line
ELF	Equivalent Lateral Force
EPRI	Electric Power Research Institute
ERDC	U.S. Army Engineer Research and Development Center
FEMA	Federal Emergency Management Agency
GERS	Generic Equipment Ruggedness Spectra
GIP	Generic Implementation Procedure
GSREB	Guidelines for Seismic Retrofit of Existing Buildings
HCAI	California Department of Health Care Access and Information (formerly California Office of Statewide Health Planning and Development or OSHPD)
HVAC	Heating, Ventilating, and Air Conditioning
IBC	International Building Code
ICC-ES	International Code Council – Evaluation Service
ICSSC	Interagency Committee on Seismic Safety in Construction
IEBC	International Existing Building Code
IEEE	Institute of Electrical and Electronics Engineers
IMF	Intermediate Moment Frame
in.	Inch
in./ft	Inches per Foot
ICBO	International Conference of Building Officials

IO	Immediate Occupancy (Performance Objective/Level)
ISAT	International Seismic Application Technologies
kg	Kilogram
kg/m³	Kilograms per Cubic Meter
km/h	Kilometers per Hour
kN	Kilonewton
kN/m	Kilonewton per Meter
kN/m²	Kilonewton per Square Meter
kPa	Kilopascal
lb/ft	Pounds per Foot
lb	Pound
LmS	Limited Safety
LRFD	Load and Resistance Factor Design
LS	Life Safety (Performance Objective/Level)
m	Meter
m/s	Meters per Second
m²	Square Meter
MC-1	Mission-Critical Level 1
MC-2	Mission-Critical Level 2
MCE_R	Risk-Targeted Maximum Considered Earthquake
mil	0.001 Inch
mm	Millimeter
mm²	Square Millimeter
MPa	Megapascal
MPa/m	Megapascal per Meter
mph	Miles per Hour

MRI	Mean Recurrence Interval
NACE	National Association of Corrosion Engineers
NAS	National Academy of Sciences
NAVFAC	Naval Facilities Engineering Command
NCMA	National Concrete Masonry Association
NDP	Nonlinear Dynamic Procedure
NEHRP	National Earthquake Hazards Reduction Program
NFPA	National Fire Protection Association
NFS	Non-Frost Susceptible
NIST	National Institute of Standards and Technology
NL	Not Limited
NMC	Non-Mission-Critical
NSP	Nonlinear Static Procedure
O&M	Operation and Maintenance
OCBF	Ordinary Concentrically Braced Frame
OMF	Ordinary Moment Frame
OP	Operational (Performance Objective/Level)
OSP	HCAI Special Seismic Certification Preapproval Program
PCI	Precast/Prestressed Concrete Institute
pci	Pounds per Cubic Inch
psf	Pounds per Square Foot
psi	Pounds per Square Inch
PSSQ	Project Specific Seismic Qualification
PTHA	Probabilistic Tsunami Hazard Analysis
PTI	Post-Tensioning Institute
RACF	Radar Approach Control Facility

RC	Risk Category
RCSC	Research Council on Structural Connections
RFP	Request for Proposal
RP	Recommended Practice (also Resource Paper)
RRS	Required Response Spectrum
SBC	Standard Building Code
SDC	Seismic Design Category
SDI	Steel Deck Institute
SDPWS	Special Design Provisions for Wind and Seismic
SEAOC	Structural Engineers Association of California
SEI	Structural Engineering Institute
SER	Structural Engineer of Record
SIOR	Special Inspector of Record
SMF	Special Moment Frame
TI	Technical Instruction
TIA	Tentative Interim Agreement; Telecommunications Industry Association
TMS	The Masonry Society
TRS	Test Response Spectrum
UBC	Uniform Building Code
UFC	Unified Facilities Criteria
UFGS	Unified Facilities Guide Specifications
URM	Unreinforced Masonry
USACE	U.S. Army Corps of Engineers
USACERL	United States Army Construction Engineering Research Laboratory (now ERDC-CERL)
UUT	Unit Under Test

V_{asd}	Allowable Stress Design Wind Speed
V_{fm}	Fastest Mile Wind Speed
V	Basic Wind Speed
WEF	Water Environment Federation

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APPENDIX A UFC 3-401-01 GLOSSARY

A-1 ABBREVIATIONS.

14 th WS	14 th Weather Squadron
ADPI	air diffusion performance index
AHU	air handling unit
ATES	aquifer thermal energy storage
BAS	building automation system
Btuh	British thermal units per hour
BTES	borehole thermal energy storage
°C	degrees Celcius
cfm	cubic feet per minute
CFR	Code of Federal Regulations
C _{LFL}	LFL conversion factor
CO ₂	carbon dioxide
C _v	valve coefficient
dB RE 1g	decibels relative to a reference value of 1 standard gravity
dBA	A-scale weighted decibels
dBC	C-scale weighted decibels
DDC	direct digital control
DOAS	dedicated outdoor air system
ECA	Energy Conservation Analysis
ECM	electrically commutated mtors
EDVC	Effective Dispersal Volume Charge
EMCS	energy management and control system
EPA	Environmental Protection Agency
ERV	energy recovery ventilator

ESC	Environmental Severity Classification
ES/EWS	emergency shower/eyewash station
EWD	Engineering Weather Data
°F	degrees Fahrenheit
ftpm	feet per minute
ft ²	square feet
G	standard gravity
gal	gallons
GCHP	ground coupled heat pump
GLHX	ground loop heat exchanger
gpf	gallons per flush
gph	gallons per hour
gpm	gallons per minute
GWHP	ground water heat pump
HAZMAT	Hazardous Material
HDPE	High Density Polyethylene
hr	hour
HVAC	Heating, Ventilation, and Air-Conditioning
Hz	Hertz
IAQP	Indoor Air Quality Procedure
IMC	International Mechanical Code
IPC	International Plumbing Code
in	inches
in/sec ²	inches per square second
in w.c.	inches water column
IP	Inch-Pound

K	Kelvin
kg	kilograms
km	kilometers
kPa	kiloPascals
Kv	valve coefficient
kW	kilowatts
lbs	pounds
L	liters
lpf	liters per flush
LFL	Lower Flammability Limit
lps	liters per second
m	meters
m ²	square meters
m ³	cubic meters
MBtuh	one thousand Btuh
MCDB	Mean coincident dry-bulb
MCWB	Mean coincident wet-bulb
MERV	Minimum Efficiency Reporting Value
mm	millimeters
NC	Noise Criteria
NEMA	National Electrical Manufacturers Association
OSHA	Occupational Safety and Health Administration
OWS	Operator workstation
Pa	Pascals
PCB	polychlorinated biphenyls
PEX	cross-linked polyethylene

PICV	pressure independent control valves
ppm	parts per million
PRV	pressure reducing valves
psi	pounds per square inch
psig	pounds per square inch – gauge
P/T	pressure/temperature
PV	photovoltaic
RC	Room Criteria
RCL	Refrigeration Concentration Limit
RPM	revolutions per minute
RT60	Reverberation time for 60 dB sound
S	second
SAR	Support Analysis Request
SDHW	solar domestic hot water
SDR	Standard Dimension Ratio
sf	square foot
SI	International System of Units
SNAP	Significant New Alternatives Policy
STC	Sound Transmission Coefficient
Sq. m	square meter
T50	air diffusion device throw at 50 fpm terminal velocity
TAB	testing, adjusting, and balancing
TES	thermal energy storage
UFAD	underfloor air distribution
UMCS	utility management and control system
UTES	underground thermal energy storage

Ve _{eff}	Effective Dispersal Volume
VAV	variable air volume
VFD	variable frequency drive
VRF	variable refrigerant flow
W	watts
“w.g.	inches water guage

APPENDIX B UFC 3-401-01 REFERENCES

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APPENDIX A UFC 3-410-01 GLOSSARY

A-1 ACRONYMS.

AFCEC	Air Force Civil Engineer Center
ASA (IE&E)	Assistant Secretary of the Army (Installations, Energy and Environment)
ACGIH	American Conference of Governmental Industrial Hygienists
AFCEC	Air Force Civil Engineer Center
AHJ	Authority Having Jurisdiction
ANSI	American National Standards Institute
ASHRAE	American Society of Heating, Refrigeration and Air Conditioning Engineers
ASN (EI & E)	Assistant Secretary of the Navy (Energy, Installations and Environment)
BAS	Building Automation System
BIA	Bilateral Infrastructure Agreement
Btuh	British Thermal Units per Hour
°C	Degrees Celsius
C	Common
CFM	Cubic Feet Per Minute
CO2	Carbon Dioxide
Cv	Valve Coefficient, English unit version
DALT	Ductwork Air Leakage Testing
DDC	Direct Digital Control
DOAS	Dedicated Outdoor Air System
DoD	Department of Defense
EMCS	Energy Monitoring Control System
ESC	Environmental Severity Classification
°F	Degrees Fahrenheit
FAR	Federal Acquisition Regulations

fpn	Feet Per Minute
ft	Feet
ft ²	Square Feet
GCHP	Ground Coupled Heat Pumps
GLHX	Ground Loop Heat Exchanger
HNFA	Host Nation Funded Construction Agreements
HQUSACE	Headquarters United States Army Corp of Engineers
HVAC	Heating, Ventilating, and Air Conditioning
Hz	Hertz
ICC	International Code Council
IMC	International Mechanical Code
km	Kilometer
kPa	Kilopascal
Kv	Valve Coefficient, Metric unit version
kW	Kilowatt
L	Liters
lbs	Pounds
LCCA	Life Cycle Cost Analysis
lps	Liters per second
m	Meters
m/s	Meters per Second
m ²	Square Meters
m ³	Cubic Meters
MBtuh	Thousand British Thermal Units per Hour
MCDB	Mean Coincident Dry-bulb
MCWB	Mean Coincident Wet-bulb

MICA	Midwest Insulation Contractors Association
MILCON	Military Construction
MIL-HDBK	Military Handbook
MIL-STD	Military Standard
mm	Millimeters
NAVFAC	Naval Facilities Engineering Systems Command
NC	Noise Criteria
NEMA	National Electrical Manufacturers Association
NFPA	National Fire Protection Association
Pa	Pascal's
PI	Pressure Independent
PICV	Pressure Independent Control Valve
ppm	Parts Per Million
PRV	Pressure Reducing Valve
psig	Pounds Per Square Inch (Gage)
RFP	Request for Proposal
RPM	Revolutions per Minute
s	Second
SAF/IEE	Secretary of the Air Force for Environment, Safety, and Infrastructure
SFPVAV	Series Fan-Powered VAV Boxes
SMACNA	Sheet Metal and Air Conditioning Contractors' National Association
SOFA	Status of Forces Agreement
STD	Standard
TAB	Testing, Adjusting and Balancing
UFC	Unified Facilities Criteria
UFGS	Unified Facilities Guide Specification

UL	Underwriters Laboratories, Inc.
UMCS	Utility Monitoring Control System
USD (AT&L)	Under Secretary of Defense for Acquisition, Technology, and Logistics
VAV	Variable Air Volume
V_{eff}	Effective Dispersal Volume
VRF	Variable Refrigerant Flow
VRLA	Valve-regulated lead-acid
W	Watts
w.g.	Water Gage

APPENDIX B UFC 3-410-01 REFERENCES

AMERICAN CONFERENCE OF GOVERNMENTAL INDUSTRIAL HYGIENISTS (ACGIH)

www.acgih.org

ACGIH Industrial Ventilation: A Manual of Recommended Practice for Design

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

www.ansi.org

ANSI/CSA C448 Series, Design and installation of ground source heat pump systems for commercial and residential buildings

ANSI/NB23, National Board Inspection Code

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS (ASHRAE)

www.ashrae.org

ASHRAE Geothermal Heating and Cooling: Design of Ground-Source Heat Pump Systems

ASHRAE Handbooks: Fundamentals, Applications, Systems and Equipment, and Refrigeration

ANSI/ASHRAE Standard 15, Safety Standard for Refrigeration Systems

ANSI/ASHRAE Standard 55, Thermal Environmental Conditions for Human Occupancy

ANSI/ASHRAE Standard 62.1, Ventilation for Acceptable Indoor Air Quality

ANSI/ASHRAE/IESNA Standard 90.1, Energy Standard for Buildings Except Low Rise Residential (Refer to UFC 1-200-02, for applicable publication date)

ANSI/ASHRAE Standard 154, Ventilation for Commercial Cooking Operations

ANSI/ASHRAE/ACCA Standard 183, Peak Cooling and Heating Load Calculations in Buildings Except Low-Rise Residential Buildings

ASHRAE Thermal Guidelines for Data Processing Environments

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

www.asme.org

ASME B31.5, Refrigeration Piping and Heat Transfer Components

INTERNATIONAL CODE COUNCIL (ICC)

www.iccsafe.org

International Mechanical Code (IMC)

(Refer to UFC 1-200-01, for applicable publication date)

MIDWEST INSULATION CONTRACTORS ASSOCIATION (MICA)

www.micainsulation.org

North American Commercial & Industrial Insulation Standards

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

www.nfpa.org

NFPA 30A, Code for Motor Fuel Dispensing Facilities and Repair Garages

NFPA 54, National Fuel Gas Code

NFPA 58, Liquefied Petroleum Gas Code

NFPA 70, National Electric Code (NEC)

NFPA 88A, Standard for Parking Structures

NFPA 90A, Standard for the Installation of Air Conditioning and Ventilating Systems

NFPA 96, Standard for Ventilation Control and Fire Protection of Commercial Cooking Operations

NFPA 1500, Standard on Fire Department Occupational Safety, Health, and Wellness Program

SHEET METAL AND AIR CONDITIONING CONTRACTORS NATIONAL ASSOCIATION, INC. (SMACNA)

www.smacna.org

HVAC Air Duct Leakage Test Manual

UNDERWRITERS LABORATORIES INC.

www.ul.com

UL 181, *Factory-Made Air Ducts and Connectors*

UL 710B, *Recirculating Systems*

UL 2075, *Gas and Vapor Detectors and Systems*

UNITED STATES DEPARTMENT OF DEFENSE

MIL-STD 3007, *Standard Practice for Unified Facilities Criteria and Unified Facilities Guide Specifications*

<https://www.wbdg.org/ffc/dod/federal-military-specifications-standards/mil-std-3007>

UNIFIED FACILITIES CRITERIA (UFC)

<https://www.wbdg.org/ffc/dod/unified-facilities-criteria-ufc>

UFC 1-200-01, *DoD Building Code*

UFC 1-200-02, *High Performance and Sustainable Building Requirements*

UFC 1-201-01, *Non-Permanent DoD Facilities in Support of Military Operations*

UFC 3-101-01, *Architecture*

UFC 3-400-02, *Design: Engineering Weather Data*

UFC 3-401-01, *Mechanical Engineering*

UFC 3-410-02, *Direct Digital Control for HVAC and Other Building Control Systems*

UFC 3-410-04, *Industrial Ventilation*

UFC 3-420-01, *Plumbing Systems*

UFC 3-430-07, *Operations and Maintenance: Inspection and Certification of Boilers and Unfired Pressure Vessels*

UFC 3-430-11, *Boiler Plant Instrumentation and Control Systems*

UFC 3-440-01, *Facility-Scale Renewable Energy Systems*

UFC 3-450-01, *Noise and Vibration Control*

UFC 3-460-01, *Design: Petroleum Fuel Facilities*

UFC 3-501-01, *Electrical Engineering*

UFC 3-520-05, *Stationary and Mission Batteries*

UFC 3-580-01, *Telecommunications Interior Infrastructure Planning and Design*

UFC 3-600-01, *Fire Protection Engineering for Facilities*

UFC 4-010-01, *DoD Minimum Antiterrorism Standards for Buildings*

UFC 4-010-06, *Cybersecurity of Facility-Related Control Systems (FRCS)*

UFC 4-179-02, *Small Arms Ranges*

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UFC 4-211-01, *Aircraft Maintenance Hangars*

UFC 4-510-01, *Design: Military Medical Facilities*

UFC 4-722-01, *Design: Dining Facilities*

UFC 4-730-10, *Fire Stations*

UFC 4-740-02, *Fitness Centers*

**UNITED STATES DEPARTMENT OF DEFENSE, UNIFIED FACILITIES GUIDE
SPECIFICATIONS (UFGS)**

<https://www.wbdg.org/ffc/dod/unified-facilities-guide-specifications-ufgs>

UFGS 23 05 93, *Testing, Adjusting, and Balancing for HVAC*

UFGS 23 07 00, *Thermal Insulation for Mechanical Systems*

UFGS 23 09 00, *Instrumentation and Control for HVAC*

UFGS 23 09 23.01, *LonWorks Digital Control for HVAC and Other Building Control
Systems*

UFGS 23 09 23.02, *BACnet Digital Control for HVAC and Other Building Control
Systems*

UFGS 25 10 10, *Utility Monitoring and Control System (UMCS) Front End and
Integration*

APPENDIX A UFC 3-410-02 GLOSSARY

A-1 ACRONYMS

BCN	Building Control Network
BCS	Building Control System
CCR	Criteria Change Request
DDC	Direct Digital Control
FPOC	Field Point of Connection
HVAC	Heating Ventilating and Air Conditioning
IDG	Installation Design Guide
IP	Internet Protocol
IT	Information Technology
LNS	LonWorks Network Services
M&C	Monitoring and Control
MCX	Mandatory Center of Expertise
NCT	Network Configuration Tool
NDAA	National Defense Authorization Act
O&M	Operations and Maintenance
PDF	Portable Document Format
PID	Proportional, Intergal, Derivative
UFC	Unified Facilities Criteria
UFGS	Unified Facilities Guide Specification
UMCS	Utility Monitoring and Control System
USACE	U.S. Army Corps of Engineers
VAV	Variable Air Volume

A-2 DEFINITION OF TERMS

Several terms are defined in this appendix for ease of reference, but UFC 4-010-06 contains a more extensive, and definitive, list of terms. Should the definitions provide here and in UFC 4-010-06 conflict, the definitions in UFC 4-0101-06 should be used, and a Criteria Change Request (CCR) should be submitted to update the definitions in this UFC.

A-2.1 Field Control System (FCS) and Field Control Network (FCN).

A Building Control System or Utility Control System. The network used by the field control system is called the field control network (FCN).

A-2.1.1 Building Automation System (BAS).

The system consisting of the UMCS Front-End and connected building control systems which provides for control of the building electrical and mechanical systems as well as a user interface and supervisory capability (i.e. the portion of the UMCS for building control and excluding any connected UCS).

A-2.1.2 Building Control System (BCS) and Building Control Network (BCN).

One type of Field Control System. A control system primarily for building electrical and mechanical systems, typically HVAC (including central plants) and lighting. Building Control Systems are generally composed of direct digital controls (DDC) and do not have a full-featured user interface. They may have some local user interface such as “local display panels” but rely on the UMCS for the full user interface functionality. The network used by the building control system is called the building control network (BCN).

A-2.1.3 Utility Control System (UCS) and Utility Control Network(UCN).

One type of Field Control System. Used for control of utility systems such as an electrical substation, sanitary sewer lift station, water pump station, etc. Building controls are excluded from a UCS, however it is possible to have a Utility Control System and a Building Control System in the same facility, and for those systems to share components such as the Field Point of Connection (FPOC). A UCS may include its own local front-end.

A-2.2 Field Point of Connection (FPOC)

The connection point between the field control network (installed by the controls contractor) and the UMCS network (generally owned and installed by the installation IT staff). The FPOC device is typically on the IP network – usually a managed switch, owned and managed by the installation IT staff, and a focal point for Cybersecurity. Note that this definition has evolved over time and may be at variance with older usage of the term.

A-2.3 Industrial Control System (ICS)

One type of control system. Most specifically a control system which controls an industrial (manufacturing) process. Sometimes also used to refer to other types of control systems, particularly utility control systems such as electrical, gas, or water distribution systems.

Note that ICS is at times used to mean “all control systems”, so care must be taken to interpret the term in the appropriate context.

A-2.4 Utility Monitoring and Control System (UMCS)

The system consisting of one or more field control systems connected to a UMCS Front-End which provides for control of the electrical and mechanical systems as well as a user interface and supervisory capability. (i.e. the complete system consisting of the UMCS Front-End with all connected BCS and UCS systems). Note that in the past the term “UMCS” has sometimes been used to mean just “the UMCS front end”, but is no longer used in this manner.

A-2.5 Utility Monitoring and Control System (UMCS) Front End

The portion of the UMCS consisting primarily of computers running software to provide a full-featured user interface. In addition to providing a full user interface, this system performs functions such as alarming, scheduling, data logging, electrical demand limiting and report generation. The front end does not directly control physical systems; it interacts with them only through field control systems.

A-2.6 Utility Monitoring and Control System (UMCS) IP Network

The IP network used by the UMCS Front End for communication between Field Control Systems. This includes the IP infrastructure components only. The UMCS IP network is often also referred to as the “platform enclave” for cybersecurity purposes.

A-3 TERMS SPECIFICALLY NOT USED BY THIS UFC

The term SCADA (Supervisory Control and Data Acquisition) is not used as the definition can vary depending on context. In general usage, however, “SCADA” can be taken to mean “UCS”.

NDAA 2010 uses the term Energy Monitoring and Utility Control System to refer to a Utility Monitoring and Control System (UMCS). This term is not a standard term used by the controls industry, so this UFC and UFGS 25 10 10 do not use this term.

A-4 OTHER TERMINOLOGY

Other terminology related to control systems is defined in UFGS 25 10 10 and in the field control system specifications (e.g. UFGS 23 09 00, UFGS 23 09.01, and UFGS 23 09.02).

APPENDIX B UFC 3-410-02 REFERENCES

UNIFIED FACILITIES CRITERIA

http://www.wbdg.org/ccb/browse_cat.php?o=29&c=4

UFC 1-200-01, *DoD Building Code (General Building Requirements)*

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UNIFIED FACILITIES GUIDE SPECIFICATIONS

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UFGS 23 09 23.01, *LonWorks Direct Digital Control for HVAC and Other Building Control Systems*

UFGS 23 09 23.02, *BACnet Direct Digital Control for HVAC and Other Building Control Systems*

UFGS 23 09 93, *Sequences of Operation for HVAC Control*

UFGS 25 05 11, *Cybersecurity for Facility-Related Control Systems Integration*

UFGS 25 10 10, *Utility Monitoring and Control System (UMCS) Front End and*

APPENDIX A UFC 3-410-04 GLOSSARY

A-1 ACRONYMS.

acfm	actual cubic feet per minute
barg	bars - gauge
cm	centimeters
ft	feet
gr/lb	grains moisture per pound air
g/kg	grains moisture per kilogram air
HEPA	High-Efficiency Particulate Air
HVLP	High volume, low pressure
LVHV	low volume-high velocity
m/s	meters per second
m ³ /min	cubic meters per minute
m ³ /s	cubic meters per second
MTV	Minimum Transport Velocity
psia	pounds per square inch – atmosphere
RH	relative humidity
scfm	standard cubic feet per minute
w.g.	water gauge

APPENDIX B UFC 3-410-04 REFERENCES

AMERICAN CONFERENCE OF GOVERNMENTAL INDUSTRIAL HYGIENISTS (ACGIH)

www.acgih.org

ACGIH IV Manual, *ACGIH Industrial Ventilation: A Manual of Recommended Practice for Design*

AIR MOVEMENT AND CONTROL ASSOCIATION, INC.

www.amca.org

AMCA 201, *Fans and Systems*

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

www.ansi.org

ANSI 01.1, *Woodworking Machinery, Safety Requirements*

ANSI/AIHA/ASSE Z9.4, *Abrasive-Blasting Operations – Ventilation and Safe Practices for Fixed Location Enclosures*

ANSI/AIHA/ASSE Z9.7, *Recirculation of Air from Industrial Process Exhaust Systems*

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS (ASHRAE)

www.ashrae.org

IEEE Standard 1635/ASHRAE Guideline 21, *Guide for the Ventilation and Thermal Management of Batteries for Stationary Applications*

ANSI/ASHRAE Standard 62.1, *Ventilation for Acceptable Indoor Air Quality*

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

www.asme.org

ASME B19.3, *Safety Standard for Compressors for Process Industries*

ASME B31.1, *Power Piping*

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ASME BPVC VIII, *Boiler and Pressure Vessel Code Section VIII*

ASME PTC-9, *Performance Test Code on Displacement Compressors, Vacuum Pumps, and Blowers*

ASME PTC-10, *Performance Test Code on Compressors and Exhausters*

COMPRESSED AIR AND GAS INSTITUTE (CAGI)

<http://www.cagi.org>

CAGI *Compressed Air and Gas Handbook*

ANSI/CAGI B19.1, *Safety Standard for Compressor Systems*

COMPRESSED GAS ASSOCIATION (CGA)

<https://www.cganet.com>

CGA G-7.1, *Commodity Specification for Air*

ENVIRONMENTAL PROTECTION AGENCY (EPA)

<http://www.epa.gov>

EPA-560-OPTS-86-001, *A Guide to Respiratory Protection for the Asbestos Abatement Industry*

INSTRUMENT SOCIETY OF AMERICA (ISA)

<http://www.isa.gov>

ANSI/ISA-7.0.01, *Quality Standard for Instrument Air*

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

www.nfpa.org

NFPA 33, *Standard Spray Application Using Flammable or Combustible Materials*

NFPA 34, *Standard for Dipping Coating and Printing Processes Using Flammable or Combustible Liquids*

NFPA 68, *Standard on Explosion Protection by Deflagration Venting*

NFPA 69, *Standard on Explosion Prevention Systems*

NFPA 70, *National Electrical Code*

NFPA 91, *Standard for Exhaust Systems for Air Conveying of Vapors, Gases, Mists, and Noncombustible Particulate Solids*

NFPA 484, *Standard for Combustible Metals*

NFPA 650, *Pneumatic Conveying Systems for Handling Combustible Particulate Solids*

NFPA 651, *Standard for the Machining and Finishing of Aluminum and the Production and Handling of Aluminum Powders*

NFPA 654, *Standard for the Prevention of Fire and Dust Explosions from the Manufacturing, Processing, and Handling of Combustible Particulate Solids*

NFPA 664, *Prevention of Fires and Explosions in Wood Processing and Woodworking Facilities*

OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION

www.osha.gov

OSHA Standard 29 CFR 1910.1001, *Asbestos, General Industry*

OSHA Standard 29 CFR 1910.94, *Ventilation*

OSHA Standard 29 CFR 1910.94(a), *Abrasive Blasting*

OSHA Standard 29 CFR 1910.95, *Occupational Noise Exposure*

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OSHA Standard 29 CFR 1910.134(d), *Respiratory Protection*

OSHA Standard 29 CFR 1910.134(i), *Breathing Air Quality and Use*

OSHA Standard 29 CFR 1915.1001, *Asbestos, Shipyards*

OSHA Standard 29 CFR 1926.403, *Battery Rooms and Battery Charging*

SHEET METAL AND AIR CONDITIONING CONTRACTORS NATIONAL ASSOCIATION, INC. (SMACNA)

www.smacna.org

Guide for Steel Stack Design and Construction

Rectangular Duct Construction Standards

Round Industrial Duct Construction Standards

UNITED STATES DEPARTMENT OF THE NAVY

OPNAVINST 5100.23, Navy Occupational Safety and Health Program Manual,
<https://www.secnave.navy.mil/doni/SECNAV%20Manuals1/5100.23%20CH-3.pdf>

**UNITED STATES DEPARTMENT OF THE NAVY, NAVAL SEA SYSTEMS
COMMAND**

<http://www.navsea.navy.mil/>

NAVSEA OP5, Volume 1, *Ammunition and Explosives Ashore Safety Regulations for
Handling, Storing, Production, Renovation, and Shipping*

NAVSEA S6340-AA-MMA-010, *Otto Fuel II Safety, Storage, and Handling Instructions*

APPENDIX A UFC 3-420-01 GLOSSARY

A-1 ACRONYMS

A/E	Architect and Engineer
ANSI	American National Standards Institute
ASHRAE	American Society of Heating, Refrigerating, and Air Conditioning Engineers, Inc.
ASME	American Society of Mechanical Engineers
ASSE	American Society of Safety Engineers
ASSE	American Society of Sanitary Engineering
ASTM	American Society of Testing and Materials
CFR	Code of Federal Regulations
DoD	Department of Defense
ES/EWS	Emergency Shower and Eyewash Station
FCCHR	Foundation for Cross-Connection Control and Hydraulic Research
FM	Factory Mutual Corporation
ICC	International Code Council
IPC	International Plumbing Code®
ISEA	International Safety Equipment Association
NFPA	National Fire Protection Association
OSHA	Occupational Safety and Health Administration
PDI	Plumbing and Drainage Institute
PPI	Plastic Pipe Institute
SWH	Service Water Heater
UEPH	Unaccompanied Enlisted Personnel Housing
UFC	Unified Facilities Criteria

UL Underwriter's Laboratory

UOPH Unaccompanied Officers Personnel Housing

APPENDIX B UFC 3-420-01 REFERENCES

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

<http://www.ansi.org>

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ANSI/ASSE Z590.3, *Prevention Through Design: Guidelines for Addressing Occupational Hazards and Risks in Design and Redesign Processes*.

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AMERICAN SOCIETY OF SAFETY ENGINEERS (ASSE)

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ASSE 1017, *Temperature Activated Mixing Valves for Hot Water Distribution Systems*

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Handbook, *Fundamentals*

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<http://www.asme.org>

ASME A112.19.2, *Vitreous China Plumbing Fixtures and Hydraulic Requirements for Water Closets and Urinals*

CSA GROUP

<http://www.csagroup.org>

CSA B125.3, *Plumbing Fittings*

DEPARTMENT OF ENERGY

<https://energy.gov/eere/femp/downloads/how-buy-energy-and-water-efficient-products-federal-government>

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ENVIRONMENTAL PROTECTION AGENCY (EPA)

<http://www.epa.gov>

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INTERNATIONAL CODE COUNCIL (ICC)

<http://www.iccsafe.org>

International Plumbing Code®

International Plumbing Code® Commentary

INTERNATIONAL SAFETY EQUIPMENT ASSOCIATION (ISEA)

<http://www.iccsafe.org>

The National Standard Plumbing Code

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

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NFPA 1, *Fire Code*

NFPA 54, *National Fuel Gas Code*

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PLASTIC PIPE INSTITUTE (PPI)

<http://plasticpipe.org/index.html>

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PLUMBING AND DRAINAGE INSTITUTE (PDI)

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UNITES STATES ACCESS BOARD

<http://www.access-board.gov>

Architectural Barriers Act, *Accessibility Standards for Department of Defense Facilities*

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<http://www.usace.army.mil/>

Engineering Manual (EM) 200-1-13, *Environmental Quality: Minimizing the Risk of Legionellosis Associated with Building Water Systems on Army Installations*

UNITED STATES DEPARTMENT OF DEFENSE, UNIFIED FACILITIES CRITERIA (UFC)

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UFC 1-200-01, *DoD Building Code (General Building Requirements)*

UFC 1-200-02, *High Performance and Sustainable Building Requirements*

UFC 3-230-02, *O&M: Water Supply Systems*

UFC 3-230-03, *Water Treatment*

UFC 3-400-02, *Engineering Weather Data*

UFC 3-410-01, *Heating, Ventilating, and Air Conditioning Systems*

UFC 3-440-01, *Facility-Scale Renewable Energy Systems*

UFC 3-501-01, *Electrical Engineering*

UFC 3-600-01, *Fire Protection Engineering for Facilities*

UFC 4-010-01, *DoD Minimum Antiterrorism Standards for Buildings*

UFC 4-010-06, *Cybersecurity of Facility-Related Control Systems*

UFC 4-440-01, *Warehouses and Storage Facilities*

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SPECIFICATIONS (UFGS)**

<http://dod.wbdg.org>

UFGS 22 00 00, *Plumbing, General Purpose*

UFGS 23 03 00.00 20, *Basic Mechanical Materials and Methods*

UFGS 23 07 00, *Thermal Insulation for Mechanical Systems*

APPENDIX A UFC 3-420-02 GLOSSARY

A-1 ACRONYMS.

ACFM	Actual Cubic Feet per Minute
AFCEC	Air Force Civil Engineer Center
ASME	American Society of Mechanical Engineers
BACnet	Building Automation and Control network
BARG	Bar (Gauge)
BAS	Building System Automation
BIA	Bilateral Infrastructure Agreement
CAGI	Compressed Air and Gas Institute
CFM	Cubic Feet per Minute
DoD	Department of Defense
DN	Diametre Nominal (Nominal Diameter)
ER	Engineering Regulation
ESC	Environmental Severity Classifications
HNFA	Host Nation Funded Construction Agreements
HQUSACE	Headquarters, U.S. Army Corps of Engineers
HVAC	Heating, Ventilation, and Air Conditioning
ISA	Instrument Society of America
LCCA	Life Cycle Cost Analysis
LCCE	Life Cycle Cost Effective
LON	Lonmark® LONtalk
m	Meter
mm	Millimeter
NEMA	National Electrical Manufacturers Association
OSHA	Occupational Safety and Health Administration

PSIA	Pounds per Square Inch, Absolute
PSIG	Pounds per Square Inch, Gauge
RH	Relative Humidity
SCFM	Standard Cubic Feet per Minute
SOFA	Status of Forces Agreements
UFC	Unified Facilities Criteria
U.S.	United States

A-2 DEFINITION OF TERMS.

Actual Cubic Feet Per Minute (Actual Cubic Meters Per Minute): The flow rate of air expressed in terms of actual conditions, that is at non-standard conditions.

Standard Conditions: For the purposes of this UFC, standard conditions are taken to be those of CAGI: Standard Pressure of 14.5 psia (1 bar); Standard Dry Bulb Temperature of 68°F (20°C); Standard Humidity of 0% Relative Humidity.

Standard Cubic Feet Per Minute (Standard Cubic Meters Per Minute): The flow rate of air expressed in terms of standard conditions.

APPENDIX B UFC 3-420-02 REFERENCES

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

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APPENDIX A UFC 3-430-09 GLOSSARY

A-1 ACRONYMS.

AFCEC	Air Force Civil Engineer Center
ASHRAE	American Society of Heating, Refrigerating, and Air Conditioning Engineers
ASME	American Society of Mechanical Engineers
BIA	Bilateral Infrastructure Agreement
CHW	Chilled Water
CL-CH	Lean and Fat Clays
DDT	Drainable-Dryable-Testable
GS	Guide Support
HQUSACE	Headquarters, U.S. Army Corps of Engineers
HNFA	Host Nation Funded Construction Agreements
HTW	High temperature hot water
LPG	Liquified Petroleum Gas
LTW	Low temperature hot water
MTW	Medium temperature hot water
NAVFAC	Naval Facilities Engineering Systems Command
NCEL	Naval Civil Engineering Laboratory
NIST	National Institute of Standards and Technology
NTIS	National Technical Information Service
OHM-CM	Ohms Per Cubic Centimeter
PVC	Polyvinyl Chloride
RTRP	Reinforced Thermosetting Resin Plastic
SOFA	Status of Forces Agreements
UFC	Unified Facilities Criteria

UFGS	Unified Facilities Guide Specifications
WSL	Water Spread Limiting

A-2 DEFINITION OF TERMS.

Designer: This term used throughout this UFC refers to the person or persons responsible for preparing contract drawings and specifications for a heating or cooling distribution design. The main engineering discipline areas of the "designer" are:

- Mechanical. Includes expansion compensation, stress analysis, piping system design (material selections, fittings, valves, insulation), equipment selection and sizing, and pipe sizing and routing.
- Structural. Includes reinforced concrete design, pipe supports, and valve manhole cover design.
- Electrical. Includes cathodic protection, electrical service and controls (for example, sump pumps).
- Civil. Includes earth work, road crossings (for buried systems), system plans and profiles, and area drainage designs.

APPENDIX B UFC 3-430-09 REFERENCES

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APPENDIX A UFC 3-440-01 GLOSSARY

A-1 ACRONYMS

AC	Alternating Current
ASHRAE Engineers	American Society of Heating, Refrigerating, and Air-Conditioning
BAS	Building Automation System
BIA	Bilateral Infrastructure Agreement
BLCC	Building Life-Cycle Cost
C	Celsius
DC	Direct Current
DoD	Department of Defense
DTC	Differential Temperature Controller
e-NTU	effectiveness-number of transfer units
F	Fahrenheit
FAA	Federal Aviation Administration
FEMP	Federal Energy Management Program
Ft	Feet
Ft ²	Square feet
ft ² -°F-h/BTU	Square foot-degree Fahrenheit-hour per British thermal unit
HNFA	Host Nation Funded Construction Agreements
IEEE	Institute of Electrical and Electronics Engineers
ISC	Integral Storage Collector
kWh	Kilowatt-Hour
km/hr	Kilometers per hour
kPa	Kilopascals
LCCA	Life Cycle Cost Analysis

LMTD	Log mean temperature difference
m	Meters
m ²	Square meters
mg/L	Milligrams per liter
mm	Millimeters
MPH	Miles per hour
m/s	Meters per second
m ² -°C/W	Square meter-degree Celsius per watt
oz	ounces
psi	Pounds per square inch
PV	Photovoltaic
UFC	Unified Facilities Criteria
UL	Underwriters Laboratories
VDC	Volts direct current
W	Watts

APPENDIX B UFC 3-440-01 REFERENCES

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AMERICAN SOCIETY FOR TESTING AND MATERIALS

www.astm.org

ASTM B813, *Standard Specification for Water Flushable Liquid and Paste Fluxes for Soldering of Copper and Copper Alloy Tube*

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UNDERWRITERS LABORATORIES INC. (UL)

<http://ulstandards.ul.com/>

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APPENDIX A UFC 3-470-01 GLOSSARY

A-1 ACRONYMS

BCN	Building Control Network
BCS	Building Control System
CAC	Common Access Card
CCR	Criteria Change Request
CoN	Certificate of Networthiness
DDC	Direct Digital Control
FCS	Field Control System
FPOC	Field Point of Connection
GIS	Geographical Information Systems
GUI	Graphical User Interface
HVAC	Heating Ventilating and Air Conditioning
IDG	Installation Design Guide
IDIQ	Indefinite Delivery Indefinite Quantity
IP	Internet Protocol
IT	Information Technology
LNS	LonWorks Network Services
M&C	Monitoring and Control
MCX	Mandatory Center of Expertise
NCT	Network Configuration Tool
NDAA	National Defense Authorization Act
O&M	Operations and Maintenance
PDF	Portable Document Format
PID	Proportional, Integral, Derivative
SCADA	Supervisory Control And Data Acquisition
UCN	Utility Control Network
UCS	Utility Control System
UFC	Unified Facilities Criteria
UFGS	Unified Facilities Guide Specification

UMCS	Utility Monitoring and Control System
USACE	U.S. Army Corps of Engineers
VAV	Variable Air Volume

APPENDIX B UFC 3-470-01 REFERENCES

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UFGS 25 05 11, *Cybersecurity for Facility-Related Control Systems*

UFGS 25 10 10, *Utility Monitoring and Control System (UMCS) Front End and Integration*

B-1 DEFINITION OF TERMS

Several terms are defined in this appendix for ease of reference, but UFC 4-010-06 contains a more extensive, and definitive, list of terms. Should the definitions provide here and in UFC 4-010-06 conflict, the definitions in UFC 4-010-06 should be used, and a Criteria Change Request (CCR) should be submitted to update the definitions in this UFC.

B-1.1 Field Control System (FCS) and Field Control Network (FCN).

A Building Control System or Utility Control System. The network used by the field control system is called the field control network (FCN).

B-1.1.1 Building Automation System (BAS).

The system consisting of the UMCS Front-End and connected building control systems which provides for control of the building electrical and mechanical systems as well as a user interface and supervisory capability (i.e. the portion of the UMCS for building control and excluding any connected UCS).

B-1.1.2 Building Control System (BCS) and Building Control Network (BCN).

One type of Field Control System. A control system primarily for building electrical and mechanical systems, typically HVAC (including central plants) and lighting. Building Control Systems are generally composed of direct digital controls (DDC) and do not have a full-featured user interface. They may have some local user interface such as “local display panels” but rely on the UMCS for the full user interface functionality. The network used by the building control system is called the building control network (BCN).

B-1.1.3 Utility Control System (UCS) and Utility Control Network(UCN).

One type of Field Control System. Used for control of utility systems such as an electrical substation, sanitary sewer lift station, water pump station, etc. Building controls are excluded from a UCS, however it is possible to have a Utility Control System and a Building Control System in the same facility, and for those systems to share components such as the Field Point of Connection (FPOC). A UCS may include its own local front-end.

B-1.2 Field Point of Connection (FPOC).

The connection point between the field control network (installed by the controls contractor) and the UMCS network (generally owned and installed by the installation IT staff). The FPOC device is typically on the IP network – usually a managed switch, owned and managed by the installation IT staff, and a focal point for Cybersecurity. Note that this definition has evolved over time and may be at variance with older usage of the term.

B-1.3 Industrial Control System (ICS).

One type of control system. Most specifically a control system which controls an industrial (manufacturing) process. Sometimes also used to refer to other types of control systems, particularly utility control systems such as electrical, gas, or water distribution systems.

Note that ICS is at times used to mean “all control systems”, so care must be taken to interpret the term in the appropriate context.

B-1.4 Utility Monitoring and Control System (UMCS)

The system consisting of one or more field control systems connected to a UMCS Front-End which provides for control of the electrical and mechanical systems as well as

a user interface and supervisory capability. (i.e. the complete system consisting of the UMCS Front-End with all connected BCS and UCS systems). Note that in the past the term “UMCS” has sometimes been used to mean just “the UMCS front end”, but is no longer used in this manner.

B-1.5 Utility Monitoring and Control System (UMCS) Front End

The portion of the UMCS consisting primarily of computers running software to provide a full-featured user interface. In addition to providing a full user interface, this system performs functions such as alarming, scheduling, data logging, electrical demand limiting and report generation. The front end does not directly control physical systems; it interacts with them only through field control systems.

B-1.6 Utility Monitoring and Control System (UMCS) IP Network

The IP network used by the UMCS Front End for communication between Field Control Systems. This includes the IP infrastructure components only. The UMCS IP network is often also referred to as the “platform enclave” for cybersecurity purposes.

B-2 TERMS SPECIFICALLY NOT USED BY THIS UFC

The term SCADA (Supervisory Control And Data Acquisition) is not used as the definition can vary depending on context. In general usage, however, “SCADA” can be taken to mean “UCS”.

NDAA 2010 uses the term Energy Monitoring and Utility Control System to refer to a Utility Monitoring and Control System (UMCS). This term is not a standard term used by the controls industry, so this UFC and UFGS 25 10 10 do not use this term.

B-3 OTHER TERMINOLOGY

Other terminology related to control systems is defined in UFGS 25 10 10 and in the field control system specifications (e.g. UFGS 23 09 00, UFGS 23 09.01, and UFGS 23 09.02).

APPENDIX A UFC 3-490-06 GLOSSARY

A-1 ACRONYMS

AFCEC	Air Force Civil Engineer Center
ATCT	Air Traffic Control Tower
ATS	Automatic Transfer Switch
BIA	Bilateral Infrastructure Agreement
DOD	Department of Defense
EMS	Emergency Medical Services
FEO	Firefighters' Emergency Operation
HQ USACE	Headquarters, U.S. Army Corps of Engineers
HNFA	Host Nation Funded Construction Agreements
MR	Machine Room
NAVFAC	Naval Facilities Engineering Command
NEI	National Elevator Industry
SME	Subject Matter Expert
SOFA	Status of Forces Agreements
UFC	Unified Facilities Criteria
UFGS	Unified Facilities Guide Specification
U.S.	United States
VTE	Vertical Transportation Equipment

APPENDIX B UFC 3-490-06 REFERENCES

AMERICAN SOCIETY OF CIVIL ENGINEERS

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AMERICAN SOCIETY OF MECHANICAL ENGINEERS

<http://www.asce.org>

ASME A17.1, Safety Code for Elevators and Escalators

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International Building Code

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NATIONAL FIRE PROTECTION ASSOCIATION

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UFC 3-501-01, Electrical Engineering

UFC 3-600-01, Fire Protection Engineering for Facilities

UFC 4-510-01, Design: Medical Military Facilities

APPENDIX A UFC 3-501-01 GLOSSARY

A-1 ACRONYMS.

AC	Alternating Current
ACSR	Aluminum Conductor Steel-Reinforced
AFI	Air Force Instruction
AL	Aluminum
ANSI	American National Standards Institute
ASHRAE	American Society of Heating, Refrigerating, and Air-Conditioning Engineers
ATS	Acceptance Test Specifications
ATS	Automatic Transfer Switch
BCO	Base Communications Officer
BEQ/BOQ	Bachelor Enlisted Quarters/Bachelor Officer Quarters
BICSI	Building Industry Consulting Services International
BIL	Basic Impulse Level
CAD	Computer Aided Drafting
cal/cm ²	Calories per Centimeter Squared
CATV	Community Antenna Television
CCTV	Closed Circuit Television
CT	Current Transformer
CU	Copper
DC	Direct Current
Degrees C	Degrees Centigrade
Degrees F	Degrees Fahrenheit
DoD	Department of Defense
DOR	Designer of Record

EIA/TIA	Electronics Industries Association/Telecommunications Industry Association
ESS	Electronic Security Systems
ESS	Electric Supply Station
FASS	Facilities Systems Safety program
ft	Feet (or Foot)
ft ²	Foot Squared
GFCI	Ground Fault Circuit Interrupter
GND	Ground
HEMP	High-Altitude Electromagnetic Pulse
HID	High Intensity Discharge
HQUSACE	Headquarters, U.S. Army Corps of Engineers
HVAC	Heating Ventilation and Air Conditioning
hp	Horsepower
ICS	Intelligence Community Standard
IEEE	IEEE (Formerly, Institute of Electrical and Electronic Engineers)
in	Inch
kcmil	Thousand circular mils
kV	Kilovolts
kVA	Kilo-Volt-Ampere
kVAR	Kilo-Volt-Ampere-Reactive
m ²	Meter Squared
mm	Millimeter
MDP	Main Distribution Panel
MVA	Mega-Volt-Ampere
NAVFAC	Naval Facilities Engineering Systems Command

NEC	National Electrical Code
NEMA	National Electrical Manufacturers Association
NESC	National Electrical Safety Code
NETA	International Electrical Test Association
NFPA	National Fire Protection Association
NMCI	Navy and Marine Corps Intranet
O&M	Operation and Maintenance
PCBs	Polychlorinated Biphenyls
PPE	Personal Protective Equipment (Clothing)
PT	Potential Transformer
RCDD	Registered Communications Distribution Designer
RFP	Request for Proposal
RUS	Rural Utility Service
SCIF	Sensitive Compartmented Information Facility
SF6	Sulfur Hexafluoride
SWD	Switch Duty
TSEWG	Tri-Service Electrical Working Group
UFC	Unified Facilities Criteria
UFGS	Unified Facilities Guide Specifications
UL	Underwriters Laboratories
UPS	Uninterruptible Power Supply
V	Volt
VA	Volt-Amp
X/R	Ratio of Reactance to Resistance

Activity: The end user of a base or facility.

Base Communications Officer (BCO): The person(s) responsible for the telecommunications and data infrastructure for a base or facility.

Closed Circuit Television (CCTV): A network of cables and equipment to monitor and transmit video signals throughout a facility.

Community Antenna Television System (CATV): A network of cables, headend and electronic components that process and amplify television and frequency-modulated (FM) radio signals for distribution from one central location to outlets throughout a facility.

Contractor: Person(s) doing actual construction portion of a project.

Corrosive Area: An area identified by the Technical Reviewing Authority as requiring special equipment corrosion mitigation methods.

Designer of Record: The engineer responsible for the actual preparation of the construction documents.

Distribution Device: A facility located within a dwelling unit for interconnection or cross connection of interior telecommunications wiring. Passive cross connect facilities enable the termination of cable elements and their interconnection or cross-connection by means of jumpers and patch cords.

Ductbank: Two or more conduits (or ducts) routed together in a common excavation with or without concrete encasement.

Electric Supply Station: A station that transforms the energy level (voltage) for further bulk distribution at medium voltage levels.

Hot Cap: A sealed insulated cover over the end of a bare energized conductor.

Low Voltage System: An electrical system having a maximum root-mean-square (rms) voltage of less than 1,000 volts.

Medium Voltage System: An electrical system having a maximum RMS AC voltage of 1,000 volts to 34.5 kV. Some documents such as ANSI C84.1 define the medium voltage upper limit as 100 kV, but this definition is inappropriate for facility applications.

Non-Linear Loads: Loads that convert AC to DC and contain some kind of rectifier.

Service: The conductors and equipment for delivering electrical energy from the serving utility or Government-owned system to the wiring system of the premises served.

Service Point: Demarcation for the “service point” (for clarification of the NFPA 70 Article 100 definition): at the building/similar support structure service equipment line side connection.

Site Electrical Utilities: Site Electrical Utilities are the primary electric power distribution to the facilities and other electrical loads, exterior lighting not attached to the building; and telecommunication services (such as fiber optic, copper cable, CATV) required by the Facilities.

Systems Furniture: Modular prewired office furniture.

Technical Reviewing Authority: The “discipline specific” person at the DoD organization who is responsible for signing the project design documents. This is different than the authority responsible for waivers and exemptions required by Mil Std 3007.

Telecommunications Room: An enclosed space for telecommunications equipment, terminations, and cross-connect wiring for horizontal cabling.

TEMPEST: The unclassified name for the studies and investigation of compromising emanations (communications security).

Utilization Electric Supply Station: Equipment such as pole or pad-mounted transformers or secondary unit substations that transforms the energy level (voltage) to a utilization voltage for consumer use.

APPENDIX B UFC 3-501-01 REFERENCES

DEPARTMENT OF DEFENSE

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UFC 3-520-01, *Interior Electrical Systems*

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AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS (ASHRAE)

<https://www.ashrae.org/>

ASHRAE Standard 90.1, *Energy Standard for Buildings Except Low-Rise Residential Buildings*

Note: Refer to UFC 1-200-02, for applicable publication date

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

<https://www.ieee.org/>

IEEE C2, *National Electrical Safety Code*

IEEE C37.06, *IEEE Standard for AC High-Voltage Circuit Breakers Rated on a Symmetrical Current Basis – Preferred Ratings and Related Required Capabilities*

IEEE C37.13.1, *IEEE Standard for Definite Purpose Switching Devices for Use in Metal-Enclosed Low-Voltage*

IEEE C37.96, *IEEE Guide for AC Motor Protection*

IEEE C57.12.00, *IEEE Standard General Requirements for Liquid-Immersed Distribution, Power, and Regulating Transformers*

IEEE Std 551, *IEEE Recommended Practice for Calculating AC Short-Circuit Currents in Industrial and Commercial Power Systems*

IEEE Std 1547, *IEEE Standard for Interconnecting Distributed Resources with Electric Power Systems*

IEEE Std 1547.4, *IEEE Guide for Design, Operation, and Integration of Distributed Resource Island Systems with Electric Power Systems*

IEEE Std 551, *IEEE Recommended Practice for Calculating Short-Circuit Currents in Industrial and Commercial Power Systems*

IEEE Std 1584, *IEEE Guide for Performing Arc-Flash Hazard Calculations, including Amendments 1 and 2*

IEEE Std 1584.1, *IEEE Guide for the Specification of Scope and Deliverable Requirements for an Arc-Flash Hazard Calculation Study in Accordance with IEEE Std 1584™*

IEEE Std 3002.3TM, *IEEE Recommended Practice for Conducting Short-Circuit Studies and Analysis of Industrial and Commercial Power Systems*

INTERNATIONAL ELECTRICAL TESTING ASSOCIATION (NETA)

NETA ATS, *Acceptance Test Specifications*

NATIONAL FIRE PROTECTION ASSOCIATION

<https://www.nfpa.org/>

NFPA 70, *National Electrical Code*

NFPA 70E, *Standard for Electrical Safety in the Workplace*

NFPA 780, *Standard for the Installation of Lightning Protection Systems*

PROFESSIONAL ELECTRICAL APPARATUS RECONDITIONING LEAGUE (PEARL)

ANSI/PEARL EERS, *Electrical Equipment Reconditioning Standard for Electrical Apparatus and Equipment used in Commercial and Industrial Applications*

UNDERWRITER'S LABORATORY

UL 489, *Molded-Case Circuit Breakers and Circuit-Breaker Enclosures*

APPENDIX A UFC 3-520-01 GLOSSARY

A-1 ACRONYMS.

A	Amperes
AC	Alternating Current
AC	Armored Cable
AFCI	Arc Fault Current Interrupter
AFPA	Air Force Pamphlet
AHJ	Authority Having Jurisdiction
ANSI	American National Standards Institute
ASD	Adjustable Speed Drive
ATS	Automatic Transfer Switch
AWG	American Wire Gauge
BEQ	Bachelor's Enlisted Quarters
BOQ	Bachelor's Officer Quarters
CCTV	Closed Circuit Television
CATV	Cable Television
CFR	Code of Federal Regulations
COPS	Critical Operations Power System
Dc	Direct Current
DCOA	Designated Critical Operations Area
EGSA	Electrical Generating Systems Association
EMT	Electrical Metallic Tubing
ENT	Electrical Non-Metallic Tubing
EV	Electric Vehicle
EVSE	Electric Vehicle Supply Equipment
FC	Flat Cable Assemblies

FCC	Flat Conductor Cable
FMC	Flexible Metal Conduit
FMT	Flexible Metallic Tubing
Ft	Feet
GFI	Ground Fault Circuit Interrupter
GRS	Galvanized Rigid Steel
HDPE	High Density Polyethylene Conduit
HID	High Intensity Discharge
Hz	Hertz
IEEE	formerly Institute of Electrical and Electronics Engineers
IMC	Intermediate Metal Conduit
kA	Kilo-Amperes
kVA	Kilo-Volt-Amperes
kW	Kilowatt
LFMC	Liquidtight Flexible Metal Conduit
LFNC	Liquidtight Flexible Nonmetallic Conduit
M	Meter
MCC	Motor Control Center
MCOV	Maximum Continuous Overvoltage Rating
Mm	Millimeter
MC	Metal-Clad Cable
MI	Mineral-Insulated, Metal-Sheathed Cable
MV	Medium Voltage Cable
NAVFAC	Naval Facilities Engineering Systems Command
NEC	National Electrical Code
NEMA	National Electrical Manufacturers Association

NETA	International Electrical Testing Association
NFPA	National Fire Protection Association
NM, NMC, NMS	Nonmetallic-Sheathed Cable
NUCC	Nonmetallic Underground Conduit with Conductors
OSHA	Occupational Safety and Health Administration
PVC	Rigid Polyvinyl Chloride Conduit
RMC	Rigid Metal Conduit
RMS	Root-Mean-Square
RTRC	Reinforced Thermosetting Resin Conduit
SCIF	Sensitive Compartmented Information Facilities
SE, USE	Service-Entrance Cable
SPD	Surge Protective Devices
SWD	Switching Duty
TC	Power and Control Tray Cable
TSEWG	Tri-Service Electrical Working Group
UF	Underground Feeder and Branch-Circuit Cable
UFC	Unified Facilities Criteria
UL	Underwriters Laboratories
UPS	Uninterruptible Power Supply
USACE	U.S. Army Corps of Engineers
V	Volts
VFD	Variable Frequency Drive (see ASD)
VRLA	Valve-Regulated Lead Acid

Note: The terms listed here are provided for clarification of the design criteria provided in this UFC. Refer to IEEE Std 100 for additional electrical-related definitions.

Automatic Transfer Switch (ATS): A switch designed to sense the loss of one power source and automatically transfer the load to another source of power.

Branch Circuit: The circuit conductors and components between the final overcurrent device protecting the circuit and the equipment.

Closed Transition Switch: Transfer switch that provides a momentary paralleling of both power sources during a transfer in either direction. The closed transition is possible only when the sources are properly interfaced and synchronized.

Existing Facility: A facility is existing if changes to be made are cosmetic or minor in nature.

Harmonic: A sinusoidal component of a periodic wave or quantity having a frequency that is an integral multiple of the fundamental frequency.

Linear Load: An electrical load device that presents an essentially constant load impedance to the power source throughout the cycle of applied voltage in steady-state operation.

Listed: Applies to equipment or materials included in a list published by an organization acceptable to the authority having jurisdiction. The organization periodically inspects production and certifies that the items meet appropriate standards or tests as suitable for a specific use.

Low Voltage System: An electrical system having a maximum root-mean-square (rms) voltage of less than 1,000 volts.

Medium Voltage System: An electrical system having a maximum RMS AC voltage of 1,000 volts to 34.5 kV. Some documents such as ANSI C84.1 define the medium voltage upper limit as 100 kV, but this definition is inappropriate for facility applications.

Molded Case Circuit Breaker: A low voltage circuit breaker assembled as an integral unit in an enclosing housing of insulating material. It is designed to open and close by nonautomatic means, and to open a circuit automatically on a predetermined overcurrent, without damage to itself, when applied properly within its rating.

Motor Control Center: A piece of equipment that centralizes motor starters, associated equipment, bus and wiring in one continuous enclosed assembly.

New Construction: A facility is considered new if changes to be made are more than cosmetic or minor, such as major renovations, additions, or new facilities.

Nonlinear Load: A steady state electrical load that draws current discontinuously or has the impedance vary throughout the input ac voltage waveform cycle. Alternatively, a load that draws a nonsinusoidal current when supplied by a sinusoidal voltage source.

Power Quality: The concept of powering and grounding sensitive equipment in a manner that is suitable to the operation of that equipment.

Service Voltage: Voltage at the facility service entrance location.

Short Circuit: An abnormal condition (including an arc) of relatively low impedance, whether made accidentally or intentionally, between two points of different potential.

Subject to Physical Damage (or Subject to Severe Physical Damage): Locations that are subject to physical damage or severe physical damage include:

- Exposed interior raceways installed less than 6 ft above finished floor elevation where personnel are operating mechanized equipment on a recurring basis. Mechanized equipment that might be operated on a recurring basis includes vehicles, carts, forklifts, and pallet-handling units.
- Exposed exterior raceways installed less than 8 ft above finished grade or 8 ft above floor elevation for raceways on elevated platforms, loading docks, or stairwells.
- Exposed raceways where personnel operate mobile or fixed-in-place hoisting equipment.

Surge Protector: A device composed of any combination of linear or nonlinear circuit elements and intended for limiting surge voltages on equipment by diverting or limiting surge current; it prevents continued flow of current and is capable of repeating these functions as specified.

Transfer Switch: A device for transferring one or more load conductor connections from one power source to another.

Uninterruptible Power Supply System: A system that converts unregulated input power to voltage and frequency controlled filtered ac power that continues without interruption even with the deterioration of the input ac power.

Utilization Voltage: The voltage at the line terminals of utilization equipment.

APPENDIX B UFC 3-520-01 REFERENCES

UNIFIED FACILITIES CRITERIA

<https://www.wbdg.org/dod/ufc>

UFC 1-200-01, *DoD Building Code*

UFC 1-200-02, *High Performance and Sustainable Building Requirements*

UFC 3-501-01, *Electrical Engineering*

UFC 3-510-01, *Foreign Voltages and Frequencies Guides*

UFC 3-520-05, *Stationary and Mission Batteries*

UFC 3-530-01, *Interior and Exterior Lighting Systems*

UFC 3-540-01, *Engine-Driven Generator Systems for Prime and Standby Power Applications*

UFC 3-550-01, *Exterior Electrical Power Distribution*

UFC 3-555-01, *Aircraft Point-of-Use Power Systems*

UFC 3-560-01, *Operation and Maintenance: Electrical Safety*

UFC 3-575-01, *Lightning and Static Electricity Protection Systems*

UFC 3-600-01, *Fire Protection Engineering for Facilities*

UFC 4-010-06, *Cybersecurity of Facility-Related Control Systems (FRCS)*

UFC 4-420-01, *Ammunition And Explosives Storage Magazines*

UNIFIED FACILITIES GUIDE SPECIFICATIONS

<https://www.wbdg.org/dod/ufgs>

UFGS 26 35 44, *70 VDC Solid State Converter*

TRI-SERVICE ELECTRICAL WORKING GROUP

<https://www.wbdg.org/dod/supp-tech-documents>

TSEWG TP-02, *Capacitors for Power Factor Correction*

TSEWG TP-04, *Stationary Battery and Charger Sizing*

TSEWG TP-05, *Interior Transformer Ratings and Installation*

TSEWG TP-08, *Equipment Enclosures and Hazardous Locations*

TSEWG TP-09, *Automatic Transfer Equipment*

TSEWG TP-19, *Static Uninterruptible Power Supply (UPS)*

DEPARTMENT OF THE AIR FORCE

AFPAM 32-1186, *Valve-Regulated Lead-Acid Batteries for Stationary Applications*

DEPARTMENT OF THE NAVY

NAVSEA OP-5, *Ammunition and Explosives Safety Ashore*

OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION

<https://www.osha.gov>

29 CFR 1910.305, *Wiring Methods, Components and Equipment for General Use — Design Safety Standards for Electrical Systems*

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

<https://www.ansi.org/>

Note: Many ANSI documents are sponsored or co-sponsored by other organizations, such as NEMA or IEEE.

ANSI C84.1, *Electric Power Systems Voltage Ratings (60 Hz)*

ANSI Z535.4, *Product Safety Signs and Labels*

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS (ASHRAE)

<https://www.ashrae.org/>

ASHRAE Standard 90.1, *Energy Standard for Buildings Except Low Rise Residential Buildings*, (Refer to UFC 1-200-02, for applicable publication date)

INTERNATIONAL ELECTRICAL TESTING ASSOCIATION (NETA)

<https://www.netaworld.org/home>

ANSI/NETA ATS, *Standard for Acceptance Testing Specifications for Electrical Power Equipment and Systems*

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

<https://www.ieee.org/>

IEEE C37.20.1, *IEEE Standard for Metal-Enclosed Low-Voltage (1000 Vac and below, 3200 Vdc and below) Power Circuit Breaker Switchgear*

IEEE C37.20.7, *IEEE Guide for Testing Switchgear Rated Up to 52 kV for Internal Arcing Faults*

IEEE C37.20.9, *IEEE Standard for Metal-Enclosed Switchgear Rated 1 kV to 52 kV Incorporating Gas Insulating Systems*

IEEE C37.23, *IEEE Standard for Metal-Enclosed Bus*

IEEE C57.110, *IEEE Recommended Practice for Establishing Liquid-Immersed and Dry-Type Power and Distribution Transformer Capability When Supplying Nonsinusoidal Load Currents*

IEEE C62.41, *IEEE Recommended Practice for Surge Voltages in Low-Voltage AC Power Circuits*

IEEE Std 100, *The Authoritative Dictionary of IEEE Standards Terms*

IEEE Std 450, *IEEE Recommended Practice for Maintenance, Testing, and Replacement of Vented Lead-Acid Batteries for Stationary Applications*

IEEE Std 484, *IEEE Recommended Practice for Installation Design and Installation of Vented Lead-Acid Batteries for Stationary Applications*

IEEE Std 485, *IEEE Recommended Practice for Sizing Lead-Acid Batteries for Stationary Applications*

IEEE Std 519, *IEEE Standard for Harmonic Control in Electrical Power Systems*

IEEE Std 1106, *IEEE Recommended Practice for Installation, Maintenance, Testing, and Replacement of Vented Nickel-Cadmium Batteries for Stationary Applications*

IEEE Std 1115, *IEEE Recommended Practice for Sizing Nickel-Cadmium Batteries for Stationary Applications*

IEEE Std 1159, *IEEE Recommended Practice for Monitoring Electric Power Quality*

IEEE Std 1184, *IEEE Guide for Batteries for Uninterruptible Power Supply Systems*

IEEE Std 1187, *IEEE Recommended Practice for Installation Design and Installation of Valve-Regulated Lead-Acid Storage Batteries for Stationary Applications*

IEEE Std 1188, *IEEE Recommended Practice for Maintenance, Testing, and Replacement of Valve Regulated Lead-Acid (VRLA) Batteries for Stationary Applications*

IEEE Std 1189, *Guide for Selection of Valve Regulated Lead-Acid (VRLA) Batteries for Stationary Applications*

IEEE Std 1375, *Guide for the Protection of Stationary Battery Systems*

NATIONAL ELECTRICAL MANUFACTURER'S ASSOCIATION (NEMA)

<https://www.nema.org/>

NEMA ICS 1, *Standard for Industrial Control and Systems: General Requirements*

NEMA ICS 2, *Industrial Control and Systems Controllers, Contactors, and Overload Relays Rated 600 Volts*

NEMA ICS 7, *Adjustable-Speed Drives*

NEMA PB 2, *Deadfront Distribution Switchboards*

NEMA ST 20, *Dry Type Transformers for General Applications*

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

<https://www.nfpa.org/>

NFPA 70, *National Electrical Code (NEC)*

NFPA 70E, *Standard for Electrical Safety in the Workplace*

NFPA 72, *National Fire Alarm and Signaling Code*

NFPA 99, *Health Care Facilities Code*

NFPA 780, *Standard for the Installation of Lightning Protection Systems*

UNDERWRITER'S LABORATORY

UL 497A, *Standard for Secondary Protectors for Communications Circuit*

UL 497B, *Protectors for Data Communications and Fire-Alarm Circuits*

UL 845, *Motor Control Centers*

UL 891, *Standard for Switchboards*

UL 1283, *Standard for Electromagnetic Interference Filters*

UL 1449, *Standard for Surge Protective Devices*

UL 1558, *Standard for Metal-Enclosed Low-Voltage Power Circuit Breaker Switchgear*

APPENDIX A UFC 3-520-02 GLOSSARY

A-1 ACRONYMS.

AC	Alternating Current
AIAG	Automotive Industry Action Group
BIT	Build-in-Test
\1\ C5ISR	Command, Control, Communications, Computer, Cyber, Intelligence, Surveillance and Reconnaissance /1/
CA	Criticality Analysis
CAIP	Critical Asset Identification Process
CBA	Cost-Benefit Analysis
CND	Cannot Duplicate
CM	Corrective Maintenance
DOD	Department of Defense
DOE	Design of Experiments
EMSG	European Maintenance System Guide
EPRI	Electric Power Research Institute
FAA	Federal Aviation Administration
FEA	Finite Element Analysis
FIT	Framework for Integrated Test
FMEA	Failure Mode and Effects Analysis
FMECA	Failure Modes, Effects and Criticality Analysis
FRACAS	Failure Reporting Analysis and Corrective Actions
FTA	Fault Tree Analysis
HFE	Human Factors Engineering
HVAC	Heating, Ventilating and Air Conditioning
IEEE	Institute of Electrical and Electronic Engineers

LRU	Line Replaceable Unit
MDT	Mean Downtime
MSG	Maintenance Steering Group
MTBF	Mean Time Between Failures
MTTF	Mean Time to Failure
MTTR	Mean Time to Repair
NASA	National Aeronautics and Space Administration
NEC	National Electrical Code
NERC	North American Electric Reliability Corporation
NDI	Nondestructive Inspection
NPRD	Nonelectronic Parts Reliability Data
O&M	Operations and Maintenance
O&S	Operating & Support
OEM	Original Equipment Manufacturer
OH	Operating Hours
OSHA	Occupational Safety and Health Administration
PC	Personal Computer
PDF	Probability Density Function
PLC	Programmable Logic Controller
PM	Preventative Maintenance
PREP	Power Reliability Enhancement Program
PREPIS	Power Reliability Enhancement Program Information System
QFD	Quality Function Deployment
R/A	Reliability/Availability
R&M	Reliability and Maintainability
R&R	Remove and Replace

RAM	Reliability, availability, and maintainability
RBD	Reliability Block Diagram
RCM	Reliability-Centered Maintenance
RGT	Reliability Growth Test
RPN	Risk Priority Number
RTOK	Retest OK
SCA	Sneak Circuit Analysis
SCADA	Supervisory Control and Data Acquisition
SE	Systems Engineering
TA	Thermal Analysis
TAAF	Test Analyze and Fix
TM	Technical Manual
UFC	Unified Facilities Criteria
UPS	Uninterruptable Power Supply
US	United States
USACE	United States Army Corps of Engineers
WCCA	Worst Case Circuit Analysis
WRA	Weapon Replaceable Assembly

A-2 DEFINITION OF TERMS.

Active Redundancy: Two or more components in a parallel combination where all are powered and active simultaneously. Not all components are required to function for the system (or next higher assembly) to function.

Active Time: That time during which an item is in an operational inventory.

Affordability: Affordability is a measure of how well customers can afford to purchase, operate, and maintain a product over its planned service life. Affordability is a function of product value and product costs. It is the result of a balanced design in which long-term support costs are considered equally with near-term development and manufacturing costs.

Alignment: Performing the adjustments that are necessary to return an item to specified operation.

Alpha (α): The probability, expressed as a decimal that a given part will fail in the identified mode. The sum of all alphas for a component will equal one (1).

Assessment: Current evaluation of a component's or system's reliability. A prediction.

Availability: The instantaneous probability that a component will be up.

Availability, Inherent (A_i): The instantaneous probability that a component will be up. A_i considers only downtime for repair due to failures. No logistics delay time, preventative maintenance, etc. is included.

Availability, Operational (A_o): A_o is the instantaneous probability that a component will be up but differs from inherent availability in that it includes ALL downtime. Included is downtime for both corrective maintenance and preventative maintenance, including any logistics delay time.

Beta (β): The conditional probability that the effect of a failure mode will occur, expressed as a decimal. If a failure is to occur, what is the probability that the outcome will occur.

Block Diagrams: Availability block diagrams and reliability block diagrams are visual representations of the interactions between contributors to reliability, availability, and maintainability. Each block tends to represent a physical component in the system and its associated reliability/availability.

Boolean Algebra: Boolean algebra is a method of calculating system availability based on logical interactions between components. AND and OR operators define mathematical operations.

Brownout: Occurs during a power failure when some power supply is retained, but the voltage level is below the minimum level specified for the system. A very dim household light is a symptom of a brownout.

Calibration: A comparison of a measuring device with a known standard and a subsequent adjustment to eliminate any differences. Not to be confused with alignment.

Cannot Duplicate (CND): A situation when a failure has been noted by the operator but cannot be duplicated by maintenance personnel attempting to correct the problem. Also see Retest OK.

Checkout: Tests or observations of an item to determine its condition or status.

Compensating Provision: Actions available or that can be taken to negate or reduce the effect of a failure on a system.

Component: A piece of electrical or mechanical equipment viewed as an entity for the purpose of reliability evaluation.

Condition-Based PM: Maintenance performed to assess an item's condition and performed as a result of that assessment. Some texts use terms such as predictive maintenance and on-condition. The definition of condition-based PM used herein includes these concepts. In summary, the objectives of condition-based PM are to first evaluate the condition of an item, then, based on the condition, either determine if a hidden failure has occurred or determine if a failure is imminent, and then take appropriate action. Maintenance that is required to correct a hidden failure is, of course, corrective maintenance.

Confidence Level/Interval: A statistical measure of the uncertainty associated with an estimate. For example, an estimate of MTBF is 103 hours. Using statistical techniques (such as the chi-square method) a 95% confidence interval of 100.1 to 105.9 is obtained. That is, 95% of the time, the actual MTBF will be between 100.1 and 105.9 hours. The confidence interval depends on sample size and variance.

Corrective Action: A documented design, process, procedure, or materials change implemented and validated to correct the cause of failure or design deficiency.

Corrective Maintenance (CM): All actions performed as a result of failure, to restore an item to a specified condition. Corrective maintenance can include any or all the following steps: Localization, Isolation, Disassembly, Interchange, Reassembly, Alignment and Checkout.

Cost: The expenditure of resources (usually expressed in monetary units) necessary to develop, acquire, or use a product over some defined period of time.

Critical Equipment/Systems: Critical equipment/systems include those items of equipment or systems that directly supply power to equipment and systems used to perform the primary mission(s) of the C5ISR site.

Criticality: A relative measure of the consequences of a failure mode and the frequency of its occurrence.

Criticality Analysis (CA): A procedure by which each potential failure mode is ranked according to the combined influence of severity and probability of occurrence.

Critical Load: That portion of the technical load used to successfully accomplish the site missions and having a requirement for 100 percent continuity in power service, such as from the Uninterruptible Power Supply (UPS) system. These loads also include any equipment which, upon loss of power, will create an unacceptable impact on the mission or mission equipment.

Dependability: A measure of the degree to which an item is operable and capable of performing its required function at any (random) time during a specified mission profile, given item availability at the start of the mission. (Item state during a mission includes the combined effects of the mission-related system R&M parameters but excludes non-mission time; see availability).

Design Agency: The agency responsible for the overall design of the facility.

Detection Method: The method by which a failure can be discovered by the system operator under normal system operation or by a maintenance crew carrying out a specific diagnostic action.

Diagnostics: The hardware, software, or other documented means used to determine that a malfunction has occurred and to isolate the cause of the malfunction. Also refers to "the action of detecting and isolating failures or faults."

Downtime: That element of time during which an item is in an operational inventory but is not in condition to perform its required function.

Effectiveness: The degree to which PM can provide a quantitative indication of an impending functional failure, reduce the frequency with which a functional failure occurs, or prevent a functional failure.

End Effect: The consequence a failure mode has upon the operation, function, or status at the highest indenture level.

Equipment: A general term designating an item or group of items capable of performing a complete function.

Failure (f): The termination of the ability of a component or system to perform a required function.

\1\ **Facility Energy:** The energy delivered to, generated by and/or consumed by an operational facility. /1/

Failure, Catastrophic: A failure that causes loss of the item, human life, or serious collateral damage to property.

Failure, Hidden: A failure that is not evident to the operator; that is, it is not a functional failure. A hidden failure may occur in two different ways. In the first, the item

that has failed is one of two or more redundant items performing a given function. The loss of one or more of these items does not result in a loss of the function. The second way in which a hidden failure can occur is when the function performed by the item is normally inactive. Only when the function is eventually required will the failure become evident to the operator. Hidden failures must be detected by maintenance personnel.

Failure, Intermittent: Failure for a limited period of time, followed by the item's recovery of its ability to perform within specified limits without any remedial action.

Failure, Random: A failure, the occurrence of which cannot be predicted except in a probabilistic or statistical sense.

Failure Analysis: Subsequent to a failure, the logical systematic examination of an item, its construction, application, and documentation to identify the failure mode and determine the failure mechanism and its basic course.

Failure Cause: The physical or chemical processes, design defects, quality defects, part misapplication or other processes which are the basic reason for failure, or which can initiate the physical process by which deterioration proceeds to failure.

Failure Effect: The consequence(s) a failure mode has on the operation, function, or status of an item. Failure effects are typically classified as local, next higher level, and end.

Failure Mechanism: The physical, chemical, electrical, thermal, or other process which results in failure.

Failure Mode: The way in which a failure is observed, describes the way the failure occurs, such as, short, open, fracture and excessive wear.

Failure Mode and Effects Analysis (FMEA): A procedure by which each potential failure mode in a product (system) is analyzed to determine the results or effects thereof on the product and to classify each potential failure mode according to its severity or risk probability number.

Failure Modes, Effects, and Criticality Analysis (FMECA): The term is used to emphasize the classifying of failure modes as to their severity (criticality).

Failure Rate (λ): The mean (arithmetic average, also known as the forced outage rate) number of failures of a component and/or system per unit exposure time. The most common unit in reliability analyses is hours (h). However, some industries use failures per year (f/y) which is denoted by the symbol (λ_y).

Failure Reporting and Corrective Action System (FRACAS): A closed-loop system for collecting, analyzing, and documenting failures and recording any corrective action taken to eliminate or reduce the probability of future such failures.

False Alarm: A fault indicated by BIT or other monitoring circuitry where no fault can be found or confirmed.

Fault: Immediate cause of failure (for example, maladjustment, misalignment, defect, etc.).

Fault Detection (FD): A process that discovers the existence of faults.

Fault Isolation (FI): The process of determining the location of a fault to the indenture level necessary to affect repair.

Fault Tree Analysis: An analysis approach in which each potential system failure is traced back to all faults that could cause the failure. It is a top-down approach, whereas the FMEA is a bottom-up approach.

Hidden Failure: See Failure, Hidden.

Hours Downtime Per Year (Hrdt/Year): Average hours the item is expected to be not functional in a one-year period, caused by both preventative maintenance and failures. This includes any logistics delay time.

Indenture Levels: The levels which identify or describe the relative complexity of an assembly or function.

Isolation: Determining the location of a failure to the extent possible, using accessory equipment.

Item: Used interchangeably in this document with product or equipment. Usually refers to the individual article rather than the inclusive class or kind of product.

Item Criticality Number (C_r): A relative measure of consequence of an item failure and its frequency of occurrence. This factor is not applicable to a qualitative analysis.

Laplace Statistic: A statistic used to determine if a data set indicates a positive or negative trend, at a given level of confidence.

Levels of Maintenance: The division of maintenance, based on different and requisite technical skill, which jobs are allocated to organizations in accordance with the availability of personnel, tools, supplies, and the time within the organization. Typical maintenance levels are organizational, intermediate, and depot.

Life Cycle Cost (LCC): The sum of acquisition, logistics support, operating, and retirement and phase-out expenses.

Line Replaceable Unit (LRU): A unit designed to be removed upon failure from a larger entity (product or item) in the operational environment, normally at the organizational level.

Local Effect: The consequence a failure mode has on the operation, function or status of the specific item being analyzed.

Localization: Determining the location of a failure to the extent possible, without using accessory test equipment.

Logistic Delay Time: That element of downtime during which no maintenance is being accomplished on the item because of either supply or administrative delay.

Logistics Support: The materials and services required to enable the operating forces to operate, maintain, and repair the end item within the maintenance concept defined for that end item.

Maintainability: The relative ease and economy of time and resources with which an item can be retained in, or restored to, a specified condition when maintenance is performed by personnel having specified skill levels, using prescribed procedures and resources, at each prescribed level of maintenance and repair. Also, the probability that an item can be retained in, or restored to, a specified condition when maintenance is performed by personnel having specified skill levels, using prescribed procedures and resources, at each prescribed level of maintenance and repair.

Maintenance: All actions necessary for retaining an item in or restoring it to a specified condition.

Maintenance Action: An element of a maintenance event. One or more tasks (such as, fault localization, fault isolation, servicing, and inspection) necessary to retain an item's condition or restore it to a specified condition.

Maintenance Concept: A description of the planned general scheme for maintenance and support of an item in the operational environment. It provides a practical basis for design, layout, and packaging of the system and its test equipment. It establishes the scope of maintenance responsibility for each level of maintenance and the personnel resources required to maintain the system.

Maintenance Event: One or more maintenance actions required to effect corrective and preventive maintenance due to any type of failure or malfunction, false alarm, or scheduled maintenance plan.

Maintenance Task: The maintenance effort necessary for retaining an item in or changing/restoring it to a specified condition.

Maintenance Time: An element of downtime that excludes modification and delay time.

Mean: Also called the expected value of a random variable, the mean is defined as follows: Let X be a continuous random variable with a probability density function = f . The expected value of X is:

$$E(X) = \int_x xf(x)dx$$

The mean, or expected value, is analogous to the concept of center of mass in mechanics.

Mean Downtime (MDT): The average downtime caused by preventative and corrective maintenance, including any logistics delay time. This is synonymous with mean time to restore system (MTTRS) as found in some publications.

Mean Time Between Failures (MTBF): The mean exposure time between consecutive failures of a component. MTBF is a required measurement used for calculating inherent availability. It can be estimated by dividing the exposure time by the number of failures in that period.

Mean Time Between Maintenance (MTBM): The average time between all maintenance events that cause downtime, both preventative and corrective maintenance, and includes any associated logistics delay time.

Mean Time To Failure (MTTF): The mean exposure time between consecutive repairs (or installations) of a component and the next failure of that component. MTTF is commonly found for nonrepairable items such as fuses or bulbs, etc.

Mean Time To Maintain (MTTM): The average downtime for preventative maintenance. This includes any logistics delay time.

Mean Time To Repair (MTTR): The mean time to replace or repair a failed component. Logistics delay time associated with the repair, such as parts acquisitions, crew mobilization, are not included. It can be estimated by dividing the summation of repair times by the number of repairs and, therefore, is practically the average repair time. The most common unit in reliability analyses is hours (h/f).

Mission Phase Operational Mode: The statement of the mission phase and mode of operation of the system or equipment in which the failure occurs.

Mission Reliability: The probability that a system will complete its intended mission. Hardware failures that do not hinder the success of the mission (for example, due to redundancy) are not counted against mission reliability.

Next Higher Level Effect: The consequence a failure mode has on the operation, functions, or status of the items in the next higher indenture level above the specific item being analyzed.

Non-Destructive Inspection (NDI): Any method used for inspecting an item without physically, chemically, or otherwise destroying or changing the design characteristics of the item. However, it may be necessary to remove paint or other external coatings to use the NDI method. A wide range of technology and methods are usually described as nondestructive inspection, evaluation, or testing (collectively referred to as non-destructive evaluation or NDE). The core of NDE is commonly thought to contain ultrasonic, visual, radiographic, eddy current, liquid penetrant, and magnetic particle inspection methods. Other methodologies include acoustic emission, use of laser interference, microwaves, NMR and MRI, thermal imaging, and so forth.

On-Condition Maintenance: See Condition-based PM.

One-Line Diagram: A one-line diagram is a drawing of an electrical or mechanical system that shows how the parts interact. It shows paths of electrical flow, water flow, gas flow, etc. It will also list system component and component sizes.

Operating and Support (O&S) Costs: Those costs associated with operating and supporting (such as, using) a product after it is purchased or fielded.

Operational Readiness: The ability of a military unit to respond to its operation plan(s) upon receipt of an operations order. (A function of assigned strength, item availability, status, or supply, training, etc.).

Operational Reliability: The reliability of a system or equipment after it is put in operation.

Parallel Combination: The combining of two or more items in such a way that not all components are required for operation – thus, the parallel combination is characterized by alternate paths of operation.

Predicted: That which is expected at some future time, postulated on analysis of past experience and tests.

Predictive Maintenance: See Condition-based PM.

Preventative Maintenance (PM): All actions performed in an attempt to retain an item in a specified condition. These actions may or may not result in downtime for the component and may or may not be performed on a fixed interval.

Probability Distribution: A formula that describes the probabilities associated with the values of a discrete random variable.

Product: An equipment, item, or hardware contracted for by a customer. Usually used to describe the inclusive class or kind of item, equipment, etc., rather than each individual entity.

Qualitative Analysis: A means of conducting an analysis without data. Team member subjectively rank probabilities of occurrence, typically 1-10, in place of failure rates.

Quantitative Analysis: An analysis that is supported with data. Data is available for assigning failure rates and failure mode probabilities.

Reassembly: Assembling the items that were removed during disassembly and closing the reassembled items.

Redundancy: The existence of more than one means for accomplishing a given function. Each means of accomplishing the function need not necessarily be identical.

Reliability (R(t)): The probability that a component can perform its intended function for a specified time interval (t) under stated conditions. This calculation is based on the exponential distribution.

Reliability-Centered Maintenance (RCM): A disciplined logic or methodology used to identify preventive and corrective maintenance tasks to realize the inherent reliability of equipment at a minimum expenditure of resources, while ensuring safe operation and use.

Reliability Prediction: An estimate of reliability based on information that includes historical data, piece parts count, complexity, and piece part failure rates.

Retest Ok (RTOK): A situation where a failure was detected on the system, either through inspection or testing, but no fault can be found in the item that was eventually removed for repair at a field or depot location. Also see Cannot Duplicate.

Risk Priority Number (RPN): The Risk Priority Number (RPN) is the product of the Severity (1-10) and the Occurrence (1-10) ranking. The Risk Priority Number is used to rank and identify the concerns or risks associated with the operation due to the design. $RPN = (S) \times (O)$.

Severity: Considers the worst possible consequence of a failure classified by the degree of injury, property damage, system damage and mission loss that could occur.

Scheduled Maintenance: Periodic prescribed inspection and/or servicing of products or items accomplished on a calendar, mileage, or hours of operation basis. Included in Preventive Maintenance.

Servicing: The performance of any act needed to keep an item in operating condition, (such as lubricating, fueling, oiling, cleaning, etc.), but not including preventive maintenance of parts or corrective maintenance tasks.

Single-Point Failure: A failure of an item that causes the system to fail and for which no redundancy or alternative operational procedure exists.

Standby Redundancy: Two or more components in a parallel combination where not all components are required at any time. The other components are disconnected, and power is applied prior to or simultaneously with switching.

Subsystem: A combination of sets, groups, etc. that performs an operational function within a product (system) and is a major subdivision of the product. (Example: Data processing subsystem, guidance subsystem).

Success: Achievement of an objective or completion of a function or set of functions.

Switch: A device that selects one component in a parallel or redundant configuration as the functioning component. Used for standby redundancy. Incorporates such provisions as logic circuits and fault detection.

System: A group of components connected or associated in a fixed configuration to perform a specified function.

System Downtime: The time interval between the commencement of work on a system (product) malfunction and the time when the system has been repaired and/or checked by the maintenance person, and no further maintenance activity is executed.

Technical Load: That portion of the operational which consists of general lighting and heating, ventilating, and air conditioning (HVAC) systems necessary to maintain normal operations and loads directly associated with the C5ISR missions at the site.

Testability: A design characteristic that allows status (operable, inoperable, or degraded) of an item to be determined and the isolation of faults within the item to be performed in a timely manner.

Total Downtime Events (Tde): The total number of downtime events (including scheduled maintenance and failures) during the Tp.

Total Failures (Tf): The total number of failures during the Tp.

Total Maintenance Actions (Tma): The total number of preventative maintenance actions which take the component down during the Tp.

Total Period (Tp): The calendar time over which data for the item was collected.

Total System Downtime: The time interval between the reporting of a system (product) malfunction and the time when the system has been repaired and/or checked by the maintenance person, and no further maintenance activity is executed.

Unscheduled Maintenance: Corrective maintenance performed in response to a suspected failure.

Uptime: That element of ACTIVE TIME during which an item is in condition to perform its required functions. (Increases availability and dependability).

Useful Life: The number of life units from manufacture to when the item has an unrepairable failure or unacceptable failure rate. Also, the period of time before the failure rate increases due to wearout.

User: The using Government Agency.

Using Government Agency: The Government Agency that will be responsible for completing the site missions and will have operational authority for the facility.

Wearout: The process that results in an increase of the failure rate or probability of failure as the number of life units increases.

Year (y): The unit of time measurement approximately equal to 8765.81277 hours (h). Any rounding of this value will have adverse effects on analyses depending on the magnitude of that rounding. 8766 is used commonly as it is the result of rounding to 365.25×24 (which accounts for a leap year every 4th year). 8760, which is 365×24 , is the most commonly used value in the power reliability field. By convention, 8760 will be used throughout this document.

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APPENDIX A UFC 3-520-05 GLOSSARY

A-1 ACRONYMS.

A	Ampere
AC	Designation for NFPA 70 Armored Cable
Ah	Ampere-hour
AFPAM	Air Force Pamphlet
ANSI	American National Standards Institute
ASCE	American Society of Civil Engineers
AWG	American wire gauge
CFM	Cubic Feet per Minute
CFR	Code of Federal Regulations
DC	Direct Current
DoD	Department of Defense
Fc	Foot-candle
FRP	Fiberglass Reinforced Plastic
HR	Hydrogen Rate
IEEE	Institute of Electrical and Electronics Engineers
IESNA	Illuminating Engineering Society of North America
In	Inches
mm	Millimeters
NEC	National Electrical Code
NFPA	National Fire Protection Association
NM, NMC	Designations for NFPA 70 Nonmetallic Sheathed Cable
O&M	Operations & Maintenance
PVC	Polyvinyl Chloride
UF	Designation for NFPA 70 Underground Feeder and Branch-Circuit Cable

UFC	Unified Facility Criteria
VR	Ventilation Rate
VRLA	Valve-Regulated Lead acid

A-2 DEFINITION OF TERMS.

Battery Cabinet: Refer to **Enclosure**.

Enclosure: A surrounding case or housing to protect the contained equipment against external conditions and to prevent personnel from accidentally contacting live parts. With respect to enclosures for stationary batteries, the enclosure might be a dedicated battery cabinet or it might contain additional equipment, such as an uninterruptible power supply.

Float Charge: The method of maintaining a battery in a charged condition by continuous, long-term charging at a level to balance self-discharge.

Full Float Operation: Operation of a dc system with the battery, battery charger, and load all connected in parallel, and with the battery charger supplying the normal dc load plus any self-discharge or charging current, or both, required by the battery. The battery will deliver current only when the load exceeds the charger output.

Primary Battery: A battery that produces electric current by electrochemical reactions without regard to the reversibility of those reactions. In the context of this UFC, a primary battery is not rechargeable, and is intended to be used once, then discarded.

Recombinant Cell: A battery cell characterized by the recombination of internally generated gases. For example, a VRLA cell is designed to recombine internally generated oxygen and suppression of hydrogen gas evolution to limit water consumption.

Room: A contained area within a facility, normally accessible through a door. With respect to stationary batteries, a room might be dedicated for the battery installation or it might contain additional equipment.

Secondary Battery: A battery that is capable of repeated use through chemical reactions that are reversible, i.e., the discharged energy can be restored by supplying electrical current to recharge the cell.

Stationary Battery: A storage battery designed for service in a permanent location.

Storage Battery: A battery consisting of one or more cells electrically connected for producing electric energy.

Valve-Regulated Lead acid (VRLA) Cell: A cell that is sealed with the exception of a valve that opens to the atmosphere when the internal gas pressure exceeds atmospheric pressure by a pre-selected amount. VRLA cells provide a means for

recombination of internally generated oxygen and the suppression of hydrogen gas evolution to limit water consumption.

Vented Battery: A battery in which the products of electrolysis and evaporation are allowed to escape freely to the atmosphere. These batteries are commonly referred to as “flooded.”

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APPENDIX A UFC 3-530-01 GLOSSARY

A-1 ACRONYMS.

ACP	Access Control Point
AFF	above finished floor
ANSI	American National Standards Institute
ASHRAE	American Society of Heating, Refrigerating, and Air-Conditioning Engineers
CCT	correlated color temperature
CRI	color rendering index
DoD	Department of Defense
ECF	Entry Control Facility
FC	Facilities Criteria
HID	high-intensity discharge
Hz	hertz
kW	kilowatts
kWh	kilowatt-hours
LEC	light-emitting capacitor
LED	light-emitting diode
LLD	lamp lumen depreciation
NEC	National Electric Code
NEMA	National Electrical Manufacturers Association
NESC	National Electrical Safety Code
NFPA	National Fire Protection Association
SF	square foot
SPD	spectral power distribution
SSL	Solid state lighting

THD	total harmonic distortion
TLED	tubular light-emitting diode
UFC	Unified Facilities Criteria
UFGS	Unified Facilities Guide Specifications
UL	Underwriters Laboratories
V	volt
W	watt

A-2 DEFINITIONS OF TERMS.

The definitions of lighting terms are from ANSI/IES LS-1-21, *Lighting Science: Nomenclature and Definitions for Illuminating Engineering*.

Adaptation: The process by which the retina becomes accustomed to more or less light than it was exposed to during an immediately preceding period. It results in a change in the sensitivity to light.

Aircraft Maintenance Area: The full area designated for aircraft parking and maintenance area. Typically includes the depth from the hangar bay door to the back wall and width from the centerline boundary of the neighboring aircraft maintenance zones or to the side wall.

Altitude: The angular distance of a heavenly body measured on the great circle that passes perpendicular to the plane of the horizon, through the body and through the zenith. It is measure positively from the horizon to the zenith, from 0 degrees to 90 degrees.

Ambient Lighting: Lighting throughout an area that produces general illumination

Area Lighting Luminaire: A complete lighting device consisting of a light source and driver, where appropriate, together with its direct appurtenances such as globe, reflector, refractor, housing, and such support as is integral with the housing. The pole, post, or bracket is not considered part of the luminaire.

Luminance: Luminance is a property of a geometric ray. Luminance as measured by conventional meters is averaged with respect to two independent variables, area and solid angle; both must be defined for a complete description of a luminance measurement.

Azimuth: The angular distance between the vertical plane containing a given line or celestial body and the plane of the meridian.

Baffle: A single opaque or translucent element to shield a source from direct view at certain angles, to absorb or block unwanted light, or to reflect and redirect light.

Ballast: A device used with an electric-discharge light source to obtain the necessary circuit conditions (voltage, current, and waveform) for starting and operating.

Bollard: A type of luminaire having the appearance of a short, thick post, used for walkway and grounds lighting. The optical components are usually top-mounted.

Bowl: An open-top diffusing glass or plastic enclosure used to shield a light source from direct view and to redirect or scatter the light.

Bracket (mast arm): An attachment to a light source post or pole from which a luminaire is suspended.

BUG (Backlight, Uplight, Glare): Backlight – the amount of percent lamp lumens or the luminaire zonal lumens distributed behind a luminaire between zero degrees vertical (nadir) and 90 degrees vertical. Uplight – the percent lamp lumens or the amount of luminaire zonal lumens distributed above a luminaire between 90 and 180 degrees vertical. Glare – the amount of percent lamp lumens or the luminaire zonal lumens distributed 60 and 90 degrees vertical.

Candela, cd: The SI unit of luminous intensity, equal to one lumen per steradian (lm/sr).

Candlepower (cp): Another term for luminous intensity,” expressed in candelas.

Clerestory: That part of a building that rises clear of the roofs or other parts and whose walls contain windows for lighting the interior.

Coefficient of Utilization (CU): The ratio of luminous flux (lumens) calculated as received on the work plane to the total luminous flux (lumens) emitted by the light sources alone. It is equal to the product of room utilization factor and luminaire efficiency.

Color Matching: The action of making a color appear the same as a given color.

Color Rendering: Effect of an illuminant on the color appearance of objects by conscious or subconscious comparison with their color appearance under a reference illuminant.

Color Rendering Index (of a light source) (CRI): A measure of the degree of color shift objects undergo when illuminated by the light source as compared with those same objects when illuminated by a reference source of comparable color temperature.

Color Temperature (of a light source): See Correlated Color Temperature.

Contrast: *See Luminance Contrast.*

Correlated Color Temperature (of a light source) (CCT): The absolute temperature of a blackbody whose chromaticity most nearly resembles that of the light source.

Daylight Availability: The luminous flux from the sun plus sky at a specific location, time, date, and sky condition.

Diffused Lighting: Lighting provided on the work plane or on an object that is not incident predominantly from any particular direction.

Dimmer: A device used to control the intensity of light emitted by a luminaire or light source by controlling the voltage or current available to it.

Direct Component: That portion of the light from a luminaire that arrives at the work plane without being reflected by room surfaces.

Direct Glare: Glare resulting from high luminances or insufficiently shielded light sources in the field of view. It is usually associated with bright areas, such as luminaires, ceilings, and windows that are outside the visual task or region being viewed. A direct glare source can also affect performance by distracting attention.

Direct-Indirect Lighting: A variant of general diffuse lighting in which the luminaires emit little or no light at angles near the horizontal.

Direct Lighting: Lighting involves luminaires that distribute 90 to 100% of the emitted light in the general direction of the surface to be illuminated. The term usually refers to light emitted in a downward direction.

Directional Lighting: Lighting provided on the workplane or on an object, predominantly from a preferred direction.

Disability Glare: The effect of stray light in the eye whereby the contrast of the retinal image is reduced and, consequently, whereby visibility and visual performance are reduced. A direct glare source that produces discomfort can also produce disability glare by introducing a measurable amount of stray light in the eye.

Discomfort Glare: Glare that produces discomfort. It does not necessarily interfere with visual performance or visibility.

Downlight: A small direct lighting unit that directs the light downward and can be recessed, surface-mounted, or suspended.

Efficacy: See *Luminous Efficacy of a Source of Light*.

Efficiency: See *Luminaire Efficiency*.

Electroluminescence: The emission of light from a phosphor excited by an electromagnetic field.

Emergency Exit: A way out of the premises that is intended to be used only during an emergency.

Emergency Lighting: Lighting designed to supply illumination essential to the safety of life and property in the event of a failure of the normal supply. The system must be capable of providing minimum required illuminance specified in NFPA 101.

Exit Sign: A graphic device including words or symbols that indicates or identifies an escape route or the location of, or direct to, an exit or emergency exit.

Floodlight: A projector designed for lighting a scene or object to a luminance considerably greater than its surroundings.

Fluorescent Light Source: A low pressure mercury electric-discharge light source in which a fluorescing coating (phosphor) transforms some of the UV energy generated by the discharge into light.

Flush-mounted or Recessed Direct/Indirect Luminaire: A luminaire that is mounted above the ceiling (or behind a wall or other surface) with the opening of the luminaire level with the surface.

Footcandle, fc: A unit of illuminance equal to 1 lm/ft².

Glare: The sensation produced by luminances within the visual field that are sufficiently greater than the luminance to which the eyes are adapted, which causes annoyance, discomfort, or loss in visual performance, and visibility. Direct glare is caused by excessive light entering the eye from a bright light source. The potential for direct glare exists anytime one has a direct view of a light source. With direct glare, the eye has a harder time seeing contrast and details. A system designed solely on lighting levels, tends to aim more light directly towards a task, thus producing more potential for glare. Direct glare can be minimized with careful equipment selection and placement.

Globe: A transparent or diffusing enclosure intended to protect a light source, to diffuse and redirect its light, or to change the color of the light.

High Ambient Temperatures: Above 50 degrees Celsius.

High-Intensity Discharge (HID) Light Source: An electric-discharge light source in which the light-producing arc is stabilized by bulb wall temperature, and the arc tube has a bulb wall loading in excess of 3 W/cm². HID light sources include groups of light sources known as mercury, metal halide, and high-pressure sodium.

High-Mast Lighting: Illumination of a large area by means of a group of luminaires that are designed to be mounted in a fixed orientation at the top of a high mast, generally 20 m (65 ft.) or higher.

High-Pressure Sodium (HPS) Light Source: A high intensity discharge (HID) light source in which light is produced by radiation from sodium vapor.

Illuminance: The areal density of the luminous flux incident at a point on a surface. Units: lux or footcandles.

Illuminance (footcandle or lux) Meter: An instrument for measuring illuminance on a plane. The instrument comprises some form of photodetector with or without a filter driving a digital or analog readout through appropriate circuitry.

Illumination: An alternative but deprecated term for illuminance. In this document, it is used in its more general sense of simply "lighting."

Incandescent Filament Light Source: A light source in which light is produced by a filament heated to incandescence by an electric current.

Indirect Component: The portion of the luminous flux from a luminaire that arrives at the workplane after being reflected by room surfaces.

Indirect Lighting: Lighting involving luminaires that distribute 90 to 100% of the emitted light upward.

Induction Lighting: Lighting technology that uses electric current to induce an electromagnetic field within the phosphor coated light source. No filaments are used. Its advantages include instant on/off operation, white light with good color rendering characteristics, and a long light source life of 100,000 hours.

Intensity (candlepower) Distribution Curve: A curve, often polar, that represents the variation of luminous intensity of a light source or luminaire in the plane through the light center.

Isolux (Isofootcandle) Line: A line plotted on any appropriate set of coordinates to show all the points on a surface where the illuminance is the same.

Kelvin: The unit of absolute temperature used to designate the color temperature or correlated color temperature of a light source. Symbol: K.

Light Source: A generic term for a source created to produce optical radiation.

Light Source Lumen Depreciation (LLD) Factor: The fractional loss of light source lumens at rated operating conditions that progressively occurs during light source operation.

Lens: A glass or plastic element used in luminaires to change the direction and control the distribution of light rays.

Light: Radiant energy that is capable of exciting the retina and producing a visual sensation.

Light-Emitting Diode (LED): A p-n junction solid state diode whose radiated output is a function of its physical construction, material used, and exciting current.

Light Loss Factor (LLF): Formerly called maintenance factor. The ratio of illuminance (or exitance or luminance) for a given area to the value that would occur if light sources operated at their (initial) rated lumens and if no system variation or depreciation had occurred.

Light Meter: A common name for an illuminance meter.

Light Source Color: The color of the light emitted by a source.

Louver: An optical-control element, usually used in multiples, to shield a source from view at certain angles, to absorb or block unwanted light, or to reflect or redirect light.

Low-Bay Lighting: Interior lighting where the roof trusses or ceiling height is approximately 6.1 m (20 ft.) or less above the floor.

Low-Pressure Mercury Vapor (LPM) Light Source: A discharge light source (with or without a phosphor coating) in which the partial pressure of mercury vapor does not exceed 100 Pa during operation.

Low-Pressure Sodium (LPS) Light Source: A discharge light source in which light is produced by radiation from sodium vapor.

Lumen, lm: SI unit of luminous flux. Radiometrically, it is determined from the radiant power (see luminous flux). Photometrically, it is the luminous flux emitted within a unit solid angle (one steradian) by a point source having a uniform luminous intensity of one candela.

Lumen Depreciation: The decrease in lumen output that occurs as a lamp is operated, until failure.

Lumen Method: A procedure used to determine the relationship between the number and types of lamps, light sources and luminaires, the room characteristics, and the average level of illuminance on the work plane. It takes into account both direct and reflected flux.

Luminaire (light fixture): A complete lighting unit consisting of a light source or light sources and driver(s) (when applicable) together with the parts designed to distribute the light, to position and protect the light sources, and to connect the light sources to the power supply.

Luminaire Dirt Depreciation (LDD): The ratio of lumens emitted from a luminaire with dirt accumulated to the lumens emitted from the same luminaire when clean.

Luminaire Efficiency: The luminous flux emitted by a luminaire, divided by the luminous flux emitted by the source(s). Sometimes called light output ratio (LOR).

Luminance: The apparent brightness of a surface, measured in candelas per foot (cd/ft^2) or square meter (cd/m^2), sometimes called nits.

Luminance Contrast: The relationship between the luminances of an object and its immediate background.

Luminance Ratio: The ratio between the luminances any two areas in the visual field.

Luminous Efficacy of a Source of Light: The quotient of the total luminous flux emitted to the total light source power input. It is expressed in lumens per watt.

Luminous Flux: The time rate of flow of radiant energy, evaluated in terms of a standardized visual response. Units: lumens.

Matte Surface: A surface from which the reflection is predominantly diffuse, with or without a negligible specular component.

Means of Egress: An unobstructed and continuous way of exit from any point in a building or structure to a public way. It consists of three distinct parts: the exit access,

the exit, and the exit discharge. A means of egress consists of the vertical and horizontal travel ways, including intervening room spaces, doorways, hallways, corridors, passageways, ramps, stairs, lobbies, horizontal exits, escalators, enclosures, courts, balconies, and yards.

Mercury Light Source: A high-intensity discharge (HID) light source in which the major portion of the light is produced by radiation from mercury operating at a partial pressure in excess of 10s Pa.

Mesopic Vision: Vision with fully adapted eyes at luminance conditions between those of photopic and scotopic vision.

Multi-level Switching: Multi-level switching allows multiple light sources within a luminaire to be switched independently. For example, a three light source luminaire would offer four light output settings: 100%, 66%, 33%, and OFF.

Orientation: The position of a building with respect to compass directions.

Overhang: In roadway lighting: the distance between a vertical line passing through a specified point (often the photometric center) of a luminaire and the curb or edge of a roadway.

Pendant Luminaire: See *Suspended* luminaire.

Peripheral Vision: The seeing of objects displaced from the primary line of sight and outside the central visual field.

Photometry: The measurement of quantities associated with light.

Photopic Vision: Vision mediated essentially or exclusively by the cones. It is generally associated with adaptation to a luminance of at least 5 cd/m².

Point-by-Point Method: A method of lighting calculation, now called the point method.

Point Method: A procedure for predetermining the illuminance at various locations in lighting installations by use of luminaire photometric data. The direct component of illuminance due to the luminaires and the interreflected component of illuminance due to the room surfaces are calculated separately. The sum is the total illuminance at a point.

Point Source: A source of radiation, whose dimensions are sufficiently small, compared with the distance between the source and the irradiated surface, that these dimensions can be neglected in calculations and measurements.

Pole (roadway lighting): A standard support generally used where luminaires are located.

Quality of Lighting: Pertains to the distribution of luminance in a visual environment. The term is used in a positive sense and implies that all luminances contribute favorably to visual performance, visual comfort, ease of seeing, safety, and aesthetics for the specific visual tasks involved.

Ramped Dimming: A gradual dimming rate that ensures occupants in the space will not perceive a sudden change in light levels.

Rated Light Source Life: The life value assigned to a particular type of light source. This is commonly a statistically determined estimate of average or of median operational life.

Reflected Glare: Glare resulting from reflections of high luminances in polished or glossy surfaces in the field of view.

Reflection: A general term for the process by which the incident flux leaves a (stationary) surface or medium from the incident side without change in frequency.

Reflector: A device used to redirect the flux from a light source by the process of reflection.

Scotopic Vision: Vision mediated essentially or exclusively by the rods. It is generally associated with adaptation to a luminance below about 0.001 cd/m^2 .

Self-Ballasted Light Sources: Any arc discharge light source of which the current limiting devices is an integral part.

Solid State Lighting: Light sources that generate light through electroluminescence rather than filaments or gas discharge. SSL sources include light emitting diodes (LEDs), organic light emitting diodes (OLEDs), and polymer light emitting diodes (PLED).

Spacing: For roadway lighting, the distance between successive lighting units, measured along the centerline of the street.

Spacing-to-Mounting-Height Ratio: The ratio of the actual distance between luminaire centers to the mounting height above the horizontal ground or work plane.

Suspended (pendant) Luminaire: A luminaire that is hung from a ceiling by supports.

Table Lamp: A portable luminaire with a short stand, suitable for mounting on furniture.

Translucent: Transmitting light diffusely or imperfectly.

Transmission: A general term for the process by which incident flux leaves a surface or medium on a side other than the incident side, without change in frequency.

Transmittance: The ratio of the transmitted flux to the incident flux.

Transmittance, Visible (T_{vis}): The percentage of the visible spectrum transmitted.

Transparent: Having the property of transmitting rays of light through its substance so that bodies situated beyond or behind can be distinctly seen.

Troffer: A long recessed direct/indirect lighting unit usually installed with the opening flush with the ceiling.

Tungsten-Halogen Light Source: A gas-filled tungsten filament incandescent light source containing a certain proportion of halogens in an inert gas whose pressure exceeds 3 atm.

Uniformity: Lighting level or illuminance uniformity is important to work surfaces where sustained tasks are performed as well as on wall and ceiling surfaces that make up a significant portion of the field of view. Poor uniformity can cause adaptation problems. Flicker or strobing of luminaires can cause annoyances as well as headaches and fatigue.

Valance: A longitudinal shielding member mounted across the top of a window or along a wall (and is usually parallel to the wall) to conceal light sources, giving both upward and downward distributions.

Valance Lighting: Lighting comprising light sources shielded by a panel parallel to the wall at the top of a window.

Veiling Reflection: Regular (specular) reflection, superimposed on diffuse reflection from an object, that partially or totally obscures the details to be seen by reducing the contrast. This sometimes is called reflected glare. Another kind of veiling reflection occurs when one looks through a pane of glass. A reflected image of a bright element or surface may be seen superimposed on what is viewed through the glass pane.

Visibility: The quality or state of being perceivable by the eye. In many outdoor applications, visibility is defined in terms of the distance at which an object can be just perceived by the eye. In indoor applications, it usually is defined in terms of the contrast or size of a standard test object when observed under standardized viewing conditions and having the same threshold as the given object.

Volt: The difference in electrical potential between two points in a circuit.

Watt: The unit of power (rate of doing work). In electrical calculation, one watt is the power produced by a current of one ampere across a potential difference of one volt.

Workplane: The plane on which a visual task is usually done, and on which the illuminance is specified and measured. Unless otherwise indicated, this is assumed to be a horizontal plane 0.76 meters (30 inches) above the floor.

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APPENDIX A UFC 3-540-01 GLOSSARY

A-1 ACRONYMS.

A	amperes
AC	alternating current
AED	automatic external defibrillator
AFI	Air Force Instruction
AHJ	authority having jurisdiction
AFMAN	Air Force Manual
AMI	advanced metering infrastructure
ASR	air surveillance radar
ATS	automatic transfer switch
AWG	American Wire Gauge
BUMED	Navy Bureau of Medicine and Surgery
C4ISR	Command, Control, Communications, Computers, Intelligence, Surveillance, and Reconnaissance
CCTV	closed circuit television
CONUS	continental United States
COPS	Critical Operations Power Systems
CSA	Canadian Standards Association
CT	current transformer
dBA	decibels, A-weighted
DC	direct current
DCS	distributed control system
DDC	direct digital control
DER	distributed energy (generation) resources
DOE	Department of Energy

EA	environmental assessment
EAID	Equipment Authorization Inventory Data
EGSA	Electrical Generating Systems Association
EIS	environmental impact study
EMF	electromagnetic field
EMI	electromagnetic interference
EMP	electromagnetic pulse
EOC	Emergency Operations Center
EPA	Environmental Protection Agency
EPRI	Electric Power Research Institute
EPSS	emergency power supply system
FACSFAC	Fleet Area Control and Surveillance Facility
HMI	human-machine interface
hr	hour
HVAC	heating, ventilating, and air conditioning
Hz	hertz
I/O	input/output
ICS	industrial control systems
IED	intelligent electronic devices
ISO	International Standards Organization
JCF	joint control facility
JRIC	Joint Reserve Intelligence Center
kBtu/hr	thousand British thermal units per hour
kV	kilovolt
kVA	kilo-volt-amperes
kW	kilowatt

kWlb/hr	kilowatt-pounds per hour
lb/in ²	pounds per square inch
LPG	liquefied petroleum gas
MP	Military Police
MEP	mobile electric power
MVA	million-volt-amperes
MWh	megawatt hour
NAVFAC	Naval Facilities Command
NEC	National Electrical Code
NEMA	National Electrical Manufacturers Association
NSPS	New Source Performance Standards
OEBGD	(DoD) Overseas Environmental Baseline Guidance Document
OSHA	Occupational Safety and Health Act
PLC	programmable logic controller
POL	petroleum, oil, and lubricants
psig	pounds per square inch gauge
PURPA	Public Utility Regulatory Policies Act
QSTAG	<i>Quadripartite Standardization Agreement</i>
QWLD	quick (connect), waterproof, type I disconnect
RAM	reliability, availability, and maintainability
RATCF	radar air traffic control facilities
RDT&E	research, development, test and evaluation
RFI	radio frequency interference
RFP	request for proposal
RPIE	real property installed equipment
rpm	revolutions per minute

SCADA	Supervisory Control and Data Acquisition
SCIF	Secure Compartmented Information Facility
SCR	selective catalytic reduction
sec	second
TEFC	totally enclosed fan cooled
TEMPEST	Telecommunications Electronics Material Protected from Emanating Spurious Transmissions
TFT	thin film transfer
UFC	Unified Facilities Criteria
UIC	Unit Identification Code
UL	Underwriters Laboratories
UPS	uninterruptible power supply
VDC	volts, direct current

A-2 DEFINITION OF TERMS.

Automatic Transfer Switch (ATS): A switch designed to sense the loss of one power source and automatically transfer the load to another source of power.

Availability: The long-term probability of success with repair and scheduled maintenance of electrical power plants, generators, and power systems. Calculate availability as the ratio in percentage of total period minus repair downtime minus maintenance downtime to total period. The calculation assumes power generated meets quality standards.

Critical Operations Power Systems: Systems that are installed in vital infrastructure facilities that, if destroyed or incapacitated, would disrupt national security, the economy, public health or safety; and where enhanced electrical infrastructure for continuity of operation has been deemed necessary by governmental authority. COPS design criteria are specified by NFPA 70 Article 708.

Emergency Systems: As specified by NFPA 70, emergency systems are those systems legally required and classed as emergency by municipal, state, Federal, or other codes, or by any governmental agency having jurisdiction. These systems are intended to automatically supply illumination, power, or both, to designated areas and equipment in the event of failure of the normal supply or in the event of accident to

elements of a system intended to supply, distribute, and control power and illumination essential for safety to human life.

Harmonic: A sinusoidal component of a periodic wave or quantity having a frequency that is an integral multiple of the fundamental frequency.

Legally Required Standby Systems: Those systems required and so classed as legally required standby by municipal, state, Federal, or other codes or by any governmental agency having jurisdiction. These systems are intended to automatically supply power to selected loads (other than those classed as emergency systems) in the event of failure of the normal source.

Linear Load: An electrical load device that presents an essentially constant load impedance to the power source throughout the cycle of applied voltage in steady-state operation.

Listed: Applies to equipment or materials included in a list published by an organization acceptable to the authority having jurisdiction. The organization periodically inspects production and certifies that the items meet appropriate standards or tests as suitable for a specific use.

Nonlinear Load: A steady state electrical load that draws current discontinuously or has the impedance vary throughout the input AC voltage waveform cycle. Alternatively, a load that draws a nonsinusoidal current when supplied by a sinusoidal voltage source.

Optional Standby Systems: Those systems intended to supply power to public or private facilities or property where life safety does not depend on the performance of the system. Optional standby systems are intended to supply on-site generated power to selected loads either automatically or manually.

Power Quality: The concept of powering and grounding sensitive equipment in a manner that is suitable to the operation of that equipment.

Transfer Switch: A device for transferring one or more load conductor connections from one power source to another.

Uninterruptible Power Supply System: A system that converts unregulated input power to voltage- and frequency-controlled filtered AC power that continues without interruption even with the deterioration of the input AC power.

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Note: The most recent edition of referenced publications applies unless otherwise specified.

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APPENDIX A UFC 3-540-08 GLOSSARY

A-1 ACRONYMS.

AC	alternating current
ACI	Activated Carbon Injection
API	American Petroleum Institute
ASCE	American Society of Civil Engineers
AWEA	American Wind Energy Association
ANSI	American National Standards Institute
ASME	American Society of Mechanical Engineers
ASHRAE	American Society of Heating, Refrigeration, and Air-Conditioning Engineers
AFI	Air Force Instruction
AGC	Automatic Generation Control
AHJ	Authority Having Jurisdiction
APC	Air Pollution Control
BACT	Best Available Control Technology
BOP	Blowout Preventer
BOS	Base Operating Support
BTU	British thermal units
C	Celsius
CAA	Clean Air Act
CCTV	closed circuit television
CEMS	Continuous Emission Monitoring System
cf	cubic foot
CFM	cubic feet per minute
CFR	Code of Federal Regulations
CH ₄	methane
cm	centimeter
CO ₂	carbon dioxide
CSAMT	Controlled Source Audio-Frequency Magnetotellurics
CTI	Cooling Technology Institute
CWA	Clean Water Act
cy	cubic yard(s)
DC	direct current
DCS	Distributed Control System
DOE	Department of Energy
DSIRE	Database of State Incentives for Renewables & Efficiencies
EA	Environmental Assessment
ECIP	Energy Conservation Investment Program
EG	Emissions Guidelines
EGS	Enhanced Geothermal System
EIS	Environmental Impact Statement
EO	Executive Order
EPA	Environmental Protection Agency
EERE	Energy Efficient Renewable Energy

EPRI	Electric Power Research Institute
EPACT	Energy Policy Act
ESPC	Energy Savings Performance Contract
EUL	Enhanced Use Lease
F	Fahrenheit
FAA	Federal Aviation Administration
FERC	Federal Energy Regulatory Commission
FEMP	Federal Energy Management Program
fps	feet per second
ft.	foot/feet
ft ²	square foot/feet
ft ³	cubic foot/feet
H ₂ S	hydrogen sulfide
HAWT	Horizontal Axis Wind Turbine
HCl	hydrogen chloride
HF	hydrogen fluoride
HEI	Heat Exchange Institute
HDPE	high density polyethylene
HVAC	Heating, ventilation, and air conditioning
GHG	Greenhouse Gas
GSU	generator setup
IAQ	indoor air quality
ICS	internet connection sharing
IEC	International Electrotechnical Commission
IEEE	Institute of Electrical and Electronics Engineers
IP	intermediate pressure
ISO/RTO	Independent Systems Operators/Regional Transmission Operators
IREC	Interstate Renewable Energy Council
ICS-CERT	Industrial Control Systems Cyber Emergency Response Team
in	inch(es)
IP	Induced Polarization
ISO	Independent System Operators
kg	kilogram
kJ	kilojoule
km	kilometer
kW	kilowatt
kWh	kilowatt-hour
lb	pound
LCOE	Levelized Cost Of Energy
LEL	Lower Explosive Limit
LFG	Landfill Gas
LIDAR	Light Detection and Ranging
l/s	liters per second
LFRT	low frequency ride through
LVRT	low voltage ride through
m	meter
m ²	square meter

m ³	cubic meter
mi	mile
MCR	Maximum Combustion Rate
M&V	Measurement & Verification
MJ	megajoule
MNS	Mass Notification System
MPa	megapascal
mph	miles per hour
m/s	meters per second
MSW	Municipal Solid Waste
MSW-DST	Municipal Solid Waste Decision Support Tool
MT	magnetotellurics
MTR	Military Training Route
MW	megawatt
MWh	megawatt-hour
NC	Noise Criteria
NCG	non-condensable gases
NAAQS	National Ambient Air Quality Standards
NFPA	National Fire Protection Act
NECA	National Electrical Contractors Association
NERC	North American Electric Reliability Corporation
NREL	National Renewable Energy Laboratory
NPSH	Net Positive Suction Head
NESHAP	National Emission Standards for Hazardous Air Pollutants
NH ₃	ammonia
NO _x	nitrous oxide
NPDES	National Pollutant Discharge Elimination System
NRCS	National Resources Conservation Service
NSPS	New Source Performance Standards
NSR	New Source Review
NTEP	National Type Evaluation Program
O ₂	oxygen
OSD	Office of the Secretary of Defense
OSHA	Occupational Safety and Health Act
OEBGD	Overseas Environmental Baseline Guidance Document
O&M	Operations and Maintenance
φ	phase
PAC	Powered Activated Carbon
pH	measure of acidity or basicity of an aqueous solution
PPA	Power Purchase Agreement
ppb	parts per billion
ppm	parts per million
PSD	Prevention of Significant Deterioration
psi	pounds per square inch
PTC	Production Tax Credits
PV	photovoltaic
PVC	polyvinyl chloride

RCRA	Resource Conservation and Recovery Act
REC	Renewable Energy Credit
REO	Renewable Energy Optimization
RO	reverse osmosis
rpm	rotations per minute
RPG	Renewable Power Generation
RPS	Renewable Portfolio Standards
RTI	Research Triangle Institute
RTU	Remote Terminal Unit
s	second
SCADA	Supervisory Control and Data Acquisition
SCR	Selective Catalytic Reduction
SGHAT	Solar Glare Hazard Analysis Tool
SIS	System Impact Study
SNCR	Selective Non-catalyst Reduction
SO ₂	sulfur dioxide
SP	Self-Potential
sq	square
TDEM	Time Domain Electromagnetics
tpd	tons per day
TSWEG	Tri-Service Electrical Working Group
UESC	Utility Energy Service Contract
UFC	Unified Facilities Criteria
UPS	uninterruptable power supply
UV	ultraviolet
V	volt(s)
VAWT	Vertical Axis Wind Turbine
VES	Vertical Sounding
VOC	volatile organic compound
wc	water column
WTE	waste to energy
yd ³	cubic yard

APPENDIX B UFC 3-540-08 REFERENCES

B-1 FEDERAL

<http://www.archives.gov/federal-register/>

10 USC Part 2667, Leases: Non-Excess Property of Military Departments and Defense Agencies

10 CFR Part 436, Federal Energy Management and Planning Programs 18

CFR Part 35, Interconnection for Wind Energy

29 CFR 1910 Occupational Safety and Health Standards 29

CFR 1910 Subpart D, Walking Working Surfaces

29 CFR 1910 Subpart I, Personal Protective Equipment

29 CFR 1910.269(g), Fall Protection for Power Generation Industry

29 CFR 1925 Safety and Health Standards for Federal Service Contracts

29 CFR 1926 Safety and Health Regulations for Construction

29 CFR 1926 Subpart E, Personal Protective and Life Saving Equipment

29 CFR 1926 Subpart M, Fall Protection

32 CFR Part 211, Mission Compatibility Evaluation Process

40 CFR Part 60, Standards of Performance for New Stationary Sources 40

CFR Part 60.51c, Definitions

40 CFR Part 61, National Emissions Standards for Hazardous Air Pollutants

40 CFR Part 70, State Operating Permit Programs

40 CFR Part 71, Federal Operating Permit Programs 40

CFR Part 77, Excess Emissions

40 CFR Part 81, Designation of Areas for Air Quality Planning Purposes

40 CFR Part 257, Criteria for Classification of Solid Waste Disposal Facilities and Practices

40 CFR Part 266-273, Standards for the Management of Specific Hazardous Wastes and Specific Types of Hazardous Waste Management Facilities - Standards for

Universal Waste Management

49 USC Part 44718, Structures Interfering with Air Commerce

50 CFR Part 13, General Permit Procedures

50 CFR Part 17, Endangered and Threatened Wildlife and Plants

E.O. 13327, Federal Real Property Asset Management

FEDERAL AVIATION ADMINISTRATION

<http://www.faa.gov/>

FAA AC 70/7460-1K, Obstruction Marking and Lighting

FAA Form 7460-1, Notice of Proposed Construction or Alteration

National Oceanographic and Atmospheric Administration (NOAA)

<https://www.ncdc.noaa.gov/data-access/severe-weather>

Severe Weather Data

NATIONAL RENEWABLE ENERGY LABORATORY

http://www.nrel.gov/analysis/tech_size.html Land

Use by System Technology

<http://www.nrel.gov/docs/fy15osti/62566.pdf>;

Making Sustainable Energy Choices

http://www.nrel.gov/analysis/tech_footprint.html

Useful Life

http://www.nrel.gov/analysis/tech_lcoe_re_cost_est.html Distributed

Generation Renewable Energy Estimate of Costs

<http://www.nrel.gov/docs/fy13osti/56266.pdf>

Land Based Wind Turbine Systems Installed Capital Costs

U.S. ENVIRONMENTAL PROTECTION AGENCY

http://cfpub.epa.gov/ols/catalog/catalog_lookup.cfm

EPA-600/R-05/047, Landfill Gas Emissions Model (LandGEM) Version 3.02 User's Guide

EPA TO 14a, Determination of Volatile Organic Compounds (VOCs) in Ambient Air Using Specially Prepared Canisters with Subsequent Analysis by Gas Chromatography

LFG Energy Project Development Handbook

<https://www3.epa.gov/lmop/publications-tools/handbook.html>

U.S. DEPARTMENT OF ENERGY

http://apps2.eere.energy.gov/wind/windexchange/what_is_wind.asp

WINDEXchange

http://apps2.eere.energy.gov/wind/windexchange/wind_maps.asp

Utility-Scale Land-Based 80-Meter Wind Maps

U.S. ENERGY INFORMATION ADMINISTRATION

http://www.eia.gov/electricity/monthly/epm_table_grapher.cfm?t=epmt_6_07_b Table

6.7.B. Capacity Factors for Utility Scale Generators Not Primarily Using Fossil Fuels, January 2013-August 2015

http://www.eia.gov/forecasts/aeo/electricity_generation.cfm

Levelized Cost and Levelized Avoided Cost of New Generation Resources in the Annual Energy Outlook 2015

B-2 DEPARTMENT OF DEFENSE

AIR FORCE

<http://www.e-publishing.af.mil/>

AFI 32-7006, Environmental Program in Foreign Countries

ARMY

<http://armypubs.army.mil/epubs>

USACE EM 200-1-22, Environmental Quality, Landfill Gas Collection and Treatment Systems

USACE ETL 1110-3-412, Engineering and Design: Transformer Application Guidance

USACE EM 200-1-22, Landfill Gas Collection and Treatment Systems

USACE EM 1110-1-1804, Geotechnical Investigations

USACE EM 1110-1-4016, Landfill Off-Gas Collection and Treatment Systems

DEPARTMENT OF DEFENSE (DOD)

<http://www.defense.gov/resources/>

DoDI 4170.11, Installation Energy Management DoDI

4270.5, Military Construction

DoDI 7041.3, Economic Analysis for Decision Making

DoDI 8500.01, Cybersecurity

DoDI 8510.01, Risk Management Framework for DoD Information Technology

JOINT SERVICE

http://www.wbdg.org/ccb/browse_cat.php?o=29&c=4

UFC 1-200-01, DoD Building Code (General Building Requirements)

UFC 3-201-01, Civil Engineering

UFC 3-230-02, Operation and Maintenance: Water Supply Systems UFC

3-240-05A, Solid Waste Incineration

UFC 3-240-10A, Sanitary Landfill

UFC 3-260-01, Airfield and Heliport Planning and Design

UFC 3-301-01, Structural Engineering

UFC 3-400-02, Design: Engineering Weather Data

UFC 3-401-01, Mechanical Engineering

UFC 3-410-01, Heating, Ventilating, and Air Conditioning Systems

UFC 3-410-04N, Industrial Ventilation

UFC 3-420-01, Plumbing Systems

UFC 3-430-02FA, Central Steam Boiler Plants

UFC 3-430-03, Air Pollution Control Systems for Boiler and Incinerators

UFC 3-440-01, Facility-Scale Renewable Energy Systems

UFC 3-440-05N, Tropical Engineering

UFC 3-450-01, Noise and Vibration Control UFC

3-501-01, Electrical Engineering

UFC 3-510-01, Foreign Voltages and Frequencies Guide

UFC 3-520-01, Interior Electrical Systems

UFC 3-520-05, Stationary Battery Areas

UFC 3-530-01, Design: Interior and Exterior Lighting and Controls

UFC 3-540-01, Engine-Driven Generator Systems for Backup Power Applications

UFC 3-550-01, Exterior Electrical Power Distribution

UFC 3-560-01, Electrical Safety, O&M

UFC 3-575-01, Lightning and Static Electricity Protection Systems

UFC 3-580-01, Telecommunications Building Cabling Systems Planning and Design UFC

3-600-01, Fire Protection Engineering for Facilities

UFC 3-730-01, Programming Cost Estimates for Military Construction UFC

3-740-05, Handbook: Construction Cost Estimating

UFC 4-010-01, DoD Minimum Antiterrorism Standards for Buildings

UFC 4-010-02, DoD Minimum Antiterrorism Standoff Distances for Buildings UFC

4-020-01, DoD Security Engineering Facilities Planning Manual

UFC 4-021-01, Design and O&M: Mass Notification Systems

**OFFICE OF THE ASSISTANT SECRETARY OF DEFENSE (ASD) FOR ENERGY,
INSTALLATIONS, AND ENVIRONMENT (OUSD (AT&L))**

http://www.acq.osd.mil/eie/IE/FEP_Policy_Program_Guidance.html

**OFFICE OF THE UNDER SECRETARY OF DEFENSE (OSD) FOR ACQUISITION,
TECHNOLOGY, AND LOGISTICS (OUSD (AT&L))**

Utilities Metering Policy, April 16, 2013

NATURAL RESOURCES DEFENSE COUNCIL | DEPARTMENT OF DEFENSE

http://www.acq.osd.mil/dodsc/library/Siting_Renewable_Energy_Primer_5SEP13_FINAL_WEB.pdf

Working with the Department of Defense: Siting Renewable Energy Development

B-3 INDUSTRY

AMERICAN CONFERENCE OF GOVERNMENTAL INDUSTRIAL HYGIENISTS (ACGIH)

<http://www.acgih.org/Store/>

ACGIH Industrial Ventilation: A Manual of Recommended Practice

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

<http://www.ansi.org/>

ANSI B16.34, Valves

ANSI C2, National Electrical Safety Code

ANSI C37.90, Standard for Relays and Relay Systems Associated with Electric Power Apparatus

ANSI C50.13, American National Standard Requirements for Cylindrical Rotor Synchronous Generators

ANSI Z359, Fall Protection Code

ANSI 60, Drinking Water Treatment Chemicals – Health Effects

AMERICAN PETROLEUM INSTITUTE (API)

<http://www.americanpetroleuminstitute.com/>

API Spec 6A, Wellhead and Christmas Trees

AMERICAN SOCIETY OF CIVIL ENGINEERS (ASCE)

<http://www.asce.org/>

ASCE/SEI 7, Minimum Design Loads for Buildings and Other Structures

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS (ASHRAE)

http://www.techstreet.com/ashrae?ashrae_auth_token=

ASHRAE 62.1, Ventilation for Acceptable Indoor Air Quality

ASHRAE Handbook - HVAC Applications

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

<http://www.asme.org/>

ASME B30.2, Overhead and Gantry Cranes (Top Running Bridge, Single, or Multiple Girder, Top Running Trolley Hoist)

ASME B31.1, Power Piping

ASME B31.3, Process Piping

ASME Boiler and Pressure Vessel Code

ASME PTC 6, Steam Turbines

AMERICAN SOCIETY FOR TESTING AND MATERIALS INTERNATIONAL (ASTM)

<http://www.astm.org/Standard/standards-and-publications.html>

ASTM D2216, Standard Test Methods for Laboratory Determination of Water (Moisture) Content of Soil and Rock by Mass

ASTM D2487, Standard Practice for Classification of Soils for Engineering Purposes (Unified Soil Classification System)

ASTM D2488, Standard Practice for Description and Identification of Soils (Visual- Manual Procedure)

ASTM D422, Standard Test Method for Particle-Size Analysis of Soils

ASTM D4318, Standard Test Methods for Liquid Limit, Plastic Limit, and Plasticity Index of Soils

ASTM D4750, Standard Test Method for Determining Subsurface Liquid Levels in a Borehole or Monitoring Well (Observation Well)

AMERICAN WIND ENERGY ASSOCIATION (AWEA)

<http://www.awea.org/>

ASCE/AWEA RP 2011, Recommended Practice Compliance Large Land-based Wind-Turbine Support Structures 2011

AWEA O&M RP, O&M Recommended Practices

COOLING TECHNOLOGY INSTITUTE (CTI)

<http://www.cti.org/pub/cticode.shtml>

CTI ATC-105, Acceptance Test Code for Water-Cooling Towers

HEAT EXCHANGE INSTITUTE (HEI)

<http://www.heatexchange.org/>

Standards for Steam Surface Condensers

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

<http://ieee.org/>

ANSI/IEEE Std C37.90, IEEE Standard for Relays and Relay Systems Associated With Electric Power Apparatus

IEEE C2, National Electrical Safety Code (NESC)

IEEE Std 142, IEEE Recommended Practice for Grounding of Industrial and Commercial Power Systems

IEEE Std 242, IEEE Recommended Practice for Protection and Coordination of Industrial and Commercial Power Systems

IEEE Std 519, IEEE Recommended Practice and Requirements for Harmonic Control in Electrical Power Systems

IEEE Std 1069, IEEE Recommended Practice for Precipitator and Baghouse Hopper Heating Systems

IEEE Std 1453, IEEE Recommended Practice for Measurement and Limits of Voltage Fluctuations and Associated Light Flicker on AC Power Systems

IEEE Std 1547, Standard for Interconnecting Distributed Resources with Electric Power Systems

IEEE Std 1613, IEEE Standard Environmental and Testing Requirements for Communications Networking Devices Installed in Electric Power Substations

IEEE Std C37.60, IEEE/IEC High-Voltage Switchgear and Controlgear – Part 111: Automatic Circuit Reclosers and Fault Interrupters for Alternating Current Systems up to 38 kV

INTERNATIONAL ELECTROTECHNICAL COMMISSION (IEC)

<http://www.iec.ch/>

IEC 61215, Crystalline Silicon Terrestrial Photovoltaic (PV) Modules – Design Qualifications and Type Approval

IEC 61400, Wind Turbine Generator Systems

IEC 61400-1, Design Requirements

IEC 61400-3, Design Requirements for Offshore Wind Turbines

IEC 61400-3-2, Design Requirements for Floating Offshore Wind Turbines

IEC 61400-4, Design Requirements for Wind Turbine Gearboxes

IEC 61400-5, Wind Turbine Rotor Blades

IEC 61400-11, Acoustic Noise Measurement Techniques

IEC 61400-12, Wind Turbine Power Performance Testing

IEC 61400-13, Measurement of Mechanical Loads

IEC 61400-14, Declaration of Apparent Sound Power Level and Tonality Values

IEC 61400-21, Measurement and Assessment of Power Quality Characteristics of Grid Connected Wind Turbines

IEC 61400-22, Conformity Testing and Certification

IEC 61400-23, Full-Scale Structural Testing of Rotor Blades IEC

61400-24, Lightning Protection

IEC 61400-25, Communication Protocol

IEC 61400 26, Time Based Availability for Wind Turbines

IEC 61400-27, Electrical Simulation Models for Wind Power Generation

IEC 61646, Thin-Film Terrestrial Photovoltaic (PV) Modules – Design Qualification and Type Approval

IEC 62727, Photovoltaic Systems - Specification for Solar Trackers

IEC 61724, Photovoltaic System Performance Monitoring - Guidelines for Measurement, Data Exchange and Analysis

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

<http://www.iso.org>

ISO 14661, Thermal Turbines for Industrial Applications (Steam Turbines, Gas Expansion Turbines) -- General Requirements

ISO 26382, Cogeneration systems -- Technical Declarations for Planning, Evaluation and Procurement

ILLUMINATING ENGINEERING SOCIETY OF NORTH AMERICA (IESNA)

<http://www.ies.org/>

The Lighting Handbook: Reference and Application

NATIONAL ELECTRICAL CONTRACTORS ASSOCIATION (NECA)

<http://www.necanet.org/neca-store/publications>

NECA/IESNA 502-2006, Standard for Installing Industrial Lighting Systems

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

<http://nfpa.org/>

NFPA 1, Fire Code

NFPA 68, Standard on Explosion Protection by Deflagration Venting NFPA

70, National Electrical Code

NFPA 72, National Fire Alarm and Signaling Code

NFPA 211, Standard for Chimneys, Fireplaces, Vents, and Solid Fuel-Burning Appliances

NFPA 214, Standard-on-Water-Cooling-Towers

NFPA 780, Standard for the Installation of Lightning Protection Systems

NFPA 850, Recommended Practice for Fire Protection for Electric Generating Plants and High Voltage Direct Current Converter Stations

NATIONAL INSTITUTE OF STANDARDS AND TECHNOLOGY (NIST)

<http://www.nist.gov/publication-portal.cfm>

NIST Handbook 44, Specifications, Tolerances, and Other Technical Requirements for Weighing and Measuring

NORTH AMERICAN ELECTRIC RELIABILITY CORPORATION (NERC)

<http://www.nerc.com/pa/stand/Pages/ReliabilityStandardsUnitedStates.aspx?jurisdiction=United States>

BAL, Resource and Demand Balancing CIP,

Critical Infrastructure Protection COM,

Communications

EOP, Emergency Preparedness and Operations

FAC, Facilities Design, Connections, and Maintenance

INT, Interchange Scheduling and Coordination

IRO, Interconnection Reliability Operations and Coordination MOD,
Modeling, Data, and Analysis

PER, Personnel Performance, Training, and Qualifications PRC,
Protection and Control

TOP, Transmission Operations

TLP, Transmission Planning VAR,

Voltage and Reactive

UNDERWRITER'S LABORATORIES (UL)

<http://www.ul.com/>

UL 98, Enclosed and Dead-Front Switches

UL 98B, Outline of Investigation for Enclosed and Dead-Front Switches for Use in
Photovoltaic Systems

UL 1703, Standard for Flat-Plate Photovoltaic Modules and Panels

UL 1741, Standard for Inverters, Converters, Controllers and Interconnection System
Equipment for Use with Distributed Energy Resources

UL 4703, Standard for Photovoltaic Wire

B-4 MISCELLANEOUS

BRITISH COLUMBIA MINISTRY OF ENVIRONMENT (BCME)

http://www.env.gov.bc.ca/epd/codes/landfill_gas/ Landfill

Gas Management Facilities Design Guidelines

DATABASE OF STATE INCENTIVES FOR RENEWABLES & EFFICIENCY

<http://www.dsireusa.org/> State

Interconnection Policies

B-4.1 Service Reachback Contacts

ARMY:

USACE Reachback Operations Center (UROC);

Email: UROC@usace.army.mil or UROC@usace.army.smil.mil NIPR

website: <https://uroc.usace.army.mil>

SIPR website: <http://uroc.usace.army.smil.mil>

NAVY/MARINE CORPS:

NAVFAC Pacific (PACOM AOR)

<https://portal.navfac.navy.mil/portal/page/portal/navfacpac/ce>

NAVFAC Atlantic (Atlantic AORs)

NFCCEL_CE_Reachback@navy.mil

Phone:

Comm: (757) 322-8302

AIR FORCE:

AFCEC Reachback Center

Email: AFCEC.RBC@us.af.mil

Phone: DSN (312) 523-6995

Comm: (850) 283-6995

B-4.2 Independent System Operators (ISOs)/Regional Transmission Organizations (RTOs)

ISO/RTO COUNCIL.

<http://www.isorto.org>

United States

California ISO

www.caiso.com

Electric Reliability Council of Texas

www.ercot.com

ISO New England

www.iso-ne.com

MISO

www.misoenergy.org

New York Independent System Operator

www.nyiso.com

PJM Interconnection

www.pjm.com

Southwest Power Pool

www.spp.org

Canada

Alberta Electric System Operator

www.aeso.ca

INDEPENDENT ELECTRICITY SYSTEM OPERATOR (IESO).

www.ieso.ca

3D Seismic Tomography: Uses seismic waves produced by explosives or vibrators to produce a 3D image. Common in oil and gas industries.

Aeromagnetics: Detects subsurface magnetic fields from aerial flights. Detects demagnetization from low temperature geothermal alteration. Much of US already surveyed.

Controlled Source Audio-Frequency Magnetotellurics (CSAMT): Similar to MT, but uses a man-made signal source. CSAMT has significantly less depth range than MT, at less than two miles (1.6 km).

DC Resistivity, Electrical Resistivity, Schlumberger, Vertical Sounding (VES): Uses electrical currents to measure resistivity/conductivity. Effective depth is proportional to distance between electrodes.

Energy Resilience: The ability to prepare for and recover from energy disruptions that impact mission assurance on military installations.

E-Scan: Proprietary DC method.

Gravity: Measures gravitational field to determine subsurface rock density.

Induced Polarization (IP): Uses direct current residual conductivity measurements. May be difficult to interpret.

Magnetotellurics (MT): Measures subsurface electricity created by naturally occurring magnetic fields. Indirectly detects temperature and permeability patterns. Can measure several miles deep and can be used to develop 3D images.

Paleomagnetics: Laboratory based measurement of magnetic field variations (rotation) in rock samples to determine crust dilations use to predict possible geothermal.

Time Domain Electromagnetics (TDEM or TEM): Uses manmade magnetic field to determine subsurface conductivity. TDEM has less distortion than other electrode based techniques.

Self-Potential (SP): Uses electrodes to measure natural subsurface electrical potentials. Useful when shallow groundwater flow is of interest

Synthetic Aperture Radar (InSAR): Uses changes in data from two separate flights, on two different dates, to detect subsidence or inflation between flight dates. Changes may indicate volume changes caused by reservoir pumping, or cooling of rock.

APPENDIX A UFC 3-550-01 GLOSSARY

A-1 ACRONYMS

AC	Alternating Current
ACSR	Aluminum Conductor Steel-Reinforced
ACSS	Aluminum Conductor Steel Supported
A/E	Architect/Engineer
AFCEC	Air Force Civil Engineer Center
AHJ	Authority Having Jurisdiction
AL	Aluminum
ASTM	American Society for Testing and Materials
AWG	American Wire Gauge
UFC	Unified Facilities Criteria
BCE	Base Civil Engineer
BIL	Basic Insulation Level
CT	Current Transformer
CU	Copper
DB	Directional Boring
DDC	Direct Digital Control
DIA	Diameter
DoD	Department of Defense
EMCS	Energy Management and Control System
ESC	Environmental Severity Classification
ESS	Electric Supply Station
Degrees F	Degrees Fahrenheit
fc	Footcandles
ft	Feet (or Foot)

ft ²	Foot Squared
GPS	Global Positioning System
HDPE	High Density Polyethylene Electrical
HQUSACE	Headquarters, US Army Corps of Engineers
HV	High Voltage
HVAC	Heating Ventilation and Air Conditioning
Hz	Hertz
IEEE	formerly Institute of Electrical and Electronic Engineers
in	Inch
kcmil	Thousand circular mils
kg	Kilograms
kV	Kilovolts
kVA	Kilo-Volt-Ampere
kVAR	Kilo-Volt-Ampere-Reactive
kW	Kilowatts
lb	Pound
LTC	Load-Tap Changing
m	Meter
m ²	Meter Squared
Max	Maximum
Min	Minimum
mm	Millimeter
MPa	Mega-Pascals
MTS	Maintenance Testing Specifications
MVA	Mega-Volt-Ampere
NAVFAC	Naval Facilities Engineering Command

NEC	National Electrical Code
NEMA	National Electrical Manufacturers Association
NETA	InterNational Electrical Testing Association
NFPA	National Fire Protection Association
OH	Overhead
O&M	Operation and Maintenance
PE	Professional Engineer
psi	Pounds per square inch
PILC	Paper Insulated Lead Covered
PPE	Personal Protective Clothing
PT	Potential Transformer
PVC	Polyvinyl Chloride
RFP	Request for Proposal
SCADA	System Control and Data Acquisition
SF6	Sodium Hexafluoride
TSEWG	Tri-Service Electrical Working Group
UESS	Utilization Electric Supply Station
UL	Underwriters Laboratories
UFC	Unified Facilities Criteria
V	Volt
VA	Volt-Amp
VFI	Vacuum Fault Interrupter
W	Watt
W/ft ²	Watts per Foot Squared
W/m ²	Watts per Meter Squared

A-2 DEFINITION OF TERMS

Activity: With respect to this UFC, approval is provided by the following:

- For the Air Force, this is the Base Civil Engineer.
- For the Navy, this is the Public Works Officer.
- For the Army, this is the Installation Director of Public Works.

Contractor: Person(s) doing actual construction portion of a project.

Corrosive Area: An area identified by the Technical Reviewing Authority as requiring special equipment corrosion mitigation methods. Corrosive areas include locations with an Environmental Severity Classification (ESC) of C3, C4, and C5. See UFC 1-200-01 for determination of ESC for project locations.

Designer of Record: The engineer responsible for the actual preparation of the construction documents.

Main Electric Supply Station: A main electric supply station is also referred to as a “switching station” and does not have power transformers to transform from the utility transmission voltage to a lower distribution voltage. The main electric supply station is the installation/utility interface point where further transmission, distribution and utilization of electrical power, the monitoring and control of such power or equipment and the protection of electrical equipment or systems usually becomes the sole responsibility of the Government or their contracted representatives.

Main Electric Supply Substation: A main electric supply station that also transforms the energy level (voltage) for further bulk distribution at medium voltage levels. A main electric supply substation includes power (substation) transformers.

Facility Core Distribution Infrastructure: System components that constitute a part of the base infrastructure system that are required to provide electrical service to multiple users.

Low Voltage System: An electrical system having a maximum root-mean-square (rms) voltage of less than 1,000 volts.

Medium Voltage System: An electrical system having a maximum RMS ac voltage of 1,000 volts to 34.5 kV. Some documents such as NEMA C84.1 define the medium voltage upper limit as 100 kV, but this definition is inappropriate for facility applications.

Primary Distribution: A system of alternating-current distribution for supplying the primary of distribution transformers from the generating station or substation distribution buses. Primary distribution can be supplied either overhead or underground and includes all associated equipment.

Project Manager: Engineer charged with the administration of the project.

Secondary Distribution: An alternating-current system that connects the secondaries of distribution transformers to the consumers' services. A secondary distribution is typically operating at low voltage, but could be operating at medium voltage.

Service: The conductors and equipment for delivering electrical energy from the serving utility or Government-owned system to the wiring system of the premises served.

Site Electrical Utilities: Site Electrical Utilities are the primary electric power distribution to the facilities and other electrical loads, all exterior lighting not attached to the building; and all telecommunication services (fiber optic, copper cable, CATV, etc.) required by the Facilities.

Switching Station: refer to Main Electric Supply Station.

Utilization Electric Supply Station: Equipment such as pole or pad-mounted transformers or secondary unit substations that transforms the energy level (voltage) to a utilization voltage for consumer use.

APPENDIX B UFC 3-550-01 REFERENCES

Note: The most recent edition of referenced publications applies, unless otherwise specified.

DEPARTMENT OF DEFENSE

AFMAN 91-118, *Safety Design and Evaluation Criteria for Nuclear Weapon Systems.*

AFMAN 91-201, *Explosive Safety Standards.*

Department of the Army Pamphlet 385-64, *Ammunition and Explosives Safety Standards.*

NAVSEA OP-5, Volume 1, *Ammunition and Explosives Ashore.*

IEEE

www.ieee.org

IEEE C2, *National Electrical Safety Code.*

IEEE C37.04, *Rating Structure for AC High-Voltage Circuit Breakers Rated on a Symmetrical Current Basis.*

IEEE C37.06, *AC High-Voltage Circuit Breakers Rated on a Symmetrical Current Basis – Preferred Ratings and Related Required Capabilities.*

IEEE C37.20.2, *Metal-Clad Switchgear.*

IEEE C37.41, *Design Tests for High-Voltage Fuses, Distribution Enclosed Single-Pole Air Switches, Fuse Disconnecting Switches, and Accessories.*

IEEE C37.46, *High Voltage Expulsion and Current-Limiting Type Power Class Fuses Automatic Circuit Reclosers and Fault Interrupters for Alternating Current Systems and Fuse Disconnection Switches.*

IEEE C37.60, *Requirements for Overhead, Pad-Mounted, Dry Vault, and Submersible Load- Interrupter Switchgear and Fused Load-Interrupter Switchgear for Alternating Current Systems up to 38 kV.*

IEEE C37.90, *Relays and Relay Systems Associated with Electric Power Apparatus.*

IEEE C37.121, *Switchgear – Unit Substations Requirements.*

IEEE C57.12.00, *General Requirements for Liquid-Immersed Distribution, Power, and Regulating Transformers.*

IEEE C57.12.01, *General Requirements for Dry-Type Distribution and Power Transformers Including Those with Solid-Cast and/or Resin-Encapsulated Windings.*

IEEE C57.12.28, *Pad-Mounted Equipment – Enclosure Integrity.*

IEEE C57.12.34, *Pad-Mounted, Compartmental-Type, Self-Cooled, Three-Phase Distribution Transformers, 10 MVA and Smaller; High-Voltage, 34.5 kV Nominal System Voltage and Below; Low-Voltage, 15 kV Nominal System Voltage and Below.*

IEEE C57.12.50, *Ventilated Dry-Type Distribution Transformers, 1 to 500 kVA, Single-Phase, and 15 to 500 kVA Three-Phase, with High-Voltage 601 to 34,500 Volts, Low-Voltage 120-600 Volts.*

IEEE C57.12.51, *Ventilated Dry-Type Power Transformers, 501 kVA and larger, Three-Phase, with High-Voltage 601 to 34,500 Volts, Low-Voltage 208Y/120 to 4160 Volts.*

IEEE C57.12.80, *Terminology for Power and Distribution Transformers.*

IEEE C57.12.90, *Test Code for Liquid-Immersed Distribution, Power, and Regulating Transformers.*

IEEE C57.12.91, *Test Code for Dry-Type Distribution and Power Transformers.*

IEEE C57.96, *Loading Dry-Type Distribution and Power Transformers.*

IEEE C57.98, *Guide for Transformer Impulse Tests.*

IEEE C57.124, *Detection of Partial Discharge and the Measurement of Apparent Charge in Dry-Type Transformers.*

IEEE 80, *IEEE Guide for Safety in AC Substation Grounding.*

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APPENDIX A UFC 3-550-04 GLOSSARY

A-1 ACRONYMS.

AFCEC	Air Force Civil Engineer Center
AFCEC/CO	AFCEC/Operations Directorate
AFCEC/CN	AFCEC/Energy Directorate
AMI	Advanced meter infrastructure
AO	Approving Official
ATO	Cybersecurity Authorization to Operate
ATS	Automatic transfer switch
BESS	Battery energy storage system
BIA	Bilateral Infrastructure Agreement
CCR	Critical Change Request
CIO	Command Information Office
CONOPS	Continuity of operations
COTS	Commercial-off-the-shelf
DA	Department of the Army
DER	Distributed energy resource
DoD	Department of Defense
DOE	Department of Energy
DSG	Distributed System Generation
EPSS	Emergency power supply systems
ERCIP	Energy Resilience and Conservation Investment Program
ESPC	Energy Savings Performance Contract
FAA	Federal Aviation Administration
FAR	Federal Acquisition Regulation
FESS	Flywheel Energy Storage System

High-VAR	High-Voltage Volt Ampere Reactive
HMI	Human machine interface
HNFA	Host Nation Funded Construction Agreements
HQUSACE	Headquarters, US Army Corps of Engineers
ICS	Industrial control system
IGSA	Intergovernmental Service Agreement
IP	Internet protocol address
IT	Information Technology
MILCON	Military construction
MTR	Military Training Route
NAVFAC	Naval Facilities Engineering Systems Command
NEPA	National Environmental Policy Act
NIPR	Non-classified Internet Protocol Router Network
NIST	National Institute of Standard and Technology
NPS	Naval Post Graduate School
NRECA	National Rural Electric Cooperative Association
OSD	Office of the Secretary of Defense
PCC	Point of common coupling
PPA	Power Purchase Agreement
PV	Photo Voltaic
PW/PWO	Public Works/Public Works Officer
RE	Renewable Energy
\1\ RICE	Reciprocating Internal Combustion Engine /1/
RMF	Risk Management Framework
SCR	Short Circuit Ratio
SOC	State of Charge

SOFA	Status of Forces Agreement
SAIDI	System Average Interruption Duration Index
SAIFI	System Average Interruption Frequency Index
SIPR	Secret Internet Protocol Router Network
UCS	Utility control systems
UPS	Uninterruptible power supply
UESC	Utility Energy Service Contract
UFC	Unified Facilities Criteria
UMCS	Utility Monitoring Control System
UP	Utility privatization
VA	Volt / Amp
VPP	Voluntary Protection Program

A-2 DEFINITION OF TERMS.

Defense-in-depth: Information security strategy integrating people, technology, and operations capabilities to establish variable barriers across multiple layers and missions of the organization. (https://csrc.nist.gov/glossary/term/defense_in_depth).

APPENDIX B UFC 3-550-04 REFERENCES

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APPENDIX A UFC 3-550-05 GLOSSARY

A-1 ACRONYMS.

AC	Alternating Current
AFCEC	Air Force Civil Engineer Center
BESS	Battery Energy Storage System
BIA	Bilateral Infrastructure Agreement
BMS	Battery Management System
CAES	Compressed Air Energy Storage System
CHP	Combined Heat and Power
DC	Direct Current
DoD	Department of Defense
EMS	Energy Management System
EPS	Electrical Power System
ESD	Emergency Shutdown
ESS	Energy Storage System
FRCS	Facility-Related Control Systems
GDL	Gas Diffusion Layer
HQUSACE	Headquarters, U.S. Army Corps of Engineers
HNFA	Host Nation Funded Construction Agreements
IBR	Inverter Based Resources
LCA	Life Cycle Assessment
MEA	Membrane Electrode Assembly
NAVFAC	Naval Facilities Engineering Systems Command
PCC	Point of Common Coupling
RFC	Regenerative Fuel Cell
SoC	State of Charge

SoH	State of Health
SOFA	Status of Forces Agreements
TOU	Time-of-Use
UCS	Utility Control Systems
UFC	Unified Facilities Criteria
UMCS	Utility Monitoring and Control System
U.S.	United States
VPP	Voluntary Protection Program
VRLA	Valve Regulated Lead Acid

A-2 DEFINITION OF TERMS.

Minimum Interoperability: The required amount of interconnection and communication which allows for real-time monitoring, control, and reporting of status, alarm generation, and operation data to be sent to and from the UMCS.

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APPENDIX A UFC 3-550-06 GLOSSARY

A-1 ACRONYMS

A	Amperage
ABA	Architectural Barriers Act
AC	Alternating Current
ADA	Americans with Disabilities Act
ADAAG	ADA Accessibility Guidelines
AFCEC	Air Force Civil Engineer Center
AFCI	Arc Fault Current Interrupter
AFV	Alternative Fuel Vehicle
AHJ	Authority Having Jurisdiction
ANSI	American National Standards Institute
ASCE	American Society of Civil Engineer
ATS	Automatic Transfer Switch
AWG	American Wire Gauge
CAN	Control Area Network
CATV	Cable Television
CFR	Code of Federal Regulations
BEV	Battery electric vehicle
BFE	Base Flood Elevation
BFG	Below Finished Grade
BPA	Blanket purchase agreement
CCS	Combined charging system
CFR	Code of Federal Register
CONUS	Contiguous United States
CPM	Charging Point Management

DC	Direct current
DCFC	Direct current fast charger
DER	Distributed Energy Resource
DLM	Dynamic Load Management
DoD	Department of Defense
DOE	U.S Department of Energy
DPW	Directorate of Public Works
EGSA	Electrical Generating Systems Association
EMP	Electro-mobility Provider
EMSP	Electro-mobility service provider
EO	Executive Order
EV	Electric vehicle
EVCF	Electric Vehicle Charging Facility / Electric Vehicle Charging Stations
FAST	Federal Automotive Statistical Tool
ft	Feet
FEMA	Federal Emergency Management Agency
FEMP	Federal Energy Management Program
FY	Fiscal year
GFCI	Ground-Fault Circuit Interrupter
GHG	Greenhouse gas
GOV	Government-Owned Vehicle
GSA	U.S. General Services Administration
HEV	Hybrid electric vehicle
HD	Heavy-Duty
HQUSACE	Headquarters, U.S. Army Corps of Engineers
HNFA	Host Nation Funded Construction Agreements

Hz	Hertz
IBC	International Building Code
ICE	Internal Combustion Engine
IEEE	formerly Institute of Electrical and Electronics Engineers
ISA	International Sign Association
kA	Kilo-Amperes
kVA	Kilovolt-ampere
KW	Kilowatt
KWH	Kilowatt-hour
LD	Light-Duty
LRC	Logistic readiness center
LSEV	Low speed electric vehicle
m	Meter
MCOV	Maximum Continuous Overvoltage Rating
MDP	Main distribution panel
MD	Medium-Duty
mm	Millimeter
MUTCD	Manual of Uniform Traffic Control Devices
MV	Medium Voltage Cable
NAVFAC	Naval Facilities Engineering Systems Command
NDAA	National Defense Authorization Act for Fiscal Year 2023
NEC	National Electrical Code
NEMA	National Electrical Manufacturers Association
NETA	International Electrical Testing Association
NFPA	National Fire Protection Association
NREL	National Renewable Energy Laboratory

OCONUS	Outside the contiguous United States
OSHA	Occupational Safety and Health Administration
PEV	Plug-in electric vehicle
PHEV	Plug-in hybrid electric vehicle
PLC	Power Line Communication
POV	Personal owned vehicle
PWD	Pulse Width Modulation
PWD	Public Works Department
RFID	Radio frequency identification
SAE	Society of Automotive Engineers
SE, USE	Service-Entrance Cable
SPD	Surge Protective Devices
SOFA	Status of Forces Agreements
RFID	Radio-frequency identification
TMP	Transportation motor pool
TSEWG	Tri-Service Electrical Working Group
TSO	Transmission system operator
UFC	Unified Facilities Criteria
UL	Underwriters Laboratories
U.S.	United States
USACE	U.S. Army Corp of Engineers
VTX	Vehicle to everything
V	Volt
ZEV	Zero Emission Vehicle

Note: The terms listed here are provided for clarification of the design criteria provided in this UFC. Refer to IEEE Std 100 for additional electrical-related definitions.

AC Level 1: An electric vehicle charging facility delivers a 120V AC power for charging Government-owned vehicles and/or personally-owned vehicles at an installation.

AC Level 2: An electric vehicle charging facility delivers either 208 or 240V AC power for charging Government-owned vehicles and/or personally-owned vehicles at an installation.

Automatic Transfer Switch (ATS): A switch designed to sense the loss of one power source and automatically transfer the load to another source of power.

Battery Electric Vehicle (BEV): A vehicle that draws motive power solely from a battery with a capacity of at least 4 kilowatt-hours; and can be recharged from an external source of electricity for motive power.

Branch Circuit: The circuit conductors and components between the final overcurrent device protecting the circuit and the equipment.

CHAdemo: Type of protocol for a connector interface between an EV and a charger. the physical, electrical, and communication requirements of the connector and mating vehicle inlet for direct-current (DC) fast charging. It is an abbreviation of “charge de move”, equivalent to “charge for moving.”

Charging Acceptance Rate: Each EV has a maximum acceptance rate for both AC and DC charging. Since this rate is the vehicles’ maximum charging speed, it should be used to calculate charging duration. While there is no current benefit for fleets and the public charging station hosts to purchase higher speed chargers, future EVs may become available with higher acceptance rates that may require installation of high-power chargers.

Charging Network: Collection of chargers located on one or more properties that are connected via digital communications to manage the facilitation of payment, the facilitation of electrical charging, and any related data requests.

Charging Network Provider: The entity that operates the digital communication network that remotely manages the chargers. Charging network providers may also serve as charging station operators and/or manufacture chargers.

Charging Port: System within a charger that charges one EV. A charging port may have multiple connectors, but it can provide power to charge only one EV through one connector at a time.

Charging Management System: A networked management system that controls the amount of power dispensed by chargers to EVs to meet user's charging needs while simultaneously responding to external power demand and/or pricing signals to provide load management, resilience, or other benefits to the electrical grid.

Combined Charging System (CCS): Standard connector interface that allows direct current fast chargers to connect to, communicate with, and charge EVs. CCS is rated to provide power of up to 350kW. It may also be called a "CCS-Type 1".

Connector: Device that attaches an EV to a charging port to transfer electricity.

Connectivity: Charging data needs to be conveyed between the chargers, controllers, and management system. Multiple connectivity options are available depending on the charger or site where the chargers are located. These include wired connections using fiber or wireless connections using radio signals such as cellular or WiFi.

DC Fast Charger (DCFC): An electric vehicle charging facility delivers 50kW-400kW DC power for charging Government-owned vehicles and/or personally-owned vehicles at an installation. DCFC is synonymous with Level 3.

Distributed Energy Resource (DER): Small-scale power generation or storage technologies (typically from 1 kW to 10,000 kW) that can provide an alternative to or an enhancement of the traditional electric power system. These can be located on an electric utility's distribution system, a subsystem of the utility's distribution system, or behind a customer meter. They may include electric storage, intermittent generation, distributed generation, demand response, energy efficiency, thermal storage or electric vehicles and their charging equipment.

Domestic fleet: means all reportable motor vehicles operated in any State, Commonwealth, territory or possession of the United States, and the District of Columbia.

Dwell Time: Since most vehicles will charge overnight while not in use, the duration of "dwell", or stationary charging time, will determine the maximum charge duration and charger size. Slower chargers (chargers with smaller kW ratings) can be considered for vehicles with longer dwell times.

Efficiency: Chargers with power load management capability are the most energy efficient; ENERGY STAR-rated chargers use 40% less energy in stand-by mode.

Electric Meter: is a device that measures the amount of electric energy consumed by an electrically powered equipment.

Electrical Switchgear: A centralized collection of circuit breakers, fuses and switches (circuit protection devices) that function to protect, control and isolate electrical equipment.

Electrical Transformer: an electrical apparatus that manipulates the level of voltage flowing through any point in a power grid. In a distribution system, the transformer decreases the voltage traveling through power lines to a level more suitable for residential and commercial use.

Electric Vehicle (EV): EVs are equipped with a battery-powered motor instead of a traditional internal combustion engine. Contrary to PHEVs and HEVs, EVs do not have a gasoline tank and output zero tailpipe emissions. They are associated with a lower carbon footprint than traditional vehicle types.

Electric Vehicle Charging Facility (EVCf): A structure that delivers energy to electric vehicles (EV) and compatible equipment, including Level I, II and DCFC (III) systems. EVCf's are considered real property when they are constructed as a charging system. EVCf Systems are permanently constructed to include charging stations (ports), electric meters, associated electrical service lines and conduits, dedicated transformers, and concrete pads/footings as required. Measurement of EVCf's unit of measurement (UM) as KW is cumulative for each charging port station associated with the asset. Example: a row of five charging stations (ports) is considered a single asset with a UM equal to the sum of each port's KW output.

Electric Vehicle Supply Equipment (EVSE): Supplies electricity to an electric vehicle (EV). Commonly called charging stations or charging docks, they provide electric power to the vehicle and use that to recharge the vehicle's batteries.

Energy Consumption: vehicle energy consumption consists of its average "daily duty cycle" (average miles traveled). For fleet managers the daily duty cycle for each vehicle type can be useful to calibrate the total charging needs for a particular site.

EV Quantity: Where multiple EVs, charger quantities and speed should be calculated to ensure reasonable charging times. This is particularly important if chargers are planned to be shared between multiple EVs, vehicles have long overnight dwell time, and/or low or medium speed chargers are planned for installation. Fleet managers should consider the frequency that they plan to rotate EVs through shared chargers and ideally plan an efficient charging schedule. Charging management software can automate or assist in rotating EVs through chargers or sharing electric capacity between chargers.

Existing Facility: A facility is existing if changes to be made are cosmetic or minor in nature.

Foreign Fleet: all reportable motor vehicles operated in areas outside any State, Commonwealth, territory or possession of the United States, and the District of Columbia.

Government motor vehicle: any motor vehicle that the Government owns or leases. This includes motor vehicles obtained through purchase, excess, forfeiture, commercial lease, or GSA Fleet lease.

Greenhouse gases (GHG): Gases in the atmosphere that trap the sun's heat and warm the Earth, contributing to climate change. The Earth's most common greenhouse gases are ozone, nitrous oxide, water vapor, methane, and carbon dioxide. Vehicles with internal combustion engines emit greenhouse gases through their tailpipe.

Hybrid electric vehicle (HEV): Vehicles that utilize both electric batteries and gasoline. More often than not, the electric motor is here to assist the internal combustion engine, during the acceleration phases, for instance. Note that HEVs cannot be plugged into regular EV charging stations. Batteries replenish themselves via the energy generated by the combustion engine or via regenerative braking.

Internal combustion engine (ICE): Internal combustion engines use liquid fuel (gasoline) to create energy to power traditional vehicles. ICE vehicles are the most common vehicle on the road.

Linear Load: An electrical load device that presents an essentially constant load impedance to the power source throughout the cycle of applied voltage in steady-state operation.

Listed: Applies to equipment or materials included in a list published by an organization acceptable to the authority having jurisdiction. The organization periodically inspects production and certifies that the items meet appropriate standards or tests as suitable for a specific use.

Low Voltage System: An electrical system having a maximum root-mean-square (rms) voltage of less than 1,000 volts.

Managed Charging: A feature provided as a part of a charge management system to balance charging load with facility energy needs.

Medium Voltage System: An electrical system having a maximum RMS AC voltage of 1,000 volts to 34.5 kV. Some documents such as ANSI C84.1 define the medium voltage upper limit as 100 kV, but this definition is inappropriate for facility applications.

Mission-Critical: Mission-critical systems are necessary to continue warfighter operations and direct mission support of warfighter operations.

Mission-Essential: Mission-essential systems are basic and necessary to accomplish an organization's mission.

Molded Case Circuit Breaker: A low voltage circuit breaker assembled as an integral unit in an enclosing housing of insulating material. It is designed to open and close by nonautomatic means, and to open a circuit automatically on a predetermined overcurrent, without damage to itself, when applied properly within its rating.

Networked Charger: Charging stations that are a part of a larger charging network.

New Construction: A facility is considered new if changes to be made are more than cosmetic or minor, such as major renovations, additions, or new facilities.

Non-networked Charger: Charging stations that are not a part of a charging network. The charger has a basic function to charge an electric vehicle when plugged in.

Open Charge Alliance (OCA): Global consortium of public and private electric vehicle infrastructure leaders that have come together to promote open standards through the adoption of the Open Charge Point Protocol (OCPP).

Open Charge Point Interface (OCPI): Open-source communication protocol that governs the communication among multiple charging networks, other communication networks, and software applications to provide information and services for users.

Open Charge Point Protocol (OCPP): Open-source communication protocol that governs the communication between chargers and the charging network that remotely manages the chargers.

Plug-in hybrid electric vehicle (PHEV): Vehicles that are propelled by both an internal combustion and heat engine and to a significant extent by an electric motor that draws electricity from a battery (with a capacity of at least four kilowatt-hours) that can be recharged from an external source. Note: PHEVs are considered to be ZEVs.

Power Sharing: Dynamically limiting the charging output of individual charging ports at the same charging station to ensure that the sum total power output to all EVs concurrently charging remains below a maximum power threshold.

Plug and Charge: Method of initiating charging, whereby an EV charger plugs a connector into their vehicle and their identity is authenticated through digital certificates define by ISO-15118, a charging session initiates, and a payment is transacted automatically, without any other action required at the point of use.

Radio-frequency identification (RFID): Type of technology that links a card to an account. It can be used in electric vehicle charging to start a charge quickly and conveniently through tapping.

Switchboard: Divide incoming power into smaller circuits using components, including panels or frames, busbars, and control or monitoring devices. They are not designed to handle high voltage and have very different safety features than switchgear.

Underground Circuit: An underground circuit is an insulated cable which is buried under the earth surface for power transmission and distribution.

Usability: The charger should be easy to use by drivers and the charge cord length should reach multiple stalls. The charger face plate should be easy to read

in any lighting condition; color-coded lights that indicate charging status increase readability.

Workplace Charging Facility: EVCF that participates in workplace charging programs, offers charging for employees of the host facility during standard work hours of operation to charge their EV.

Workplace Charging Program: A set of guidelines that regulate how EVCF is utilized within the workplace for POV charging.

Zero emission Vehicle (ZEV): A vehicle that when operating produces zero tailpipe exhaust emissions of any criteria pollutant (or precursor pollutant) or greenhouse gas.

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FEDERAL ENERGY MANAGEMENT PROGRAMS (FEMP)

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Federal Fleet ZEV Ready Center

<https://www.energy.gov/sites/default/files/2020/11/f80/federal-workplace-charging-guide.pdf>

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Federal Workplace Charging Fee Calculator

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Sample Employee Survey for Workplace Charging Planning

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[Federal Highway Administration \(dot.gov\)](https://www.fhwa.dot.gov)

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<https://evitp.org/>

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[EO 14057, Executive Order on Catalyzing Clean Energy Industries and Jobs Through Federal Sustainability](#)

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INTERNATIONAL ELECTRICAL TESTING ASSOCIATION

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ISO

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SAE J3072, SAE Interconnection Requirements for Onboard, Grid Support Inverter Systems (AC)

SAE J3105, SAE Electric Vehicle Power Transfer System Using Conductive Automated Connection Devices

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UNITED STATES OF THE ARMY CORPS OF ENGINEERS (USACE)

PCASE

APPENDIX A UFC 3-555-01 GLOSSARY

A-1 ACRONYMS

AC	Alternating Current
AFCEC	Air Force Civil Engineer Center
AWG	American Wire Gage
BIA	Bilateral Infrastructure Agreement
BOD	Basis of Design
CCR	Criteria Change Request
CE	Civil Engineering
DC	Direct Current
DoD	Department of Defense
DOR	Designer of Record
FDS	Functional Data Sheets
FLEDS	Flight Line Exterior power Distribution Systems
FPUS	Fixed Point Utility Systems
FRD	Facility Requirements Document
ft	Feet (or Foot)
GPU	Ground Power Unit
GSE	Ground Support Equipment
HQUSACE	Headquarters, U.S. Army Corps of Engineers
HNFA	Host Nation Funded Construction Agreements
Hz	Hertz
kVA	Kilo-Volt-Ampere
LANT	NAVFAC Atlantic
lb	Pound
m	Meter

mm	Millimeter
MCM	kcmil, or thousands of circular mils
MILCON	Military Construction
NAVAIR	Naval Air Systems Command
NAVFAC	Naval Facilities Engineering Systems Command
NAWCAD	Naval Air Warfare Center Aircraft Division
NEMA	National Electrical Manufacturers Association
NFPA	National Fire Protection Association
O&M	Operation and Maintenance
POU	Point-of-Use
RFP	Request for Proposals
RPIE	Real Property Installed Equipment
SE	Service Equipment
SOFA	Status of Forces Agreements
SOPs	Standard Operating Procedures
UFC	Unified Facilities Criteria
UFGS	United Facilities Guide Specifications
U.S.	United States
V	Volt
VAC	Volts Alternating Current
VDC	Volts Direct Current

APPENDIX B UFC 3-555-01 REFERENCES

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B-2 NON-GOVERNMENT

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NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION

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APPENDIX A UFC 3-570-01 GLOSSARY

A-1 ACRONYMS.

AC	Alternating Current
A-E or A/E	Architect-Engineer
AFCEC	Air Force Civil Engineer Center
ANSI	American National Standards Institute
AST	Aboveground (on-grade) storage tank
ASTM	American Society for Testing and Materials
AWG	American Wire Gage
CFR	Code of Federal Regulations
CP	Cathodic Protection
CPC	Corrosion Prevention and Control
CSE	Copper-copper sulfate (reference) electrode
DC	Direct Current
DoD	Department of Defense
EPA	Environmental Protection Agency
ER (probes)	Electrical Resistance (probes)
ERDC/CERL	Engineering Research and Development Center/Construction Engineering Research Laboratory
FGS	Final Governing Standards
GCP	Galvanic (Sacrificial) cathodic protection
HMWPE	High molecular weight polyethylene
HQUSACE	Headquarters, U.S. Army Corps of Engineers
ICCP	Impressed current cathodic protection
IR (Drop)	Voltage (drop)
OEBGD	Overseas Environmental Baseline Guidance Document

pH	A measure of hydrogen ion activity
POL	Petroleum, Oil and Lubricants
PRC	Permanent reference electrode
NACE International	National Association Corrosion Engineers International
NAVFAC	Naval Facilities Engineering Command (NAVFACENGCOM)
NAVFAC ESC	NAVFAC Engineering Service Center (now NAVFAC EXWC)
NAVFAC EXWC	NAVFAC Engineering and Expeditionary Warfare Center
NEC	National Electrical Code
NSF	National Sanitation Foundation
O&M	Operation and Maintenance
RFP	Request for Proposal
SCE	Silver-silver chloride (reference) electrode
S/E	Structure-to Electrolyte
UFC	Unified Facilities Criteria
UFGS	Unified Facilities Guides Specifications.
U.S.	United States
USACE	United States Army Corps of Engineers
UST	Underground Storage Tank
ZRE	Zinc reference electrode

Amphoteric material: Material subject to attack by both acid and alkaline environments; include aluminum, zinc, and lead.

Anode: The electrode of an electrochemical cell at which oxidation occurs. (The anode is usually the electrode where corrosion occurs and metal ions enter the solution).

Bimetallic corrosion: (See galvanic corrosion).

Bond: See continuity bond.

Cathode: The electrode of an electrochemical cell at which reduction occurs.

Cathodic protection: A technique to prevent the corrosion of a metal surface by making that surface the cathode of an electrochemical cell.

Coating: A dielectric material applied to a structure to separate it from its environment.¹

Conductivity: The measurement of a material's ability to conduct electrical current.

Continuity bond: Metallic connection that provides electrical continuity.

Corrosion: The deterioration of a material or its properties due to a reaction of that material with its chemical environment.

Corrosion potential: The potential of a corroding metal surface relative to a reference electrode under specific conditions in an electrolyte.

Corrosion rate: The rate at which corrosion proceeds.¹

Crevice Corrosion: Localized corrosion resulting from a concentration cell formed between two metal surfaces or between a metal and non-metallic surface.

Current density: The current per unit area.

Electrical isolation: Condition of being electrically separated from other metallic structures or the environment.¹

Electrode: A conductor used to establish electrical contact with an electrolyte and through which current is transferred to or from an electrolyte.¹

Electrolyte: A chemical substance or mixture containing ions that migrate in an electric field. Examples are soil and seawater.

Fiber Reinforced Plastics (FRP): A broad group of composite materials composed of fibers embedded in a plastic resin matrix.

Foreign structure: Any structure that is not intended as a part of the system of interest.

Galvanic anode. A metal that, because of its relative position in the galvanic series, provides sacrificial protection to metal or metals that are more noble in the series, when coupled in an electrolyte. These anodes are the current source in one type of cathodic protection.¹

Galvanic cell: A corrosion cell in which anode and cathode are dissimilar conductors, producing corrosion because of their innate difference in potential.

Galvanic corrosion: Corrosion resulting from the coupling of dissimilar metals in an electrolyte.

Holiday: A discontinuity in a coating that exposes the metal surface to the environment.

Hydrogen embrittlement: The severe loss of ductility of a metal when hydrogen has been introduced into the metal structure.

Hydrogen overvoltage: Voltage characteristic for each metal-environmental combination above which hydrogen gas is liberated.

Impressed current: Direct current supplied by a power source external to the electrode system. For the purposes of this manual, direct current for cathodic protection.

Interference: Any electrical disturbance on a metallic structure as a result of stray current.

Interference bond: A metallic connection designed to control electrical current interchange between metallic systems.

IR drop: Voltage across a resistance according to Ohm's Law. ¹

pH: A measure of hydrogen ion activity defined by: $\text{pH} = \log_{10} (1/a\text{H}^+)$ where $a\text{H}^+$ = hydrogen ion activity = molal concentration of hydrogen ions multiplied by the mean ion activity coefficient (= 1 for simplified calculations).

Pipe-to-soil potential: The potential difference between the pipe metallic surface and electrolyte that is measured with reference to an electrode in contact with the electrolyte. See also structure-to-electrolyte potential.¹

Polarization: The deviation from the open circuit potential of an electrode resulting from the flow of current.

Reference electrode: A reversible electrode with a potential that may be considered constant under similar conditions of measurement.

Resistivity: The measurement of a material's ability to oppose the flow of electric current.

Rust: A reddish-brown corrosion product of iron that is primarily hydrated iron oxide.

Sacrificial anode: See galvanic anode.

Stray current: Current flowing through paths other than the intended circuit.

Stray current corrosion: Corrosion resulting from stray current flow through paths other than the intended circuit.

Structure-to-electrolyte potential (also structure-to-soil potential): The potential difference between a buried metallic structure surface and electrolyte that is measured with reference to an electrode in contact with the electrolyte. See also pipe-to-soil potential.

Structure-to-structure voltage (also structure-to-structure potential): Difference in voltage between metallic structures in a common electrolyte.

Surge arrestors: A protective device for limiting surge voltages by discharging or bypassing surge current, and it also prevents continued flow of follow current while remaining capable of repeating these functions. Designed primarily for connection between a conductor of an electrical system and ground to limit the magnitude of transient (surge) overvoltages on equipment. Also known as arrestor or arrester, surge arrester, lightning arrestor or arrester.

Uniform corrosion: Corrosion attack of a metal that is essentially the same at all exposed areas of its surface.

Voltage: An electromotive force, or a difference in electrode potentials expressed in volts.

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NACE International Standard Practice SP0196 (latest edition): *Galvanic Anode Cathodic Protection of Internal Submerged Surfaces of Steel Water Storage Tanks*

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APPENDIX A UFC 3-575-01 GLOSSARY

A-1

ACRONYMS.

AFMAN	Air Force Manual
AFI	Air Force Instruction
AHJ	Authority Having Jurisdiction
AWG	American Wire Gauge
BCE	Base Civil Engineer
DoD	Department of Defense
EES	earth electrode system
EMI	electromagnetic interference
EMP	electromagnetic pulse
EPDM	ethylene propylene diene monomer (m-class) rubber feet (or foot)
ft ²	foot squared or square feet
HQUSACE	Headquarters, US Army Corps of Engineers
IEEE	formerly Institute of Electrical and Electronics Engineers
in	inch, inches
ILS	instrument landing system
LPS	lightning protection system
m	meter
m ²	meter squared or square meter
mm	millimeter
NAVFAC	Naval Facilities Engineering Command
NFPA	National Fire Protection Association
POL	petroleum oil lubricants
USACE	US Army Corps of Engineers
UFC	Unified Facilities Criteria

UFGS	Unified Facilities Guide Specifications
UL	Underwriters Laboratories

A-2

DEFINITION OF TERMS.

Activity: The end use of a facility.

Approved: Acceptable to the Authority Having Jurisdiction.

Bonding: An electrical connection between an electrically conductive object and a component of a lightning protection system intended to significantly reduce potential differences created by lightning currents.

Catenary System: A lightning protection system consisting of one or more overhead wires. Each overhead wire forms a catenary between masts, and serves the function of both a strike termination device and a main conductor.

Conductor, Bonding: A conductor used for equalizing potential between metal bodies and the lightning protection subsystem.

Contractor: Person(s) doing actual construction portion of a project.

Copper Clad Steel: Steel with a coating of copper bonded on it.

Designer of Record: The engineer responsible for the actual preparation of the construction documents.

Down Conductor, Lightning: The conductor connecting the roof conductors or overhead ground wire to the earth ground subsystem.

Labeled: Equipment or materials to which has been attached a label, symbol, or other identifying mark of an organization that is acceptable to the authority having jurisdiction and concerned with product evaluation, that maintains periodic inspection of production of labeled equipment or materials, and by whose labeling the manufacturer indicates compliance with appropriate standards or performance in a specified manner.

Listed: Equipment, materials, or services included in a list published by an organization that is acceptable to the authority having jurisdiction and concerned with evaluation of products or services, that maintains periodic inspection of production of listed equipment or materials or periodic evaluation of services, and whose listing states that either the equipment, material, or service meets appropriate designated standards or has been tested and found suitable for a specified purpose.

Power Ground: Driven electrode at service entrance.

Qualified: Compliant with NEC/NFPA definitions and, for Air Force, experience and training meeting AFI 32-1065.

User: Facility occupants or Facility Manager.

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APPENDIX A UFC 3-580-01 GLOSSARY

A-1 ABBREVIATIONS AND ACRONYMS.

μm	micrometer
10G-EPON	10 Gigabit per second Ethernet passive optical network
10G-PON	10 Gigabit per second passive optical network
8P8C	8-position, 8-contact
A	ampere
A/C	air conditioning
A/E	architect/engineer
A/V	audio visual
AASHTO	American Association of State Highway and Transportation Officials
ABF	air blown fiber
AC	alternating current
ACMS	Automated Control Management System
ACPA	American Concrete Pipe Association
ADA	Americans with Disabilities Act
ADNCon	area distribution node
AFCEC	Air Force Civil Engineer Center
AFF	above the finished floor
AFH	Army Family Housing
AHJ	Authority Having Jurisdiction
ANSI	American National Standards Institute
APC	angle polished connector
APWA	American Public Works Association
ASHRAE	American Society of Heating, Refrigeration, and Air Conditioning Engineers

ASTM	American Society for Testing and Materials
AV	audio visual
AWG	American Wire Gauge
B/P/C/S	base/post/camps/stations
BAS	building automation systems
BATB	base area transport boundary
BBC	backbone bonding conductor
BCO	Base Communications Officer
BCT	bonding conductor for telecommunications
BD	building distributor
BEQ	Bachelor Enlisted Quarters
BET	building entrance terminals
BIA	Bilateral Infrastructure Agreements
BICSI	Building Industry Consulting Service, International, Inc.
BIM	Building Information Modeling
BMS	Building Management System
BNC	bayonet navy connector
BOQ	Bachelor Officers Quarters
BTS	base transceiver system
BTU/hr	British thermal units per hour
C4I	Command, Control, Communications, Computers, and Intelligence
CAT6A	Category 6A
CATV	community antenna or cable television
C-C	center-to-center
CCTV	closed-circuit television

CD	chromatic dispersion
CE	compromising emanations
CIC	cable in conduit
CNIC	Commander Navy Installations Command
CNSSAM	Committee on National Security Systems Advisory Memorandum
CNSSI	Committee on National Security Systems Instruction
CONUS	Continental United States
CP	consolidation point
CTTA	Certified TEMPEST Technical Authority
CV	cable vault
Cx	commissioning
CXA	Commissioning Agent
CXM	Commissioning Manager
DA	Department of the Army
DAA	Designated Accreditation Authority
DAQ	delivered audio quality
DAS	distributed antenna system
dB	decibel
DB	direct buried
dBmV	decibel millivolt
DC	direct current
DCO	Dial/Digital Central Office
DDC	direct digital controller
DIN	Deutsche Industrie-Normen
DISN	Defense Information System Network

DMFO	Defense Medical Facilities Office
DODIN	Department of Defense Information Network
DoN	Department of Navy
DOR	Designer of Record
DPW	Directorate of Public Works
DSN	Defense Switched Network
DWDM	dense wave division multiplexing
ED	edge device
EDFA	erbium-doped fiber amplifier
EES	earth electrode subsystem
EF	entrance facility
EIA	Electronic Industries Alliance
EOC	Emergency Operations Center
EPO	emergency power off
EPON	Ethernet passive optical network
ER	equipment room
ESS	enterprise survivable server
EUB	end user building
FACP	fire alarm control panel
fc	foot candle
FCC	Federal Communications Commission
FD	floor distributor
FEC	Facilities Engineering Command
FO	fiber-optic
FO/GO	Flag Officer/General Officer

FOC	fiber optic cable
FOCIS	Fiber Optic Connector Intermateability Standard
FOPP	fiber optic patch panel
FOUO	For Official Use Only
FRCS	Facility Related Control System
ft/s	feet per second
Gb	gigabit
GbE	gigabit Ethernet
Gbps	gigabit per second
GE	grounding equalizer
GEC	grounding electrode conductor
GFCI	ground fault circuit interrupter
GHz	gigahertz
GPON	gigabit passive optical network
GPR	ground penetrating radar
GSA	General Services Administration
HC	horizontal crossconnect
HCDS	hardened carrier distribution systems
HCP	horizontal connection point
HDCP	high definition copy protocol
HDD	horizontal directional drilling
HDMI	high definition multimedia interface
HDPE	high density polyethylene
HNFA	Host Nation Funded Construction Agreements
hp	horsepower

HVAC	heating, ventilation, and air conditioning
I3A	Installation Information Infrastructure Architecture
IC	integrated circuit
IC	intermediate crossconnect
ICAN	Installation and Campus Area Network
ICD	Intelligence Community Directive
ICDS	installation communications distribution system(s)
ICG	installation communications grid
ICS	industrial control system
ICT	information and communications technology
IDC	insulation displacement connector
IDF	intermediate distribution frame
IDS	intrusion detection system
IEC	International Electrotechnical Commission
IEEE	Institute of Electrical and Electronics Engineers
IFC®	International Fire Code®
IMA	Information Mission Area
IMC	intermediate metallic conduit
IoT	Internet of Things
IP	internet protocol
IPN	installation processing node
IPS	intrusion prevention system
IPTV	internet protocol television
IS	information system
ISDN	Integrated Services Digital Network

ISN	installation servicing node
ISO	International Organization for Standardization
ISP	inside plant wiring
IT	information technology
ITE	information technology equipment
ITS	information transport system
ITU	International Telecommunications Union
ITU-T	International Telecommunication Union - Telecommunication Standardization Sector
JWICS	Joint Worldwide Intelligence Communications System
Km	kilometer
kPa	kilopascal
LAN	local area network
LATB	local area transport boundary
lb	pound
LC	lucent connector
LPS	limited power source
LSA	line-sharing adapter
LSA+	line-sharing adapter plus
LSPM	light source and power meter
LVDC	low voltage direct current
m/s	meters per second
mA	milliampere
MAC	moves, adds, and changes
MB	megabyte
Mb/s	megabit per second

MC	main crossconnect
MCEN	Marine Corps Enterprise Network
MCN	main communications node
MDF	main distribution frame
MH	maintenance holes
MHz	megahertz
MILCON	Military Construction
MIL-DTL	Military Detail
MIL-HDBK	Military Handbook
MIL-STD	Military Standard
MM	multi-mode
MPa	megapascal
MPD	metallic pedestal disconnect
MPTL	male plug terminated link
MTP®	Multi-fiber Termination Push-on
MT-RJ	mechanical transfer registered jack
MUTOA	multi-user telecommunication outlet assembly
N	Newton
NAVFAC	Naval Facilities Engineering Command
NCTAMS	Naval Computer and Telecommunications Area Master Station
NEC®	National Electrical Code®
NEC	network enterprise center
NEMA	National Electrical Manufacturers Association
NEN	Naval Enterprise Network
NESC®	National Electrical Safety Code®

NETCOM	U.S. Army Network Enterprise Technology Command
NFPA	National Fire Protection Association
NGEN	Next Generation Enterprise Network
nm	nanometer
NMCI	Navy and Marine Corps Intranet
NPCA	National Precast Concrete Association
NRTL	Nationally Recognized Testing Laboratory
NSANet	National Security Agency Intranet
NSC	network service center
NSI	National Security Information
NZDSF	non-zero dispersion-shifted fiber
OCONUS	outside of the continental United States
OLT	optical line terminal
OLTS	optical loss test sets
ONT	optical network terminal
OSHA	Occupational Safety and Health Administration
OSP	outside plant
OT	operational technology
OTDR	optical time domain reflectometer
OTN	optical transport node
P2P	peer-to-peer
PBB	primary bonding busbar
PBX	private branch exchange
PC	physical contact
PDS	protected distribution system

PE	polyethylene
PET	protected entrance terminal
PF	fluorinated ethylene propylene
PIC	plastic insulated cable
PM	Program Manager
PMD	polarization mode dispersion
POC	point of contact
PoDL	power over data line
PoE	power over Ethernet
POH	Power over HDBaseT
POL	passive optical local area network
PON	passive optical network
POTS	plain old telephone service
pr	pair
ps	picosecond
PSI	physical site identifier
psi	pounds per square inch
psi	pounds per square inch
PVC	polyvinyl chloride
PWD	Public Works Department
PWO	Public Works Office
QA	quality assurance
QC	quality control
RBB	rack bonding busbar
RC	Regional Coordinator

RCDD	Registered Communications Distribution Designer
RF	radio frequency
RGB	rack grounding busbar
RMC	rigid metallic conduit
ROICC	Resident Officer in Charge of Construction
RSC	rigid steel conduit
RTRC	red threaded fiberglass conduit/reinforced thermosetting resin conduit
RU	rack unit
RUS	Rural Utilities Service
SAPF	Special Access Program Facility
SBB	secondary bonding busbar
SC	standard connector
SCADA	Supervisory Control and Data Acquisition
SCIF	Sensitive Compartmented Information Facility
ScTP	screened twisted pair
SDAN	software defined access network
SDR	standard dimensional ratio
SEBQ	Senior Enlisted Bachelor Quarters
SFP	small form pluggable
SIDR	standard internal dimension ratio
SIPRNET	Secret Internet Protocol Router Network
SM	single-mode
SMF	single mode fiber
SOFA	Status of Forces Agreements
SPC	super personal computer

ST	subscriber terminal
STP	shielded twisted pair
TBB	telecommunications bonding backbone
TBC	telecommunications bonding conductor
TDM	time domain multiplexing
TDMM	Telecommunications Distribution Methods Manual
TE	telecommunications enclosure
TEBC	Telecommunications Equipment Bonding Conductor
TEF	telecommunications entrance facility
TEMPEST	Telecommunications Electronics Material Protected from Emanating Spurious Transmissions
TGB	telecommunications grounding busbar
TIA	Telecommunications Industry Association
TMGB	telecommunications main grounding busbar
TR	telecommunications room
TS	telecommunications space
TSB	Telecommunications Systems Bulletin
UCR	Unified Capabilities Requirements
UFC	Unified Facilities Criteria
UFGS	Unified Facilities Guide Specification
UL	Underwriters Laboratory
UPC	ultra physical contact
UPS	uninterruptable power systems
USACE	United States Army Corps of Engineers
USDA	United States Department of Agriculture
USG	U.S. Government

UTP	unshielded twisted pair
UXO	unexploded ordnance
VA	volt-amp
VAC	volts alternating current
VDC	volts direct current
VDE	Verband Deutscher Elektrotechniker
VTC	video teleconference
WAO	work area outlet
WAP	wireless access point
WIDS	wireless intrusion detection system
WLAN	wireless local area network

A-2 DEFINITION OF TERMS.

core node: On an installation (B/P/C/S), a core node (also referred to as a main communications node or area distribution node) is a physical location that typically hosts the commercial point of presence and serves as the aggregation point for the area distribution nodes.

intra-building backbone: Connectivity for the voice, video, and data networks between the entrance facility or equipment room, to a telecommunications room.

inter-building backbone: Connectivity between buildings, also referred to as part of outside plant (OSP).

ps: chromatic dispersion coefficient (Table 3-5)

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APPENDIX A UFC 3-600-01 GLOSSARY

A-1 ACRONYMS.

°F	degrees Fahrenheit
°C	degrees Celsius
A&E	Architectural and Engineering Services
ABA	Architectural Barriers Act
ABAAG	Architectural Barriers Act Accessibility Guidelines
ACA	American Correctional Association
ADP	Automatic Data Processing
AFCEC	Air Force Civil Engineer Center
AFFF	Aqueous Film-Forming Foam
AFMAN	Air Force Manual
AFTO	Air Force Technical Order
ANSI	American National Standards Institute
AR	Army Regulation
ASCE	American Society of Civil Engineers
ASME	American Society of Mechanical Engineers
ASTM	American Society for Testing and Materials
ATFP	Anti-Terrorism Force Protection
AWG	American Wire Gauge
AWWA	American Water Works Association
BIA	Bilateral Infrastructure Agreement
CAD	Computer Aided Drafting
CDC	Child Development Center
CFPE	Component Fire Protection Engineer
cm ²	square centimeter(s)

CO	Carbon Monoxide
CO ₂	Carbon Dioxide
CRAC	Computer Room Air Conditioning
CSS	Central Security Service
DA	Department of the Army
DESR	Defense Explosives Safety Regulation
DFPE	Designated Fire Protection Engineer
DLA	Defense Logistics Agency
DoD	Department of Defense
DoDI	Department of Defense Instruction
DPDT	Double-Pole, Double-Throw
ECB	Engineering Construction Bulletins
EM	Engineering Manual
EMCS	Energy Monitoring and Control System
ESFR	Early Suppression Fast-Response Sprinklers
ETL	Engineering Technical Letters
FAAA	Fire Administration Authorization Act
FACP	Fire Alarm Control Panel
FC	Facilities Criteria
FM	Factory Mutual Global
FPE	Fire Protection Engineer
FRT	Fire Retardant Treated Plywood
FS	Flame Spread Rating
ft	feet; foot
ft ²	square feet; square foot
gal	gallon(s)

GH ₂	Gaseous hydrogen
GOX	Gaseous Oxygen
gpm	gallons per minute
HQUSACE	Headquarters, U.S. Army Corps of Engineers
HNFA	Host Nation Funded Construction Agreements
HP	Horsepower
HVAC	Heating, Ventilating and Air Conditioning
HVLS	High Volume Low Speed
HZ	Hertz
IBC	International Building Code
ICC	International Code Council
IEEE	Institute of Electrical and Electronics Engineers
IFC	International Fire Code
in	inch(es)
in ²	square inch(es)
IRC	International Residential Code
kPa	kilopascal
L	liter(s)
LCD	Liquid Crystal Display
LEC	Electroluminescence
LED	Light Emitting Diode
LH ₂	Liquid Hydrogen
LOX	Liquid Oxygen
m	meter(s)
m ²	square meter(s)
MAF	Missile Alert Facilities

MIL-HDBK	Military Handbook
min	minute
mm	millimeter(s)
ms	milliseconds
NARA	National Archives and Records Administration
NATOPS	Naval Air Training & Operating Procedures Standardization
NAVAIR	Naval Air Systems Command
NAVFAC	Naval Facilities Engineering Systems Command
NAVSEA	Naval Sea Systems Command
NCEES	National Council of Examiners for Engineering & Surveying
NFPA	National Fire Protection Association
NGA	National Geospatial-Intelligence Agency
NGR	National Guard Regulation
NPSH	Net Positive Suction Head
NRO	National Reconnaissance Office
NRTL	Nationally Recognized Testing Laboratory (as defined on the OSHA website https://www.osha.gov/dts/otpc/nrtl/nrtllist.html)
NSA	National Security Agency
OHES	Office of Occupational Health, Environmental & Safety Services
OPNAVINST	Naval Operations Instructions
P.E.	Registered Professional Engineer
PIV	Post Indicator Valve
POL	Petroleum Oil Lubricant
PRVs	Pressure-Regulating Valves
psi	pounds per square inch
PV	Photovoltaic

QFPE	Qualified Fire Protection Engineer
RPA	Remotely Piloted Aircraft
s	second(s)
SAE	Society of Automotive Engineers
SD	Smoke Developed Rating
SFPE	Society of Fire Protection Engineers
SOFA	Status of Forces Agreements
SPD	Surge Protection Device
UAS	Unmanned Aerial System
UAV	Unmanned Aerial Vehicle
UFAS	Uniform Federal Accessibility Standard
UFC	Unified Facilities Criteria
UL	Underwriters Laboratories Inc.
USC	United States Code
Vac	volts alternating current
Vdc	volts direct current
WHS	Washington Headquarters Services

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ASTM E136, Standard Test Method for Behavior of Materials in a Vertical Tube Furnace at 750°C

ASTM E814, Standard Test Method for Fire Tests of Penetration Firestop Systems

DEPARTMENT OF DEFENSE, WASHINGTON HEADQUARTERS SERVICE

<http://www.dtic.mil/whs/directives/>

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DEPARTMENT OF THE AIR FORCE

<http://www.e-publishing.af.mil/>

AFMAN 91-201, Explosives Safety Standard

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DEPARTMENT OF THE ARMY

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AR 385-10, The Army Safety Program

AR 420-90, Facilities Engineering Fire and Emergency Services

DA PAM 385-61, Toxic Chemical Agent Safety Standards

EM-385-1-1, Safety and Health Requirements Manual

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DEPARTMENT OF DEFENSE

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[MIL-STD-3007](#), Standard Practice for Unified Facilities Criteria and Unified Facilities Guide Specifications

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FM Global Data Sheet 8-34, Automatic Storage and Retrieval Systems

INTERNATIONAL CODE COUNCIL (ICC)

<http://www.iccsafe.org>

IBC, International Building Code®

IFC, International Fire Code®

IRC, International Residential Code®

IWUIC, International Wildland-Urban Interface Code®

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

www.nfpa.org

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NFPA 24, Standard for the Installation of Private Fire Service Mains and Their Appurtenances

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NFPA 30A, Code for Motor Fuel Dispensing Facilities and Repair Garages

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NFPA 45, Standard on Fire Protection for Laboratories Using Chemicals

NFPA 51, Standard for the Design and Installation of Oxygen-Fuel Gas Systems for Welding, Cutting, and Allied Processes

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NFPA 76, Standard for the Fire Protection of Telecommunications Facilities

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NFPA 99, Health Care Facilities Code

NFPA 99B, Standard for Hypobaric Facilities

NFPA 101, Life Safety Code®

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NFPA 204, Standard for Smoke and Heat Venting

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NFPA 241, Standard for Safeguarding Construction, Alteration, and Demolition Operations

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NFPA 303, Fire Protection Standard for Marinas and Boatyards

NFPA 307, Standard for the Construction and Fire Protection of Marine Terminals, Piers, and Wharves

NFPA 312, Standard for Fire Protection of Vessels During Construction, Conversion, Repair, and Lay-Up

NFPA 400, Hazardous Materials Code

NFPA 409, Standard on Aircraft Hangars

NFPA 704, Standard System for the Identification of the Hazards of Materials for Emergency Response

NFPA 750, Standard on Water Mist Fire Protection Systems

NFPA 850, Recommended Practice for Fire Protection for Electric Generating Plants and High Voltage Direct Current Converter Stations

NFPA 855, Standard for the Installation of Stationary Energy Storage Systems

NFPA 909, Protection of Cultural Resource Properties - Museums, Libraries, and Places of Worship

NFPA 1141, Standard for Fire Protection Infrastructure for Land Development in Wildland, Rural, and Suburban Areas

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NFPA 2001, Standard on Clean Agent Fire Extinguishing Systems

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UFC 4-151-10, General Criteria for Waterfront Construction

UFC 4-152-01, Design: Piers and Wharves

UFC 4-159-01N, Design: Hyperbaric Facilities

UFC 4-211-01, Aircraft Maintenance Hangars

UFC 4-211-02, Aircraft Corrosion Control and Paint Facilities

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UFC 4-740-14, Design: Child Development Centers

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APPENDIX A UFC 3-810-01 GLOSSARY

A-1 ACRONYMS AND ABBREVIATIONS.

ACM	Asbestos Containing Materials
AFCEC	Air Force Civil Engineer Center
AST	Aboveground Storage Tank
BIA	Bilateral Infrastructure Agreement
C&D	Construction and Demolition
CAA	Clean Air Act
CDC	Centers for Disease Control & Prevention
CFR	Code of Federal Regulations
DHHS	Department of Health and Human Services
DoD	Department of Defense
DOR	Designer of Record
EFH	Essential Fish Habitat
EIAP	Environmental Impact Analysis Process
ELPAT	Environmental Lead Proficiency in Analytical Testing
EO	Executive Order
EPA	United States Environmental Protection Agency
EPF	Environmental Planning Function
EV	Environmental
FC	Facilities Criteria
FFDCA	Federal Food, Drug, and Cosmetics Act of 1938
FGS	Final Governing Standards
FR	Federal Register
HID	High Intensity Discharge
HQUSACE	Headquarters, U.S. Army Corps of Engineers

HNFA	Host Nation Funded Construction Agreements
HUD	U.S. Department of Housing and Urban Development
ID	Identification
IR	Installation Restoration
kg	Kilogram
LLR	Low-Level Radioactive
MBTA	Migratory Bird Treaty Act
mg/cm ²	Milligrams per Square Centimeter
mg/L	Milligrams per Liter
NAAQS	National Ambient Air Quality Standards
NAGPRA	Native American Graves Protection and Repatriation Act of 1990
NAVFAC	Naval Facilities Engineering Systems Command
NAVRAMP	Navy Radon Assessment and Mitigation Program
NAVSEADET	Naval Sea Systems Command Detachment
NELAC	National Environmental Laboratory Accreditation Conference
NEPA	National Environmental Policy Act
NESHAP	National Emissions Standards for Hazardous Air Pollutants
NHPA	National Historic Preservation Act of 1966
NIST/NVLAP	National Institute of Standards and Technology/National Voluntary Laboratory Accreditation Program
NLLAP	National Lead Laboratory Accreditation Program
NPDES	National Pollutant Discharge Elimination System
NRHP	National Register of Historic Places
NSPS	New Source Performance Standards
OSHA	Occupational Safety and Health Administration
PCB or PCBs	Polychlorinated Biphenyls

PFAS	Per-and Polyfluoroalkyl Substances
RACM	Regulated Asbestos Containing Materials
RCRA	Resource Conservation and Recovery Act
RMS	Radon Mitigation Standards
RASO	Radiological Affairs Support Office
RFP	Request for Proposal
RRP	Renovation, Repair, and Painting
SHPO	State Historic Preservation Office
SOFA	Status of Forces Agreements
SSG	Soil Screening Guidance
SSL	Soil Screening Level
TCLP	Toxicity Characteristic Leaching Procedure
TNI	The NELAC Institute
TSCA	Toxic Substance Control Act
UFC	Unified Facilities Criteria
UFGS	Unified Facility Guide Specifications
UN	United Nations
U.S.	United States
USACE	U.S. Army Corps of Engineers
U.S.C.	United States Code
UST	Underground Storage Tank
XRF	X-Ray Fluorescence

DEFINITION OF TERMS.

Aerosol Cans: A non-refillable receptacle containing a gas compressed, liquefied, or dissolved under pressure, the sole purpose of which is to expel a liquid, paste, or powder and fitted with a self-closing release device allowing the contents to be ejected by the gas.

Batteries: A device consisting of one or more electrically connected electrochemical cells that are designed to receive, store, and deliver electric energy. An electrochemical cell is a system consisting of an anode, cathode, and an electrolyte, plus such connections (electrical and mechanical) as may be needed to allow the cell to deliver or receive electrical energy. The term battery also includes an intact, unbroken battery from which the electrolyte has been removed.

Hazardous waste: a solid waste that exhibits a hazardous characteristic (such as ignitability, corrosivity, reactivity, or toxicity) as specified in 40 CFR 261, Subpart C, or waste that contains a listed hazardous waste as identified in 40 CFR 261, Subpart D.

Lamps: the bulb or tube portion of an electrical lighting device that may contain hazardous materials such as mercury or lead. Examples of common types of lamps containing hazardous materials include fluorescent, high-intensity discharge (HID), neon, mercury vapor, high-pressure sodium, and metal halide.

Mercury-containing equipment: a device or part of a device (including thermostats but excluding batteries and lamps) that contains elemental mercury integral to its function.

Nonhazardous waste: a solid waste that does not meet the RCRA or state criteria for hazardous waste.

Pesticides: any substance or mixture of substances intended for preventing, destroying, repelling, or mitigating any pest, or intended for use as a plant regulator, defoliant, or desiccant, other than any article that: is a new animal drug under Federal Food, Drug and Cosmetic Act (FFDCA) section 201(w), or is an animal drug that has been determined by regulation of the Secretary of Health and Human Services not to be a new animal drug, or is an animal feed under FFDCA section 201(x) that bears or contains any substances described above.

Polychlorinated biphenyls: commonly referred to as PCB or PCBs, are a group of toxic substances with chlorinated compounds that are either oily liquids or solids. PCB are typically colorless or light yellow. Some common PCB manufacturer product trade names are as follows: Apirolio, Aroclor, Asbestol, Bakola 131, Chlorextol, Clophen, Inerteen, Kanechlor, No-Flamol, Pyralene, Pyranol, Saf-T- Kuhl and Sovol.

Radon: a colorless, odorless, tasteless, radioactive gas that is a byproduct of naturally decaying uranium.

Solid waste: a solid, liquid, semi-solid, or contained gaseous waste. A solid waste may be a hazardous waste or a nonhazardous waste.

Universal waste: a subset of hazardous waste that includes five types of materials: batteries, pesticides, mercury-containing equipment, aerosol cans, and lamps.

Used oil: any petroleum-based or synthetic fluid that has been used and is regulated under 40 CFR 279.

APPENDIX B UFC 3-810-01 REFERENCES

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36 CFR 800, *Protection of Historic Properties*

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40 CFR 261, SUBPART C, *Characteristics of Hazardous Waste*

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40 CFR 262, *Standards Applicable to Generators of Hazardous Waste*

40 CFR 264, *Standards for Owners and Operators of Hazardous Waste Treatment, Storage and Disposal Facilities*

40 CFR 264 Subpart J, *Tank Systems*

40 CFR 265, *Interim Status Standards for Owners and Operators of Hazardous Waste Treatment, Storage, and Disposal Facilities*

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40 CFR 270, EPA Administered Permit Programs: *The Hazardous Waste Permit Program*

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40 CFR 745, *Lead-Based Paint Poisoning Prevention in Certain Residential Structures*

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40 CFR 763, *Asbestos*

40 CFR Parts 1500-1508, *Regulations for Implementing the Procedural Provisions of the National Environmental Policy Act*

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UFGS 02 65 00, *Underground Storage Tank Removal*

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APPENDIX A UFC 4-010-01 GLOSSARY

A-1 ACRONYMS.

ACHP	Advisory Council on Historic Preservation
ANSI	American National Standards Institute
ARPA	Archaeological Resources Protection Act
ASTM	ASTM International (Not an abbreviation formerly, American Society of Testing and Materials)
AT	Antiterrorism
BIA	Bilateral Infrastructure Agreements
C	Tributary width increase factor
CCSD	Conventional Construction Standoff Distance
CFM	Cubic Feet per Minute
cm	Centimeter
CONEX	Container Express
DIN	Deutsches Institut für Normung (German Institute for Standardization)
DBT	Design Basis Threat
DoD	Department of Defense
EIFS	Exterior Insulation and Finish System
ESC	Expandable Shelter Containers
FCU	Fan Coil Unit
FPCON	Force Protection Condition
FSTFS	Frame-Supported Tensioned Fabric Structures
ft.	Feet
GCC	Geographic Combatant Commander
GP	General Purpose
GSA	General Services Administration (U.S.)

HNFA	Host Nation Funded Construction Agreement
HQUSACE	Headquarters, US Army Corps of Engineers
HVAC	Heating, Ventilating, and Air Conditioning
IGU	Insulating Glass Units
in.	Inches
ISC	Interagency Security Committee
JFOB	Joint Forward Operations Base
Kg/m²	Kilograms per square meter
Kg/m³	Kilograms per cubic meter
LB	Load Bearing
LOC	Local Operating Console
m	Meters
MILCON	Military Construction
mm	Millimeters
MPa	MegaPascals
MSS	Medium Shelter Systems
M_{cw}	Moment, conventional wall
M_{sse}	Moment, Supporting Structural Element
NATO	North Atlantic Treaty Organization
NAVFAC	Naval Facilities Engineering Command
NCR	National Capital Region
NFPA	National Fire Protection Association
NHPA	National Historic Preservation Act
NLB	Non-Load Bearing
OPORD	Operations Orders
OSD	Office of the Secretary of Defense

Pa	Pascals
PCF	Pounds per cubic foot
PDC	Protective Design Center
PSF	Pounds per square foot
PSI	Pounds per square inch
RFP	Request for Proposal
R_M	Minimum Standoff Distance
SHAPE	Supreme Headquarters Allied Powers Europe
SOFA	Status of Forces Agreement
SSS	Small Shelter System
TEMPER	Tent, Extendable, Modular, Personnel
TNT	Trinitrotoluene
UFC	Unified Facilities Criteria
USD (AT&L)	Undersecretary of Defense for Acquisition, Technology, and Logistics
V_{CW}	Shear, conventional wall
V_{SSE}	Shear, Supporting Structural Element

A-2 DEFINITIONS OF TERMS.

Access control. For the purposes of these standards, any combination of barriers, gates, electronic security equipment, and/or guards that can limit entry or parking of unauthorized personnel or vehicles.

Access road. Any roadway such as a maintenance, delivery, service, emergency, or other special limited use road that is necessary for the operation of a building or structure.

Analysis. For the purposes of evaluating compliance with standoff distance and supporting structure requirements of these standards, evaluation of structural components using commonly accepted analysis methodologies such as single degree of freedom or finite element analysis.

Breezeway. A covered passage that passes between two buildings or portions of buildings or covered areas underneath or attached to buildings.

Building. A structure, usually enclosed by walls and a roof, constructed to provide support or shelter for an intended occupancy. Note that other structures, such as canopies or gazebos, are not considered buildings for the purposes of these standards.

Building hardening. Enhanced conventional construction that mitigates threat hazards where standoff distance is limited. Building hardening may also be considered to include the prohibition of certain building materials and construction techniques.

Building occupancy. For the purposes of these standards, the planned occupancy of a building or the allowable occupancy calculated in accordance with life safety codes where the occupancy is not known.

Building overhangs. Any structural configuration in which the outer walls or columns of the ground floor are set back from the outer walls or column lines of floors above.

Building separation. The distance between closest points on the exterior walls of adjacent buildings or structures.

Clear Zone. Areas commonly associated with perimeters that are free of all obstacles, topographical features, and vegetation exceeding 8 in. \1\ (200 mm) /1/ in height that could impede observation or provide cover and concealment of an aggressor.

Collateral damage. Injury to personnel or damage to buildings that are not the primary targets of attacks.

Command vehicles. Government owned or leased vehicles operated by installation or senior mission commanders, exclusive of privately owned vehicles

Commercial facilities. Facilities that are not DoD owned or operated and that support commercial activities other than food service and retail activities such as banks.

Container structures. Structures built using shipping containers that are designed to withstand structural loadings associated with shipping, including Container Express (CONEX) and International Organization for Standardization (ISO) containers. Testing has shown that these structures behave similarly to buildings for the purposes of these standards.

Controlled parking. For the purposes of these standards, parking that is limited to authorized vehicles that is enforced through physical security measures such as card operated gates, identification or vehicle checks by personnel or similar measures that are acceptable to physical security personnel.

Controlled perimeter. For the purposes of these standards, a physical boundary at which vehicle access is controlled with sufficient means to channel vehicles to the access control points. At a minimum, access control at a controlled perimeter requires the demonstrated capability to search for and detect explosives. Where the controlled perimeter includes a shoreline and there is no defined perimeter beyond the shoreline, the boundary for measuring standoff distances will be at the mean high water mark or the elevation associated with top of bank (associated with a flood recurrence interval of 1.2 years).

Conventional construction. Building construction that is not specifically designed to resist weapons or explosives effects. Conventional construction is designed only to resist common loadings and environmental effects such as wind, seismic, and snow loads. Note that for the purposes of these standards, conventional construction may still require special windows, structural reinforcement around windows, and progressive collapse resistant construction.

Conventional construction standoff distance. The standoff distances at which conventional construction may be used for building components other than doors and windows without a specific analysis of blast effects, except as otherwise required in these standards.

Change of occupancy. Change of occupancy level as defined in these standards. It does not relate to conversions of facility category code. Examples include occupancy changing from low occupancy to inhabited.

Design basis threat. The threat (aggressors, tactics, and associated weapons, tools or explosives) against which assets within a building must be protected and upon which the security engineering design of the building is based.

DoD building. Any building or portion of a building (permanent, temporary, or expeditionary) owned, leased, privatized, or otherwise occupied, managed, or controlled by or for DoD. DoD buildings other than leased buildings are categorized within these standards as low occupancy, inhabited, and high occupancy family housing.

DoD components. The Office of the Secretary of Defense (OSD); the Military Departments (including their National Guard and Reserve Components); the Chairman, Joint Chiefs of Staff and Joint Staff; the Combatant Commands; the Office of the

Inspector General of the Department of Defense; the Defense Agencies; the DoD Field Activities; and all other organizational entities within DoD.

DoD Installation. A base, camp, post, yard, center, homeport facility for any ship, or other activity under the jurisdiction of the Department of Defense

DoD personnel. Any U.S. military, DoD civilian or family member thereof, host-nation employees working for DoD, or contractors occupying DoD buildings. For the purposes of these standards, non-DoD visitors to DoD owned or controlled visitor centers, visitor control centers, museums, and similar facilities will be included in DoD personnel populations of those facilities. Visitor counts will be based on routine visitor levels.

Door. A building component for opening or closing an opening in a wall that allows normal access and passage.

Emergency vehicles. Vehicles such as fire trucks and ambulances and other vehicles that are critical to emergency response and for which close proximity to inhabited buildings or containment therein is essential.

Enhanced use lease. Out leases of non-excess DoD land or facilities to a public or private entities for development under the authority of 10 US Code Section 2667.

Equivalent level of protection. Performance of building components that results in building damage or door and glazing hazards similar to that required for the required level of protection as described in **Error! Reference source not found.** or as specified in PDC Technical Report 06-08.

Expeditionary structures. \2\ Non-permanent DoD facilities that are erected in support of military operations in accordance with UFC 1-201-01. /2/ This group of structures typically includes but is not limited to tents, Small and Medium Shelter Systems, Expandable Shelter Containers (ESC), ISO and CONEX containers, General Purpose (GP) Medium tents and GP Large tents, trailers, and modular and light wood framed structures.

Fabric covered structures. A construction type that can be identified by wood or metal (usually aluminum) posts or load-bearing frames with some type of fabric (such as canvas) stretched or pulled over the posts or frames. Examples of the types of structures that should be considered under this classification of structures include Frame-Supported Tensioned Fabric Structures (FSTFS); Tent, Extendable, Modular, Personnel (TEMPER Tents); and Small and Medium Shelter Systems (SSS and MSS); General Purpose (GP) Medium tents and GP Large tents; and air supported fabric structures. Testing has shown that for these fabric structures, the posts and frames are what cause hazards.

Family housing. DoD buildings used as quarters for DoD personnel and their dependents. For the purposes of these standards, family housing will be considered to include Morale, Welfare, and Recreation housing (cottages) and temporary family lodging of similar occupancies.

Fan Coil Unit (FCU). A device consisting of a heating and / or cooling heat exchanger (coil) and a fan.

Final denial active vehicle barrier. Vehicle barriers that can be raised and lowered or otherwise moved to block traffic lanes to stop the motion of threat vehicles. In Entry Control Facilities / Access Control Points they are located at the end of the response zone (see UFC 4-022-01).

Fisher Houses. Houses constructed by the Fisher House Foundation at military medical centers for lodging families of military personnel while the military personnel are hospitalized.

Force Protection Condition (FPCON). A DoD-approved system that standardizes the Departments' identification and recommended preventive actions and responses to terrorist threats against U.S. personnel and facilities. This system is the principle means for a commander to apply an operational decision on how to protect against terrorism and facilitates inter-Service coordination and support for antiterrorism activities.

Glazing. The part of a window, skylight, or door assembly that is transparent or translucent and transmits light, but not air.

High occupancy family housing. Family housing with 13 or more units per building.

Identification check point. The location in an Entry Control Facility / Access Control Point at which driver identification is checked to control access into controlled perimeters

Inhabited building. Buildings or portions of buildings routinely occupied by 11 or more DoD personnel and with a population density of greater than one person per 430 gross square feet (40 gross square meters). This density generally excludes industrial, maintenance, and storage facilities, except for more densely populated portions of those buildings such as administrative areas. The inhabited building designation also applies to expeditionary structures with similar population densities. In a building that meets the criterion of having 11 or more personnel with low occupancy portions that do not have sufficient population densities to qualify as inhabited buildings, those portions that have sufficient population densities will be considered inhabited buildings while the remainder of the building may be considered low occupancy, subject to provisions of these standards. An example would be a hangar, warehouse, or maintenance facility with an administrative area within it. The administrative area would be treated as an inhabited building while the remainder of the facility could be treated as low occupancy. External stairwells and covered or enclosed walkways are not part of the inhabited space of a building. (Note: This definition differs significantly from the definition for inhabited building used by DoD 6055.09-M and is not to be construed as authorization to deviate from criteria of DoD 6055.09-M.)

Installation: For the purposes of these standards, the installation is an area or locality subject to the custody, jurisdiction, or administration of the Secretary of a Military Department or the Secretary of Defense, in the case of an activity in a foreign country,

under the operational control of the Secretary of a Military Department or the Secretary of Defense. This term includes but not limited to, military reservations, bases, posts, camps, stations, or arsenals.

Installation Perimeter: For the purposes of these standards, the installation perimeter is defined as any demarcation identifying the limit of DoD property and directly or indirectly indicating that unauthorized access is prohibited. The landside perimeter may be established with fences, walls, signage, natural barriers or other means. The waterside perimeter will be at the mean high water mark or the elevation associated with top of bank (associated with a flood) recurrence interval of 1.2 years) or be established with channel markers, buoys, float lines, signage, or boat barriers.

Laminated glass. Multiple sheets of glass bonded together by a bonding interlayer.

Level of protection. The degree to which an asset (person, equipment, object, etc.) is protected against injury or damage from an attack.

Low occupancy building. Any building or portion of a building routinely occupied by fewer than 11 DoD personnel or with a population density of less than one person per 430 gross square feet (40 gross square meters).

Low occupancy family housing. Family housing with 12 or fewer units per building.

Mail room. A facility operated by or for the Department of Defense for the receipt and delivery of mail for military units or other authorized organizations and agencies by entities outside the DoD. This does not include mail rooms that receive mail distribution that was initially received at a central DoD mail handling facility.

Mass notification. Capability to provide real-time information to all building occupants or personnel in the immediate vicinity of a building during emergency situations.

Military protective construction. Military facilities designed to resist military conventional and nuclear weapons to the NATO (or equivalent) standards of hardened, protected, semi-hardened, collaterally protected, or splinter protected.

Minimum standoff distance. The smallest permissible standoff distance regardless of any analysis results or hardening of the building.

Mobile ground tactical platforms. Vehicle mounted tactical ground station for posting, processing, and distributing real-time intelligence, surveillance, and reconnaissance information.

Operations support vehicles. Vehicles such as airfield support equipment or material handling equipment whose purpose is direct support to operations, and which are operated only within a restricted access area.

Parking areas. Designated areas where vehicles may be left unattended, including parking lots, designated parking areas along roadways, and roadways within or accessing parking areas.

Progressive collapse. The spread of an initial local failure from building element to building element, eventually resulting in the collapse of an entire structure or a disproportionately large part of it.

Punched window. A window installed as a punched opening surrounded by cladding, as opposed to being arranged in vertical or horizontal strips.

Relocatable building. \3\ A facility that is specially designed and constructed to be readily erected, disassembled, transported, stored, and re-used. Examples of relocatable facilities include, but are not limited to, trailers, CONEX boxes, sheds on skids, tension fabric structures, and air supported domes. A relocatable facility is not constructed as a part of any other military vehicle, DoD tactical equipment (vehicle mounted or wheeled and towable) or equipment which is already accounted for in a designated accountable property system of record. /3/

Replacement Cost. The cost to design and construct a facility to current standards and building codes to replace an existing facility at the same location calculated in accordance with UFC 3-701-01.

Ribbon window. Windows installed in vertical or horizontal strips with no building wall elements between them but surrounded by cladding around the overall opening perimeter.

Roadways. Any surface intended for routine motorized vehicle traffic, including driving lanes of parking areas.

Routinely occupied. For the purposes of these standards, an established or predictable pattern of activity within a building that terrorists could recognize and exploit.

Scaled Range. A relationship based on cube-root scaling that allows comparisons to be made among blast wave properties created by detonations of different explosive quantities.

Security engineering. The process of identifying practical, risk managed short and long-term solutions to reduce and/or mitigate dynamic manmade hazards by integrating multiple factors, including construction, equipment, manpower, and procedures.

Skylight. Sloped or horizontal application of a fenestration product that allows for natural day lighting and that may be either fixed (non-operable) or venting (operable).

Spandrel Glass. Glass used in non-vision areas of building exteriors

Specific threat. Known or postulated aggressor activity focused on targeting a particular asset.

Standoff distance. A distance maintained between a building or portion thereof and the potential location for an explosive detonation.

Structural glazed window systems. Window systems in which glazing is bonded to both sides of the window frame using an adhesive such as a high-strength, high-performance structural silicone.

Superstructure. The supporting elements of a building above the foundation.

Supporting structural elements. Structural elements that support windows and that are not in direct contact with the glass, such as walls.

Temporary buildings. For the purposes of these standards, those buildings, other than expeditionary structures, that are real property facilities and are designed and constructed with a life expectancy of \3\ seven /3/ years or less.

Testing. For the purposes of these standards, experiments performed in accordance with standardized procedures that prove that building components meet the performance required to meet a specific level of protection.

Town Centers. Mixed use small scale retail, health, or community services and family housing facilities in the same buildings.

TNT equivalent weight. The weight of TNT (trinitrotoluene) that has an equivalent energetic output to that of a different weight of another explosive compound.

Transitional structures and spaces. Structures or spaces within buildings that are used to temporarily relocate DoD occupants of buildings while those buildings or other buildings to which they will relocate undergo renovations, modifications, repairs, or restorations or are being constructed. (Also known as swing space.)

Uncontrolled Public Access. Spaces within and beneath buildings where there is insufficient positive access control to preclude unauthorized access. For the purposes of these standards, positive access control will be considered to include (but not be limited to) electronic access control on all exterior doors or providing personnel to control visitors.

Unobstructed space. Space around inhabited buildings in which there are no opportunities for concealment from observation of explosive devices of no less than a 6 in. (150 mm) cube.

Window or Skylight Replacement. The removal of an existing window or skylight assembly and replacement with a new window assembly. For the purposes of this definition a "window assembly" is considered to be the entire system of glazing, framing and anchorage components that fill in and fit within the opening in the wall or roof structure.

APPENDIX B UFC 4-010-01 REFERENCES

ADVISORY COUNCIL ON HISTORIC PRESERVATION

<http://www.achp.gov/>

36 CFR Part 800, *Protection of Historic Properties*

ASTM INTERNATIONAL

<http://www.astm.org>

ASTM E1300, *Standard Practice for Determining Load Resistance of Glass in Buildings*

ASTM E1996, *Standard Specification for Performance of Exterior Windows, Curtain Walls, Doors, and Impact Protective Systems Impacted by Windborne Debris in Hurricanes*

ASTM F1642, *Standard Test Method for Glazing and Glazing Systems Subject to Airblast Loadings*

ASTM F2247, *Standard Test Method for Metal Doors Used in Blast Resistant Applications (Equivalent Static Method)*

ASTM F2248, *Standard Practice for Specifying an Equivalent 3-Second Duration Design Loading for Blast Resistant Glazing Fabricated with Laminated Glass*

ASTM F2912, *Standard Specification for Glazing and Glazing Systems Subject to Airblast Loadings*

ASTM F2927, *Standard Test Method for Door Systems Subject to Airblast Loadings*

DEPARTMENT OF DEFENSE

<http://www.dtic.mil/whs/directives/index.html>

\3\ DOD Instruction 4165.56, *Relocatable Facilities* /3/

DoD 6055.09-M, *DoD Ammunition and Explosive Safety Standards*

DoD Instruction 2000.12, *DoD Antiterrorism (AT) Program*

DoD Instruction O-2000.16, *DoD Antiterrorism (AT) Program Implementation (Volumes 1 and 2)*

GTA 90-01-011, \1\ *Deployed Forces Protection Handbook (JFOB Handbook)* \1\ (For Official Use Only [FOUO])

The Deputy Secretary of Defense, 7 December 2012, *Memorandum, Subject: Antiterrorism Building Standards for Leased Space*

DEPARTMENT OF DEFENSE, UNIFIED FACILITIES CRITERIA PROGRAM

<http://dod.wbdg.org/>

Military Standard (MIL-STD) 3007, *Department Of Defense Standard Practice Standard Practice For Unified Facilities Criteria and Unified Facilities Guide Specifications*

UFC 1-200-01, *DoD Building Code \1V1/*

UFC 4-020-01, *DoD Security Engineering Facilities Planning Manual*

UFC 4-020-02FA, *DoD Security Engineering Facilities Design Manual*

UFC 4-021-01, *Design and O&M: Mass Notification Systems*

UFC 4-022-01, *Security Engineering: Entry Control Facilities/Access Control Points*

UFC 4-023-03, *Design of Buildings to Resist Progressive Collapse*

UFC 4-023-07, *Design to Resist Direct Fire Weapons Effects*

UFC 3-701-01, *DoD Facilities Pricing Guide*

UFC 3-301-01, *Structural Engineering*

DEPARTMENT OF HOMELAND SECURITY

<http://www.dhs.gov/interagency-security-committee>

Interagency Security Committee Standards

FEDERAL HIGHWAY ADMINISTRATION

<http://mutcd.fhwa.dot.gov/>

Manual on Uniform Traffic Control Devices

NATIONAL FIRE PROTECTION ASSOCIATION

<http://www.nfpa.org>

NFPA 30, *Flammable and Combustible Liquids Code*

SUPREME HEADQUARTERS ALLIED POWERS EUROPE

SHAPE Document 6160/SHLOFA-059/82, *NATO Approved Criteria and Standards for Tactical and Transport Airfields (6th Addition) (NATO Restricted)*

UNITED STATES ARMY

<https://pdc.usace.army.mil>

PDC Technical Report 06-08, *Single Degree of Freedom Structural Response Limits for Antiterrorism Design*

PDC Technical Report 10-01, *Conventional Construction Standoff Distances for the Low and Very Low Levels of Protection*

PDC Technical Report 10-02, *Blast Resistant Design Methodologies for Window Systems Designed Staticallly and Dynamically*

PDC Technical Report 15-01, *Minimum Standoff Distances for Non-Load-Bearing Steel Stud In-Fill Walls*

APPENDIX A UFC 4-010-06 GLOSSARY

A-1 ACRONYMS

A-1.1 General Acronyms

ACL	Access Control List
AO	Authorizing Official
BAS	Building Automation System
BCS	Building Control System
CCTV	Closed Circuit Television
CNSSI	Committee on National Security Systems Instruction
CCI	Control Correlation Identifier
COTS	Commercial Off The Shelf
CS	Control System
DoD	Department of Defense
ESS	Electronic Security System
FCN	Field Control Network
FCS	Field Control System
FIPS	Federal Information Processing Standards
FISMA	Federal Information Security Management Act
FPOC	Field Point of Connection
GFE	Government Furnished Equipment
ICS	Industrial Control System
IDS	Intrusion Detection System
ISSM	Information System Security Manager
ISSO	Information System Security Officer
IP	Internet Protocol
IT	Information Technology
MOA	Memorandum Of Agreement
MOU	Memorandum Of Understanding
NIST	National Institute of Standards and Technology
OS	Operating System
PIT	Platform Information Technology
PKI	Public Key Infrastructure

SO	System Owner
UCS	Utility Control System
UFC	Unified Facilities Criteria
UFGS	Unified Facilities Guide Specification
UMCS	Utility Monitoring and Control System
UPS	Uninterruptible Power Supply
USACE	U.S. Army Corps of Engineers

A-1.2 Security Control Family Acronyms

<u>Acronym</u>	<u>Term</u>
AC	Access Control
AT	Awareness and Training
AU	Audit and Accountability
CA	Security Assessment and Authorization
CM	Configuration Management
CP	Contingency Planning
IA	Identification and Authorization
IR	Incident Response
MA	Maintenance
MP	Media Protection
PE	Physical and Environmental Protection
PL	Planning
PM	Program Management
PS	Personnel Security
RA	Risk Assessment
SA	System and Services Acquisition
SC	System and Communications Protection
SI	System and Information Integrity

Note that UFGS 25 05 11, "Cybersecurity for Facility-Related Control Systems" also has definitions that complement those provided here.

<u>Term</u>	<u>Definition</u>
Authorizing Official (CNSSI No. 4009)	A senior (federal) official or executive with the authority to formally assume responsibility for operating an information system at an acceptable level of risk to organizational operations (including mission, functions, image, or reputation), organizational assets, individuals, other organizations, and the Nation. Within the DoD, they are presumed to be SES or flag-level officers.
Building Automation System (BAS)	The system consisting of a UMCS Front End, connected Building Control Systems which control building electrical and mechanical systems, and user interfaces for building control supervision. The BAS is a subsystem of the Utility Monitoring and Control System. This term is being phased out in favor of UMCS.
Building Control System (BCS)	A system that controls building electrical and mechanical systems such as HVAC (including central plants), lighting, vertical transport systems, and irrigation systems. Building Control Systems generally do not have a full-featured user interface; they may have "local display panels" but typically rely on the UMCS front end for full user interface functionality. BCS is a subsystem of the Utility Monitoring and Control System and is a class of Field Control System.
Closed Circuit Television System (CCTV)	An ESS that allows video assessment of alarm conditions via remote monitoring and recording of video events. Video monitoring may also be incorporated into other systems which are not CCTV.
Control Correlation Identifier (CCI)	The Control Correlation Identifier (CCI) provides a standard identifier and description for each of the singular, actionable statements that comprise a security control.

<u>Term</u>	<u>Definition</u>
Control System (CS)	Control systems are a combination of special purpose controlling devices (controllers), control components (e.g. electrical, mechanical or pneumatic sensors and actuators), and generally (but not always) network infrastructure (which may contain standard IT components) that act together upon underlying mechanical and/or electrical equipment to achieve an objective. A control system (CS) typically consists of sensors and actuators connected to networked digital controllers, includes a user interface, and is used to control and monitor equipment. Some control systems are “monitor only” (they do not exercise control) but are still considered control systems.
Controller	An electronic device – usually having internal programming logic and digital and analog input/output capability – which performs control functions.
Distributed Control System	This term describes a system architecture (not system functionality) and appears in some reference material (such as a NIST publications) but is not used by this UFC. (It’s included here due to use in some reference material.)
Electronic Security System (ESS)	The integrated electronic system that encompasses interior and exterior (physical) intrusion detection systems (IDS), CCTV systems for assessment of alarm conditions, access control systems, data transmission media, and alarm reporting systems for monitoring, control, and display.
Engineering Tool Software	Software that is used to perform device or network management for a control system, including network configuration, controller configuration and controller programming.
Facility-Related Control System	A control system which controls equipment and infrastructure that is part of a DoD building, structure, or linear structure.
Field Control System (FCS)	A Building Control System, Utility Control System, Electronic Security System, etc. within the Facility and "downstream" of the FPOC.
Field Control Network (FCN)	The network used by the Building Control System, Utility Control System, etc., within a facility "downstream" of the FPOC. This includes IP, Ethernet, RS-485, TP/FT-10 and other network infrastructure that support control system(s) in a given facility.

<u>Term</u>	<u>Definition</u>
Field Point of Connection (FPOC)	The FPOC is the point of connection between the Level 4 control system IP network and the Level 2 field control IP network. Note the Level 2 IP network here may consist entirely of a single patch cable from the FPOC to a Level 1-to-Level 2 device. The hardware which provides the connection at this location is an IT device such as a switch, IP router, or firewall.
[UMCS, ESS, etc.] Front End	The portion of the control system consisting primarily of IT equipment, such as computers and related equipment, intended to perform operational functions and run monitoring and control/engineering tool application software. The front end does not directly control physical systems; it interacts with them only through field control systems (FCS). The front end is a component of the [UMCS, ESS, etc.] infrastructure (see definition).
Impact	The effect on organizational operations, organizational assets, or individuals due to a loss of Confidentiality, Integrity, or Availability in the control system. Impact is categorized as one of three levels: • LOW: limited adverse effect • MODERATE: serious adverse effect • HIGH: severe or catastrophic adverse effect The impact level of a system is generally written in ALL CAPS for clarity. Further discussion of Impact level is in Error! Reference source not found.
Incident (FIPS PUB 200)	An occurrence that actually or potentially jeopardizes the confidentiality, integrity, or availability of an information system or the information the system processes, stores, or transmits or that constitutes a violation or imminent threat of violation of security policies, security procedures, or acceptable use policies
Industrial Control System (ICS)	One type of control system. Most specifically a control system which controls an industrial (manufacturing) process. Sometimes also used to refer to other types of control systems, particularly utility control systems such as electrical, gas, or water distribution systems.

<u>Term</u>	<u>Definition</u>
Information System (CNSSI No. 4009)	A discrete set of information resources organized for the collection, processing, maintenance, use, sharing, dissemination, or disposition of information. Note: Information systems also include specialized systems such as industrial/process controls systems, telephone switching and private branch exchange (PBX) systems, and environmental control systems.
Information Technology (IT)	Any equipment or interconnected system or subsystem of equipment that is used in the automatic acquisition, storage, manipulation, management, movement, control, display, switching, interchange, transmission, or reception of data or information by the executive agency.
[UMCS, ESS, ...] Infrastructure	The portion of a control system (such as a UMCS or ESS) which includes all components that are not part of a field control system. These components include the FPOC, the Platform Enclave, and the front end (i.e., its architecture Levels 3, 4 and 5)
Intrusion Detection System (IDS) [Physical/ESS]	A system consisting of interior and exterior sensors, surveillance devices, and associated communication subsystems that collectively detect an intrusion of a specified site, facility, or perimeter and annunciate an alarm.
Intrusion Detection System (IDS) [Cyber]	A device or software application that monitors network or system activities for malicious activities or policy violations and produces reports to management.
Least Privilege (CNSSI No. 4009)	The principle that a security architecture should be designed so that each entity is granted the minimum system resources and authorizations that the entity needs to perform its function.
Mobile Code (NIST SP 800-53r4)	Software programs or parts of programs obtained from remote information systems, transmitted across a network, and executed on a local information system without explicit installation or execution by the recipient.
Mobile Code Technology (NIST SP 800-53r4)	Software technologies that provide the mechanisms for the production and use of mobile code (e.g., Java, JavaScript, ActiveX, VBScript). Note that use of Mobile Code Technology does not necessarily imply the use of Mobile Code.

<u>Term</u>	<u>Definition</u>
Network	A network is a group of two or more devices that can communicate using a network protocol. Network protocols must provide a method for addressing devices on the network; a communication method that does not provide an addressing scheme is not a networked form of communication. Devices that communicate using a method of communication that does not support device addressing are not using a network.
Non-Local Maintenance (NIST SP 800-53)	Maintenance activities conducted by individuals communicating through a network; either an external network (e.g., the Internet) or an internal network.
Operational Architecture (OA)	Those components of a control system that represent the purely operational components of the system such as controllers, Front End software, and other devices which support operational functions. When the "Platform Enclave" approach to authorizing a control system is used, the "non-standard IT" portions of the control system are authorized as the Operational Architecture and the overall system has two authorizations: Platform Enclave and Operational Architecture.
[UMCS, ESS, ...] Platform Enclave	Those components of the control system that are standard IT components and can be secured in a standard manner independent of the type of control system. These components serve only the control system and include the IP network, network management and security devices (e.g., switches, routers), software, computers and/or other devices which provide management and security of the network.
Platform IT (PIT)	IT, both hardware and software, which is physically part of, dedicated to, or essential in real time to the mission performance of special purpose systems.
Remote Access (NIST SP 800-53)	Remote access is access to organizational information systems by users (or processes acting on behalf of users) communicating through external networks (e.g., the Internet). See definition of "User Interface, Remote" for critical distinction in use of terminology

<u>Term</u>	<u>Definition</u>
Risk (NIST SP 800-53)	A measure of the extent to which an entity is threatened by a potential circumstance or event, and typically a function of: (i) the adverse impacts that would arise if the circumstance or event occurs; and (ii) the likelihood of occurrence. Information system-related security risks are those risks that arise from the loss of confidentiality, integrity, or availability of information or information systems and reflect the potential adverse impacts to organizational operations (including mission, functions, image, or reputation), organizational assets, individuals, other organizations, and the Nation.
Risk Management (NIST SP 800-53)	The process of managing risks to organizational operations (including mission, functions, image, reputation), organizational assets, individuals, other organizations, and the Nation, resulting from the operation of an information system, and includes: (i) the conduct of a risk assessment; (ii) the implementation of a risk mitigation strategy; and (iii) employment of techniques and procedures for the continuous monitoring of the security state of the information system.
Supervisory Control and Data Acquisition (SCADA)	This term has a variety of meanings depending on context and is therefore not used in this UFC.
System Owner (SO) (CNSSI No. 4009)	Person or organization having responsibility for the development, procurement, integration, modification, operation and maintenance, and/or final disposition of an information system.
User Interface	A User Interface is something that allows a person to interact with the control system. For purposes of cybersecurity, a User Interface does not include simple analog and/or binary inputs such as HOA switches, dials, etc. ("Manual Local Input" in UFGS 25 05 11) or simple Card Readers. Proper addressing of User Interfaces is critical for cybersecurity. Note that User Interfaces are defined in a more exhaustive and prescriptive manner in UFGS 25 05 11 to enable specification of user interface requirements.

Term**Definition**

User Interface,
Remote or Local

A Remote User Interface implements a Client/Server model where the physical hardware the user interacts with (Client) is physically distinct from the device being affected (Server). Most or all of the security and functionality characteristics of the user interface are defined by the Server, not the Client. A common example of a remote user interface is a web-based interface where the browser (client) is generally on different hardware than the web server (server). A Remote UI remains a Remote UI even if the user happens to be at a Client on the same hardware as the Server. What is important is that a) the Client may be on different hardware than the Server and b) the majority of the security and functional characteristics of the interface are defined at the Server.

A Local User Interface is a user interface where the physical hardware the user interacts with (keyboard, buttons, display, etc.) is physically part of the device being affected. All of the relevant characteristics of the user interface are embodied within a single device.

Note that this definition of "remote" is consistent with that generally used in the control industry but is not aligned with the NIST 800-53 definition of "Remote" or "Remote Access", which refers to "outside the system". The term "Remote" here better aligns with the NIST 800-53 definition of "Network" (remote from within the system) Access.

User Interface,
Limited or Full

A Limited User Interface is a user interface that - by design - can only alter information local to the user interface. Note that the determination of "alter" includes only direct interactions, it explicitly excludes interactions that might occur as secondary effects. For example, an interface changing the flow setpoint in a pump controller is a direct interaction, the subsequent change in flow (as well as any subsequent downstream changes in valve position) are not direct interactions. For example, a variable speed drive has a Limited Local User Interface which allows the user to change properties within the drive but does not allow affecting things outside the drive.

A Full User Interface can alter information in devices outside the device with the user interface. For example, a typical Local Display Panel is a Full Local User Interface while a browser-based front end is a Full Remote User Interface.

Term**Definition**

User Interface,
Privileged

A Privileged UI is a UI that has sufficient capabilities or functionality that it requires specific cybersecurity measures to be put in place to limit its unauthorized use. Ultimately, whether a specific user interface is considered a Privileged User Interface must be determined by usage. In general however, user interfaces that do not offer significant capabilities above and beyond those available at that location via other means (e.g., a disconnect switch, breaker, hand-off-auto switch, or direct physical attack) are not privileged.

Note that UFGS 25 05 11 contains additional information and requirements on identifying when a UI is a Privileged UI to enable specification of UI requirements.

Utility Control
System (UCS)

A type of field control system used for control of utility systems such as electrical distribution and generation, sanitary sewer collection and treatment, water generation and pumping, etc. Building controls are excluded from a UCS, however it is possible to have a Utility Control System and a Building Control System in the same facility, and for those systems to share components such as the FPOC. A UCS is a subsystem of a Utility Monitoring and Control System (UMCS) and is a class of Field Control System (FCS).

Utility Monitoring and
Control System
(UMCS)

The system consisting of one or more building control systems and/or utility control systems and the associated UMCS Infrastructure. In other words, it is the complete utility monitoring system – from the front end to the controllers. At the highest level the UMCS is composed of a UMCS Platform Enclave and UMCS Front End (jointly referred to as UMCS Infrastructure) and connected Field Control System(s). There are many different names for a UMCS – Energy Monitoring and Control System being a common one – but regardless of the name systems that perform this function are a type of UMCS.

Vulnerability
(NIST SP 800-53)

Weakness in an information system, system security procedures, internal controls, or implementation that could be exploited or triggered by a threat source.

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APPENDIX A UFC 4-021-02 GLOSSARY

A-1 ACRONYMS AND ABBREVIATIONS.

AA&E	Arms, Ammunition, and Explosives
ABA	Architectural Barriers Act
ACS	Access Control System
ANSI	American National Standards Institute
AO	Accrediting Official
APL	Approved Products List
AR	Army Regulation
Bio	Biometric
CAC	Common Access Card
CAD	Computer Aided Design
CCDE	Command and Control Display Equipment
CHUID	Card Holder Unique Identifier
CIIC	Controlled Inventory Item Code
CNSSI	Committee on National Security Systems Instruction
DAFI	Department of the Air Force Instruction
dB	Decibel
DC	Direct Current
DESR	Defense Explosives Safety Regulation
DFPE	Designated Fire Protection Engineer
DoD	Department of Defense
DoDI	Department of Defense Instruction
DoDM	Department of Defense Manual
DOR	Designer of Record
DPDT	Double Pole Double Throw

DPS	Door Position Switch
DTM	Data Transmission Media
EMI	Electromagnetic Interference
ESS	Electronic Security System
ET	Endurance Testing
FAR	False Alarm Rate
FAR	Federal Acquisition Regulation
FDB	Field Distribution Box
FIPS	Federal Information Processing Standards
FOUO	For Official Use Only
FOV	Field of View
FPAT	Functional Pre-Acceptance Test
fps	Frames Per Second
FSAT	Functional System Acceptance Test
GSA	General Services Administration
HD	High Definition
HDMI	High-Definition Multimedia Interface
HSS	High Security Switch
Hz	Hertz
ICIDS	Integrated Commercial Intrusion Detection System
IDB	Interior Distribution Box
IDS	Intrusion Detection System
IEEE	Institute of Electrical and Electronics Engineers
IEL	Integrated Electronic Lock
IP	Internet Protocol
IR	Infrared

IT	Information Technology
LED	Light Emitting Diode
m	Meter
Mbps	Megabits Per Second
MCESS	Marine Corps Electronic Security System
MCO	Marine Corps Order
MILCON	Military Construction
MJPEG	Motion Joint Photographic Experts Group
MP	Megapixel
NAR	Nuisance Alarm Rate
NEC	National Electrical Code
NEMA	National Electrical Manufacturer's Association
NFPA	National Fire Protection Association
NIST	National Institute of Standards and Technology
NVR	Network Video Recorder
O&M	Operations and Maintenance
ONVIF	Open Network Video Interface Forum
OPNAV	Office of the Chief of Naval Operations
PAT	Pre-Acceptance Test
PC	Personal Computer
PCU	Premise Control Unit
PIN	Personal Identification Number
PIR	Passive Infrared
PIT	Platform Information Technology System
PIV	Personal Identity Verification
Pixels	Picture Elements

PKI	Public Key Infrastructure
POC	Point of Contact
PoE	Power over Ethernet
PTZ	Pan/Tilt/Zoom
REX	Request-to-Exit
RFP	Request For Proposal
SAPCO	Special Access Program Central Office
SAPF	Special Access Program Facility
SAT	System Acceptance Testing
SCI	Sensitive Compartmented Information
SCIF	Sensitive Compartmented Information Facility
SHDSL	Single-Pair High-Speed Digital Subscriber Line
SRM	Sustainment, Restoration, and Modernization
TCP	Transmission Control Protocol
UL	Underwriters Laboratory
UPS	Uninterruptible Power Supply
USACE	U.S. Army Corps of Engineers
VAC	Volts Alternating Current
VMS	Video Management System
VoIP	Voice Over Internet Protocol
VSS	Video Surveillance System

A-2 DEFINITION OF TERMS.

Access Control System (ACS): An automated system that interfaces with locking mechanisms that momentarily permit access (for example, by unlocking doors or gates) after verifying entry credentials (for example, using a card reader). Other DoD documents may refer to the ACS as an automated access control system, electronic entry control system, automated access control system (AACS), electronic access control system, physical access control system (PACS), and electronic entry control.

Anti-Passback: A functional characteristic employed within ACS that is used to eliminate/mitigate the risk of someone giving their credential (passing it back) to another person after that credential is used to access a secure area.

Authorization to Operate (ATO): The official management decision given by a senior organizational official to authorize operation of an information system and explicitly accept the risk to organizational operations (including mission, functions, image, or reputation), organizational assets, individuals, other organizations, and the nation, based on the implementation of an agreed-upon set of security controls.

Balanced Magnetic Switch (BMS): A legacy term superseded by the current term High Security Switch (HSS). A door position switch using a switch held in a balanced or center position by interacting magnetic fields when not in an alarm condition.

Base Level Information Infrastructure: Information technology (IT) infrastructure that exists on DoD proprietary or leased property.

Biometric (Bio): A measurable, physical characteristic or personal behavioral trait used to recognize the identity, or verify the claimed identity, of an applicant. Facial images, fingerprints, and iris image samples are all examples of biometrics.

Card Reader: A device that communicates with a card or "smart card" using either radio frequency (RF) signals via a circuit chip or wires via a contact pad on the card. The readers consist of various types, including contactless, contact, or a hybrid with the option of a combination of a keypad or biometric reader.

Central Monitoring Station: The space that serves as a central monitoring and assessment facility for the ACS, video, and IDS systems. The key components of a central monitoring station include consoles, monitors, and printers. Normally, the central monitoring station is staffed 24 hours a day, seven days a week by trained personnel. Other names for the central monitoring station include security operations center (SOC), security command center and security control center (SCC), dispatch center, data transmission center (DTC), and alarm control center (ACC).

Common Access Card (CAC): The CAC, a "smart" card about the size of a credit card, is the standard identification for active-duty military personnel, Selected Reserve, DoD civilian employees, and eligible contractor personnel. It is also the principal card used to enable physical access to buildings and controlled spaces, and provides access to defense computer networks and systems.

Controlled Access Area (CAA): The complete building or facility area under direct physical control within which unauthorized persons are denied unrestricted access and are either escorted by authorized personnel or are under continuous physical or electronic surveillance.

Data Transmission Media (DTM): The system that allows electronic security systems (ESS) data transmission and communication between system nodes and also back to the central monitoring station. The DTM is the security communications system and can consist of fiber, dedicated conductors, wireless networks, leased T-1 lines, or virtual private networks. DTM includes both base level information infrastructure (BLII: on-base) and defense information infrastructure (DII: inter-base).

Defense Information Infrastructure: The information technology (IT) infrastructure that is not on DoD proprietary or leased property and requires transmission of information across property boundary lines, for example, inter-base communications.

Electronic Security System (ESS): The integrated electronic system that encompasses interior and exterior intrusion detection systems (IDS), video surveillance systems (VSS) for assessment of alarm conditions, access control systems (ACS), data transmission media (DTM), and alarm reporting systems for monitoring, control, and display.

Electromagnetic Interference (EMI): A naturally occurring phenomenon when the electromagnetic field of one device disrupts, impedes, or degrades the electromagnetic field of another device by coming into proximity with it. With ESS, devices are susceptible to EMI because electromagnetic fields are a byproduct of passing electricity through a wire. Data lines that have not been properly shielded are susceptible to EMI. A good example of an ESS application is using shielded wiring from a field card reader back to the local ACS panel.

False Acceptance Rate (FAR): The rate or percentage at which a false credential is inaccurately accepted as being valid by an ACS. A sample FAR for a product could be 0.1%.

False Alarm: An alarm when there is no alarm stimulus.

False Rejection Rate (FRR): The rate or percentage at which an ACS product or system rejects an authorized credential holder.

Federal Information Processing Standard (FIPS): A standard for adoption and use by federal departments and agencies that has been developed within the Information Technology Laboratory and published by NIST, a part of the U.S. Department of Commerce. A FIPS covers some topic in information technology (IT) to achieve a common level of quality or some level of interoperability.

Field Distribution Box (FDB): A field distribution box (FDB) serves as a critical node for sensor processing, data transmission, and electrical power along a site perimeter.

Frame Rate Per Second (fps): When referring to a video image, this term refers to how often the visual still image is updated.

Identity Assurance: A category that conveys the degree of confidence that a person's claimed identity is their real identity, as defined in [NIST SP 800-63-3] in terms of three levels: IAL 1 (Some confidence), IAL 2 (High confidence), IAL 3 (Very high confidence).

Intrusion Detection System (IDS): A system consisting of interior and exterior sensors, surveillance devices, and associated communication subsystems that collectively detect an intrusion of a specified site, facility, or perimeter and annunciate an alarm.

Local Area Network (LAN): A geographically limited data communication system for a specific user group consisting of a group of interconnected computers sharing applications, data, and peripherals.

Nuisance Alarm: An alarm resulting from the detection of an appropriate alarm stimulus, or failure to use established entry control procedures, but which does not represent an attempt to intrude into the protected area. Examples of nuisance alarms would be improper opening of a monitored exit door or activation of an exterior intrusion detection system by a DoD maintenance crew. Animal activation of detection systems is a potential cause of nuisance alarms. Another example would be a wind-generated alarm of a fence monitoring system caused by flexing of the fence. Numerous nuisance alarms can cause complacency.

Premise Control Unit (PCU): A PCU is an electronic device that continuously monitors the alarm status of local intrusion detection sensors and duress devices and transmits alarm conditions to a remote monitoring station. The PCU allows authorized personnel to place the alarm zone in an "armed" or "disarmed" state via a local keypad, credential reader, or biometric device. The term "PCU" is generally synonymous with the terms "ACS local processor", "access control panel", "IDS local processor," and "intrusion panel."

Personal Identification Number (PIN): An identification string used as a password to authenticate identity and gain access to a location or computer resource. Although there are alphanumeric product options, most hardware entry devices make use of a numeric keypad. Many computer resource programs require an alphanumeric string.

Personal Identity Verification (PIV) Card: A physical artifact (for example, identity card, "smart" card) issued to an individual that contains a PIV card application that stores identity credentials (for example, photograph, cryptographic keys, digitized fingerprint representation) so the claimed identity of the cardholder can be verified against the stored credentials by another person (human readable and verifiable) or an automated process (computer readable and verifiable).

Physical Protection System/Physical Security System: A system to prevent unauthorized physical access to a protection and/or security system, such as fences, walls, locks, sensors, and surveillance.

Probability of Detection (Pd): A measure of an intrusion detection sensor's performance in detecting an intruder within its detection zone.

Proprietary Security Network: A completely self-contained dedicated local area network (LAN) with security system software installed and run on a host server (computer). Proprietary security networks are dedicated to the ESS, with no outside (Internet, LAN, or WAN) connections.

Regional Dispatch Center (RDC): A centralized security command center for multiple bases and facilities within a geographic region. This location is typically staffed 24 hours a day by staff trained to assess and initiate responses for ESS alarms. The RDC requires interface and communication systems to different bases and facilities. The RDC concept is a trend of economically consolidating different base ESS at one centralized location to save money and infrastructure rather than having different discrete base operations centers.

Security Equipment Integration Working Group (SEIWG): A working group responsible for a standard (SEIWG-012) pertaining to information encoded on an access control card. This standard is generally referred to as "SEIWG," although there are other SEIWG specifications as well.

Sensitive Compartmented Information (SCI): Classified information concerning or derived from intelligence sources, methods, or analytical processes that is required to be handled within formal access control systems established by the Director of National Intelligence.

Sensitive Compartmented Information Facility (SCIF): A facility capable of storing or processing Sensitive Compartmented Information (SCI) material. Requirements for these facilities are defined in *Technical Specifications for Construction and Management of Sensitive Compartmented Information Facilities, IC Tech Spec – for ICD/ICS 705*.

Special Access Program (SAP): A program established for a specific class of classified information that imposes safeguarding and access requirements that exceed those typically required for information at the same classification level.

Stand-alone System: A system that is not connected to any other network and does not transmit, receive, route, or exchange information outside of the system's authorization boundary.

Time Domain Reflectometry (TDR): Process of sending an electronic signal down a conductor or fiber (wiring or cabling) and measuring the time it takes for the signal or part of the signal to return to determine the location of a flaw or disturbance. The signal's reflection begins at the flaw or disturbance point.

Uninterruptible Power Supply (UPS): A power supply system that includes a rectifier, battery, and inverter to maintain power in the event of a power outage. UPS systems are specified by hours of operation to sustain power during an outage (six hours, ten

hours, or 24 hours). UPS systems can be standby power systems or on-line systems. Typically, a centralized UPS is not a mandated requirement for an ESS project.

Video Analytics: Software that allows the user to input a specific set of rules for each scene of interest, which, if violated, generate visual cues on the monitor, thus drawing the operator's attention to suspicious objects or behaviors.

Video Management System (VMS): Software that allows the user to conveniently view the video from all camera and recording devices.

Video Surveillance System (VSS): A system that is primarily used for surveillance, collection of forensic evidence, or assessment during alarm conditions. VSS was formerly known as closed circuit television (CCTV).

Wide Area Network (WAN): An internetwork that uses telecommunication links to connect geographically distant networks.

Workstation: A computer that allows a user to interact with an IDS, ACS, or VSS with the necessary software for each.

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APPENDIX A UFC 4-022-03 GLOSSARY

A-1 DEFINITION OF TERMS

Active Sensors: A sensor that can receive signals from either itself or other devices.

Attack Side: The outer side of the perimeter fence for a protected area. The side from which to expect attacks or forced entry.

Bollards: Reinforced concrete shapes or sleeves, or steel pipe filled with concrete, used to slow down vehicles and/or deny vehicle access.

Clear Zone: Area free of obstacles, topographical features and vegetation which reduce the effectiveness of the physical barrier, impede observation or provide cover and concealment of an intruder.

Color Polymer Coating: A coating on chain link fences to further protect and add color.

Concrete Sill or Curb: A concrete section to create a non-passable connection between a fence and the existing ground.

Corrosive or Salt Laden Environment/Atmosphere: Atmosphere which contains enough salt or corrosive elements to corrode a fence enough to decrease the intended strength. These areas are typically bordering oceans or other water bodies. For the application of coating requirements in this UFC, the corrosive atmosphere is considered 10 miles (16 kilometers) from coastal mean high water line.

Deadman Anchor : An anchor for fence or gate reinforcement cables which transfer the force from the fencing or gate fabric/surface to the ground. Deadman anchors are typically concrete blocks buried in the ground.

Electronic Security System (ESS): The integrated electronic system that encompasses interior and exterior Intrusion Detection Systems (IDS), Closed Circuit Television (CCTV) systems for assessment of alarm conditions, Automated Access Control Systems (ACS), Data Transmission Media (DTM) and alarm reporting systems for monitoring, control and display.

Eye Ends: Center space created when looping a wire for termination or reinforcement.

Fabric Diamonds: The opening formed by the woven wires in chain link fence fabric.

Fabric Tension Bar: A bar that the fencing fabric is attached to and it serves to hold the fabric tight to decrease ability to deform.

Fencing Accessories: Any fasteners, ties, wires or other objects used to attaching the fencing fabric, posts or top guards.

Fence Fabric: As used in this document means (chain link, ornamental, welded wire mesh, expanded metal).

Gate Leaf: A panel of a gate (There is one gate leaf for single fences and two for double fences).

Gates Openings: In the barrier system to allow authorized entry and exit.

Intrusion Detection System (IDS): A system consisting of interior and exterior sensors, surveillance devices, and associated communication subsystems that collectively detect an intrusion of a specified site, facility or perimeter and annunciate an alarm.

Line Posts: intermediate posts spaced a maximum of 10 feet (3 m) apart and considered the backbone of the fence line.

Mesh Openings: The opening formed by the woven wires.

Moving Vehicle Bomb Tactic: A forced entry tactic to place an explosive or other damaging terrorist device in a moving vehicle to create a mobile vehicular weapon.

Natural Boundaries: Natural formations such as bodies of water, rough terrain, or densely wooded areas that may act as a barrier to define and protect the restricted area's perimeter.

Outriggers: Metal top guards which are placed on the top of the barrier and may consist of at least one vertical arm, angled arm facing inside, angles arm facing outside or two angles arms facing both inside and outside. In addition, there may be more than one arm on each side if desired. The number and direction of arms depends on the nature of the barrier.

Passive Barriers: A passive barrier has no moving parts. Passive barrier effectiveness relies on its ability to absorb energy and transmit the energy to its foundation. Highway medians (Jersey), bollards or posts, tires, guardrails, ditches and reinforced fences are examples of passive barriers.

Passive Sensors: A sensor that receives signals through light or radio waves.

Protected Side: The inside of the fence that is being protected through various security measures.

Sally Ports: Sally ports are used to control entry into highly protected and restricted areas and used as part of an entry control facility/access control point for vehicle inspection.

Secure Side: The inside of the fence that is being protected through various security measures.

Special Nuclear Material (SNM): Nuclear material that requires extensive protection.

Swaged Loop: A loop created by looping a wire and fastening the loop with a swaged on clamp.

Swing Gates: Gates for entrance or exit that swing open either manually or automatically when controlled either electronically or manually.

Taut Wire: Wire that is continuously under tension through a series of spring connections.

Tension Bar: The bar that is threaded through the last vertical link of fabric. It attaches the fabric to the terminal post.

Top Guard: Additional protection of fences or other barriers that are placed on top of the barrier to prevent climbing or jumping. Possible top guards include, but are not limited to, outriggers with barbed wire, concertina wire, or barbed tape.

Truss Rods: A rod used in brace assemblies to draw and hold the line post firmly to the brace rail. The truss rod uses an adjustable turnbuckle to maintain proper tension. Truss rod details can be seen in the "Non Reinforced Chain-Link Fence" drawings in Appendix C.

Turnbuckles: A metal coupling device consisting of an oblong piece internally threaded at both ends into which the corresponding sections of two threaded rods are screwed in order to form a unit that can be adjusted for tension or length.

Turnstile Gates: Pedestrian gates which rotate around a central pole to allow only one entrance at a time.

Welded Bar Grill: A series of vertical and horizontal bars that are placed over an opening and welded at all bar crossing points.

Additional Terms relating to fences are found in:

- ASTM F552 - Standard Terminology Relating to Chain Link Fencing
- ASTM F2814 - Standard Guide for Design and Construction of Ornamental Steel Picket Fence Systems for Security Purposes
- ASTM F2780 - Standard Guide for Design and Construction of Expanded Metal Security Fences and Barriers
- ASTM F1379 - Standard Terminology Relating to Barbed Tape
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APPENDIX B UFC 4-022-03 REFERENCES

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ASTM A121, Standard Specification for Metallic- Coated Carbon Steel Barbed Wire

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ASTM A500/A500M, Standard Specification for Cold-Formed Welded and Seamless Carbon Steel Structural Tubing in Rounds and Shapes

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ASTM F1911, Standard Practice for Installation of Barbed Tape

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ASTM F2408, Standard Specification for Ornamental Fences Employing Galvanized Steel Tubular Pickets

ASTM F2453/F2453M, Standard Specification for Welded Wire Mesh Fence Fabric (Metallic-Coated or Polymer Coated) for Meshes of 6 in2 [3871mm2] or Less, in Panels or Rolls, with Uniform Meshes

ASTM F2548, Standard Specification for Expanded Metal Fence Systems for Security Purposes

ASTM F2611, Standard Guide for Design and Construction of Chain Link Security Fencing

ASTM F2780, Standard Guide for Design and Construction of Expanded Metal Security Fences and Barriers

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APPENDIX A UFC 4-025-01 GLOSSARY

A-1 ACRONYMS.

ACP	Access Control Point
AFCEC	Air Force Civil Engineer Center
AT	Antiterrorism
ATON	Aids to Navigation
BIA	Bilateral Infrastructure Agreement
COCOM	Combatant Command
DBT	Design Basis Threat
EHSS	Electronic Harbor Security System
ESS	Electronic Security System
ESSC	Electronic Security System Console
FP	Force Protection
FPCON	Force Protection Condition
GCC	Geographic Combatant Commander
HQUSACE	Headquarters, U.S. Army Corps of Engineers
HNFA	Host Nation Funded Construction Agreements
IDS	Intrusion Detection System
LOD	Line of Demarcation
LOP	Level of Protection
NAVFAC	Naval Facilities Engineering Systems Command
RAM	Random Antiterrorism Measures
SID	Security in Depth
SOFA	Status of Forces Agreements
UFC	Unified Facilities Criteria
UPS	Uninterruptable Power Systems

A-2 DEFINITION OF TERMS.

Controlled Area: A space extending upward and outward from a specified point. This area is typically designated by a commander or director, wherein sensitive information or operations occur and requires limitations of access.

Design basis threat (DBT): The threat against which an asset must be protected and upon which the protective system's design is based. It is the baseline type and size of threat that buildings or other structures are designed to withstand. The DBT includes the tactics aggressors will use against the asset and the tools, weapons, and explosives employed in these tactics.

Dispatch Center: The space that serves as a central monitoring and assessment facility for the ACS, Video, and IDS systems. The key components of a Dispatch Center include consoles, monitors, and printers. Normally, the Dispatch Center is staffed 24 hours a day, seven days a week by trained personnel. Other names for the Dispatch Center include Regional Dispatch Center (RDC), Security Operations Center (SOC), Security Command Center and Security Control Center (SCC), Central Monitoring Station, Data Transmission Center (DTC), and Alarm Control Center (ACC).

Duress Alarm: A silent alarm signal generated by the manual activation of a device requiring a security force response.

Electronic Harbor Security System (EHSS): An electronic system that integrates surface detection, subsurface detection, and video systems to detect, assess, track, and archive waterside threats.

Electronic Security System (ESS): The integrated electronic system that encompasses interior and exterior Intrusion Detection Systems (IDS), video systems for assessment of alarm conditions, Automated Access Control Systems (ACS), Data Transmission Media (DTM), and alarm reporting systems for monitoring, control, and display.

Electronic Security System Console (ESSC): While not always specifically referred to as the ESSC, most security systems have a console that houses monitoring and server interface equipment. Generally, this console is located in the Dispatch Center.

Foot of pier: The landside end of the pier

Force Protection (FP): Actions taken to prevent or mitigate hostile actions against DoD personnel (including family members), resources, facilities, and critical information.

Force Protection Condition (FPCON): A DoD-approved system standardizing the Department's identification, recommended preventive actions, and responses to terrorist threats against U.S. personnel and facilities.

Head of pier: The waterside end of the pier sometimes referred to as the pier head.

Installations: Real DoD properties including bases, stations, forts (including National Guard and Federal Reserve Centers), depots, arsenals, plants (both contractor and Government operated), hospitals, terminals, and other special mission facilities, used primarily for military purposes.

Level of protection: The degree to which an asset is protected against compromise.

Physical Security: That part of security concerned with physical measures designed to safeguard personnel; to prevent unauthorized access to equipment, installations, material, and documents; and to safeguard them against espionage, sabotage, damage, and theft.

Ports of Call: A port not located on a DoD installation where ships dock in the course of deployment to load or unload cargo, obtain supplies, liberty, or undergo repairs.

Restricted Area: An area under military jurisdiction in which special security measures are employed to prevent unauthorized entry.

Waterside Security: Measures or actions taken to prevent or guard against the use of a waterside approach to a waterfront facility or vessel by persons or vessels intent on theft, sabotage, terrorism, and/or belligerent acts.

APPENDIX B UFC 4-025-01 REFERENCES

UNITED STATES ACCESS BOARD

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DEPARTMENT OF DEFENSE

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APPENDIX A UFC 4-860-01 GLOSSARY

A-1 ACRONYMS.

AASHTO	American Association of State Highway and Transportation Officials
AHJ	the Authority Having Jurisdiction
ASTM	American Society of Testing and Materials
AREMA	American Railway Engineering and Maintenance-of-Way Association
AWPA	American Wood Preservers Association
AFCEC	Air Force Civil Engineer Center
BIA	Bilateral Infrastructure Agreement
CAPY	capacity
CCR	criteria change request
CONUS	The Continental U.S.
CWR	continuous welded rail
DoD	Department of Defense
ESQD	Explosive Safety Quantity Distances
FEMA	Federal Emergency Management Agency
HQUSACE	Headquarters, U.S. Army Corps of Engineers
HNFA	Host Nation Funded Construction Agreements
ITO	Installation Transportation Officer
MUTCD	Manual of Uniform Traffic Control
NAVFAC	Naval Facilities Engineering Systems Command
OTM	other track materials
OCONUS	Outside the Continental U.S.
POL	petroleum, oil and lubricants
Q-D	quantity-distance
RTCH	Rough Terrain Container Handler

SOFA	Status of Forces Agreements
TI	technical instructions
TM	technical manual
UFC	Unified Facilities Criteria
U.S.	United States
USACE	U.S. Army Corps of Engineers

APPENDIX B UFC 4-860-01 REFERENCES

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